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Effect of Tanezumab on Joint Pain, Physical Function, and Patient Global Assessment of Osteoarthritis Among Patients With Osteoarthritis of the Hip or Knee

A Randomized Clinical Trial

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Visual Abstract

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IMPORTANCE Patients with osteoarthritis (OA) may remain symptomatic with traditional OA treatments.

OBJECTIVE To assess 2 subcutaneous tanezumab dosing regimens for OA.

DESIGN, SETTING, AND PARTICIPANTS A randomized, double-blind, multicenter trial from January 2016 to May 14, 2018 (last patient visit). Patients enrolled were 18 years or older with hip or knee OA, inadequate response to OA analgesics, and no radiographic evidence of prespecified joint safety conditions.

INTERVENTIONS Patients received by subcutaneous administration either tanezumab, 2.5 mg, at day 1 and week 8 (n = 231); tanezumab, 2.5 mg at day 1 and 5 mg at week 8 (ie, tanezumab, 2.5/5 mg; n = 233); or placebo at day 1 and week 8 (n = 232).

MAIN OUTCOMES AND MEASURES Co-primary end points were change from baseline to week 16 in Western Ontario and McMasters Universities Osteoarthritis Index (WOMAC) Pain (0-10, no to extreme pain), WOMAC Physical Function (0-10, no to extreme difficulty), and patient global assessment of osteoarthritis (PGA-OA) (1-5, very good to very poor) scores.

RESULTS Among 698 patients randomized, 696 received 1 or more treatment doses (mean [SD] age, 60.8 [9.6] years; 65.1% women), and 582 (83.6%) completed the trial. From baseline to 16 weeks, mean WOMAC Pain scores decreased from 7.1 to 3.6 in the tanezumab, 2.5 mg, group; 7.3 to 3.6 in the tanezumab, 2.5/5 mg, group; and 7.3 to 4.4 in the placebo group (least squares mean differences [95% CI] vs placebo were −0.60 [−1.07 to −0.13; P = .01] for tanezumab, 2.5 mg, and −0.73 [−1.20 to −0.26; P = .002] for tanezumab, 2.5/5 mg). Mean WOMAC Physical Function scores decreased from 7.2 to 3.7 in the 2.5-mg group, 7.4 to 3.6 in the 2.5/5-mg group, and 7.4 to 4.5 with placebo (differences vs placebo, −0.66 [−1.14 to −0.19; P = .007] for tanezumab, 2.5 mg, and −0.89 [−1.37 to −0.42; P < .001] for tanezumab, 2.5/5 mg). Mean PGA-OA scores decreased from 3.4 to 2.4 in the 2.5-mg group, 3.5 to 2.4 in the 2.5/5-mg group, and 3.5 to 2.7 with placebo (differences vs placebo, −0.22 [−0.39 to −0.05; P = .01] for tanezumab, 2.5 mg, and −0.25 [−0.41 to −0.08; P = .004] for tanezumab, 2.5/5 mg). Rapidly progressive OA occurred only in tanezumab-treated patients (2.5 mg: n = 5, 2.2%; 2.5/5 mg: n = 1, 0.4%). The incidence of total joint replacements was 8 (3.5%), 16 (6.9%), and 4 (1.7%) in the tanezumab, 2.5 mg; tanezumab, 2.5/5 mg; and placebo groups, respectively.

CONCLUSIONS AND RELEVANCE Among patients with moderate to severe OA of the knee or hip and inadequate response to standard analgesics, tanezumab, compared with placebo, resulted in statistically significant improvements in scores assessing pain and physical function, and in PGA-OA, although the improvements were modest and tanezumab-treated patients had more joint safety events and total joint replacements. Further research is needed to determine the clinical importance of these efficacy and adverse event findings.

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Nerve growth factor (NGF) is a neurotrophin involved in pain signaling and nociceptor gene expression.[1-3] NGF has been shown to contribute to the clinical symptom of pain hypersensitivity, which is often seen during inflammation and chronic pain conditions.[3-6] NGF is expressed in the subchondral bone of patients with osteoarthritis (OA),[7] consistent with a role in symptomatic OA pain. Additionally, inhibiting NGF blocks increased pain-related behaviors in several animal models of pain, including an arthritis pain model.[4-6,8]

Tanezumab is an IgG2Δa monoclonal antibody that inhibits NGF binding to its receptors, and is under investigation for the treatment of chronic pain conditions, such as OA and chronic low back pain.[9-13] The US Food and Drug Administration placed a partial clinical hold on NGF antibodies in 2010 due to joint safety findings in humans. Adverse changes observed in the sympathetic nervous system of mature animals led to a second hold in 2012. The holds were lifted after detailed investigation of the events (joints) and additional nonclinical studies (neurological). Because some patients in previous tanezumab studies reported transient abnormalities in cutaneous sensation (commonly paresthesia or hypoesthesia), subsequent studies implemented an overall risk minimization strategy and comprehensive assessment of joint and neurological adverse events (AEs).[14,15] These strategies included prohibiting chronic concomitant nonsteroidal anti-inflammatory drug (NSAID) use, prohibiting NGF antibody doses that have no demonstrated benefit over lower doses for the particular condition studied, excluding patients with evidence of or risk factors for rapidly progressive OA, and excluding patients who are not suitable candidates for total joint replacement.

This study assessed the efficacy and AEs associated with tanezumab administered subcutaneously as a fixed dosing regimen or a forced titration dosing regimen in patients with moderate to severe OA who had not responded to or were unable to receive standard pharmacological OA pain treatments.

Key Points

Question Among patients with moderate to severe osteoarthritis of the knee or hip and inadequate treatment response to standard analgesics, what is the effect of subcutaneous tanezumab on joint pain, physical function, and patient global assessment of osteoarthritis?

Findings In this randomized clinical trial that enrolled 698 patients, subcutaneous tanezumab administered with fixed doses at 8-week intervals or with a forced titration at week 8, compared with placebo, resulted in statistically significant improvements in joint pain, physical function, and patient global assessment of osteoarthritis over 16 weeks, although the improvements were modest and tanezumab-treated patients had more joint safety events and total joint replacements.

Meaning Further research is needed to determine the clinical importance of these efficacy and adverse event findings with regard to use of tanezumab for treatment of osteoarthritis.

Methods

Study Design

This was a randomized, double-blind, placebo-controlled, multicenter, parallel-group, dose-titration study (16-week treatment period, 24-week follow-up period) assessing the efficacy and AE profile of subcutaneous tanezumab in patients with moderate to severe hip or knee OA conducted at 89 clinical research sites in the United States, Canada, and Puerto Rico from January 2016 to May 2018. The study was approved by the institutional review board or independent ethics committee at each study center. All patients provided written informed consent before participating. The study was conducted in compliance with the Declaration of Helsinki and all International Conference on Harmonisation Good Clinical Practice guidelines.[16,17] The study protocol and statistical analysis plan can be found in Supplement 1 and Supplement 2, respectively.

The primary objective was to evaluate the efficacy of 2 subcutaneous tanezumab treatment regimens at week 16—fixed dosing (2.5 mg administered at baseline and week 8) and forced dose titration (2.5 mg administered at baseline and 5 mg at week 8)—compared with placebo treatment. The secondary objectives evaluated (1) the efficacy of tanezumab titrated from 2.5 mg to 5 mg at week 8 compared with 2 administrations of tanezumab, 2.5 mg, 8 weeks apart and (2) the AE profile of both tanezumab dosing regimens.

The study was divided into 3 periods: screening (≤37 days), treatment (16 weeks), and follow-up (24 weeks). Screening procedures included a washout period of prohibited medications and an initial pain assessment period (3-7 days prior to randomization/baseline).

Study Population

Patients were aged 18 years and older and diagnosed as having hip or knee OA according to American College of Rheumatology criteria with radiographic confirmation at screening (Kellgren-Lawrence grade ≥2). Entry criteria included an index joint Western Ontario and McMasters Universities Osteoarthritis Index (WOMAC) Pain subscale score of 5 or greater (on an 11-point numerical rating scale from 0 = no pain to 10 = extreme pain) at both screening and baseline, a baseline WOMAC Physical Function subscale score of 5 or greater (on an 11-point numerical rating scale from 0 = no difficulty to 10 = extreme difficulty), and a baseline patient global assessment of osteoarthritis (PGA-OA) rating of fair, poor, or very poor (on a scale from 1 = very good to 5 = very poor). Index joint was defined as the most painful hip or knee at screening. Patients had a documented history of (1) insufficient pain relief from acetaminophen; (2) insufficient pain relief, intolerance to, or contraindication to NSAIDs; and (3) insufficient pain relief, intolerance to, or contraindication to either tramadol or opioids (or were unwilling to take opioids). To characterize the study population, race (4 fixed categories with an open-ended option for a response of "other") and ethnicity (2 fixed categories) data were provided to the study site staff by patients.

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Main exclusion criteria included evidence of prespecified joint safety conditions (eg, rapidly progressive OA, subchondral insufficiency fracture, osteonecrosis, pathologic fracture) in any major joint on screening radiographs as determined by the central reader; a history of diseases that could confound index joint efficacy assessments (eg, rheumatoid arthritis, seronegative spondyloarthropathies, gout, chondrocalcinosis/pseudogout); significant trauma or surgery in a hip, knee, or shoulder in the previous year; any planned surgery during the study; any recent intra-articular corticosteroid or hyaluronic acid injection in the index joint; a history of neurological conditions (eg, peripheral or autonomic neuropathy, Alzheimer disease, multiple sclerosis); a Survey of Autonomic Symptoms[18] score greater than 7 at screening; and pregnancy or breastfeeding.

Intervention

Using a computer-generated randomization code, patients were randomized in equal allocation to 1 of 3 subcutaneous treatment regimens: tanezumab, 2.5 mg, at baseline and week 8 (ie, tanezumab, 2.5 mg); tanezumab, 2.5 mg at baseline and 5 mg at week 8 (ie, tanezumab, 2.5/5 mg); and placebo at baseline and week 8 (ie, placebo) (eFigure in Supplement 3). Randomization was stratified by index joint and highest Kellgren-Lawrence grade. Placebo (identical in composition [color, volume, pH] to the drug product, but without the active pharmaceutical ingredient) was administered in a manner matching subcutaneous tanezumab.

Permitted concomitant treatments included aspirin less than or equal to 325 mg/d for cardiovascular prophylaxis and stable doses of medications for other non-OA, nonpain conditions. Analgesics were prohibited except as follows: NSAIDs for non-OA conditions were permitted for up to 10 days per 8-week period between baseline and week 24, but not within 48 hours of a study visit. Rescue medication (acetaminophen) was allowed up to 3000 mg/d and for 3 or fewer days per week during the treatment period, but not within 24 hours of a study visit, and usage was recorded by the patient via a handheld electronic diary (eDiary; Trial Collector version 3.2.0.1; CRF Health). Standard of care treatment for OA pain was permitted 16 weeks after the last study drug dose.

Efficacy Assessments

The co-primary efficacy end points were change from baseline to week 16 in WOMAC Pain subscale, WOMAC Physical Function subscale, and PGA-OA scores.[19,20] WOMAC Pain and WOMAC Physical Function subscales measured symptoms within the last 48 hours using an 11-point numeric rating scale (0 = no pain/no difficulty to 10 = extreme pain/extreme difficulty). For PGA-OA, patients answered the question, "Considering all the ways your osteoarthritis in your hip/knee affects you, how are you doing today?" on a scale from 1 = very good to 5 = very poor.

A key secondary efficacy end point was the WOMAC Pain responder rate of 50% or greater at week 16, defined as the proportion of patients with a 50% or greater reduction from baseline in WOMAC Pain score at week 16. Other secondary efficacy end points were WOMAC Pain responder rates (≥30%, ≥70%, and ≥90% reductions from baseline) at week 16 and patient-reported rescue medication use during weeks 2, 4, 8, 12, and 16. All prespecified secondary end points can be found in eTable 1 in Supplement 3.

A recent systematic review of minimal clinically important difference values for improvement in OA by the OMERACT Rasch group reported results that ranged from −29.9 to −7.5 for WOMAC Pain and from −33.5 to −5.3 for WOMAC function, both reported on a scale of 0 to 100.[21] The minimal clinically important difference for improvement in PGA has been reported as 0.4 on a Likert scale of 0 to 4 and −15.2 (hip) and −18.3 (knee) on a 0 to 100 visual analog scale.[22,23]

Adverse Event Assessments

AE assessments included AE reporting; laboratory testing; 12-lead electrocardiogram; sitting vital signs; orthostatic blood pressure assessments; physical examinations; musculoskeletal examinations; neurological examinations reported using the Neuropathy Impairment Score[24]; Survey of Autonomic Symptoms scores[18]; adjudication of joint safety events including total joint replacements; and antidrug antibody assessments. AEs were coded using the Medical Dictionary for Regulatory Activities version 21.0 (International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use). AE severity and relationship to study treatment were assessed by the investigator.

Neurological Assessments

Patients meeting protocol-specified criteria were further evaluated for peripheral neuropathy and/or sympathetic autonomic neuropathy. Neurological examinations were conducted at all clinic visits by investigators or designated physicians who received protocol-required training, and findings were recorded using the Neuropathy Impairment Score.[24] Patients were referred to a consulting neurologist if an AE of peripheral neuropathy or abnormal peripheral sensation was reported as (1) serious, (2) of severe intensity, (3) resulted in study withdrawal, or (4) was ongoing at the end of study participation. Patients with reported AEs suggestive of sympathetic autonomic neuropathy (ie, bradycardia, orthostatic hypotension, syncope, anhidrosis, or hypohidrosis) of any seriousness or severity were further evaluated by a cardiologist or neurologist.

Joint Safety Events

Investigators performed musculoskeletal examinations of all major joints at each study visit. Radiographs of both knees, hips, and shoulders, as well as other major joints exhibiting signs or symptoms of OA, were obtained at screening and week 40 (or early discontinuation). Images were evaluated by a central reader for the presence of exclusionary conditions. Patients recorded daily average index joint pain and weekly average nonindex joint(s) pain via eDiary during the treatment period using an 11-point numeric rating scale (0 to 10, indicating no to worst possible pain). Weekly average pain was recorded for all joints during the safety follow-up period.

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Investigators evaluated patients reporting increased severe (score of 7-10 out of 10) persistent pain via eDiary lasting 2 or more weeks to determine whether additional follow-up was needed. Postbaseline radiographs (scheduled or for cause) were assessed by the central reader for possible or probable rapidly progressive OA, subchondral insufficiency fracture, primary osteonecrosis, or pathologic fracture. If warranted, magnetic resonance imaging scans were performed and/or patients were referred to an orthopedic surgeon. All cases of possible or probable joint safety events or cases of total joint replacements for any reason were adjudicated by an external, blinded adjudication committee consisting of orthopedic surgery, rheumatology, orthopedic pathology, and musculoskeletal radiology experts.

Statistics

A sample size of approximately 230 patients per treatment group was determined to provide 90% power to achieve statistical significance at the 5% 2-sided level for comparisons of tanezumab, 2.5 mg, and tanezumab, 2.5/5 mg, vs placebo across all 3 co-primary end points. This was based on estimates from a combined analysis of 2 previous studies (treatment difference of −1.0 for WOMAC Pain and Physical Function subscales and −0.32 for PGA-OA, and SDs of 2.73, 2.58, and 0.92, respectively).[12,13] Co-primary end points were analyzed using an analysis of covariance model, with terms for baseline score, baseline patient diary mean pain, index joint, Kellgren-Lawrence grade, and treatment group, and study site as a random effect. The efficacy analysis set was analyzed by randomization group and included all patients who received 1 or more study medication doses. Missing data were handled with a multiple imputation strategy dependent on the reason for discontinuation. Missing data at week 16 due to discontinuation for lack of efficacy, AE, or death were based on sampling from a normal distribution using a mean value equal to the patient's baseline efficacy value; missing data at week 16 due to other reasons were based on sampling from a normal distribution using a mean value of the patient's last observed efficacy value. Final results were calculated using the combined sets of results from each imputation data set analysis. All statistical testing was 2-sided.

The co-primary end points used a stepdown strategy designed to maintain type I error at 5% or less for multiple comparisons, defined as first testing tanezumab, 2.5/5 mg, vs placebo and, if statistically significant at the 5% level, testing tanezumab, 2.5 mg, vs placebo at the 5% level. Tanezumab treatment groups were considered more effective than placebo if all 3 co-primary end points were statistically significant. The key secondary efficacy end point was tested using the Hochberg procedure for both tanezumab regimens vs placebo, contingent on successful primary comparisons. Comparisons for other secondary end points were unadjusted. Comparisons between the tanezumab dose regimens were descriptive only. A planned exploratory analysis assessed WOMAC Pain response (percentage of patients with a reduction from baseline of ≥15%, ≥30%, and ≥50%) at week 16 according to response at week 8, using a logistic regression model including baseline WOMAC Pain subscale, baseline diary average pain and classification variables of index joint, highest Kellgren-Lawrence grade, and treatment. AEs were summarized by treatment group as percentages of the treatment group population. Because of the potential for type I error due to multiple comparisons, findings for analyses of secondary end points should be interpreted as exploratory. Statistical software used was SAS version 9.4 (SAS Institute).

Results

Patients

Randomization included 698 patients, and 696 received 1 or more study treatment doses (2 randomized patients who did not meet eligibility criteria were discontinued prior to dosing; **Figure 1**). All patients who received 1 or more study treatment doses were analyzed for efficacy and AEs. Eighty-three percent of patients had observations at all planned time points between baseline and week 16. An additional 6% of patients had observations at least at baseline and week 16. Patient baseline characteristics were similar across treatment groups (**Table 1**).

Efficacy

Baseline mean (SD) WOMAC Pain scores were 7.1 (1.2), 7.3 (1.3), and 7.3 (1.2) for the tanezumab, 2.5 mg; tanezumab, 2.5/5 mg; and placebo groups, respectively. Baseline mean (SD) WOMAC Physical Function scores were 7.2 (1.1), 7.4 (1.2), and 7.4 (1.1) for the tanezumab, 2.5 mg; tanezumab, 2.5/5 mg; and placebo groups, respectively. Baseline mean (SD) PGA-OA scores were 3.4 (0.6), 3.5 (0.6), and 3.5 (0.6) in the tanezumab, 2.5 mg; tanezumab, 2.5/5 mg; and placebo groups, respectively. Both tanezumab 2.5 mg and 2.5/5 mg demonstrated statistically significant improvement in WOMAC Pain, WOMAC Physical Function, and PGA-OA scores compared with placebo at week 16 (**Table 2** and **Figure 2**); thus, both tanezumab dosing regimens met the co-primary end points.

At 16 weeks, mean (SD) WOMAC Pain scores were 3.6 (2.6) in the tanezumab, 2.5 mg, group; 3.6 (2.7) in the tanezumab, 2.5/5 mg, group; and 4.4 (2.7) in the placebo group. Mean (SD) WOMAC Physical Function scores were 3.7 (2.7) in the 2.5-mg group, 3.6 (2.7) in the 2.5/5-mg group, and 4.5 (2.7) with placebo. Mean (SD) PGA-OA scores were 2.4 (0.9) in the 2.5-mg group, 2.4 (1.0) in the 2.5/5-mg group, and 2.7 (0.9) with placebo. Least squares mean differences vs placebo were as follows: WOMAC Pain scores were −0.60 (95% CI, −1.07 to −0.13; $P = .01$) for tanezumab, 2.5 mg, and −0.73 (95% CI, −1.20 to −0.26; $P = .002$) for tanezumab, 2.5/5 mg; WOMAC Physical Function scores were -0.66 (95% CI, −1.14 to −0.19; $P = .007$) for tanezumab, 2.5 mg, and −0.89 (95% CI, −1.37 to −0.42; $P < .001$) for tanezumab, 2.5/5 mg; and PGA-OA scores were -0.22 (95% CI, −0.39 to −0.05; $P = .01$) for tanezumab, 2.5 mg, and −0.25 (95% CI, −0.41 to −0.08; $P = .004$) for tanezumab, 2.5/5 mg.

At week 16, a greater proportion of patients in each tanezumab regimen (54.5% and 57.1% in the tanezumab,

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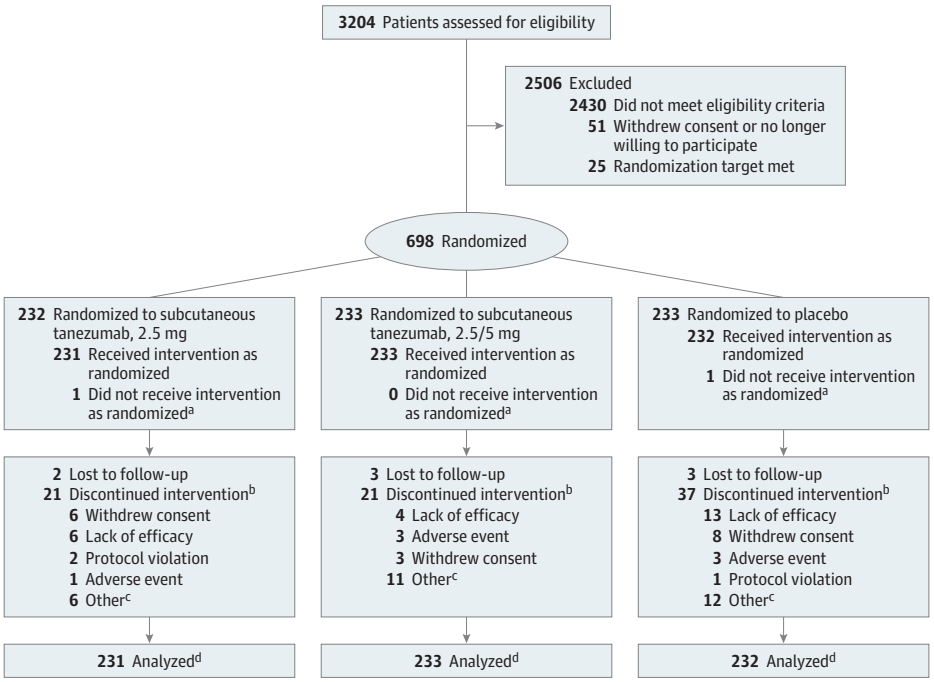
Figure 1. CONSORT Diagram



[a] Two randomized patients did not meet eligibility criteria and therefore did not receive the study drug.

[b] During the 16-week treatment period.

[c] Other reasons for treatment discontinuation were typically personal such as patient moving, loss of transportation to visits, and family illness.

[d] Efficacy and safety analysis sets were identical and included all patients who received at least 1 dose of study treatment.

2.5 mg, and tanezumab, 2.5/5 mg, groups, respectively) reported 50% or more reduction from baseline in WOMAC Pain subscale score compared with placebo (37.9%, $P = .001$ for tanezumab, 2.5 mg, vs placebo; $P < .001$ for tanezumab, 2.5/5 mg, vs placebo; **Table 3**). A greater proportion of patients in both tanezumab groups also reported 30% or more and 70% or more (but not ≥90%) reductions from baseline in WOMAC Pain subscale scores compared with placebo at week 16 (Table 3). Both tanezumab dosing regimens demonstrated statistically significant differences in WOMAC Pain and WOMAC Physical Function at first assessment (week 2) vs placebo (WOMAC Pain mean changes from baseline: -2.87 [95% CI, -3.28 to -2.46], -2.89 [95% CI, -3.29 to -2.48], and -2.20 [95% CI, -2.62 to -1.79] in the tanezumab, 2.5 mg; tanezumab, 2.5/5 mg; and placebo groups, respectively, P. = 002 for tanezumab, 2.5 mg, vs placebo, $P = .001$ for tanezumab, 2.5/5 mg, vs placebo; WOMAC Physical Function mean changes: -2.89 [95% CI, -3.30 to -2.48], -3.05 [95% CI, -3.46 to -2.65], and -2.14 [95% CI, -2.55 to -1.72] in the tanezumab, 2.5 mg; tanezumab, 2.5/5 mg; and placebo groups, respectively, $P < .001$ for each tanezumab group vs placebo).

In a planned exploratory analysis, most patients who did not achieve 15% or greater, 30% or greater, or 50% or greater reduction from baseline at week 8 also did not respond at week 16 (eTable 2 in Supplement 3). However, of patients who did not achieve a 50% or greater reduction from baseline in WOMAC Pain at week 8, a higher proportion (33%) experienced 50% or greater improvement relative to baseline at week 16 after receiving tanezumab, 5 mg, at week 8 compared with those receiving another 2.5-mg dose (22%) or those treated with placebo (19%).

The proportion of patients who took rescue medication was not significantly different between the 2 tanezumab treatment groups and placebo, except at week 2, in which 160 placebo-treated patients (69.0%) reported taking rescue medication compared with 138 patients (59.2%) in the tanezumab, 2.5/5 mg, group ($P = .03$) and at week 4, in which 143 placebo-treated patients (61.6%) reported taking rescue medication vs 119 patients (51.5%) in the tanezumab, 2.5 mg, group and 110 patients (47.2%) in the tanezumab, 2.5/5 mg, group ($P = .03$ for tanezumab, 2.5 mg, vs placebo; $P = .001$ for tanezumab, 2.5/5 mg, vs placebo). The total amount of rescue medication used was not significantly different in the tanezumab groups vs placebo at all weeks tested (eTable 3 in Supplement 3).

Adverse Events

During the treatment period, 55.4%, 46.8%, and 49.6% of patients experienced a treatment-emergent AE in the tanezumab, 2.5 mg; tanezumab, 2.5/5 mg; and placebo groups, respectively (**Table 4**). A total of 0.9%, 2.6%, and 2.2% of patients in the tanezumab, 2.5 mg; tanezumab, 2.5/5 mg; and placebo groups, respectively, had severe AEs. Nasopharyngitis, pain in extremity, and paresthesia occurred in 3% or more of patients in any treatment group and more frequently in tanezumab-treated patients compared with placebo during the treatment period. Seven patients were discontinued due to AEs. One death due to non-small-cell lung cancer stage IV and 1 due to suicide were reported during the follow-up

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Table 1. Patient Demographic Characteristics and Baseline Measurements

Characteristic	No. (%)		
	Tanezumab		
	2.5 mg (n = 231)	2.5/5 mg (n = 233)	Placebo (n = 232)
Sex			
Men	86 (37.2)	82 (35.2)	75 (32.3)
Women	145 (62.8)	151 (64.8)	157 (67.7)
Age, mean (range), y	60.9 (27-84)	61.2 (32-83)	60.4 (31-85)
Race			
White	178 (77.1)	170 (73.0)	156 (67.2)
Black or African American	43 (18.6)	50 (21.5)	60 (25.9)
Asian	5 (2.2)	8 (3.4)	13 (5.6)
Other[a]	5 (2.2)	5 (2.1)	3 (1.3)
Ethnicity			
Not Hispanic or Latino	188 (81.4)	193 (82.8)	196 (84.5)
Hispanic or Latino	43 (18.6)	40 (17.2)	36 (15.5)
History of inadequate pain relief from or intolerance to classes of pain medication			
Acetaminophen/paracetamol	230 (99.6)	232 (99.6)	232 (100)
Inadequate pain relief	230 (99.6)	231 (99.1)	232 (100)
Intolerability	0	1 (0.4)	0
Oral nonsteroidal anti-inflammatory drugs	230 (99.6)	233 (100)	232 (100)
Inadequate pain relief	209 (90.5)	211 (90.6)	211 (90.9)
Intolerability	16 (6.9)	22 (9.4)	23 (9.9)
Contraindication	12 (5.2)	7 (3.0)	5 (2.2)
Opioids	172 (74.5)	180 (77.3)	179 (77.2)
Unwilling to take	78 (33.8)	89 (38.2)	90 (38.8)
Inadequate pain relief	69 (29.9)	58 (24.9)	58 (25.0)
Intolerability	28 (12.1)	33 (14.2)	33 (14.2)
Contraindication	5 (2.2)	3 (1.3)	1 (0.4)
Tramadol	73 (31.6)	79 (33.9)	71 (30.6)
Inadequate pain relief	53 (22.9)	65 (27.9)	62 (26.7)
Intolerability	19 (8.2)	14 (6.0)	9 (3.9)
Contraindication	1 (0.4)	0	0
Time since osteoarthritis diagnosis, median (IQR), y	6.4 (3.2-11.7)	7.2 (3.5-11.8)	6.9 (3.4-12.6)
Index joint			
Knee	197 (85.3)	198 (85.0)	199 (85.8)
Hip	34 (14.7)	35 (15.0)	33 (14.2)
WOMAC Pain subscale score, median (IQR)[b]	7.0 (6.2-7.8)	7.2 (6.4-8.2)	7.4 (6.4-8.0)
WOMAC Physical Function subscale score, median (IQR)[c]	7.2 (6.4-7.9)	7.2 (6.5-8.3)	7.5 (6.5-8.1)
Patient global assessment of osteoarthritis[d]			
Good (2)	1 (0.4)	0	0
Fair (3)	144 (62.3)	125 (53.6)	134 (57.8)
Poor (4)	74 (32.0)	92 (39.5)	89 (38.4)
Very poor (5)	12 (5.2)	16 (6.9)	9 (3.9)
Mean (range)	3.42 (2-5)	3.53 (3-5)	3.46 (3-5)
Kellgren-Lawrence grade of index joint[e]			
1 (least affected)	1 (0.4)	0	0
2	60 (26.0)	59 (25.4)	65 (28.0)
3	101 (43.7)	105 (45.3)	98 (42.2)
4 (most affected)	69 (29.9)	68 (29.3)	69 (29.7)

Abbreviations: IQR, interquartile range; WOMAC, Western Ontario and McMasters Universities Osteoarthritis Index.

[a] Other races self-reported by patients were mixed race, American Indian, Middle Eastern/Indian, Hispanic, Pacific Islander, African American, Spanish, white and Native American, biracial, and Mexican.

[b] WOMAC Pain subscale on an 11-point numerical rating scale, from 0 = no pain to 10 = extreme pain.

[c] WOMAC Physical Function subscale on an 11-point numerical rating scale, from 0 = no difficulty to 10 = extreme difficulty.

[d] Patient global assessment of osteoarthritis scale ranged from 1 = very good to 5 = very poor.

[e] Kellgren-Lawrence grades ranged from 0 = no radiographic features of osteoarthritis to 4 = complete loss of joint space with large osteophytes, marked joint space narrowing, severe sclerosis, and definite deformity of bone contour.

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Table 2. Change From Baseline to Week 16 in WOMAC Pain Subscale, WOMAC Physical Function Subscale, and Patient Global Assessment of Osteoarthritis (Co-primary End Points)[a]

Scale	Tanezumab 2.5 mg (n = 231)	Tanezumab 2.5/5 mg (n = 233)	Placebo (n = 232)
WOMAC Pain[b]			
Baseline Pain score, mean (range)	7.1 (4.8 to 10.0)	7.3 (5.0 to 10.0)	7.3 (4.2 to 10.0)
Least squares change from baseline, mean (95% CI)	−3.23 (−3.67 to −2.79)	−3.37 (−3.81 to −2.93)	−2.64 (−3.08 to −2.19)
Difference of least squares vs placebo, mean (95% CI)	−0.60 (−1.07 to −0.13)	−0.73 (−1.20 to −0.26)	
P value	.01	.002	
WOMAC Physical Function[c]			
Baseline Physical Function score, mean (range)	7.2 (5.1 to 9.9)	7.4 (3.2 to 9.9)	7.4 (4.4 to 10.0)
Least squares change from baseline, mean (95% CI)	−3.22 (−3.66 to −2.79)	−3.45 (−3.88 to −3.03)	−2.56 (−3.00 to −2.12)
Difference of least squares vs placebo, mean (95% CI)	−0.66 (−1.14 to −0.19)	−0.89 (−1.37 to −0.42)	
P value	.007	<.001	
Patient Global Assessment of Osteoarthritis[d]			
Baseline score, mean (range)	3.4 (2 to 5)	3.5 (3 to 5)	3.5 (3 to 5)
Least squares change from baseline, mean (95% CI)	−0.87 (−1.02 to −0.72)	−0.90 (−1.05 to −0.75)	−0.65 (−0.80 to −0.50)
Difference of least squares vs placebo, mean (95% CI)	−0.22 (−0.39 to −0.05)	−0.25 (−0.41 to −0.08)	
P value	.01	.004	

Abbreviation: WOMAC, Western Ontario and McMasters Universities Osteoarthritis Index.

[a] Analysis of the co-primary end points used a stepdown strategy (described in the Methods section) designed to maintain type I error at ≤5% for multiple comparisons.

[b] WOMAC Pain subscale on an 11-point numerical rating scale, from 0 = no pain to 10 = extreme pain.

[c] WOMAC Physical Function subscale on an 11-point numerical rating scale, from 0 = no difficulty to 10 = extreme difficulty.

[d] Patient global assessment of osteoarthritis scale ranged from 1 = very good to 5 = very poor.

period in the tanezumab, 2.5/5 mg, group; neither was considered treatment related.

Neurological Assessments

The incidences of patients with AEs of abnormal peripheral sensation during the treatment period were 6.1%, 3.0%, and 2.2% in the tanezumab, 2.5 mg; tanezumab, 2.5/5 mg; and placebo groups, respectively, and all events were mild or moderate in severity. A total of 90.9%, 94.1%, and 91.7% of new or worsened abnormalities in the tanezumab, 2.5 mg; tanezumab, 2.5/5 mg; and placebo groups, respectively, at the final neurological examination were deemed not clinically significant. There was no significant difference in Neuropathy Impairment Score change from baseline between tanezumab-treated patients and placebo at any time point (eTable 4 in Supplement 3). No diagnoses of sympathetic neuropathy were reported by the principal investigator in patients evaluated by cardiology or neurology specialists.

Joint Safety Events

Thirty-seven patients had adjudicated joint safety events. Most patients (30/37, 81%) had joint safety events adjudicated as normal OA progression. Adjudicated rapidly progressive OA cases were classified by the predefined terms: type 1 (defined as a significant loss of joint space width ≥2 mm [predicated on optimal joint positioning] within approximately 1 year, without gross structural failure; n = 4) or type 2 (defined as abnormal bone loss or destruction, including limited or total collapse of at least 1 subchondral surface that is not normally present in conventional end-stage OA; n = 2), and were seen only in tanezumab-treated patients (6/464; 1.3%; Table 4).[25] Events adjudicated as rapidly progressive OA (types 1 and 2) occurred more frequently in the tanezumab, 2.5 mg, group (2.2%) compared with tanezumab, 2.5/5 mg (0.4%). Both rapidly progressive OA type 2 events occurred in index joints that were Kellgren-Lawrence grade 4 at screening, and neither patient reported NSAID use during the study. In 1 rapidly progressive OA type 2 case, the screening radiographs adjudicated after study completion suggested that this patient had atrophic OA and possible osteonecrosis before tanezumab treatment.

Most adjudicated joint safety events were total joint replacements (28/37 patients; 75.7%). Eight tanezumab, 2.5 mg-treated patients (3.5%); 16 tanezumab, 2.5/5 mg-treated patients (6.9%); and 4 placebo-treated patients (1.7%) underwent total joint replacements. All total joint replacements occurred in joints that were Kellgren-Lawrence grade 3 to 4 at screening. Most patients underwent total joint replacements that were of the index joint (26/28 patients; 92.9%), elective (ie, there was no associated AE and the total

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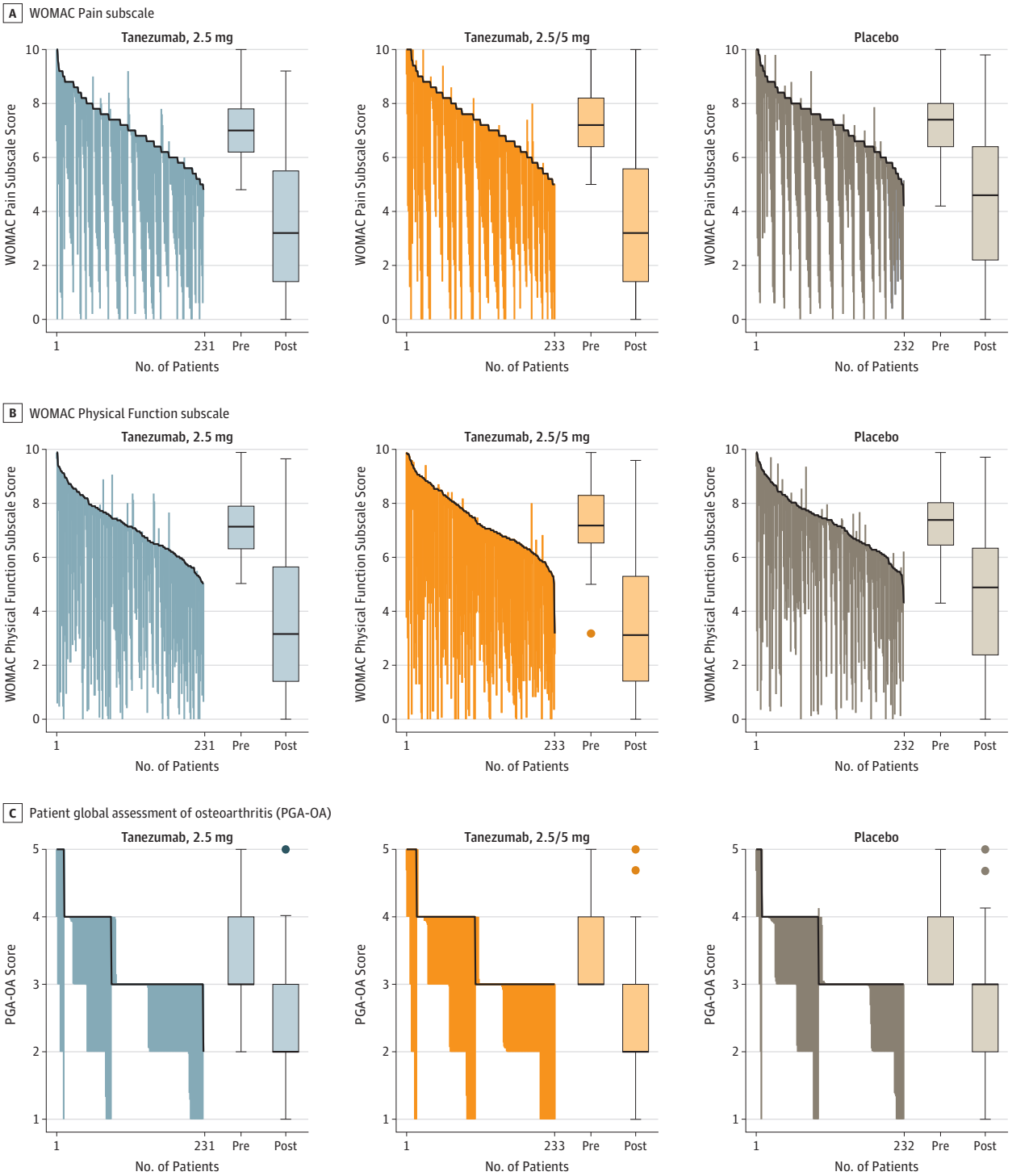
Figure 2. Change From Baseline to Week 16 in WOMAC Pain, WOMAC Physical Function, and Patient Global Assessment of Osteoarthritis at the Patient Level



The black lines represent baseline values for each individual patient, sorted by severity. Individual colored lines represent the change from baseline at week 16 for each individual patient (colored lines extending below the black line indicate lower score and, thus, improvement at week 16). The pre (baseline) and post (week 16) boxplots show median (middle horizontal line in the box), and quartiles 1 and 3 (bottom and top lines of the box). Lines extend from the box to the smallest and largest observations no further than 1.5 times the interquartile range from quartile 1 and quartile 3, respectively, and any data beyond this range are plotted individually. WOMAC Pain subscale ranges from 0 = no pain to 10 = extreme pain. WOMAC Physical Function subscale ranges from 0 = no difficulty to 10 = extreme difficulty. PGA-OA scale ranges from 1 = very good to 5 = very poor. WOMAC indicates Western Ontario and McMasters Universities Osteoarthritis.

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Table 3. Proportion of Patients With ≥30%, ≥50%, ≥70%, and ≥90% Reduction From Baseline in WOMAC Pain at Week 16

	Tanezumab		
	2.5 mg (n = 231)	2.5/5 mg (n = 233)	Placebo (n = 232)
≥30% Reduction			
Patients, No. (%)	157 (68.0)	164 (70.4)	127 (54.7)
Difference from placebo, %	13.2	15.6	
Odds ratio (95% CI) vs placebo[a]	1.72 (1.17-2.52)	1.95 (1.33-2.88)	
P value	.006	<.001	
≥50% Reduction			
Patients, No. (%)	126 (54.5)	133 (57.1)	88 (37.9)
Difference from placebo, %	16.6	19.2	
Odds ratio (95% CI) vs placebo[a]	1.89 (1.29-2.76)	2.17 (1.48-3.16)	
P value	.001	<.001	
≥70% Reduction			
Patients, No. (%)	80 (34.6)	85 (36.5)	58 (25.0)
Difference from placebo, %	9.6	11.5	
Odds ratio (95% CI) vs placebo[a]	1.53 (1.02-2.31)	1.72 (1.14-2.57)	
P value	.04	.009	
≥90% Reduction			
Patients, No. (%)	34 (14.7)	33 (14.2)	22 (9.5)
Difference from placebo, %	5.2	4.7	
Odds ratio (95% CI) vs placebo[a]	1.60 (0.89-2.86)	1.56 (0.87-2.79)	
P value	.11	.14	

Abbreviation: WOMAC, Western Ontario and McMasters Universities Osteoarthritis Index.

[a] Odds ratio and *P* values vs placebo using a logistic regression model including baseline WOMAC Pain subscale, baseline diary average pain and classification variables of index joint, highest Kellgren-Lawrence grade, and treatment.

joint replacement was adjudicated as normal OA progression; 21/28 patients; 75%), adjudicated as normal OA progression (26/28 patients; 92.9%), and occurred after the treatment period (19/28 patients; 67.9%). Two tanezumab-treated patients had total joint replacements and adjudicated rapidly progressive OA.

Discussion

In this clinical trial of patients with moderate to severe OA of the knee or hip and inadequate treatment response to standard analgesics, subcutaneous tanezumab demonstrated statistically significant improvements over placebo in pain, physical function, and PGA-OA at week 16; approximately 85% of patients had knee index joints. Forced titration of tanezumab from 2.5 mg to 5 mg resulted in nominally greater improvements compared with patients who continued taking tanezumab, 2.5 mg.

A significantly greater proportion of patients treated with tanezumab, 2.5 mg and 2.5/5 mg, had an improvement of 50% or greater reduction in WOMAC Pain score from baseline to week 16 compared with placebo.

The efficacy results at week 16 in this study were similar to the prior OA study in patients with index knee OA treated with intravenous tanezumab, 2.5 mg, or intravenous tanezumab, 5 mg, every 8 weeks[12] and in patients with index knee or hip OA (approximately 80% knee and 20% hip) treated with tanezumab, 5 mg, every 8 weeks.[26] Importantly, patients in the current dose-titration study had more radiographically severe OA and failed more treatments than patients in prior tanezumab OA studies.[12,26]

Across treatment groups, more AEs occurred during the treatment period vs the follow-up period. The observed pattern for paresthesia and pain in extremity, which were among the most common AEs in this study, is consistent with data from previous controlled phase 3 tanezumab studies of OA.[27] In the present study, several of the most common AEs overall (eg, arthralgia and paresthesia) were also among the most common AEs in previous tanezumab studies of OA.[12,13,28] However, in the present study, arthralgia was less common in tanezumab-treated patients than in placebo-treated patients, a pattern that differed from previous tanezumab studies.[12,13,28]

In this study, few patients overall (37/696; 5.3%) experienced joint safety events that warranted adjudication, and most events (30/37; 81.1%) were adjudicated as normal OA progression. Overall, rapidly progressive OA (types 1 and 2) occurred in 6 tanezumab-treated patients (1.3%) and no placebo-treated patients.[25] No joint safety events were adjudicated as osteonecrosis, subchondral insufficiency fracture, or pathologic fracture. Rapidly progressive OA (types 1 and 2) occurred more frequently in the tanezumab, 2.5 mg, group (2.2%) compared with tanezumab, 2.5/5 mg (0.4%). There was no consistent pattern of pain relief or of severe increase in pain associated with rapidly progressive OA cases.

All total joint replacements occurred in joints that were considered radiographically moderate to severe at screening (Kellgren-Lawrence grade 3-4). Most total joint replacements occurred after the treatment period (67.9% of patients) and were elective (75.0% of patients), ie, the total joint replacement was not associated with an AE and the events were adjudicated as normal OA progression. Substantially more total joint replacements occurred in the tanezumab groups than in the placebo group. Most previous studies with tanezumab have

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Table 4. Summary of Overall Adverse Events, Neurological Adverse Events (AEs), and Joint Safety Outcomes

	Treatment Period (0-16 wk), No. (%)			Up to End of Study (0-40 wk), No. (%)		
	Tanezumab			Tanezumab		
	2.5 mg (n = 231)	2.5/5 mg (n = 233)	Placebo (n = 232)	2.5 mg (n = 231)	2.5/5 mg (n = 233)	Placebo (n = 232)
Overall Treatment-Emergent AEs						
Patients with AEs	128 (55.4)	109 (46.8)	115 (49.6)	156 (67.5)	143 (61.4)	145 (62.5)
Patients with treatment-related AEs	29 (12.6)	22 (9.4)	24 (10.3)	40 (17.3)	33 (14.2)	31 (13.4)
Patients discontinued treatment due to AEs	1 (0.4)	3 (1.3)	3 (1.3)	1 (0.4)	3 (1.3)	3 (1.3)
Patients with serious AEs	4 (1.7)	4 (1.7)	4 (1.7)	7 (3.0)	11 (4.7)	9 (3.9)
Most Common Treatment-Emergent AEs[a]						
Arthralgia	19 (8.2)	22 (9.4)	29 (12.5)	34 (14.7)	34 (14.6)	40 (17.2)
Nasopharyngitis	12 (5.2)	11 (4.7)	8 (3.4)	20 (8.7)	17 (7.3)	16 (6.9)
Fall	11 (4.8)	5 (2.1)	6 (2.6)	13 (5.6)	10 (4.3)	9 (3.9)
Back pain	10 (4.3)	6 (2.6)	7 (3.0)	13 (5.6)	8 (3.4)	9 (3.9)
Joint swelling	8 (3.5)	4 (1.7)	4 (1.7)	9 (3.9)	6 (2.6)	5 (2.2)
Paresthesia	8 (3.5)	3 (1.3)	1 (0.4)	8 (3.5)	3 (1.3)	1 (0.4)
Upper respiratory tract infection	7 (3.0)	3 (1.3)	6 (2.6)	11 (4.8)	10 (4.3)	9 (3.9)
Musculoskeletal pain	7 (3.0)	2 (0.9)	8 (3.4)	17 (7.4)	8 (3.4)	11 (4.7)
Headache	6 (2.6)	7 (3.0)	7 (3.0)	9 (3.9)	9 (3.9)	10 (4.3)
Pain in extremity	4 (1.7)	7 (3.0)	2 (0.9)	9 (3.9)	9 (3.9)	3 (1.3)
Treatment-Emergent Neurological AEs						
Abnormal peripheral sensation AEs occurring in ≥2% in any treatment group[b]						
Paresthesia	8 (3.5)	3 (1.3)	1 (0.4)	8 (3.5)	3 (1.3)	1 (0.4)
Hypoesthesia	5 (2.2)	3 (1.3)	3 (1.3)	5 (2.2)	6 (2.6)	6 (2.6)
Sympathetic nervous system AEs occurring in ≥1% in any treatment group[c]						
Diarrhea	5 (2.2)	5 (2.1)	3 (1.3)	5 (2.2)	8 (3.4)	4 (1.7)
Nausea	3 (1.3)	1 (0.4)	3 (1.3)	4 (1.7)	1 (0.4)	4 (1.7)
Orthostatic hypotension	3 (1.3)	1 (0.4)	1 (0.4)	3 (1.3)	1 (0.4)	1 (0.4)
Urinary incontinence	2 (0.9)	1 (0.4)	3 (1.3)	2 (0.9)	3 (1.3)	4 (1.7)
Sinus bradycardia	1 (0.4)	3 (1.3)	2 (0.9)	1 (0.4)	3 (1.3)	4 (1.7)
Joint Safety Outcomes						
Patients with adjudicated joint safety events						
Normal osteoarthritis progression				8 (3.5)	17 (7.3)	5 (2.2)
Rapidly progressive OA type 1[d]				3 (1.3)	1 (0.4)	0
Rapidly progressive OA type 2[e]				2 (0.9)	0	0
Other (eg, preexisting subchondral insufficiency fracture)[f]				1 (0.4)	0	0
Patients with ≥1 total joint replacement				8 (3.5)	16 (6.9)[g]	4 (1.7)
Joints replaced						
Knee				3	10	4
Hip				5	7	0

[a] Those occurring in ≥3% in any treatment group during the treatment period.

[b] AEs of abnormal peripheral sensation include the following terms: allodynia, axonal neuropathy, burning sensation, carpal tunnel syndrome, decreased vibratory sense, demyelinating polyneuropathy, dysesthesia, formication, hyperesthesia, hyperpathia, hypoesthesia, hypoesthesia oral, intercostal neuralgia, neuralgia, neuritis, neuropathy peripheral, paresthesia, paresthesia oral, peripheral sensorimotor neuropathy, peripheral sensory neuropathy, polyneuropathy, polyneuropathy chronic, sciatica, sensory disturbance, sensory loss, tarsal tunnel syndrome, and thermohypoesthesia.

[c] Sympathetic nervous system AEs included abdominal discomfort, anal incontinence, anhidrosis, blood pressure orthostatic decreased, bradycardia, diarrhea, dizziness postural, early satiety, ejaculation delayed, ejaculation disorder, ejaculation failure, heart rate decreased, hypertonic bladder, hypohidrosis, micturition urgency, nausea, orthostatic hypotension, nocturia, pollakiuria, presyncope, respiratory distress, respiratory failure, sinus bradycardia, syncope, urinary hesitation, urinary incontinence, and vomiting.

[d] Rapidly progressive osteoarthritis (OA) type 1 is defined as a significant loss of joint space width ≥2 mm (predicated on optimal joint positioning) within approximately 1 year, without gross structural failure. One patient with rapidly progressive OA type 1 in the tanezumab, 2.5/5 mg, treatment group had a total joint replacement.

[e] Rapidly progressive OA type 2 is defined as abnormal bone loss or destruction, including limited or total collapse of at least 1 subchondral surface, that is not normally present in conventional end-stage osteoarthritis. One patient with rapidly progressive OA type 2 in the tanezumab, 2.5 mg, treatment group had a total joint replacement.

[f] A condition was adjudicated as "preexisting" if it was not identified by the central reader at screening but the adjudication committee determined it to be preexisting after reviewing all available postbaseline clinical and imaging information for the joint safety event in question.

[g] One patient had 2 joint replacements.

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not demonstrated such differences in total joint replacement rates when compared with active comparators or placebo, though the current study used a longer posttreatment observation period.[12,13] However, in a recently completed 1-year, NSAID-controlled study of tanezumab in patients with OA, the incidence of rapidly progressive OA was 6.3% in the tanezumab, 5 mg, group; 3.2% in the tanezumab, 2.5 mg, group; and 1.2% in the NSAID group. The incidence of total joint replacement was 8.0% in the tanezumab, 5 mg, group; 5.3% in the tanezumab, 2.5 mg, group; and 2.6% in the NSAID group (ClinicalTrial.gov Identifier: NCT02528188).[29]

Limitations

This study has several limitations. First, the 16-week treatment period is too short to assess the ability to maintain efficacy in treating symptomatic OA pain with repeated tanezumab dosing over longer periods. Second, while the study population is appropriate for evaluating efficacy, larger patient populations studied over longer durations are required for more precise estimates of adverse events. Other recently completed OA studies of tanezumab, 2.5 mg and 5 mg, subcutaneously over 6- and 12-month treatment periods will provide additional efficacy and AE data (ClinicalTrial.gov Identifiers: NCT02709486 and NCT02528188).

Conclusions

Among patients with moderate to severe OA of the knee or hip and inadequate response to standard analgesics, tanezumab, compared with placebo, resulted in statistically significant improvements in scores assessing pain and physical function, and in PGA-OA, although the improvements were modest and tanezumab-treated patients had more joint safety events and total joint replacements. Further research is needed to determine the clinical importance of these efficacy and AE findings.

ARTICLE INFORMATION

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Author Contributions: Dr Schnitzer had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.
Concept and design: Pixton, Viktrup, Davignon, Brown, West, Verburg.
Acquisition, analysis, or interpretation of data: Schnitzer, Easton, Pang, Levinson, Pixton, Viktrup, Davignon, Brown, West.
Drafting of the manuscript: Easton, Levinson, Pixton, Viktrup, Brown, West, Verburg.
Critical revision of the manuscript for important intellectual content: Schnitzer, Easton, Pang, Pixton, Viktrup, Davignon, Brown, West.
Statistical analysis: Pixton.
Administrative, technical, or material support: Easton, Levinson.
Supervision: Easton, Pang, Viktrup, Davignon, Brown, West, Verburg.

Conflict of Interest Disclosures: Dr Schnitzer reported receiving nonfinancial support from Pfizer during the conduct of the study and grants, personal fees, and nonfinancial support from Pfizer, Regeneron, AbbVie, and Kolon TissueGene; personal fees from Vertex, Sanofi, Astellas, and Calibr; grants and personal fees from Flexion; personal fees and nonfinancial support from GlaxoSmithKline and Aptinyx; and grants from Galapagos and Grunenthal outside the submitted work. Dr Schnitzer also reported performing clinical research for Pfizer Inc and Eli Lilly and Company and has received consulting fees from Pfizer Inc and Eli Lilly and Company. Dr Easton reported receiving compensation for participation in multiple Pfizer-sponsored clinical trials. Dr Pang reported receiving consulting fees from Pfizer Inc. Dr Levinson reported receiving grants from Pfizer during the conduct of the study and grants from Amgen, Regeneron, and AbbVie outside the submitted work. Mr Pixton and Drs Davignon, Brown, Verburg, and West are employees of Pfizer Inc and own stock. Dr Viktrup is an employee of Eli Lilly and Company. Drs Brown and West reported having tanezumab method of treatment patents pending. No other disclosures were reported.

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Role of the Funder/Sponsor: Pfizer Inc and Eli Lily and Company contributed to the study design; Pfizer contributed to the management and collection of data. In their role as authors, employees of Pfizer and Eli Lilly were involved in the interpretation of data, preparation, review, and approval of the manuscript and the decision to submit for publication, along with their co-authors. The study sponsors approved the manuscript from an intellectual property perspective but had no right to veto the publication.

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Data Sharing Statement: See Supplement 4.

Additional Information: Pfizer Inc, in collaboration with Eli Lilly and Company, is the manufacturer of tanezumab, which is being investigated for the treatment of chronic pain.

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Tanezumab
Protocol A4091056
Final Protocol Amendment 1, 23 September 2015

Note:

Certain information has been redacted from the protocol for public disclosure purposes.
Redacted information is as follows:

- **Biomarkers**: The specified serum and urine biomarkers (pages 92 and 148 to 151) have not previously been disclosed publicly for this study and are considered proprietary company confidential information prior to such disclosure. Please note that these were exploratory endpoints and their redaction will not affect the reader's ability to understand the conduct of the study and interpret the results.

- **Protocol Stopping Rules**: The specifics of the stopping rule (page 113) were established pursuant to an agreement with FDA and have not previously been disclosed publicly. They are to remain confidential until such disclosure. The stopping rule was not met and the study was not stopped prematurely.

- **WOMAC® instrument:** The WOMAC® instrument (Appendix 4, pages 130 to 135) is protected by copyright and confidentiality restrictions that do not permit disclosure to a 3rd party.

- **Survey of Autonomic Symptom (SAS):** The table displayed in the protocol (Appendix 11, page 145) is protected by a 3rd party copyright. The original paper is referenced and can be accessed by the reader.

CLINICAL PROTOCOL

A PHASE 3 RANDOMIZED, DOUBLE-BLIND, PLACEBO-CONTROLLED, MULTICENTER STUDY OF THE ANALGESIC EFFICACY AND SAFETY OF A DOSE TITRATION REGIMEN FOR THE SUBCUTANEOUS ADMINISTRATION OF TANEZUMAB IN SUBJECTS WITH OSTEOARTHRITIS OF THE HIP OR KNEE

Compound:	PF-04383119
Compound Name:	Tanezumab
US IND Number:	BB-IND 11,680
European Clinical Trial Database (EudraCT) Number:	2013-002222-23
Protocol Number:	A4091056
Phase:	3

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Document History

Document	Version Date	Summary of Changes and Rationale
Final Protocol Amendment 1	23 September 2015	Changes resulting from FDA guidance: • Exclusion of patients with autonomic neuropathy and orthostatic hypotension (current criteria #20, 27 and 28). • Addition of orthostatic hypotension assessments and sitting vital signs at all study visits (removed temperature). • Addition of ECGs at end of treatment and end of study. • Addition of Added Survey of Autonomic Symptoms (SAS). • Addition of summaries of decreased sympathetic function adverse events. • Added requirement for immediate referral for consult of subjects with symptomatic confirmed orthostatic hypotension • Removed "topical NSAIDs" from inclusion criterion #4. • Allowed acetaminophen up to 24 hours prior to each study visit. • SAE stopping rule section updated (section 9.6.1.1). • Stopping rule for Hy's law updated (section 9.6.1.2). Adjudication Committee and External Radiologist Central Radiology Reader recommended changes: • Stratification by Kellgren-Lawrence Grade. • Screening NRS pain score on bilateral shoulders, hips and knees plus any joint imaged at baseline. • Modifications to joint-related exclusion criteria #3, 4, 5. • Addition of exclusion for history of osteoporotic fracture (current #6). Update WOMAC version to most recent. • WOMAC NRS 3.1 updated to NRS 3.1 USA-V5. Addition and updates to endpoints and outcomes research instruments to provide additional information to characterize the efficacy response: • Addition of WPAI-OA. • Addition of HCRU. • Addition of WOMAC Physical function-related endpoints. • Change from EQ-5D-3L to EQ-5D-5L v2.0.0. • EQ-5D-5L added at Week 8. • Timing of biomarker sampling reduced to Day 1 and Week 8 and Week 24. Clarification of existing procedures and updates for consistency with other tanezumab program protocols or most recent investigator brochure: • Added weekly collection of NSAID use via IRT from IPAP to End of Study. • Extended weekly assessment of pain in the non-index major joints and weekly report of rescue medication use

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		via IRT to the end of the follow up period. • Restricted use of rescue medications to 3000 mg/day until Week 24. • Duration of Screening period and timing of Screening X-rays. • Timing of selection of the index joint. • Rapidly progressive OA in subject with underlying osteoarthritis added to Table 2 (adverse drug reaction). • IPAP changed to 7 days (minimum 3 days) instead of 3 days (throughout the protocol, example Section 3, study design). • Replaced 3 exclusion criteria (related to aspirin use >=325 mg/day, use of biologics, and use of investigational medications with one criterion (current #14) referring to the prohibited medication section. • Adjusted definition of End of Study to completion of the Safety Follow-up period. • Removed description of patient care in case of allergic reaction (Section 6). • Removed contraception instructions at Week 24 because after complete washout of study medications. Changes resulting from preclinical data obtained after protocol finalization: • Removal of requirement for male subjects to follow contraceptive requirement. • Update to Introduction language to include preclinical data related to sympathetic nervous system. Changes resulting from updates to clinical supply since protocol finalization: • Replaced use of vials with use of prefilled syringes. • Because prefilled syringes are considered medical devices, addition of text relevant to the use of medical devices. • Deleted any reference to latex in vials which is not relevant to prefilled syringes. Changes to statistics section for clarification and/or alignment with endpoints and consistency with other program protocols: • Added OMERACT-OARSI responder criteria to analyses. • Updated time points around analyses of rescue medication. • Added baseline WOMAC pain subscale score to incidence of discontinuation due to lack of efficacy. • Removed CMH test for analysis of WPAI:OA. • Specified there would be no multiplicity adjustments for significance tests performed for AEs of interest. Updates related to Pfizer Protocol Template update (current version 16 Feb 2015). • Use of “subjects” rather than “patients’. • Use of “investigational product” rather than “study medication” and “study drug” replaced with “investigational product”.

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		• Use of E-DMC instead of DMC throughout to reflect use of an "external" DMC. • Table of content moved to before Protocol Summary. • Template standard wording updates: Schedule of activities (list of abbreviations used in table added in table footnote), Section 4 Paragraph 1, inclusion criterion #9, exclusions #36-37; Section 4.4 (childbearing potential definition, post-menopausal status and requirements for contraceptives), Section 4.5 (new section "Sponsor Qualified Medical Personnel"), Section 5 (section restructured and language updated, definition of investigational product), Section 5.6 (new wording regarding storage conditions and reporting of protocol deviations), Section 5.7.1 (destruction of investigational product), Section 8 (updated language throughout), Medication Error section moved to Section 8.4, Section 10 (site to notify sponsor of regulatory inspections), Section 12.3 (subject privacy), Section 12.4 (sponsor review of recruitment materials), Section 15 (Publication of Study Results language updated). • Moved Patient-Level stopping rules from 3.1 to new Section: 7.4 "Triggered Events". • Female of child-bearing potential wording updated. • Serum FSH testing Section 7.3.3.4 adjusted for consistency with template. • Word "rationale" added to column header in Summary of Changes "and Rationale". • Added list of abbreviations (Appendix 13). Other administrative changes for clarification and consistency across protocols and the tanezumab program (eg, adjustment to wording and descriptions, removed redundancies, removed instructions for PK, NGF, biomarkers and ADA sample collection and processing, moved inclusion criteria assessed at baseline to 4.3 randomization criteria, deleted Appendix 4, spelling out of abbreviations, removed reference to worksheets, added EudraCT number and protocol summary.
Final Protocol	27 July 2012	N/A

This amendment incorporates all revisions to date, including amendments made at the request of country health authorities, institutional review boards/ethics committees (IRBs/ECs), etc.

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PROTOCOL SUMMARY

Background

Tanezumab is a monoclonal antibody that binds to and inhibits the actions of nerve growth factor (NGF). The Nerve Growth Factor Inhibitor (NGFI) class may offer an important breakthrough in the treatment of chronic pain and is under clinical investigation for the treatment of pain associated with osteoarthritis or other chronic pain conditions. The completed Phase 2 and Phase 3 studies conducted to date have demonstrated that tanezumab is efficacious and generally safe and well tolerated for the treatment of pain due to osteoarthritis and chronic low back pain. In addition, completed Phase 1/2 studies suggest tanezumab is also efficacious and generally safe for the treatment of neuropathic, visceral, and cancer pain.

In 2010, the US Food and Drug Administration's (FDA) Division of Analgesia, Anesthetic, and Addiction Products (DAAAP) placed tanezumab (June/July 2010) and subsequently the entire NGFI class (December 2010) on partial clinical hold due to adverse events initially described by investigators as osteonecrosis that in some cases resulted in total joint replacement. Pfizer voluntarily imposed the partial clinical hold on study conduct in all countries.

Extensive analyses of the reports of osteonecrosis and other total joint replacements were conducted.[1] On March 12, 2012, the FDA Arthritis Advisory Committee reviewed these results as well as those prepared by the FDA.[2,3] The committee endorsed continued clinical development of the NGFI class of compounds with additional measures to minimize the risk and further protect subject safety. On August 28, 2012, the FDA lifted the partial clinical hold on tanezumab allowing the resumption of clinical studies for osteoarthritis and all other chronic pain conditions.

Risk mitigation measures were developed using recommendations from discussions with European agencies [United Kingdom's Medicines and Healthcare products Regulatory Agency (MHRA), Germany's Paul Ehrlich Institute (PEI) and Spain's Agency on Medicinal Products and Medical Devices (AEMPS)] as well as the FDA Arthritis Advisory Committee and interactions with FDA. They are incorporated in the design and objectives of the current study.

The FDA placed another partial clinical hold on the tanezumab clinical development program as well as all anti-NGF antibody studies in December 2012 due to concerns about adverse changes in the sympathetic nervous system of mature animals. Only studies in patients with cancer pain were allowed to continue.

In animal studies in rats and non-human primates, tanezumab treatment for up to 6 months, with doses producing greater systemic exposure than observed with clinical doses, was associated with lower sympathetic ganglion volume and lower average size of post-ganglionic sympathetic neurons when compared to control animals. All effects were completely reversible following a dosing-free recovery period. In a separate cardiovascular

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function study in non-human primates, functional changes in the cardiovascular system controlled by the sympathetic nervous system were not observed.

Although evidence of clinically important effects on the sympathetic nervous system have not been identified in previously completed tanezumab studies, per agreement with the FDA, this and other clinical studies of tanezumab will incorporate additional safety measures to monitor for and manage subjects who may develop evidence of clinically important sympathetic nervous system dysfunction.

The primary objective of this study is to gain additional experience with the 2.5 mg dose which was shown to provide efficacy benefits with a favorable safety profile in previous Phase 3 clinical trials. In addition, the study will explore the potential additional benefits of a dose titration regimen in which subjects would begin treatment at the 2.5 mg dose then titrate up to a 5 mg dose.

In order to optimize the potential benefit-risk relationship, the study will be conducted in subjects with moderate to severe osteoarthritis who have had inadequate pain relief with previous conventional pharmacological treatment options for osteoarthritis or are unable to take these pain medications due to contraindication or inability to tolerate them and are seeking effective treatment options other than/or prior to joint replacement surgery.

STUDY OBJECTIVES AND ENDPOINTS

Primary Objective

- Demonstrate superior efficacy of tanezumab 2.5 mg administered subcutaneously (SC) and tanezumab 2.5 mg SC titration to 5 mg SC versus placebo at Week 16.

Secondary Objectives

- Evaluate treatment benefit (efficacy) of a tanezumab 2.5 mg SC titration to 5 mg SC dosing regimen relative to tanezumab 2.5 mg SC alone (descriptive analyses);
- Evaluate the safety of tanezumab 2.5 mg SC and tanezumab 2.5 mg SC titration to 5 mg SC regimens.

Primary Endpoints

The co-primary efficacy endpoints are:

- Change from Baseline to Week 16 in the Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) Pain subscale;
- Change from Baseline to Week 16 in the WOMAC Physical Function subscale;
- Change from Baseline to Week 16 in the Patient's Global Assessment of Osteoarthritis.

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Secondary Endpoints

Efficacy:

- WOMAC Pain subscale change from Baseline to Weeks 2, 4, 8, 12 and 24;
- WOMAC Physical Function subscale change from Baseline to Weeks 2, 4, 8, 12 and 24;
- Patient's Global Assessment of Osteoarthritis (5-point Likert scale) change from Baseline to Weeks 2, 4, 8, 12 and 24;
- OMERACT-OARSI responder index at Weeks 2, 4, 8, 12, 16 and 24;
- Treatment Response: Reduction in the WOMAC Pain subscale of ≥30%, ≥50%, ≥70% and ≥90% at Weeks 2, 4, 8, 12, 16 and 24;
- Cumulative distribution of percent change from Baseline in the WOMAC Pain subscale score to Week 16 (endpoint for summary only);
- Treatment Response: Reduction in the WOMAC Physical Function subscale of ≥30%, ≥50%, ≥70% and ≥90% at Weeks 2, 4, 8, 12, 16 and 24;
- Cumulative distribution of percent change from Baseline in the WOMAC Physical Function subscale score to Week 16 (endpoint for summary only);
- Treatment Response: Improvement of ≥2 points in Patient's Global Assessment of Osteoarthritis at Weeks 2, 4, 8, 12, 16 and 24;
- Average pain score in the index joint change from Baseline to Weeks 1, 2, 3, 4, 6, 8, 10, 12, 16, 20 and 24;
- WOMAC Stiffness subscale change from Baseline to Weeks 2, 4, 8, 12, 16 and 24;
- WOMAC Average change from Baseline to Weeks 2, 4, 8, 12, 16 and 24;
- WOMAC Pain Subscale Item: Pain When Walking on a Flat Surface, change from Baseline to Weeks 2, 4, 8, 12, 16 and 24;
- WOMAC Pain Subscale Item: Pain When Going Up or Down Stairs, change from Baseline to Weeks 2, 4, 8, 12, 16 and 24;
- Work Productivity and Activity Impairment Questionnaire for Osteoarthritis (WPAI:OA) impairment scores change from Baseline to Week 16;
- EuroQol 5 Dimension (EQ-5D-5L™) dimensions and overall health utility score at Baseline, Week 8 and Week 16;

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- Health Care Resource Utilization (HCRU) at Baseline, Week 24 and Week 40;
- Incidence and time to discontinuation due to Lack of Efficacy;
- Usage of rescue medication (incidence, number of days) during Weeks 2, 4, 8, 12, 16 and 24;
- Usage of rescue medication (amount taken) during Weeks 2, 4, 8, 12, and 16.

Safety:

- Adverse events;
- Standard safety assessments (safety laboratory testing [chemistry, hematology], sitting vital signs, electrocardiogram (ECG; 12-lead);
- Joint Safety Adjudication outcomes;
- Total joint replacements;
- Neurologic examination (Neuropathy Impairment Score [NIS]);
- Orthostatic (supine/standing) blood pressure assessments;
- Survey of Autonomic Symptom (SAS) scores;
- Anti-drug antibody (ADA) assessments;
- Physical examinations.

Tertiary Endpoints

Pharmacokinetics and Pharmacodynamics:

- Plasma tanezumab concentrations;
- Serum NGF assessments;
- Serum and urine osteoarthritis biomarker concentrations.

STUDY DESIGN

This is a randomized, double-blind, placebo-controlled, multicenter, parallel-group, Phase 3 study of the efficacy and safety of tanezumab when administered by SC injection for 16 weeks compared to placebo in subjects with osteoarthritis of the knee or hip. Approximately 690 subjects (approximately 230 per treatment group) will be randomized to

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one of 3 treatment groups in a 1:1:1 ratio. Subjects will receive a total of two SC injections, separated by 8 weeks:

- tanezumab 2.5 mg (Day 1) and tanezumab 2.5 mg (Week 8);
- tanezumab 2.5 mg (Day 1), and tanezumab 5 mg (Week 8);
- placebo to match tanezumab (Day 1) and placebo to match tanezumab (Week 8).

This study is designed with a total post-randomization duration of 40 weeks and will consist of three periods: Screening (up to a maximum of 37 days), Double-blind Treatment (16 weeks, 6 in-clinic visits), and Safety Follow-up (24 weeks).

The Screening Period (to begin up to 37 days prior to Randomization) includes a Washout Period (lasting a minimum of 2 days for all prohibited pain medications) if required, and an Initial Pain Assessment Period (IPAP, 7 days prior to Randomization/Baseline; minimum 3 days).

Rescue medication (acetaminophen) will be provided and may be taken if necessary, up to the maximum daily dose of 3000 mg per day but must be discontinued at least 24 hours prior to any in-clinic visit where efficacy assessments will be collected.

The Treatment Period ends and the Safety Follow-Up Period begins at the Week 16 visit. Subjects who are discontinued from treatment prior to Week 16, either at their request or at the decision of the investigator, will be required to undergo 24 weeks of follow-up (referred to as Early Termination Follow-Up). The 24 weeks of follow-up will be obtained through 3 clinic visits and monthly phone calls to yield 24-weeks of post-treatment follow-up.

Subjects who have undergone or plan to undergo total joint replacement or other arthroplasty procedure during the study will be discontinued from study treatment. Follow-up procedures for these subjects are described in Section 6.16.2. In addition, subjects who undergo total knee, hip or shoulder joint replacement surgery during the study (Treatment Period or Follow-up Period) will be followed for 24 weeks after the procedure as part of a separate protocol,-provided the subject consents (refer to Section 6.17).

STATISTICAL METHOD

A sample size of approximately 230 subjects per treatment group is needed to provide approximately 90% power to achieve statistical significance (at the 5% two-sided level) for the two comparisons of tanezumab 2.5 and 2.5/5 mg SC versus placebo, over all three co-primary endpoints. The total sample size will be approximately 690 subjects.

The primary efficacy population will be the ITT (intent to treat) population, defined as all randomized subjects who received SC investigational product (either tanezumab or matching placebo). All treatment comparisons will use the two-sided 5% significance level.

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The co-primary efficacy endpoints will be analyzed using an Analysis of Covariance (ANCOVA) model, with model terms for Baseline score, Baseline Diary Average Pain, index joint (knee or hip), Kellgren-Lawrence (KL) grade and treatment group, and study site as a random effect. The assessment of significance for the tanezumab SC versus placebo treatment contrasts will use a step-down testing strategy within each of the co-primary efficacy endpoints defined as first testing tanezumab 2.5/5 mg versus placebo, and if statistically significant ($p \leq 0.05$) to then test tanezumab 2.5 mg versus placebo. Finally, a tanezumab treatment group is declared as superior to placebo if the corresponding treatment contrast is significant over all three co-primary endpoints. This testing procedure will maintain the Type I error to 5% or less within each of the co-primary efficacy endpoints, and to less than 5% for all three co-primary efficacy endpoints.

The primary analysis of the co-primary endpoints will use multiple imputations for missing data, to account for uncertainty around the subject response. The basis for imputing missing values will be dependent on the reasons for missing data.

Standard safety reporting tables will summarize and list the safety data.

DATA MONITORING COMMITTEE

An independent, external Data Monitoring Committee (E-DMC) has been reinstituted for the tanezumab clinical program. This committee will be composed of at least one rheumatologist, neurologist, statistician, and epidemiologist. The E-DMC will review unblinded safety data including (but not limited to) adverse events and serious adverse events on a regular basis throughout the trial. The E-DMC will have written operating procedures and a Charter, including a specific description of the scope of their responsibilities.

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Table 1. SCHEDULE OF ACTIVITIES

The Schedule of Activities table provides an overview of the protocol visits and procedures. Refer to STUDY PROCEDURES and ASSESSMENTS sections of the protocol for detailed information on each procedure and assessment required for compliance with the protocol.

The investigator may schedule visits (unplanned visits) in addition to those listed on the schedule of activities, in order to conduct evaluations or assessments required to protect the well-being of the subject.

Visit Identifier	Screening[b] Day -37 to Day -1			Double-Blind Treatment					End of Treat-ment	Safety Follow-Up[r]		
										Phone Calls	16 weeks post last dose Visit	End of Study/ Final visit
	Screening visit	Washout and X-rays	Initial Pain Assessment Period	Baseline[c]	Week 2	Week 4	Week 8 [c]	Week 12	Week 16	Weeks 20, 28, 32, 36	Week 24	Week 40
Visit Window/ Study Activities[a]			Day -7 to -1	Day 1	Day 15 ±3 days	Day 29 ±3 days	Day 57 ±7 days	Day 85 ±7 days	Day 113 ±7 days	Days 141, 197, 225, 253 ±7 days	Day 169 ±7 days	Day 280 ±7 days
Informed Consent	X											
Inclusion/Exclusion Criteria; subject eligibility	X			X								
Demographics, General and Musculoskeletal Specific Medical History and Prior/Current Medication Use[d]	X											
Primary Diagnosis/Selection of Index Joint	X											
Weight/Height/BMI/Smoking Status/Female Hormonal Status/Alcohol Use	X								X[e]			
General Physical Examination	X								X			
Musculoskeletal Physical Examination[f]	X			X	X	X	X	X	X		X	X
Screening Numeric Pain Scale rating (NRS)[g]	X											

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Visit Identifier	Screening[b] Day -37 to Day -1			Double-Blind Treatment					End of Treat-ment	Safety Follow-Up[r]		
										Phone Calls	16 weeks post last dose Visit	End of Study/ Final visit
	Screening visit	Washout and X-rays	Initial Pain Assessment Period	Baseline[c]	Week 2	Week 4	Week 8 [c]	Week 12	Week 16	Weeks 20, 28, 32, 36	Week 24	Week 40
Visit Window/ Study Activities[a]			Day -7 to -1	Day 1	Day 15 ±3 days	Day 29 ±3 days	Day 57 ±7 days	Day 85 ±7 days	Day 113 ±7 days	Days 141, 197, 225, 253 ±7 days	Day 169 ±7 days	Day 280 ±7 days
Radiograph Assessment of Hips, Knees and Shoulders	X[b]											X
Neurologic Exam Neuropathy Impairment Score (NIS)[h]	X			X	X	X	X	X	X		X	X
Vital Signs (sitting; BP, HR)	X			X	X	X	X	X	X		X	X
Orthostatic Blood Pressure (supine/standing)	X			X	X	X	X	X	X		X	X
Electrocardiogram (ECG 12-lead)	X								X			X
Laboratory												
Hepatitis Screen (Hepatitis B & Hepatitis C); HIV test, Urine Toxicology screen	X											
Hemoglobin A1c	X											
Serum FSH Test[i]	X											
Serum/Urine Pregnancy Test[i]	X			X			X		X		X	
Hematology	X			X							X	
Blood Chemistry	X			X							X	
Urinalysis	X											
Serum and Plasma Retention Samples				X					X		X	
Plasma Pharmacokinetic sample				X	X[j]	X[j]	X	X[j]	X		X	
Serum NGF Assessment				X	X[j]	X[j]	X	X[j]	X		X	
Serum Anti-Drug Antibody				X			X		X		X	X

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Visit Identifier	Screening[b] Day -37 to Day -1			Double-Blind Treatment					End of Treat-ment	Safety Follow-Up[r]		
										Phone Calls	16 weeks post last dose Visit	End of Study/ Final visit
	Screening visit	Washout and X-rays	Initial Pain Assessment Period	Baseline[c]	Week 2	Week 4	Week 8 [c]	Week 12	Week 16	Weeks 20, 28, 32, 36	Week 24	Week 40
Visit Window/ Study Activities[a]			Day -7 to -1	Day 1	Day 15 ±3 days	Day 29 ±3 days	Day 57 ±7 days	Day 85 ±7 days	Day 113 ±7 days	Days 141, 197, 225, 253 ±7 days	Day 169 ±7 days	Day 280 ±7 days
Serum and Urine Biomarkers[k]				X			X				X	
Banked biospecimen (whole blood)				X								
Discontinue Current Pain Medication (washout)	X											
Subject Daily/Weekly Assessments (IRT)												
Pain NRS and Rescue Medication use[l]	(X)	(X)	----------X (daily)----------							----------X (weekly)----------		
Pain in non-index joints and NSAID use[m]			----------X (weekly)----------									
Randomization				X								
Trial Treatment												
SC Investigational product				X			X					
Subject Reported Assessments Completed at Study Visits[n]												
WOMAC Subscales	X			X	X	X	X	X	X		X	
Patient's Global Assessment of Osteoarthritis				X	X	X	X	X	X		X	
Survey of Autonomic Symptoms (SAS) questionnaire	X										X	X
Health Care Resource Utilization (HCRU)				X							X	X
WPAI:OA				X					X			
EQ-5D-5L				X			X		X			
Other Assessments/Procedures Conducted at Study Visits												
Adverse event assessment				X	X	X	X	X	X	X	X	X
Concomitant medication review				X	X	X	X	X	X	X	X	X

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Visit Identifier	Screening[b] Day -37 to Day -1			Double-Blind Treatment					End of Treat-ment	Safety Follow-Up[r]		
										Phone Calls	16 weeks post last dose Visit	End of Study/ Final visit
	Screening visit	Washout and X-rays	Initial Pain Assessment Period	Baseline[c]	Week 2	Week 4	Week 8 [c]	Week 12	Week 16	Weeks 20, 28, 32, 36	Week 24	Week 40
Visit Window/ Study Activities[a]			Day -7 to -1	Day 1	Day 15 ±3 days	Day 29 ±3 days	Day 57 ±7 days	Day 85 ±7 days	Day 113 ±7 days	Days 141, 197, 225, 253 ±7 days	Day 169 ±7 days	Day 280 ±7 days
Review compliance with subject IRT assessments[o]			X	X	X	X	X	X	X	X	X	
Rescue medication return/compliance[o]				X	X	X	X	X	X		X	
Contraceptive requirement instruction/reminder[p]	X		X	X	X	X	X	X	X	X (week 20)		
Dispense rescue medication	X			X	X	X	X	X	X		X	
Assign standard of care treatment as needed[q]											X	

Abbreviations: → or ---X--- = ongoing/continuous event, BMI = body mass index, BP = blood pressure, ECG = electrocardiogram, EQ-5D-5L = Euroqol – 5 dimensions – 5 levels, FSH = follicle stimulating hormone, HIV= human immunodeficiency virus, HR = heart rate, HCRU = healthcare resource utilization, IRT = interactive response technology, NRS = numerical rating scale, NSAID = non-steroidal anti-inflammatory agent, SAS = survey of autonomic symptoms, WPAI:OA = work productivity and activity index: osteoarthritis

a. Day relative to start of study treatment (Day 1).

b. The Screening Period begins up to 37 days prior to randomization and should allow for a minimum 2 day washout prior to entering the Initial Pain Assessment Period (IPAP) unless no washout is required. Screening x-rays should be obtained as early as possible in the Screening period to ensure that Central Reader confirmation of eligibility can be obtained within the 37-day Screening period (see Section 7.3.7 for radiographic assessments). Subjects should preferably begin the IPAP once x-ray confirmation of radiographic eligibility has been obtained from the Central Reader. During the IPAP, diary entries (index joint pain and rescue medication use) must be provided for a minimum of 3 days. See Section 6 for the order in which study procedures are to be conducted.

c. All study activities at dosing visits (Baseline [Day 1] and Week 8), including sample collection, are performed prior to dosing, unless otherwise noted.

d. At Screening, a comprehensive musculoskeletal medical history is obtained; see Section 7.3.1.2 for details. History of insufficient pain relief from, inability to tolerate or contraindication to take acetaminophen, NSAIDs and tramadol or opioids should be collected. Any prior use of medications used to treat osteoarthritis and osteoarthritis pain and prior 30 day use for all other medications should also be collected.

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e. Only body weight is collected at Week 16 (End of Treatment).

f. A musculoskeletal directed physical examination will be performed at each clinic visit; findings will be recorded on a case report form and findings considered clinically significant will be reported as adverse events. This physical examination is described in Section 7.3.1.2).

g. Average pain in the shoulders, hips and knees or any other painful major joint with potential signs and symptoms of osteoarthritis that will be imaged at Screening.

h. A neurological examination (NIS) will be performed by the Investigator (or designated physician) at each study visit and assessed for clinically significant changes from Baseline. Requirements for neurological consultation at which a full neurological examination is to be performed by a neurologist are detailed in Section 7.3.8.

i. FSH testing in female subjects as described in Section 7.3.3.4. For female subjects of child-bearing potential, urine pregnancy tests are performed prior to dosing at dosing visits. Serum pregnancy tests are performed at Screening, Week 16 and Week 24 (or Early Termination, see Section 6.16).

j. Blood samples for pharmacokinetics (PK) and pharmacodynamics (PD, NGF) at Weeks 2, 4 and 12 will only be collected in a subset of subjects.

k. Biomarker samples in serum and urine should be collected prior to dosing and at approximately the same time of the day at all scheduled time points, and as much as possible, following a fasting period of at least 8 hours. Urine sample should not be taken from the first void of the day.

l. Pain in the index joint and rescue medication use are collected daily from the beginning of the IPAP to Week 16 (End of Treatment visit) though they may start as early as at the Screening visit, then are collected weekly until Week 40 (End of Study); see Sections 7.1.1 and 7.1.2. Compliance with IRT assessments is to be reviewed at each study visit, including Phone visits.

m. Assessment of pain in the non-index joints and NSAID use are collected weekly via IRT from the IPAP to the Week 40 visit (Sections 7.1.1 and 7.1.3). Post-baseline, subjects who indicate that they have experienced increased pain in a non-index major joint will rate the pain in that joint for the remainder of the study (see Section 7.1.1). The subject will also report any NSAID use weekly via IRT from the IPAP period to the Week 40 visit. NSAID, dosage and reason for use information will be obtained via phone calls or at clinic visits.

n. The SAS questionnaire, WOMAC subscales, PGA of OA, WPAI:OA, HCRU and EQ-5D-5L will be administered using IRT at Study visits.

o. Review of compliance and entries for joint pain scores, rescue medication and NSAID use entered by the subject via IRT. Rescue medication should be discontinued 24 hours prior to any clinic visit up to and including Week 24 (including during the Early Termination Follow-Up period, see Section 6.16). Subjects should return rescue medication bottles at each study visit for assessment of compliance.

p. For female subjects of child-bearing potential; up to Week 24.

q. If needed, following completion of the visit at which final efficacy assessments are collected (Week 24, 16 weeks after the last dose of investigational product administered), standard of care treatment (as described in Section 5.8.2) may be initiated, and usage recorded on the concomitant medication case report form (CRF). NSAID and acetaminophen use will still be recorded weekly via IRT during that period.

r. The 24-week Safety Follow-Up period begins after completion of the End of Treatment visit. Subjects discontinuing the study at their request or at the decision of the Investigator prior to Week 16 should be withdrawn from treatment and begin the 24-week Early Termination Follow-Up period. As described in Section 6.16 and 6.16.1, the follow-up period for subjects who terminate early includes 3 clinic visits and 2 telephone visits). Subjects who undergo total joint replacement will be followed for 24 weeks after the procedure as part of a separate protocol described briefly in Section 6.17.

1. INTRODUCTION

1.1. Mechanism of Action/Indication

Tanezumab (PF-04383119, formerly RN624) is an anti-nerve growth factor monoclonal antibody under development for the relief of signs and symptoms of Osteoarthritis (OA). Full details can be found in the Investigator's Brochure.[11]

1.2. Background and Rationale

1.2.1. Role of Nerve Growth Factor in the Modulation of Pain

During mammalian development, nerve growth factor (NGF) is required for the survival and growth of several populations of neurons. In adults, the effect of NGF signaling shifts from the regulation of neuronal survival to the regulation of neuronal phenotype and function. The role of NGF in the adult mammal appears to principally be as a modulator of nociceptive neuronal activity. Thus, NGF plays an important role in modulation of the pain response.[4,5] Many non-clinical studies employing a variety of antibodies to NGF or tropomyosin receptor kinase A (trkA)-IgG fusion protein have demonstrated that blocking NGF bioactivity normalizes pain sensitivity, particularly in states of allodynia and hypersensitivity, following a variety of insults such as Freund's adjuvant, carrageenan, or surgical incision.[5,6]

Both interleukin (IL)-1β and tumor necrosis factor alpha (TNF-α) have been shown to induce synthesis of NGF. Inhibition of NGF in turn blocks the hyperalgesia experienced after administration of these cytokines.[7,8] Together these observations suggest that NGF may play a role in pain secondary to inflammation or injury.

1.2.2. Description of Investigational Product

Tanezumab is a humanized immunoglobulin G Type 2 (IgG_2) monoclonal antibody, derived from a murine precursor by grafting the murine complementarity determining regions onto a human antibody framework, followed by extensive site-directed mutagenesis using proprietary technology to improve binding affinity and specificity. A mutation was performed in the Fc portion of the antibody to decrease its ability to activate complement or to support antibody dependent cell-mediated cytotoxicity.[9,10]

Tanezumab is highly potent in sequestering NGF and preventing interaction with the trkA or p75 receptors. Tanezumab and/or the murine precursor have been shown to be an effective analgesic in nonclinical animal models of pathological pain including arthritis, cancer pain, and post-surgical pain models.[11]

1.2.3. Overview of Clinical Studies

Over 11,000 subjects participated in the tanezumab clinical program comprising 32 studies conducted as of September 2014. Approximately 9810 healthy volunteers or patients have been treated with tanezumab in non-cancer pain clinical studies. In patients treated with tanezumab monotherapy or tanezumab + NSAID in completed non cancer pain studies, treatment experience with tanezumab approximates 5900 patient years of treatment exposure.

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Most of these studies were conducted in patients with osteoarthritis of the knee or hip. A total of 17 clinical studies (4 Phase 2 studies and 13 Phase 3 studies), were initiated to provide evidence of efficacy and safety of tanezumab with intravenous (IV) or SC administration for the relief of the signs and symptoms of osteoarthritis alone or in combination with non-steroidal anti-inflammatory agents (NSAIDs). Both IV and SC routes of administration with tanezumab at fixed dose levels of 2.5 mg, 5 mg, and 10 mg every 8 weeks were evaluated in Phase 3 clinical studies of osteoarthritis subjects. In the 4 Phase 2 studies of tanezumab in patients with osteoarthritis, weight based doses of tanezumab ranging from 3 μg/kg to 1000 μg/kg approximating fixed doses of 0.3 mg to 100 mg were administered.

In addition to the osteoarthritis studies, 11 Phase 1/2 studies were conducted to examine the efficacy and safety of tanezumab in other musculoskeletal, neuropathic, and visceral pain conditions, and 2 Phase 2 studies were conducted in subjects with metastatic bone pain. In these studies, tanezumab was administered by IV or SC administration every 8 weeks at fixed doses ranging from 1 mg to 20 mg or equivalent body-weight adjusted doses up to 100 mg.

In 2010, the US Food and Drug Administration's (FDA) Division of Analgesia, Anesthetic, and Addiction Products (DAAAP) placed tanezumab (June/July 2010) and subsequently the entire NGFI class (December 2010) on partial clinical hold due to adverse events initially described by investigators as osteonecrosis that in some cases resulted in total joint replacement. Pfizer voluntarily imposed the partial clinical hold on study conduct in all countries. The conduct of Phase 2/3 studies in osteoarthritis or other chronic pain conditions was impacted to varying extents by the partial clinical hold placed on the tanezumab clinical development program in June/July 2010.

Extensive analyses of the reports of osteonecrosis and other total joint replacements were conducted.[1] On March 12, 2012, the FDA Arthritis Advisory Committee reviewed these results as well as those prepared by the FDA.[2,3] The committee endorsed continued clinical development of the NGFI class of compounds with additional measures to minimize the risk and further protect subject safety. On August 28, 2012, the FDA lifted the partial clinical hold on tanezumab allowing the resumption of clinical studies for osteoarthritis and all other chronic pain conditions.

The FDA placed another partial clinical hold on the tanezumab clinical development program as well as all anti-NGF antibody studies in December 2012 due to concerns about adverse changes in the sympathetic nervous system of mature animals. Only studies in patients with cancer pain were allowed to continue. During 2013-2014, Pfizer conducted a comprehensive series of nonclinical studies to investigate the nonclinical effects on the sympathetic nervous system which led to the partial clinical hold (described in Section 5.3 of the tanezumab Investigators' Brochure).

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In animal studies in rats and non-human primates (described in Section 5.3 of the tanezumab Investigators' Brochure), tanezumab treatment for up to 6 months, with doses producing greater systemic exposure than observed with clinical doses, was associated with lower sympathetic ganglion volume and lower average size of post-ganglionic sympathetic neurons when compared to control animals. All effects were completely reversible following a dosing-free recovery period. In a separate cardiovascular function study in non-human primates (described in Section 5.1 of the tanezumab Investigators' Brochure), functional changes in the cardiovascular system controlled by the sympathetic nervous system were not observed.

Although evidence of clinically important effects on the sympathetic nervous system have not been identified in previously completed tanezumab studies, per agreement with the FDA, this and other clinical studies of tanezumab will incorporate additional safety measures to monitor for and manage subjects who may develop evidence of clinically important sympathetic nervous system dysfunction.

1.2.3.1. Overview of Efficacy in Osteoarthritis Clinical Studies

Across the 3 coprimary measures of efficacy in four completed Phase 3 studies of tanezumab monotherapy (IV administration) doses of 2.5 mg, 5 mg, and 10 mg provided significant improvement over placebo treatment. All of the tanezumab doses tested were consistently efficacious. In Studies A4091011[13] and A4091014,[14] the degree of mean improvement across the three efficacy domains was similar and generally dose ordered with tanezumab 10 mg providing the greatest mean response in each of the 3 co-primary comparisons to placebo treatment although only small differences in the magnitude of response were evident among the three doses of tanezumab. In Studies A4091015[15] and A4091018,[16] tanezumab 5 mg provided modestly greater mean improvement over tanezumab 10 mg across most of the co-primary endpoints.

In each of these four studies, the mean Baseline WOMAC[12] Pain subscale scores exceeded 7 on a scale of 0 to 10 indicating a subject population with severe pain on average prior to study entry. The mean reduction in pain with tanezumab treatment was typically 3 points or greater yielding an improvement in pain on average from severe to nearly mild in severity. This reduction in pain was associated with equivalent improvements in function and global well-being.

The clinical significance of the reduction in pain with tanezumab treatment was assessed by the evaluation of subject responder rates using a categorical assessment of the WOMAC Pain subscale results, the OMERACT-OARSI[17] (Outcomes Measures in Arthritis Clinical Trials-Osteoarthritis Research Society International) Responder Index and a 2-grade or larger categorical improvement in the Patient's Global Assessment of Osteoarthritis. All responder analyses were determined with Baseline Observation Carried Forward (BOCF) imputation unless otherwise noted.

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In studies A4091011 and A4091014, statistically significant response rates compared to placebo treatment were demonstrated at Week 16 with all tanezumab doses for the percentage of subjects with reductions in pain ≥30%, ≥50%, ≥70%, and ≥90% on the WOMAC Pain subscale in both studies with one exception (A4091011 ≥70% reduction with tanezumab 2.5 mg vs placebo p-value was 0.053).

A prospectively defined analysis of subjects participating in Study A4091011, A4091014, A4091015, or A4091018 was performed to evaluate the efficacy of tanezumab with severe symptomatic osteoarthritis. Subjects defined with severe pain were those with a Baseline WOMAC Pain score ≥7 and a WOMAC Physical Function score of ≥7 and a score of "poor" or "very poor" in the Patient's Global Assessment of Osteoarthritis. Of the 2979 subjects enrolled across the 4 studies, 742 (25.1%) met these criteria for severe disease. Tanezumab 5 mg and 10 mg provided significant and clinically meaningful benefit in this severe subject cohort when compared to placebo treatment.

A key objective of Study A4091015 and A4091018 was to compare the efficacy of tanezumab 5 mg and 10 mg versus naproxen 500 mg twice daily (BID) for the symptomatic treatment of osteoarthritis. To control the type 1 error rate for multiple comparisons, fixed sequence (step down) testing and the Hochberg procedure were predefined to begin with the highest dose (tanezumab 10 mg) followed by tanezumab 5 mg. As a result, despite replicate statistically significant differences with tanezumab 5 mg versus naproxen across 3 co-primary endpoints, a total of 6 comparisons, it is only possible to conclude statistically that tanezumab was superior to naproxen with respect to WOMAC Physical Function in both studies and the Patient's Global Assessment of Osteoarthritis in one study.

In Study A4091030,[18] a key objective was to demonstrate that tanezumab 5 mg and tanezumab 10 mg were superior, or at a minimum non-inferior, to oxycodone controlled release (CR) 10-40 mg every 12 hours in the treatment of osteoarthritis. This objective was achieved. All four comparisons were statistically significant at Week 8; non-inferiority of tanezumab 10 mg versus oxycodone CR, superiority of tanezumab 10 mg versus oxycodone CR (p=0.018), non-inferiority of tanezumab 5 mg versus oxycodone CR, and superiority of tanezumab 5 mg versus oxycodone CR (p<0.001). In this study, oxycodone CR failed to separate from placebo treatment across all response measures at Week 8. Treatment with tanezumab 10 mg did not provide additional efficacy benefit above that observed with tanezumab 5 mg treatment.

In the largest safety and efficacy study, Study A4091025,[19] the addition of tanezumab 5 mg or 10 mg to NSAID treatment (ie, celecoxib 100 mg BID or naproxen 500 mg BID) did not provide substantial benefit over tanezumab 5 mg or 10 mg monotherapy, respectively. Only two statistically significant treatment differences were observed among the coprimary endpoint comparisons.

1.2.3.2. Overview of Safety in Clinical Studies

Based on data from all subject populations who have received tanezumab in completed clinical studies to date, the adverse drug reactions listed in Table 2 are considered to be expected in subjects who are treated with tanezumab.

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The majority of information regarding the safety profile of tanezumab comes from studies conducted in subjects with osteoarthritis. The safety profile observed in tanezumab to date in other chronic pain subject populations does not differ markedly from the results described below.

Table 2. Adverse Drug Reactions in Subjects Receiving Tanezumab

System Organ Class	Adverse Drug Reaction	Frequency[2]
Nervous system disorders	Burning sensation Carpal tunnel syndrome Hyperesthesia Hypoesthesia Paresthesia	Common
	Allodynia Peripheral neuropathy	Uncommon
Musculoskeletal and connective tissue disorders	Rapidly Progressive Osteoarthritis *(in subjects with underlying osteoarthritis[1])*	Uncommon
	Arthralgia Joint swelling Myalgia Pain in extremity	Common
General disorders and administration site conditions	Peripheral edema	Common

[1] Rapidly Progressive Osteoarthritis may occur in subjects with underlying osteoarthritis. The frequency is estimated from adjudicated events of rapidly progressive osteoarthritis in historic clinical studies of tanezumab, which did not include specific risk minimization measures for this adverse reaction.
[2] Common (≥1% and <10%); Uncommon (≥0.1% and <1%).

A total of 7491 subjects were treated in 9 controlled Phase 3 osteoarthritis studies. The majority of these studies were conducted using IV administration of tanezumab; however, over 900 subjects were treated using SC administration in 2 studies. The adverse event profile of SC administration of tanezumab is comparable to that observed with the IV route based on the results of study A4091027[20] which compared SC versus IV administration in subjects with osteoarthritis. The incidence of adverse events, withdrawals due to adverse events, and serious adverse events in subjects treated with tanezumab monotherapy (5-10 mg) was similar to subjects receiving active comparator treatment and increased over placebo-treated subjects. In the tanezumab 2.5 mg monotherapy treatment group, the incidence of adverse events was similar to active comparator while the incidence of withdrawals due to adverse events and serious adverse events was similar to that of the placebo treatment group. Across the tanezumab monotherapy doses, the rates of adverse events, withdrawals due to adverse events, and serious adverse events were similar with tanezumab 5 mg and 10 mg, and elevated in comparison to tanezumab 2.5 mg. Tanezumab/NSAID combination therapy was associated with higher overall adverse event rates. The relationship of incidence to the dose of tanezumab administered was similar to that observed with tanezumab monotherapy.

Based on data from the Phase 3 osteoarthritis studies and results of an independent adjudication of investigator-reported adverse events of osteonecrosis and total joint

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replacements, the risk of rapidly progressive osteoarthritis with tanezumab treatment was greater than with placebo or active comparator treatment.

Among the most frequently reported adverse events in the controlled Phase 3 osteoarthritis studies, the incidence of peripheral edema, upper respiratory tract infection, fall, arthralgia, back pain, joint swelling, pain in extremity, hypoesthesia, and paresthesia tended to be higher in subjects receiving tanezumab monotherapy than subjects receiving either placebo or active comparator treatment. The incidence of peripheral edema, arthralgia, joint swelling, pain in extremity, and paresthesia increased with increasing doses of tanezumab monotherapy. The adverse events with increased incidence observed with active comparator over tanezumab monotherapy included the following: constipation, nausea, urinary tract infection, nasopharyngitis, osteoarthritis, and hypertension.

The most common adverse events reported in the non-controlled, long-term Phase 3 osteoarthritis studies were similar to those seen in the controlled Phase 3 osteoarthritis studies with the exception of the inclusion of musculoskeletal pain and exclusion of hypertension and nasopharyngititis and all gastrointestinal-related adverse events. Dose-related increases in the incidence of peripheral edema, joint swelling, osteoarthritis and paresthesia were observed.

1.2.3.2.1. Sympathetic Nervous System

In completed Phase 3 osteoarthritis studies, the incidence and discontinuation rates due to adverse events consistent with decreased sympathetic function associated with tanezumab monotherapy (combined doses of 2.5 to 10 mg) were less than or equal to rates with placebo or active comparator. No evidence of dose related elevations in the frequency of adverse events suggestive of decreased sympathetic nervous system function were observed, at doses of 2.5 to 10 mg in subjects with osteoarthritis. Tanezumab 20 mg in chronic low back pain had marginally higher event rates compared to placebo and active comparator treatment groups.

Based on completed osteoarthritis studies where orthostatic blood pressure, heart rate deep breathing, or autonomic symptoms captured with the Neuropathy Symptom Change (NSC) questionnaire were specifically assessed, data are not suggestive of an adverse effect of tanezumab on autonomic function.

Although evidence of clinically important effects on the sympathetic nervous system have not been identified in previously completed tanezumab studies, per agreement with the FDA, this and other clinical studies of tanezumab will incorporate additional safety measures to monitor for and manage subjects who may develop evidence of clinically important sympathetic nervous system dysfunction.

1.2.3.3. Subcutaneous Administration of Tanezumab in Clinical Studies

The formulation of the tanezumab drug product for SC injection is identical to that used for IV administration and is administered as a 1 mL SC injection in the thigh or abdomen. The safety and efficacy of tanezumab when administered by SC injection has been evaluated in OA subjects primarily in two studies, A4091027 and A4091043, and in subjects with chronic

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low back pain in one study (A4091039), all of which were impacted by the FDA imposed clinical hold.[20,21,56] A total of 1905 subjects were treated in these studies. The observed efficacy and safety profile of tanezumab administered SC was similar to IV administration.

Study A4091027[20] was designed as a randomized, placebo-controlled study to demonstrate the efficacy and safety of tanezumab 2.5 mg, 5 mg, or 10 mg SC administered at 8-week intervals and the therapeutic equivalence of tanezumab 10 mg SC and tanezumab 10 mg IV administered at 8-week intervals in subjects with OA of the knee. Due to the clinical hold, enrollment was stopped prematurely and, therefore, was insufficient to yield adequate power to fulfill the primary objectives. The overall incidence of adverse events was comparable to that observed in previous tanezumab IV OA studies, and with the exception of injection-site reactions, the adverse event profile was comparable to that of previous tanezumab IV studies in subjects with OA. The higher incidence of injection-site reaction compared to other studies is likely due to the implementation of directed injection-site assessment – a procedure that was not a component of earlier tanezumab studies, which had included only IV administration of tanezumab. The majority of reported adverse events of injection-site reaction were mild, and none were severe. Across treatment groups, the incidence of serious adverse events and discontinuation due to adverse events was low.

Study A4091043[21] was a randomized, double-blind, parallel-group, multicenter long-term safety study of tanezumab when administered by SC injection in subjects with OA of the knee or hip. Approximately 600 subjects were to be entered into the study. All subjects were randomized to receive SC administration of tanezumab 2.5, 5, or 10 mg at 8-week intervals (7 doses total) for approximately 1 year (56 weeks). This study was impacted by the FDA clinical hold in June 2010. A total of 231, 222, and 226 subjects were assigned to tanezumab 2.5, 5, and 10 mg treatment groups, respectively; all of these subjects were treated with study drug except for 1 subject in the tanezumab 2.5 mg treatment group who was randomized but not treated. No subjects completed the study as a result of the clinical hold. The overall incidence of adverse events (all causality) was dose responsive and was highest in the tanezumab 10 mg (80.1%) treatment group which was followed by successively lower incidences in the tanezumab 5 mg (76.1%) and 2.5 mg (68.7%) treatment groups. Overall and based upon a tendency toward greater incidences of adverse events, serious adverse events, all-cause total joint replacement, abnormalities in neurological examinations, and neurological consultation findings thought to be consistent with new or worsened peripheral neuropathy, the tanezumab 10 mg treatment group appeared to be less well tolerated relative to the tanezumab 2.5 and 5 mg dose levels.

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1.2.3.4. Joint Safety

Following the imposition of the clinical hold, a comprehensive investigation and analyses related to joint-safety were conducted, based on tanezumab monotherapy exposure in over 6400 subjects and tanezumab/NSAID combination therapy in 3400 subjects. There were over 5000 subjects who received tanezumab treatment alone or in combination with NSAIDs for 6 months or longer. The program was sufficient to define and characterize the adverse event of concern – rapidly progressive osteoarthritis – and evaluate the risk of rapidly progressive osteoarthritis in the context of the overall benefit-risk profile of tanezumab compared to standard of care. The results and conclusions regarding tanezumab and the other anti-NGF therapies are provided in detail elsewhere.[1]

After careful investigation, no evidence was found to substantiate that tanezumab is associated with an increased risk of osteonecrosis, a disease process quite distinct from osteoarthritis. A risk of rapidly progressive osteoarthritis was identified. The risk of rapidly progressive osteoarthritis with tanezumab monotherapy was well below that observed with tanezumab/NSAID combination therapy but greater than with placebo or active comparator treatment. A majority of subjects identified with rapidly progressive osteoarthritis had advanced osteoarthritis of the affected joint prior to treatment. The event rate of all-cause joint replacements in subjects with osteoarthritis was comparable among placebo, active comparator, and tanezumab monotherapy treatment groups.

On March 12, 2012 an FDA Arthritis Advisory Committee reviewed these results as well as those prepared by the FDA.[2,3] The Committee endorsed continued clinical development of the anti-nerve growth factor class of compounds with additional measures taken to minimize the risk and further protect subject safety. On August 28, 2012, the US FDA removed the partial clinical hold on tanezumab allowing the resumption of clinical studies for osteoarthritis and all other chronic pain conditions as related to joint safety.

Risk mitigation measures have been developed as an outgrowth of the joint-related safety analyses to reduce the risk of rapidly progressive osteoarthritis using recommendations from discussions with European agencies [United Kingdom's Medicines and Healthcare products Regulatory Agency (MHRA), Germany's Paul Ehrlich Institute (PEI) and Spain's Agency on Medicinal Products and Medical Devices (AEMPS)] as well as the FDA Arthritis Advisory Committee and interactions with FDA. These risk mitigation measures have been included in the tanezumab clinical program and those applicable to this study are outlined as follows:

Risk Minimization: (1) exclude chronic concomitant NSAID use, (2) exclude tanezumab doses that have been explored and do not demonstrate benefit over lower doses in the condition under study, (3) exclude subjects with evidence of rapidly progressive osteoarthritis or risk factors for such from participating in clinical studies, and (4) exclude subjects who are <u>not</u> suitable candidates for total joint replacement from study participation.

Risk Identification and Management: (1) evaluation and follow-up for severe persistent joint pain, (2) extended post-treatment follow-up, (3) a program-level Central Radiograph Reader and subject-level stopping criteria, (4) an Adjudication Committee, and (5) a Data Monitoring Committee (DMC) and protocol-level stopping rules.

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Risk Characterization: (1) Comprehensive evaluation of osteoarthritis medical history prior to study entry, (2) scheduled radiographic assessments during the studies, (3) surgical and post-operative total joint replacement outcomes, and (4) biomarker determinations.

1.2.3.4.1. Rapidly Progressive Osteoarthritis

Rapidly progressive osteoarthritis of the hip was first described by Forestier in 1957 and subsequently described in a number of studies as atrophic osteoarthritis, rapidly destructive osteoarthritis, rapidly destructive arthropathy, rapidly progressive hip disease, or rapidly destructive coxarthrosis.[23,24,25,26,27,28,29,30,31,32,33,34,35] Rapidly progressive hip osteoarthritis is characterized by subjects who typically present with hip pain, often severe, with radiographs that show rapid joint space narrowing as a result of chrondrolysis from a prior radiograph and, subsequently, an osteolytic phase with severe progressive atrophic bone destruction involving the femoral head and the acetabulum. There can be marked flattening of the femoral head and loss of subchondral bone in the weight bearing area and in some cases the femoral head appears sheared off. Osteophytes are typically conspicuously small or absent. Bone sclerosis is often present at sites of impaction of the femoral head and the acetabulum, subchondral detritus is invariably present and bone fragmentation and debris are commonly observed that can lead to synovitis. Lequesne proposed that subjects with 2 mm/year or greater of joint space narrowing or loss of more than 50% of the joint space within 1 year should be considered to have rapidly progressive osteoarthritis.[25] Due to a lack of longitudinal studies, it is not clear what proportion of subjects with rapid loss of joint space (chondrolysis) will progress to have bone destruction. Rapid progression of osteoarthritis has also been described in the shoulder and the knee.[36,37]

The incidence of rapidly progressive osteoarthritis in the overall osteoarthritis population is not well defined. For rapid progression of hip osteoarthritis, the prevalence ranges from approximately 2% to 18% based on clinical case series analyses.[26,27,29,30,31,38]

The pathophysiology of rapidly progressive osteoarthritis is not understood. Various mechanisms have been proposed including; ischemia, venous stasis, local nutritional deficiencies, synovitis, mechanical overloading, NSAID or corticosteroid use, intra articular deposition of hydroxyapatite or pyrophosphate crystals and subchondral insufficiency fractures.[28,31,35,39,40.]

1.2.4. Dose Selection Rationale

Intravenous and subcutaneous administration of tanezumab at doses of 2.5 mg, 5 mg and 10 mg was shown to reduce pain and improve function in a dose-related manner in Phase 3 studies of osteoarthritis. As one of the risk mitigation features identified through analysis of orthopedic safety and efficacy data, no further study of the tanezumab 10 mg dose will be conducted in subjects with osteoarthritis as this dose did not provide sufficient additional efficacy benefit over the 5 mg dose. Unlike the 10 mg dose which did not provide added efficacy benefit over the 5 mg dose, the 5 mg dose is expected to provide added efficacy benefit over the 2.5 mg dose. Based upon prior studies, this additional efficacy benefit is likely to be most evident in subjects considered to have severe symptomatic osteoarthritis as defined above in Section 1.2.3.1.

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The current study will investigate the efficacy and safety of a fixed dose of tanezumab 2.5 mg administered twice at an 8-week interval and a titration regimen in which a first dose of tanezumab 2.5 mg administered on Day 1 will be followed by a dose of tanezumab 5 mg at Week 8, versus placebo. The current study will provide additional experience with the 2.5 mg dose which was shown to provide efficacy benefits with a favorable safety profile in previous Phase 3 clinical trials. In addition, the study will explore the potential additional benefits of a dose titration regimen in which subjects would begin treatment at the 2.5 mg dose then titrate up to a 5 mg dose. All study treatments will be administered by SC injection.

1.2.5. Rationale for Placebo Treatment

In this study, placebo treatment is used as comparator which is considered the gold-standard for assessing efficacy in short-term osteoarthritis studies. This is particularly relevant considering the intended subject population for this study consists of subjects who have had inadequate pain relief with pharmacologic treatments commonly used as treatments in osteoarthritis pain or are unable to take these pain medications due to contraindication or inability to tolerate them. Utilization of placebo as a comparator also necessitates a smaller sample size thus demonstrating the study objectives of efficacy more efficiently than in an active comparator study. In addition, the use of a placebo is most rigorous when the trial endpoints are subjective measures such as those used in this study because of the often great variation in the way individuals perceive and report subject-reported outcomes. This was reported to be particularly relevant for studies involving pain relief.[22] Rescue medication (acetaminophen) will be provided to all subjects for use in the event of inadequate pain relief.

1.2.6. Rationale for Population

The population selected for this study is of subjects with moderate to severe osteoarthritis who have had inadequate pain relief with previous conventional pharmacological treatment options for osteoarthritis or are unable to take these pain medications due to contraindication or inability to tolerate them and are seeking effective treatment options other than /or prior to joint replacement surgery. The rationale for choice of this population is to optimize the potential benefit-risk relationship for subjects entering the study by selecting subjects who have pain that is more severe or treatment-resistant and who have limited treatment options remaining.

Complete information for this compound may be found in the Single Reference Safety Document, which for this study is the Investigator's Brochure.

2. STUDY OBJECTIVES AND ENDPOINTS

2.1. Objectives

Primary Objective

- Demonstrate superior efficacy of tanezumab 2.5 mg administered subcutaneously (SC) and tanezumab 2.5 mg SC titration to 5 mg SC versus placebo at Week 16.

Secondary Objectives

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- Evaluate treatment benefit (efficacy) of a tanezumab 2.5 mg SC titration to 5 mg SC dosing regimen relative to tanezumab 2.5 mg SC alone (descriptive analyses);
- Evaluate the safety of tanezumab 2.5 mg SC and tanezumab 2.5 mg SC titration to 5 mg SC regimens.

2.2. Endpoints

2.2.1. Primary Endpoints

The co-primary efficacy endpoints are:

- Change from Baseline to Week 16 in the WOMAC Pain subscale;
- Change from Baseline to Week 16 in the WOMAC Physical Function subscale;
- Change from Baseline to Week 16 in the Patient's Global Assessment of Osteoarthritis.

2.2.2. Secondary Endpoints

Efficacy Measures:

- WOMAC Pain subscale change from Baseline to Weeks 2, 4, 8, 12 and 24;
- WOMAC Physical Function subscale change from Baseline to Weeks 2, 4, 8, 12 and 24;
- Patient's Global Assessment of Osteoarthritis (5-point Likert scale) change from Baseline to Weeks 2, 4, 8, 12 and 24;
- OMERACT-OARSI responder index at Weeks 2, 4, 8, 12, 16 and 24;
- Treatment Response: Reduction in the WOMAC Pain subscale of ≥30%, ≥50%, ≥70% and ≥90% at Weeks 2, 4, 8, 12, 16 and 24;
- Cumulative distribution of percent change from Baseline in the WOMAC Pain subscale score to Week 16 (endpoint for summary only);
- Treatment Response: Reduction in the WOMAC Physical Function subscale of ≥30%, ≥50%, ≥70% and ≥90% at Weeks 2, 4, 8, 12, 16 and 24;
- Cumulative distribution of percent change from Baseline in the WOMAC Physical Function subscale score to Week 16 (endpoint for summary only);
- Treatment Response: Improvement of ≥2 points in Patient's Global Assessment of Osteoarthritis at Weeks 2, 4, 8, 12, 16 and 24;

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- Average pain score in the index joint change from Baseline to Weeks 1, 2, 3, 4, 6, 8, 10, 12, 16, 20 and 24;
- WOMAC Stiffness subscale change from Baseline to Weeks 2, 4, 8, 12, 16 and 24;
- WOMAC Average change from Baseline to Weeks 2, 4, 8, 12, 16 and 24;
- WOMAC Pain Subscale Item: Pain When Walking on a Flat Surface, change from Baseline to Weeks 2, 4, 8, 12, 16 and 24;
- WOMAC Pain Subscale Item: Pain When Going Up or Down Stairs, change from Baseline to Weeks 2, 4, 8, 12, 16 and 24;
- Work Productivity and Activity Impairment Questionnaire for Osteoarthritis (WPAI:OA) impairment scores change from Baseline to Week 16;
- EuroQol 5 Dimension (EQ-5D-5L™) dimensions and overall health utility score at Baseline, Week 8 and Week 16;
- Health Care Resource Utilization at Baseline, Week 24 and Week 40;
- Incidence and time to discontinuation due to Lack of Efficacy;
- Usage of rescue medication (incidence, number of days) during Weeks 2, 4, 8, 12, 16 and 24;
- Usage of rescue medication (amount taken) during Weeks 2, 4, 8, 12, and 16.

Safety Measures

- Adverse events;
- Standard safety assessments (safety laboratory testing [chemistry, hematology], sitting vital signs, ECG [12-lead]);
- Joint safety adjudication outcomes;
- Total joint replacements;
- Neurologic examination (Neuropathy Impairment Score [NIS]);
- Orthostatic (supine/standing) blood pressure assessments;
- Survey of Autonomic Symptom scores;
- Anti-drug antibody (ADA) assessments;

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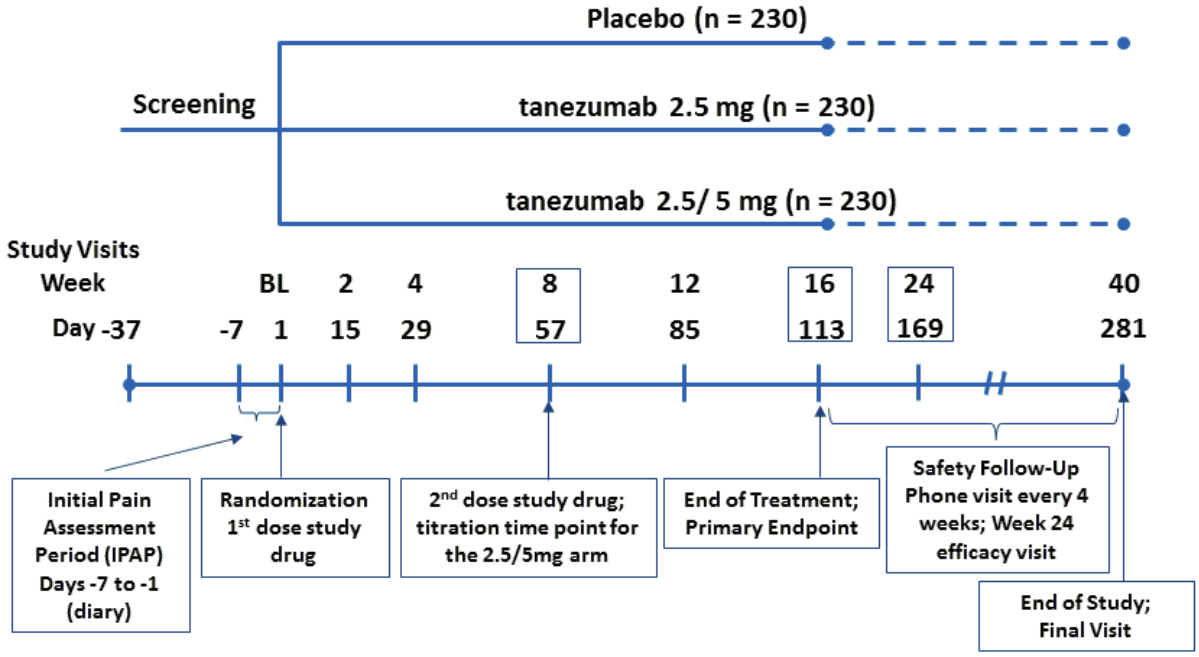
- Physical examinations.

2.2.3. Tertiary Endpoints

Pharmacokinetics and Pharmacodynamics

- Plasma tanezumab concentrations;
- Serum NGF assessments;
- Serum and urine osteoarthritis biomarker concentrations.

3. STUDY DESIGN



This is a randomized, double-blind, placebo-controlled, multicenter, parallel-group, Phase 3 study of the efficacy and safety of tanezumab when administered by SC injection for 16 weeks compared to placebo in subjects with osteoarthritis of the knee or hip. Approximately 690 subjects (approximately 230 per treatment group) will be randomized to one of 3 treatment groups in a 1:1:1 ratio. Subjects will receive a total of two SC injections, separated by 8 weeks:

- tanezumab 2.5 mg (Day 1) and tanezumab 2.5 mg (Week 8);
- tanezumab 2.5 mg (Day 1), and tanezumab 5 mg (Week 8);
- placebo to match tanezumab (Day 1) and placebo to match tanezumab (Week 8).

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The randomization will be stratified by the factors of index joint and highest Kellgren-Lawrence grade in the knee and hip joints. This will result in the following 6 randomization strata:

1. Kellgren-Lawrence Grade 2 and Knee index joint;
2. Kellgren-Lawrence Grade 3 and Knee index joint;
3. Kellgren-Lawrence Grade 4 and Knee index joint;
4. Kellgren-Lawrence Grade 2 and Hip index joint;
5. Kellgren-Lawrence Grade 3 and Hip index joint;
6. Kellgren-Lawrence Grade 4 and Hip index joint.

This study is designed with a total (post-randomization) duration of 40 weeks and will consist of three periods: Screening (up to a maximum of 37 days), Double-blind Treatment (16 weeks, 6 in-clinic visits), and Safety Follow-up (24 weeks). The Screening Period (to begin up to 37 days prior to Randomization) includes a Washout Period (lasting a minimum of 2 days for all prohibited pain medications) if required, and an Initial Pain Assessment Period (IPAP; 7 days prior to Randomization/Baseline; minimum 3 days).

To be eligible for the study, subjects must have documented history indicating that previous treatment with acetaminophen, NSAIDs and either tramadol or opioids have not provided adequate pain relief, or that they could not be taken by the subject due to contraindication or inability to tolerate.

During the Screening Period all subjects will undergo knee, hip, and shoulder x-rays. Other major joints exhibiting signs or symptoms suggestive of osteoarthritis should also be imaged. A major joint is defined as a mobile synovial joint in the limbs such as the shoulders, elbows, wrists, hips, knees, ankle and excluding the joints of the toes and hands. All x-rays will be assessed by a Central Reader to determine subject radiographic eligibility for study participation. A pain score will be collected on an 11-point NRS via IRT for all joints for which a radiograph is obtained. Screening X-rays should be obtained as early as possible in the Screening period to facilitate completion of all required screening procedures within the 37-day Screening period.

In addition, the Screening Period will include the discontinuation and washout of all prohibited pain medications for at least 5 times the elimination half-life. The minimum washout period is 2 days for all prohibited medications that have an elimination half-life of less than 10 hours. Subjects who do not require a washout of prohibited pain medications may begin the IPAP preferably after x-ray confirmation of radiographic eligibility of the subject has been received from the Central Reader.

The Initial Pain Assessment Period, during which the daily average pain score in the index joint (scored with an 11-point numerical rating scale [NRS]) will be collected via Interactive

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Response Technology (IRT), will begin up to 7 days prior to the Baseline (Day 1) randomization visit, and subjects must complete at least 3 diary entries during the IPAP period. Collection of other daily and weekly assessments via IRT is also performed during the IPAP.

During the Washout Period and the Initial Pain Assessment Period, subjects will be provided with rescue medication (acetaminophen) that may be taken if necessary, up to a maximum daily dose of 3000 mg per day, but must be discontinued at least 24 hours prior to the Baseline (Randomization) Visit.

Those subjects who qualify at the Baseline Visit (Day 1) will be randomized to 1 of 3 treatment groups.

Tanezumab 2.5 mg or placebo will be administered as a SC injection and the subject will be observed for adverse events including signs and symptoms of hypersensitivity in the clinic for a minimum of 1 hour after administration of investigational product. Subjects will return at Weeks 2 and 4 for interim visits and at Week 8 for the second administration of investigational product (placebo SC, tanezumab 2.5 mg SC or tanezumab 5 mg SC). The subjects will then return for clinic visits at Weeks 12 and 16.

The Week 16 visit marks both the end of the treatment period and the beginning of the 24-week Follow-up period. After completion of the Week 16 visit, subjects will be asked to return to the clinic for 2 additional study visits at Study Weeks 24 and 40. Between the clinic visits in the Follow-up period, subjects will be contacted by clinical research site staff at approximately monthly intervals to collect adverse event, concomitant drug and concomitant non-drug information.

At the Week 24 visit, (16 weeks after the last dose of investigational product which corresponds to more than 5 times the elimination half-life of tanezumab) general safety assessments and the last study efficacy assessments will be collected. Following this visit, standard of care treatment may be initiated. Standard of care treatment refers to approved analgesics used to relieve the pain of OA; these include opioids, topical analgesics, NSAIDs, capsaicin products, injectable corticosteroids and viscosupplementation (eg, hyaluronan) and are prescribed at the discretion of the investigator. Pre-specified analgesics will be reimbursed by Pfizer.

Subjects will then return for the Week 40 End of Study visit which marks completion of the Follow-Up period. At this visit, all End of Study procedures are to be completed, including repeat knee, hip, and shoulder x-rays as well as any additional major joint that was imaged at Screening or any at risk major joint (See Section 7.4.4) identified during the study period. The window for obtaining end of study x-rays is 30 days from the nominal time point of the visit. The Week 40 (End of Study) X-rays should be obtained as close as possible to the Week 40 visit, preferably before the visit and not more than 14 days after the Week 40 visit.

Continuing on from the Initial Pain Assessment Period, subjects will provide a daily assessment of their index joint pain and rescue medication use through Week 16 (End of Treatment) then weekly between Weeks 16 and 40 (End of Study) via IRT, using a 24-hour

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recall period. Subjects will also provide a weekly assessment of pain in a major, non-index joint via IRT using a 24-hour recall period from the Initial Pain Assessment Period to Week 40 (as described in Section 7.1.1). In addition, on a weekly basis from the IPAP to Week 40, subjects will record use of concomitant NSAID via IRT.

The treatment period ends and the 24 week Safety Follow-up period begins at the Week 16 visit. Subjects who complete the Week 16 visit will be considered to have completed the Double-blind Treatment Period and will enter the 24-week Safety Follow-up period. Subjects who leave the study prior to completing the Week 16 visit will not be considered to have completed the Double-blind Treatment Period.

Subjects who are discontinued from treatment prior to Week 16, either at their request or at the decision of the investigator, will also be required to undergo 24 weeks of follow-up (referred to as Early Termination Follow-Up).

Subjects who complete the 24-week post treatment follow-up period (safety follow-up or Early Termination Follow-up) will be considered to have completed the study, whether the subject completed the treatment period or discontinued prior to completing the treatment period. Subjects who do not complete the 24 week post-treatment follow-up period will not be considered to have completed the study.

The 24 weeks of follow-up will be obtained through 3 clinic visits and monthly phone calls to yield 24-weeks of post-treatment follow-up, as described in the Schedule of Activities and Section 6.16.1 and subsections. In addition, subjects will be asked to continue to enter pain scores for index joints (and major, non-index joints when applicable) via IRT, weekly, through the end of the 24-week follow-up period. Subjects who have entered the Early Termination Follow-up period will also be asked to record weekly rescue medication and concomitant NSAID use via IRT.

X-rays of the shoulders, knees and hips (both shoulders, both knees and both hips) and any other joint imaged at Screening or identified as at risk during the study) should be performed as soon as possible after the decision to withdraw from the study has been made, provided at least 30 days have passed since the last set of X-rays were collected. The remainder of efficacy and safety assessments should be done at the scheduled first visit which is to occur 8 weeks after the last dose of investigational product (as described in 6.16.1.1). The site should also schedule the subject for two additional clinic visits. The second visit should be scheduled to occur approximately 16 weeks after the subject's last dose of investigational product (which corresponds to more than 5 times the elimination half-life of tanezumab) to collect safety and efficacy data (as described in 6.16.1.3). Once the clinic visit 16-weeks after the last administration of investigational product has been completed and final efficacy assessments have been collected, standard of care treatment will be offered to subjects for the remaining 8 weeks of the required Follow-Up period. Standard of care treatment may be initiated as needed and should be recorded on the concomitant medication CRF. The third and final clinic visit should be scheduled to take place approximately 24-weeks after the subject received the last dose of investigational product. That visit (described in 6.16.1.5), will include repeat shoulder, knee and hip radiographs (and any other joint imaged at

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Screening or identified as at risk during the study) providing at least 30 days have elapsed since the last radiographs were obtained. The window for obtaining end of Safety Follow-Up X-rays is 30 days from the nominal time of the visit however should be obtained as close as possible to the End of Safety Follow-Up visit (Early Termination Visit 3), preferably before and not more than 14 days after the Visit. Every effort should be made to have the subject agree to complete the entire 24 week Early Termination Safety Follow-Up described above.

In the event a subject refuses the Early Termination Safety Follow-up, or chooses to discontinue during the regularly planned Safety Follow-up period (after Week 16 and through Week 40), a complete early termination visit should be performed. This early termination visit should include all procedures scheduled for the Week 24 and Week 40 visits, unless the Week 24 visit has already been completed; in that case, only Week 40 procedures will be required. In addition, if the Week 16 visit was not completed prior to termination, a general physical examination, body weight, WPAI:OA and EQ-5D-5L will also be obtained. Female subjects of child-bearing potential will be advised to continue their contraception regimen during a period of 112 days (16 weeks) after the last dose of investigational product.

Subjects who have undergone or plan to undergo total joint replacement or other arthroplasty procedure during the study will be discontinued from study treatment. Follow-up procedures for these subjects are described in Section 6.16.2. In addition, subjects who undergo total knee, hip or shoulder joint replacement surgery during the study (Treatment Period or Follow-up Period) will be followed for 24 weeks after the procedure as part of a separate protocol provided the subject consents (refer to Section 6.17).

4. SUBJECT SELECTION

This study can fulfill its objectives only if appropriate subjects are enrolled. The following eligibility criteria are designed to select subjects for whom participation in the study is considered appropriate. All relevant medical and non-medical conditions should be taken into consideration when deciding whether a particular subject is suitable for this protocol.

4.1. Inclusion Criteria

Subject eligibility must be reviewed and documented by an appropriately qualified member of the investigator's study team before subjects are included in the study.

Subjects must meet all of the following inclusion criteria to be eligible for enrollment into the study:

1. Evidence of a personally signed and dated informed consent document indicating that the subject (or a legally acceptable representative) has been informed of all pertinent aspects of the study.

2. Male or female of any race, ≥18 years of age.

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3. Diagnosis of osteoarthritis of the hip or knee in the index joint based on American College of Rheumatology criteria[41,53] with x-ray confirmation (a Kellgren-Lawrence[42] x-ray grade of ≥2 as diagnosed by the Central Reader; Appendix 1).

4. Subjects must meet the following criteria:

   - Documented history indicating that acetaminophen therapy has not provided sufficient pain relief;

   - Documented history indicating that oral NSAID therapy has not provided adequate pain relief or subject is unable to take NSAIDs due to contraindication or inability to tolerate;

     AND at least 1 of the following criteria:

   - Documented history indicating that tramadol treatment has not provided adequate pain relief or subject is unable to take tramadol due to contraindication or inability to tolerate;

   - Documented history indicating that opioid treatment has not provided adequate pain relief or subject is unwilling to take opioids, or unable to take opioids due to contraindication or inability to tolerate.

5. WOMAC Pain subscale NRS ≥5 in the index joint at Screening.

6. Subjects must be willing to discontinue all pain medications for osteoarthritis except rescue medication (acetaminophen) and not use prohibited pain medications throughout the duration of the study except as permitted per protocol.

7. Female subjects of childbearing potential and at risk for pregnancy must agree to use 2 highly effective methods of contraception throughout the study and for 112 days (16 weeks) after the last dose of assigned treatment.

8. Female subjects who are not of childbearing potential (ie, must meet at least 1 of the following criteria):

   - Have undergone a documented total hysterectomy and/or bilateral oophorectomy;

   - Have medically confirmed ovarian failure; or

   - Achieved post-menopausal status defined as follows: cessation of regular menses for at least 12 consecutive months with no alternative pathological or physiological cause and have a serum FSH level confirming the post-menopausal state.

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9. Subjects who are willing and able to comply with lifestyle guidelines, scheduled visits, treatment plan, laboratory tests, and other study procedures through the End of Study visit.

4.2. Exclusion Criteria

Subjects presenting with any of the following characteristics/conditions will not be included in the study:

1. Subjects who are investigational site staff members directly involved in the conduct of the study and their family members, site staff members otherwise supervised by the Investigator, or subjects who are Pfizer employees directly involved in the conduct of the study.

2. Body Mass Index (BMI) of >39 kg/m^2.

3. History of other disease that may involve the index joint including inflammatory joint disease such as rheumatoid arthritis, seronegative spondyloarthropathy (eg, ankylosing spondylitis, psoriatic arthritis, inflammatory bowel disease related arthropathy), crystalline disease (eg, gout or pseudogout), endocrinopathies, metabolic joint diseases, lupus erythematosus, joint infections, Paget's disease, or tumors.

4. Radiographic evidence of any of the following conditions in any screening radiograph as determined by the central radiology reviewer and as defined in the tanezumab program imaging atlas: excessive malalignment of the knee, severe chondrocalcinosis; other arthropathies (eg, rheumatoid arthritis), systemic metabolic bone disease (eg, pseudogout, Paget's disease; metastatic calcifications), large cystic lesions, primary or metastatic tumor lesions, stress or traumatic fracture.

5. Radiographic evidence of any of the following conditions as determined by the central radiology reviewer and as defined in the tanezumab program imaging atlas at Screening: 1) rapidly progressive osteoarthritis, 2) atrophic or hypotrophic osteoarthritis, 3) subchondral insufficiency fractures, 4) spontaneous osteonecrosis of the knee (SPONK), 5) osteonecrosis, or 6) pathologic fracture.

6. A history of osteonecrosis or osteoporotic fracture (ie, a subject with a history of osteoporosis and a minimally traumatic or atraumatic fracture).

7. History of significant trauma or surgery to a knee, hip, or shoulder within the previous year.

8. Planned surgical procedure during the duration of the study.

9. Largely or wholly incapacitated, (eg, subject bedridden or confined to a wheelchair, permitting little or no self-care).

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10. Fibromyalgia, regional pain caused by lumbar or cervical compression with radiculopathy or other moderate to severe pain that may confound assessments or self-evaluation of the pain associated with osteoarthritis. Subjects with a present (current) history of sciatica are not eligible for participation. Subjects with a past history of sciatica who have been asymptomatic for at least one year and who have no evidence of radiculopathy or sciatic neuropathy on thorough neurologic examination are eligible participation.

11. Past history of carpal tunnel syndrome (CTS) with signs or symptoms of CTS in the one year prior to Screening.

12. Subjects considered unfit for surgery, defined as Grade >3 on the American Society of Anesthesiologists (ASA) physical classification system for surgery (see Appendix 2), or subjects who would not be willing to undergo joint replacement surgery if required.

13. History of intolerance or hypersensitivity to acetaminophen (paracetamol) or any of its excipients or existence of a medical condition or use of concomitant medication for which the use of acetaminophen is contraindicated (refer to product labeling).

14. Use of prohibited medications without the appropriate washout period (if applicable) prior to Screening or Initial Pain Assessment Period (refer to Section 5.8.1).

15. Oral or intramuscular corticosteroids within 30 days prior to the Initial Pain Assessment Period.

16. Intra-articular corticosteroid injection in the index joint within 12 weeks, or to any other joint within 30 days prior to the Initial Pain Assessment Period.

17. Intra-articular hyaluronic acid injection in the index joint within 30 days (or within 18 weeks for long-acting formulations such as Synvisc) prior to the Initial Pain Assessment Period.

18. History of cancer within 5 years prior to Screening, except for cutaneous basal cell or squamous cell cancer resolved by excision.

19. Signs and symptoms of clinically significant cardiac disease including but not limited to:

    - Ischemic cardiac disease (eg, unstable angina, myocardial infarction) in the 6 months prior to Screening;

    - Surgery or stent placement for coronary artery disease in the 6 months prior to Screening;

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- New York Heart Association (NYHA) Class III or IV congestive heart failure or known left ventricular dysfunction with ejection fraction ≤35%, cardiomyopathy, myocarditis in the 6 months prior to Screening;
- Resting tachycardia (heart rate ≥120) or resting bradycardia (heart rate ≤45) on ECG at Screening;
- QTcF interval >500 msec in the absence of confounding factors like bundle branch block or paced rhythm at Screening;
- Any other cardiovascular illness that in the opinion of the investigator would render a subject unsuitable to participate in the study.

  Subjects with a history of heart block now controlled by a functioning cardiac pacemaker are eligible.

20. Diagnosis of a transient ischemic attack in the 6 months prior to Screening, diagnosis of stroke with residual deficits (eg, aphasia, substantial motor or sensory deficits), that would preclude completion of required study activities.

21. History, diagnosis, or signs and symptoms of clinically significant neurological disease, including but not limited to:

    - Alzheimer's disease or other types of dementia;
    - Clinically significant head trauma within the past year;
    - Peripheral or autonomic neuropathy;
    - Multiple sclerosis;
    - Epilepsy or seizure disorder with history of seizure within the last 2 years;
    - Myopathy.

22. History, diagnosis, signs or symptoms of any clinically significant psychiatric disorder, including but not limited to:

    - Psychotic disorders;
    - Somatoform disorders;
    - Bipolar disorders;
    - Hospital admission for depression or suicide attempt within 5 years of Screening, or active severe major depression at Screening (determined from medical history: if needed, severity of depression may be assessed using the Patient Health

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Questionnaire [PHQ-9].[43] A score of ≥15 on questions 1-9 of the PHQ-9 corresponds to severe depression. See Appendix 9);

- Any other psychiatric illness that in the opinion of the investigator would render a subject unsuitable to participate in the study.

23. History of known alcohol, analgesic or drug abuse within 2 years of Screening.

24. Previous exposure to exogenous NGF or to an anti-NGF antibody.

25. History of allergic or anaphylactic reaction to a therapeutic or diagnostic monoclonal antibody or IgG-fusion protein.

26. Resting, sitting blood pressure (BP) ≥160 mm Hg in systolic pressure or ≥100 mm Hg in diastolic pressure at Screening. If a subject is found to have untreated significant hypertension at Screening and antihypertensive treatment is initiated, assessment for study eligibility should be deferred until BP and antihypertensive medication have been stable for at least one month. For subjects with previously diagnosed hypertension, antihypertensive medications must be stable for at least 1 month prior to Screening.

27. Subjects who have evidence of orthostatic hypotension based upon replicate orthostatic blood pressure measurements (See 7.3.4.1). If orthostatic blood pressure change cannot be determined (eg, unable to establish a stable supine systolic and diastolic blood pressure) then that subject is not eligible for the study.

28. Subjects with a total impact score of >7 on the Survey of Autonomic Symptoms (SAS, see Appendix 11).

29. Alanine aminotransferase (ALT) or aspartate aminotransferase (AST) ≥3.0 times the upper limit of normal, or creatinine exceeding 1.7 mg/dL (150 μmol/L) in men or 1.5 mg/dL (133 μmol/L) in women, or hemoglobin A1c ≥10% at Screening. Repeat confirmatory tests may be performed.

30. Presence of drugs of abuse (including prescription medications without a valid prescription), or illegal drugs in the urine toxicology screen obtained at Screening.

31. Positive Hepatitis B, Hepatitis C, or Human Immunodeficiency Virus (HIV) tests at Screening indicative of current infection.

32. Participation in other studies involving investigational drug(s) (Phases 1-4) within 30 days (or 90 days for biologics) before Screening and/or during study participation.

33. Pregnant female subjects; breastfeeding female subjects; female subjects of childbearing potential who are unwilling or unable to use two (2) highly effective methods of contraception as outlined in this protocol for the duration of the study and for 112 days (16 weeks) after last dose of investigational product.

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34. Other severe acute or chronic medical or psychiatric condition or laboratory abnormality that may increase the risk associated with study participation or investigational product administration or may interfere with the interpretation of study results and, in the judgment of the investigator, would make the subject inappropriate for entry into this study.

4.3. Randomization Criteria

In addition to meeting all inclusion and exclusion criteria requirements listed above, the following randomization criteria must be met before randomization can be called into the IRT system at the Baseline visit:

1. Subject must have completed appropriate washout of analgesics.
2. Subject must have made at least 3 pain diary entries in the 7 days prior to the Baseline (Day 1) visit.
3. Subject must have abstained from taking rescue medication (acetaminophen) within the 24 hours that precede dosing.
4. WOMAC Pain subscale NRS ≥5 in the index joint at Baseline.
5. WOMAC Physical Function subscale NRS ≥5 in the index joint at Baseline.
6. Patient's Global Assessment of Osteoarthritis must be "fair", "poor" or "very poor" at Baseline.
7. Review of the ECG and laboratory results and confirmation that there are no clinically significant or exclusionary findings.
8. Radiographic eligibility must have been confirmed by the Central Reader, including KL grade ≥2 in the index joint.

4.4. Life Style Guidelines

Subjects should maintain their normal daily routine, including stable doses of permitted medications and exercise program. Subjects are also permitted to continue with non-pharmacologic activities (eg, massage, physical therapy) during the trial. Subjects should be cautioned against initiating or altering strenuous exercise regimens during the study as this may influence efficacy and laboratory results.

Subjects will be advised to avoid elective surgery (eg, oral surgery) during the course of the study if possible; the study clinician should be contacted for discussion prior to the surgery whenever possible. Subjects who undergo joint replacement or arthroplasty will be discontinued from study treatment as described in 6.16.2 and followed as part of a separate protocol as described in Section 6.17.

Refer to Section 5.8 and Appendix 3 for guidance on permitted and prohibited medications.

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All female subjects who are of childbearing potential and are sexually active and at risk of pregnancy must agree to use two (2) methods of highly effective contraception consistently and correctly throughout the duration of the active treatment period and for 112 days (16 weeks) after the last dose of investigational product. The investigator or his or her designee, in consultation with the subject, will confirm that the subject has selected 2 appropriate forms of contraception for the individual subject from the permitted list of contraception methods (see below), and instruct the subject in their consistent and correct use. Subjects need to affirm that they meet the criteria for correct use of at least two of the selected methods of contraception. The investigator or his/her designee will discuss with the subject the need to use highly effective contraception consistently and correctly according to the Schedule of Activities (SOA) and document such conversation in the subject's chart. In addition, the investigator or his or her designee will instruct the subject to call immediately if a selected contraceptive method is discontinued or if pregnancy is known or suspected in the subject or the subject's partner.

Highly effective methods of contraception are those that, alone or in combination, result in a failure rate of less than 1% per year when used consistently and correctly (ie, perfect use) and include the following:

1. Established use of oral, inserted, injected, implanted or transdermal hormonal methods of contraception is allowed provided the subject plans to remain on the same treatment throughout the entire study and has been using that hormonal contraceptive for an adequate period of time to ensure effectiveness.

2. Correctly placed copper-containing intrauterine device (IUD).

3. Male condom or female condom used WITH a spermicide (ie, foam, gel, film, cream, or suppository). For countries where spermicide is not available or condom plus spermicide is not accepted as highly effective contraception, this option is not appropriate.

4. Male sterilization with absence of sperm in the post-vasectomy ejaculate.

5. Bilateral tubal ligation/bilateral salpingectomy or bilateral tubal occlusive procedure (provided that occlusion has been confirmed in accordance with the device's label).

4.5. Sponsor Qualified Medical Personnel

The contact information for the sponsor's appropriately qualified medical personnel for the trial is documented in the study contact list located in the study manual.

To facilitate access to appropriately qualified medical personnel on study related medical questions or problems, subjects are provided with a contact card. The contact card contains, at a minimum, protocol and investigational compound identifiers, subject study numbers, contact information for the investigational site and contact details for a contact center in the event that the investigational site staff cannot be reached to provide advice on a medical question or problem originating from another healthcare professional not involved in the

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subjects participation in the study. The contact number can also be used by investigational staff if they are seeking advice on medical questions or problems, however it should be used only in the event that the established communication pathways between the investigational site and the study team are not available. It is therefore intended to augment, but not replace the established communication pathways between the investigational site and study team for advice on medical questions or problems that may arise during the study. The contact number is not intended for use by the subject directly and if a subject calls that number he or she will be directed back to the investigational site.

5. STUDY TREATMENTS

For the purposes of this study, and per International Conference on Harmonization (ICH) guidelines investigational product is defined as a pharmaceutical form of an active ingredient or placebo being tested or used as a reference in a clinical trial, including a product with a marketing authorization when used or assembled (formulated or packaged) in a way different from the approved form, or when used for an unapproved indication, or when used to gain further information about an approved use (ICH E6 1.33).

5.1. Allocation to Treatment

Subjects will be randomized at Baseline to one of the following treatments:

Treatment Group
1. Placebo SC at Baseline and Week 8
2. Tanezumab SC 2.5 mg at Baseline and Week 8
3. Tanezumab SC 2.5 mg at Baseline and tanezumab SC 5 mg at Week 8

Subjects will be randomly assigned in a 1:1:1 ratio to the above treatment regimens according to a computer generated randomization code. As detailed in Section 3, randomization will be stratified by index joint and most severe Kellgren-Lawrence grade (of any knee or hip joint). Randomization will be coordinated centrally through Interactive Response Technology (IRT). The system will provide subject identification numbers at Screening, which are subsequently linked to the treatment assignments at Randomization. The randomization code will be maintained by a person(s) who is independent of the trial conduct and produces the randomization code. It is the responsibility of the Principal Investigator to ensure that the subject is eligible for participation in the study before requesting Randomization. The study site will obtain the subject's randomization number and dispensable unit identification numbers (ie, the drug supply to be administered) from the IRT. Further details are provided in the Pharmacy Manual.

5.2. Breaking the Blind

This is a randomized, double-blind, parallel group study. The subjects, investigators, Study Coordinators, clinical site staff, Clinical Research Associate (CRA), and staff directly involved with the study at Pfizer and its designees will be blinded to subject assignment.

Blinding should only be broken in emergency situations for reasons of subject safety. Whenever possible, the investigator or sub-investigator consults with a member of the study

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team prior to breaking the blind. When the blinding code is broken, the reason must be fully documented and entered on the CRF.

5.3. Subject Compliance

Tanezumab and placebo SC dosing will be recorded on the appropriate CRF. Because investigational product is administered by site staff, subject compliance with SC treatment is not anticipated to be an issue.

Compliance with the use of rescue medication will be reviewed and reconciled at each study visit. Refer to Section 5.9 for rules governing rescue medication use.

5.4. Investigational Product Supplies

5.4.1. Dosage Form(s) and Packaging

Tanezumab and placebo for tanezumab will be supplied by the Sponsor or designee.

5.4.1.1. Tanezumab

Tanezumab 2.5 mg is presented as a sterile solution for subcutaneous administration, in a glass pre-filled syringe (PFS). Each PFS contains a sufficient amount of tanezumab to provide the intended dose of drug at a concentration of 2.5 mg/mL.

Tanezumab 5 mg is presented as a sterile solution for subcutaneous administration, in a glass pre-filled syringe (PFS). Each PFS contains a sufficient amount of tanezumab to provide the intended dose of drug at a concentration of 5 mg/mL.

Each prefilled syringe is packed in an individual carton. Each prefilled syringe has a unique container number.

5.4.1.2. Placebo for Tanezumab

Placebo for tanezumab is presented as a sterile solution for subcutaneous administration, in a matching glass prefilled syringe (PFS). Each prefilled syringe has a unique container number.

5.4.1.3. Acetaminophen (rescue medication)

Acetaminophen will be issued by the study sites in its approved marketed product dress, (carton, bottle, documents). Any approved commercial product of acetaminophen tablet/caplet/capsule is permitted.

5.4.2. Preparation and Dispensing

See the Drug Administration Instructions (DAI) for instructions on how to prepare tanezumab 2.5 mg SC, 5 mg SC and placebo SC for administration. Investigational product should be prepared and dispensed by an appropriately qualified and experienced member of the study staff (eg, physician, nurse, physician's assistant, practitioner, or pharmacist) as allowed by local, state, and institutional guidance.

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5.5. Administration

Tanezumab or matching placebo will be administered via SC injection by an appropriately qualified and experienced member of the study staff (eg, physician, nurse, physician's assistant, practitioner, or pharmacist) as allowed by local, state, and institutional guidance and where facilities to handle allergic reactions are available. All subjects will receive 1 mL of investigational product administered as a SC injection. Subcutaneous injections are to be administered in the abdomen or anterior aspect of the thigh. Selection of the SC injection site for each injection will be at the discretion of the investigator taking into account subject preferences when possible. The SC injection should not be administered in areas where the skin is burned, reddened, inflamed, swollen, or scarred.

5.6. Investigational Product Storage

Tanezumab and placebo will be shipped and stored at a temperature between 2° and 8°C and protected from light.

Acetaminophen should be stored according to the product's directions.

The investigator or an approved representative (eg, Pharmacist) will ensure that all investigational product is stored in a secure area with controlled access under required storage conditions and in accordance with applicable regulatory requirements.

Investigational product should be stored in its original container and in accordance with the drug label. See the Pharmacy Manual for storage conditions of the product.

Storage conditions stated in the single reference safety document (SRSD) (ie, investigator's brochure [IB]), will be superseded by the storage conditions stated in the labeling.

Site systems must be capable of measuring and documenting (for example, via a log), at a minimum, daily minimum and maximum temperatures for all site storage locations (as applicable, including frozen, refrigerated and/or room temperature products). This should be captured from the time of investigational product receipt throughout the study. Even for continuous monitoring systems, a log or site procedure that ensures active daily evaluation for excursions should be available. The operation of the temperature monitoring device and storage unit (for example, refrigerator), as applicable, should be regularly inspected to ensure it is maintained in working order.

Any excursions from the product label storage conditions should be reported upon discovery. The site should actively pursue options for returning the product to the storage conditions, described in the labeling, as soon as possible. Deviations from the storage requirements, including any actions taken, must be documented and reported to the sponsor.

Once an excursion is identified, the investigational product must be quarantined and not used until the sponsor provides documentation of permission to use the investigational product. It will not be considered a protocol deviation if the sponsor approves the use of the investigational product after the temperature excursion. Use of the investigational product

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prior to sponsor approval will be considered a protocol deviation. Specific details regarding information the site should report for each excursion will be provided to the site.

5.7. Investigational Product Accountability

The investigative site must maintain adequate records documenting the receipt, use, loss, or other disposition of the investigational product(s).

5.7.1. Destruction of Investigational Product Supplies

The Sponsor or designee will provide guidance on the destruction of unused investigational product (eg, at the site). If destruction is authorized to take place at the study site, the investigator must ensure that the materials are destroyed in compliance with applicable environmental regulations, institutional policy, and any special instructions provided by Pfizer, and all destruction must be adequately documented.

5.8. Concomitant Medication(s)

5.8.1. Prohibited Medications

Use of analgesics (including marijuana) other than acetaminophen is prohibited throughout Week 24 of the study beginning 48 hours prior to the start of the Initial Pain Assessment Period or at the period of time prior to the start of the Initial Pain Assessment Period that is ≤5 times the half-life of the particular analgesic used, whichever is greater. Refer to Appendix 3: Half-Lives of Prohibited Prior and Concomitant Medications, for a detailed washout schedule for prohibited medications. This is not an exhaustive list and Pfizer study management should be consulted for assistance if needed, in determining whether or not specific medications are permitted. Sites must consult product labeling and conduct a taper according to the product instructions if a taper is required.

Use of prescription or over-the-counter (OTC) NSAIDs (including COX-2 selective inhibitors) is also prohibited except in the following circumstances. Limited concomitant use of prescription or OTC NSAID may be allowed on an occasional basis for self-limiting conditions not related to osteoarthritis however, they may not be taken within 48 hours (or 5 half-lives, whichever is greater) of a study visit where efficacy assessments are being collected. The study monitor or Pfizer clinician should be contacted for approval prior to use whenever possible, and all doses and days of use will be recorded on the concomitant analgesic CRF. NSAID use should not exceed a total of 30 days between Day 1 (Baseline) and Week 24. Beginning at Week 24 (16-weeks after the last dose of investigational product), subjects may use NSAIDs at their discretion.

Subjects will be instructed that many over-the-counter medications contain NSAIDs and to guard against overuse. Subjects will be instructed to keep a recording of NSAID usage and amount. The number of days of NSAID use will be collected weekly via IRT from Baseline (Day 1) through the Week 40 visit. Additional information regarding NSAID use such as medication names, dosage and reason for use will be collected by the site on a CRF.

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NSAID usage will be monitored at 2 levels: the cumulative use from Baseline to Week 24 (a maximum of 30 days of use) and the aggregate days of use during each 8 week dosing interval (a maximum of 10 days of use).

Subjects who exceed the allowable concomitant NSAID use limit of 30 days of use during the treatment phase of the study will be discontinued from the study and entered in the Early Termination Follow-up Period. Subjects taking greater than 10 days of aggregate use of NSAIDs per 8-week dosing interval (any dosage level of NSAIDs) should be interviewed by study site personnel to determine reason for use and if the subject anticipates being able to take NSAIDs according to protocol requirements in the future. The discussion should be noted in the subject's source documents. Subjects who indicate that they anticipate being able to take NSAIDs no more than 10 days per 8-week period going forward will be allowed to continue in the treatment period. However, if these subjects continue to take NSAIDs more than 10 days per 8-week period, they should be withdrawn from study treatment and entered into the Early Termination Follow-Up period (See Section 6.16). Subjects who indicate they are taking NSAIDs because of insufficient osteoarthritis pain relief or those who indicate they cannot or will not follow the protocol specified restrictions for NSAID use due to any other condition, should be withdrawn from study treatment and entered in the Early Termination Follow-Up period (See Section 6.16).

Herbal, homeopathic, and naturopathic remedies should not be initiated during the study; however, subjects who have taken a stable dose of these products for at least 30 days prior to the Initial Pain Assessment Period will be allowed to continue their regimen. Subjects should be advised that St. John's Wort and other inducers of CYP3A4/5 such as carbamazepine and rifampin may interfere with the efficacy of hormonal contraceptive products. Biologics (for example: TNF-α inhibitors such as adalimumab, etanercept, infliximab) other than investigational product, and live attenuated vaccines must not have been taken within 3 months prior to the Initial Pain Assessment Period and are prohibited during the study. Flumist® Influenza Virus Vaccine Live, Intranasal or other inhaled influenza vaccines are the only exception of a live attenuated vaccine that will be permitted during the study.

5.8.2. Permitted Medications

Daily low dose aspirin (≤325 mg) therapy for cardiovascular prophylaxis is permitted without restriction.

Medications for other (non-OA, non-pain) conditions are permitted provided the subject has received a stable dose for at least 30 days before the Initial Pain Assessment Period (30 days prior to Screening for antihypertensive medications) and the dose is not expected to change during the study. Note however that dose adjustments (including starting a new needed therapy) during the study can be made if required, and recorded on the concomitant medication CRF.

Occasional use of pain medications for pain is permitted in situations such as outpatient diagnostic procedures (eg, colonoscopy, dental procedures) or limited accidental injury (eg, ankle sprains, minor fractures, minor burns/sunburns). The subject should be counseled

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to avoid scheduling prospective procedures such that pain medications would be needed within 48 hours of a study visit (24 hours for acetaminophen). Contact study management for guidance/approval regarding the use of prohibited medications for other self-limiting conditions, accidental injury or other surgical procedures as the extent of the condition, injury or procedure and the resulting pain medication usage may require termination from the study. Any use of NSAID must be consistent with the allowed limit described in the Prohibited Medications Section 5.8.1 above.

For subjects who have discontinued investigational product, the investigator may prescribe standard of care treatment for subjects once the last efficacy assessments have been obtained at the visit occurring 16 weeks after the last dose of investigational product. In this study, standard of care treatment refers to analgesics approved by FDA (for US subjects) or another applicable Health Authority (for non-US subjects) and used to relieve the pain of OA. These medications include opioids, topical analgesics, NSAIDs, capsaicin products, injectable corticosteroids and viscosupplementation (eg, hyaluronan) and are prescribed at the discretion of the investigator. Pre-specified analgesics are not considered investigational product but will be reimbursed by Pfizer while the subject is participating in the Safety Follow-Up Period, beginning at Week 24. Their use will be recorded on the concomitant medication CRF.

5.9. Rescue Medication

Acetaminophen will be issued to the subject by the study sites in its approved marketed product dress, (carton, bottle, documents) for use as rescue medication.

During the Washout Period and the Initial Pain Assessment Period, subjects may take rescue medication as needed for osteoarthritis or other types of pain or illness up to a maximum daily dose of 3000 mg per day. Rescue medication must be discontinued at least 24 hours prior to the Baseline (Randomization) visit.

In the event of inadequate pain relief for osteoarthritis during the treatment period (ie, after the Baseline [Randomization] visit through Week 16), subjects may take up to 3000 mg of acetaminophen per day but only up to 3 days per week. Subjects must discontinue rescue medication within 24 hours of any scheduled site visit at which efficacy assessments are collected.

From the Baseline (Randomization) visit through Week 16, subjects taking greater than 3 days per week of rescue medication (any level of acetaminophen used specifically for osteoarthritis pain) must be interviewed by study site personnel to determine if this is due to lack of efficacy or other reasons, and the discussion should be noted in the subject's source documents. Up to Week 16, subjects who have taken rescue medication more frequently than specified in the protocol and indicate that they cannot or will not follow the rescue medication protocol requirements because of insufficient osteoarthritis pain relief should be withdrawn from study treatment due to lack of efficacy and entered in the Early Termination Follow-Up period (See Section 6.16). Subjects who indicate that they anticipate being able to take rescue medication no more than 3 days per week going forward will be allowed to continue in the treatment period. However, if these subjects continue to take rescue

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medication more than 3 days per week, they should be withdrawn from study treatment and entered into the Early Termination Follow-Up period (See Section 6.16).

After the Week 16 visit subjects may take acetaminophen rescue medication daily, up to the maximum permitted dose of 3000 mg per day. After Week 24, subjects may be started on standard of care. After Week 24, subjects may continue to use acetaminophen as needed up to the maximum dose per day as permitted by local or national labeling.

The same principle applies during the Early Termination Follow-Up period (described in Section 6.16); up to 16 weeks after the last dose of SC investigational product, the subject will be advised not to exceed 3000 mg of acetaminophen per day and will be asked not to take rescue medication in the 24 hours preceding in clinic Early Termination Follow-Up period visits (48 hours for any other analgesics; see Section 6.16). After the Early Termination Follow-up visit that occurs approximately 16 weeks after the last dose of investigational product, subjects may be started on standard of care treatments until the end of the follow-up period. Subjects may continue to use acetaminophen as needed up to the maximal permitted dose of acetaminophen per day per local or national labeling.

All rescue medication must be discontinued at least 24 hours prior to any scheduled study visit at which efficacy data is collected (ie, up to the Week 24 visit or the Early Termination visit that occurs 16 weeks after the last dose of investigational product).

Subjects will be asked to return rescue medication bottles at each study visit for assessment of compliance.

Subjects will be instructed that many over-the-counter medications contain acetaminophen, and to guard against overuse. Subjects will be instructed to keep a record of their acetaminophen rescue medication daily usage via the IRT through Week 16. After Week 16 and up to the Week 40 visit, usage of acetaminophen rescue medication will be recorded once weekly via the IRT. The dose and reason for acetaminophen use in instances other than as rescue medication must be recorded on the appropriate concomitant medication case report form (CRF).

6. STUDY PROCEDURES

The site monitor should be consulted in the event that site layout, logistics, or equipment require adjustment to the ordering of study procedures or resolution of technical difficulties to enable performance of the Protocol. Such changes will be implemented administratively and documented in the appropriate venue (eg, site trial documentation and/or clinical study report). As much as possible, each subject's clinic visit should be conducted at approximately the same time of day throughout their participation in the study.

As a general rule, scales/instruments should be completed by the subject first, upon arrival at the clinic; vital signs should be assessed prior to blood draws.

Study visit windows are ±3 days for Weeks 2 and 4, and ±7 days for Weeks 8, 12, 16, 24 and 40 as well as for telephone visits at Weeks 20, 28, 32 and 36. In the event the subject

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requires a visit within the extremes of the visit windows, following study visits should be scheduled with reference to the original baseline visit date. Subject scheduling issues should be brought to the attention of study management for resolution. Dosing visits should be targeted to occur no earlier than 8 weeks ±7 days from the previous injection. Visit window for the Week 40 (End of study) x-rays is ±30 days from the nominal time point of the visit however they should preferably be obtained prior to the Week 40 visit and if possible, less than 14 days after the visit.

Subjects will be reminded to abstain from taking rescue medication 24 hours prior any study visit at which efficacy data is collected.

6.1. Screening

The Screening Period allows for completion of Screening procedures, including washout of prohibited medications, and completion of the Initial Pain Assessment Period (IPAP). The IPAP may begin once washout has been completed (if needed) and once x-ray confirmation that the selected index joint meets the KL grade criteria, and that no exclusionary conditions are present on X-ray has been obtained from the Central Reader. The total duration of the Screening period should not exceed 37 days.

Written informed consent will be obtained from each subject prior to any trial assessments. Each subject will be assessed as to his/her suitability per inclusion/exclusion criteria review.

Subject demographics and the date of OA diagnosis for every affected joint will be obtained. A comprehensive medical history and concomitant medication review will be performed for each subject; in addition, a comprehensive evaluation of musculoskeletal history and musculoskeletal physical exam will be performed (see 7.3.1.2). The index joint will be selected at the Screening visit based on pain scores. If bilateral knee and/or hip pain is present, the investigator will select the more painful joint as the study (index) joint. X-ray confirmation of eligibility (Kellgren-Lawrence Grade ≥2 for selected index joint and absence of exclusionary conditions) will be obtained prior to the Baseline visit from the imaging Central Reader based on x-rays of each knee, hip and shoulder (and other major joint x-ray, if relevant) obtained at Screening. Anticipate that it may take up to two weeks to obtain X-ray confirmation from the central reader prior to the Baseline visit.

History of insufficient pain relief from, inability to tolerate or contraindication to take, acetaminophen, NSAID, and tramadol or opioid treatments will be clearly documented on the appropriate CRF page. The required level of evidence to establish that subjects meet this inclusion criterion will be based upon the investigator's judgment. Investigators should rely upon available medical records that he or she may already have access to, prescription medication records (eg, retail pharmacy records), records or information provided by referring physicians and/or subject historical recall if the investigator is satisfied with the level of detail subjects are able to provide on past medication use. Investigators should clearly document in source records the information used to establish whether a subject does or does not meet this inclusion criterion. As a guide, investigators should document medication names, medication doses, reasons for use, dates of use and reason for discontinuation. If one or more of the above medications could not be used due to

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contraindication or if the subject refuses to take the medication due to fear of known side effects, this should also be clearly documented with supporting detail.

Activities at Screening (Initial Screening Visit):

- Informed consent;
- Numeric Pain Scale rating (IRT) for major joints (shoulders, hips and knees) and any painful major joint to be imaged at Screening;
- Primary Diagnosis and selection of index hip or knee by the investigator;

  NOTE: Based on pain scores. If the subject experiences pain in more than one joint, then the most painful joint should be selected as the index joint. Radiographic eligibility of the selected index joint will need to be confirmed by the Central Reader;
- WOMAC Subscales (index joint);

  NOTE: Subjects should be thoroughly instructed on completion of WOMAC scales, no coaching or other interpretative assistance should be given to subjects during completion of these questionnaires;
- Assess and document history of insufficient pain relief from, inability to tolerate, or contraindications to acetaminophen, NSAIDs and tramadol or opioids;
- Demographics and General medical history;
- Assessment of depression by medical history (use of PHQ-9 is optional and suggested as a tool to assess seriousness of depression if needed); if the PHQ-9 is utilized, the completed questionnaire should be archived in the subject's source documents.
- Comprehensive musculoskeletal/joint related medical history;
- Review of prior medication (record any prior medication used to treat osteoarthritis and osteoarthritis pain, and prior 30 day use for all other medications);
- Weight and Height with BMI calculation, Smoking Status, Alcohol Use/Dependency, Female Hormonal Status (if known or pending laboratory results);
- Vital signs after sitting for at least 5 minutes, (blood pressure and heart rate);
- Orthostatic blood pressure (supine/standing) measurement;
- General Physical Examination;
- Musculoskeletal Physical Examination;

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- Neurologic exam/Neuropathy Impairment Score;
- Survey of Autonomic Symptoms (SAS);
- ECG (12-lead);
- Clinical laboratory tests (blood chemistry, hematology, urinalysis, Hepatitis B, C, and HIV screen, urine toxicology screen, HbA1c, serum pregnancy testing/FSH testing if needed);
- Subject eligibility and inclusion/exclusion review (pending results of laboratory tests, ECG and central reading of x-rays). If a subject qualifies other than pending results they may begin the Washout Period. Sites will contact subjects who are found to be ineligible subsequent to the receipt of disqualifying laboratory, ECG or x-ray results may be asked to return requested study related materials and exit the Screening Period (Screen Failure);
- Female subjects of child bearing potential will be instructed on the contraception requirements for this study; the investigator or designee will confirm that female subjects of child-bearing potential have selected 2 highly effective forms of contraception from the list of acceptable methods and instruct the subject in their consistent and correct use. The conversation will be documented in the subject chart;
- Dispense rescue medication; subjects will be instructed on the permissible amounts of rescue medication during the washout period, during the IPAP and during the treatment period, as well as the need to refrain from rescue medication use 24 hours prior to a study visit (Refer to Sections 6.2 and 6.3 below);
- Subjects will be instructed in the use of the IRT system to record daily pain score and rescue medication use, weekly joint pain assessments, weekly concomitant NSAID use entries with specific instructions as when to test the system and when to begin entering data;
- Subjects will be provided with a washout schedule for current pain medications;
- Subjects will be scheduled for x-rays of both knees, both hips and both shoulders (and any other painful major joint if relevant). It is recommended that the x-rays be scheduled as soon as possible and at least 2 weeks prior to the scheduled Baseline visit) to allow sufficient time to obtain confirmation of eligibility from the imaging Central Reader prior to the Baseline visit.

6.2. Washout Period

The beginning of the Washout Period will preferably be scheduled based on the planned IPAP so as to minimize the time spent without analgesic medications prior to Randomization. The Washout Period will include the discontinuation and washout of all prohibited medications pain for at least 5 half-lives or 48 hours (whichever is greater) prior to the Initial

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Pain Assessment Period and will be a minimum of 2 days (Refer to Appendix 3). Subjects experiencing pain during the Washout Period may take acetaminophen as needed up to 3000 mg per day, but must discontinue rescue medication for at least 24 hours prior to the Baseline (Randomization) Visit.

If necessary, the Screening/Washout Period may be adjusted due to individual subject circumstances (eg, stabilization of a concomitant medication), contact study management for guidance. However, the total duration of the Screening period is not to exceed 37 days.

6.3. Initial Pain Assessment Period (IPAP)

The Initial Pain Assessment Period will begin up to 7 days prior to the Randomization/Baseline Visit (Day 1) to allow subjects to complete at least 3 diary entries during the IPAP period.

During this time, the subjects will record their pain scores in the index joint and rescue medication use via the IRT.

Study sites will monitor the IRT reports for compliance with diary recordings and rescue medication use and reschedule those subjects who fail to provide 3 diary days or fail to refrain from rescue medication use 24 hours prior to baseline.

Assessment of pain in the major, non-index joints will be performed once during the IPAP. Subject will also be reminded of contraception requirements.

6.4. Baseline Visit (Day 1)

6.4.1. Pre-Randomization: Assessment of Randomization Criteria

Subjects must continue to satisfy Inclusion/Exclusion Criteria (general criteria and those specific to the Baseline visit - See Section 4.3) to be eligible for Randomization. Full eligibility, including (but not limited to) confirmation of appropriate washout of concomitant medication, abstinence from acetaminophen in the previous 24 hours, completion of at least 3 diary entries in the past 7 days, subject meets required WOMAC Pain subscale Baseline score, subject meets required Baseline WOMAC Physical Function subscale and Patient's Global Assessment of Osteoarthritis scores, and that no adverse events occurred since the Screening visit that would render the subject ineligible for randomization, should be assessed before carrying out Randomization in the IRT system.

Subjects will undergo the following assessments prior to randomization:

- Inclusion/exclusion review (including results of ECG, laboratory and x-ray obtained at the Screening visits);
- WOMAC (all subscales);
- Patient's Global Assessment of Osteoarthritis;

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NOTE: Subjects should be thoroughly instructed on completion of Patient's Global Assessment of Osteoarthritis questionnaires, no coaching or other interpretative assistance should be given to subjects during completion of these questionnaires;

- Musculoskeletal physical examination;
- Vital signs after sitting for at least 5 minutes, (blood pressure, and heart rate).
- Orthostatic blood pressure (supine/standing) measurement;
- Review of concomitant medication;
- Rescue medication review;
- Instruction/review of subject compliance with diary assessments and operation of the IRT;
- Confirm with female subjects of child bearing potential that they understand and are willing to follow the contraceptive requirements;
- Adverse Events that occurred after signing the Informed Consent Document (pretreatment Adverse Events);
- Urine pregnancy test for females of childbearing potential (must be negative).

6.4.2. Randomization

- If still eligible, the subject can be randomized. Randomization will be accomplished through an IRT. The randomization number assigned to the subject will be provided by the system.

6.4.3. Pre-dosing (Day 1)

Randomized subjects will undergo the following assessments prior to the first dose of investigational product. Some of these may be performed prior to randomization for convenience:

- WPAI:OA;
- EQ-5D-5L;
- Health Care Resource Utilization (HCRU);
- Neurologic exam/Neuropathy Impairment Score;
- Clinical laboratory tests (blood chemistry, hematology);
- Serum and plasma retention samples;

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- Blood sample for Anti-Drug Antibody assessment (see Assessments; Anti-Drug Antibodies Section 7.3.9);
- Blood samples for PK and PD (NGF) analyses (see Assessments; Pharmacokinetics Section 7.5);
- Blood and urine samples for biomarkers (fasting if possible; urine sample should not be taken from the 1st void of the day; see Section 7.6);
- Blood sample for banked biospecimens (see Section 7.7.1, Markers of Drug response and Additional Research) (if subject has specifically consented; see Section 7.7.2);
- Dispense rescue medication;
- Female subjects of child-bearing potential to be reminded of contraceptive requirements.

6.4.4. Dosing (Day 1)

- Subjects will receive a single SC injection of blinded investigational product according to the treatment assigned by the IRT system (refer to Section 5.1).

The administration of investigational product must be performed by medical staff and where facilities to handle allergic reactions are available. Should a subject experience symptoms typical of an allergic reaction, then investigational product administration should be discontinued immediately and permanently. No other dosage modifications are allowed.

6.4.5. Post Dosing (Day 1)

Subjects will be observed in clinic for at least 1 hour after dosing. The following assessments will be completed at 1-hour post-dose:

1-hour post-dose

- Review and record Adverse Events;

Each subject should be reminded to seek medical care and/or contact the investigator if the subject experiences symptoms of an acute or severe hypersensitivity reaction (eg, shortness of breath, anaphylaxis, urticaria, angioedema) after leaving the clinic.

6.5. Week 2

- WOMAC (all subscales);
- Patient's Global Assessment of Osteoarthritis;
- Vital signs after sitting for at least 5 minutes, (blood pressure and heart rate);
- Orthostatic blood pressure (supine/standing) measurement;

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- Musculoskeletal physical examination;
- Neurologic exam/Neuropathy Impairment Score (referral for neurological consult if required, see Section 7.3.8);
- Concomitant medication review/update;
- Adverse event review;
- Subject diary entries and compliance review (daily index joint pain score, weekly non-index joint pain assessment and as applicable, daily rescue medication and weekly NSAID use entries);
- Rescue medication use and compliance review/Dispense;
- Blood samples for PK and PD (NGF) analyses in a subset of subjects;
- Female subjects of child-bearing potential subjects to be reminded of contraceptive requirements.

6.6. Week 4

- WOMAC (all subscales);
- Patient's Global Assessment of Osteoarthritis;
- Vital signs after sitting for at least 5 minutes, (blood pressure and heart rate);
- Orthostatic blood pressure (supine/standing) measurement;
- Musculoskeletal physical examination;
- Neurologic exam/Neuropathy Impairment Score (referral for neurological consult if required);
- Concomitant medication review/update;
- Adverse event review;
- Subject diary entries and compliance review (daily index joint pain score, weekly non-index joint pain assessment and as applicable, daily rescue medication and weekly NSAID use entries);
- Rescue medication compliance review/Dispense;
- Blood samples for PK and PD (NGF) analyses in a subset of subjects;

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- Female subjects of child-bearing potential to be reminded of contraceptive requirements.

6.7. Week 8

6.7.1. Predosing (Week 8)

- WOMAC (all subscales);
- Patient's Global Assessment of Osteoarthritis;
- EQ-5D-5L;
- Vital signs after sitting for at least 5 minutes, (blood pressure and heart rate);
- Orthostatic blood pressure (supine/standing) measurement;
- Musculoskeletal Physical Examination;
- Neurologic exam/Neuropathy Impairment Score (referral for neurological consult if required);
- Concomitant medication review/update;
- Adverse event review;
- Subject diary entries and compliance review (daily index joint pain score, weekly non-index joint pain assessment and as applicable, daily rescue medication and weekly NSAID use entries);
- Rescue medication compliance review/Dispense;
- Blood samples for PK and PD (NGF) analyses;
- Blood sample for Anti-Drug Antibody assessment;
- Blood and urine samples for biomarkers (fasting if possible; urine sample should not be taken from the 1st void of the day. See Section 7.6);
- Urine pregnancy test for females of childbearing potential (must be negative prior to dosing);
- Female subjects of child-bearing potential to be reminded of contraceptive requirements.

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6.7.2. Dosing (Week 8)

Subjects will receive a single SC injection of blinded investigational product according to the treatment assigned by the IRT system (see Section 5).

The administration of investigational product will be performed by medical staff and where facilities to handle allergic reactions are available. Should a subject experience symptoms typical of an allergic reaction then investigational product administration should be discontinued immediately and permanently. No other dosage modifications are allowed. Subjects will receive appropriate treatment at the discretion of the investigator.

6.7.3. Post-Dosing (Week 8)

Subjects will be observed in clinic for at approximately 1 hour after dosing.

1-hour post-dose

- Review and record adverse events;

Each subject should be reminded to seek medical care and/or contact the investigator if the subject experiences symptoms of an acute or severe hypersensitivity reaction (eg, shortness of breath, anaphylaxis, urticaria, angioedema) after leaving the clinic.

6.8. Week 12

- WOMAC (all subscales);
- Patient's Global Assessment of Osteoarthritis;
- Vital signs after sitting for at least 5 minutes, (blood pressure and heart rate);
- Orthostatic blood pressure (supine/standing) measurement;
- Musculoskeletal physical examination;
- Neurologic exam/Neuropathy Impairment Score (referral for neurological consult if required);
- Concomitant medication review/update;
- Adverse event review;
- Subject diary entries review and compliance review (daily index joint pain score, weekly non-index joint pain assessment and as applicable, daily rescue medication and weekly NSAID use entries);
- Rescue medication compliance review/Dispense;
- Blood samples for PK and PD (NGF) analyses in a subset of patients;

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- Female patients of child-bearing potential to be reminded of contraceptive requirements.

6.9. Week 16 (End of Treatment; Beginning of Safety Follow-Up Period)

- WOMAC (all subscales);
- Patient's Global Assessment of Osteoarthritis;
- WPAI:OA;
- EQ-5D-5L;
- Weight;
- Vital signs after sitting for at least 5 minutes, (blood pressure and heart rate);
- Orthostatic blood pressure (supine/standing) measurement;
- ECG (12-lead);
- General Physical Examination;
- Musculoskeletal assessment;
- Neurologic exam/Neuropathy Impairment Score (referral for neurological consult if required);
- Concomitant medication review/update;
- Adverse event review;
- Subject diary entries review and compliance review (daily index joint pain score, weekly joint pain assessment, and rescue medication and NSAID use entries if applicable);
- Rescue medication compliance review/Dispense;
- Serum pregnancy test for females of childbearing potential;
- Plasma and serum retention samples);
- Blood samples for PK and PD (NGF) analyses;
- Blood sample for Anti-Drug Antibody assessment;

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- Instruct subject on visit schedule for the Follow-Up Period, including (but not limited to) maintenance of concomitant medication rules and changes to schedule of diary entries via IRT;
- Reminder to female subjects of child-bearing potential to continue with contraceptive requirements.

6.10. Week 20 (Safety Follow-Up Period Phone Visit)

- Review/Update of concomitant medication;
- Subject diary entries review and compliance review (weekly joint pain, rescue medication and NSAID use entries);
- Adverse events review;
- Female subjects of child-bearing potential to be reminded of contraceptive requirements.

6.11. Week 24 (Safety Follow-Up Period Efficacy Visit)

- WOMAC (all subscales);
- Patient's Global Assessment of Osteoarthritis;
- Survey of Autonomic Symptoms (SAS);
- Health Care Resource Utilization (HCRU);
- Vital signs after sitting for at least 5 minutes, (blood pressure and heart rate);
- Orthostatic blood pressure (supine/standing) measurement;
- Musculoskeletal physical examination;
- Neurologic exam/Neuropathy Impairment Score;
- Subject diary entries review and compliance review (weekly joint pain, rescue medication and NSAID use entries);
- Concomitant medication review/update;
- Rescue medication compliance review/dispense;
- Adverse events review;
- Clinical laboratory tests (blood chemistry, hematology);

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- Serum and plasma retention samples;
- Blood samples for PK and PD (NGF) analyses;
- Blood sample for Anti-Drug Antibody;
- Blood and urine samples for biomarkers (fasting if possible; urine sample should not be taken from the 1st void of the day; see Section 7.6);
- Serum pregnancy test for females of childbearing potential;
- Assignment of standard of care treatment at the discretion of the investigator.

6.12. Week 28 (Safety Follow-Up Period Phone Visit)

- Review/Update of concomitant medication;
- Subject diary entries review and compliance review (weekly joint pain scores, rescue medication/acetaminophen and NSAID use);
- Adverse events review.

6.13. Week 32 (Safety Follow-Up Period Phone Visit)

- Review/Update of concomitant medication;
- Subject diary entries review and compliance review (weekly joint pain scores, rescue medication/acetaminophen and NSAID use);
- Adverse events review.

6.14. Week 36 (Safety Follow-Up Period Phone Visit)

- Review/Update of concomitant medication;
- Subject diary entries review and compliance review (weekly joint pain scores, rescue medication/acetaminophen and NSAID use);
- Schedule Week 40 X-rays, to occur preferably prior to the nominal time point of the Week 40 visit and if possible, no more than 14 days after the Week 40 visit;
- Adverse events review.

6.15. Week 40 (End of Safety Follow-Up/End of Study Visit)

- Radiographic assessment (x-rays) of both knees, both hips and both shoulders and any other joint for which a radiograph was obtained at the Screening visit or any at risk joint identified during the study period; final study x-ray should be collected as close as possible to the nominal time point of the visit, within a ±30 day window,

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preferably prior to the Week 40 visit and no more than 14 days after the visit. Refer to Section 6.16 for Early Termination visit description;

- Health Care Resource Utilization (HCRU);
- Survey of Autonomic Symptoms (SAS);
- Musculoskeletal physical examination;
- Vital signs after sitting for at least 5 minutes, (blood pressure and heart rate);
- Orthostatic blood pressure (supine/standing) measurement;
- ECG (12-lead);
- Neurologic exam/Neuropathy Impairment Score (referral for neurological consult if required);
- Concomitant medication review/update;
- Subject diary entries review (weekly joint pain scores, rescue medication/ acetaminophen and NSAID use);
- Adverse events review;
- Blood sample for Anti-Drug Antibody assessment.

6.16. Subject Withdrawal/Early Termination Follow-Up Period

Subjects may withdraw from the study at any time at their own request, or they may be withdrawn at any time at the discretion of the investigator or Sponsor for safety, behavioral, or the inability of the subject to comply with the protocol required schedule of study visits or procedures at a given study site.

If a subject does not return for a scheduled visit, every effort should be made to contact the subject. A subject thought lost to follow-up, must be contacted through a minimum of 3 documented phone call attempts and, if phone calls are unsuccessful, a certified letter sent to the subject. In any circumstance, every effort should be made to document subject outcome, if possible. The investigator should inquire about the reason for withdrawal, follow-up with the subject regarding any unresolved adverse events and request that the subject return for follow-up visits as indicated in the schedule below. Female subjects of child-bearing potential should be reminded to continue contraceptive measures at least 112 days (16 weeks) after the last dose of investigational product.

If the subject withdraws from the study, and also withdraws consent for disclosure of future information, no further evaluations should be performed and no additional data should be

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collected. Pfizer may retain and continue to use any data collected before such withdrawal of consent.

Subjects discontinuing from treatment (prior to Week 16), either at their request or at the decision of the investigator, will be required to undergo 24 weeks of post-treatment follow-up (as described in Section 6.16.1). The 24 weeks of follow-up will be obtained through 3 clinic visits and monthly phone calls to yield 24-weeks of post-treatment follow-up, as described in Section 6.16.1. Subjects will be asked to continue to enter pain scores for index joints and provide an assessment of pain in their major, non-index joints (as described in Section 7.1.1) via IRT, weekly, through the end of the 24-week follow-up period. On a weekly basis during the 24 week safety follow-up period, subjects will also be asked to continue to record rescue medication and NSAID use via IRT.

X-rays of both knees, both shoulders and both hips (and any other joint imaged at Screening or identified as at risk during the study) should be performed as soon as possible after the decision to withdraw from the study has been made, provided at least 30 days have passed since the last set of X-rays were collected. The remainder of efficacy and safety assessments should be done at the scheduled first visit which is to occur 8 weeks after the last dose of investigational product (as described in 6.16.1.1). The site should also schedule the subject for two additional clinic visits. The second visit should be scheduled to occur approximately 16 weeks after the subject's last dose of investigational product (which corresponds to more than 5 times the elimination half-life of tanezumab) to collect safety and efficacy data (as described in 6.16.1.3). Once the clinic visit 16-weeks after the last administration of investigational product has been completed and final efficacy assessments have been collected, standard of care treatment will be offered to subjects who are completing the remaining 8 weeks of the required follow-up period. The third and final clinic visit should be scheduled to take place approximately 24-weeks after the subject received the last dose of investigational product. That visit, (described in 6.16.1.5), includes repeat bilateral shoulder, knee and hip radiographs as well as any additional joint that was imaged at Screening or any joint identified as at risk during the study, providing at least 30 days have elapsed since the last radiographs were obtained. The window for obtaining end of study X-rays is 30 days from the nominal time of the visit. Every effort should be made to have the subject agree to complete the entire 24 week Early Termination Safety Follow-Up described above.

In the event a subject refuses the Early Termination safety follow-up, or chooses to discontinue during the safety follow-up (after Week 16 and through Week 40), a complete early termination visit should be performed. This early termination visit should include all procedures scheduled for the Week 24 and Week 40 visits unless the Week 24 visit has already been completed, in which case only Week 40 procedures will be required. In addition, if the Week 16 visit was not completed prior to termination, a general physical examination, body weight, WPAI:OA and EQ-5D-5L will also be obtained. Subjects will be advised to continue their contraception regimen during a period of 112 days (16 weeks) after the last dose of investigational product.

Subjects entered in the Early Termination Follow-Up period will be able to take acetaminophen daily up to the Early Termination Follow-Up period Visit 2 (which occurs

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16 weeks after the last dose of investigational product), but will be advised not to exceed 3000 mg per day. Subjects will be requested not to take acetaminophen (or any other analgesic) in the 24 hours that precede clinic visits at which efficacy assessments are collected (Early Termination Follow-Up period Visits 1 and 2, which occur 8 and 16 weeks after the last dose of investigational product, respectively).

6.16.1. Early Termination Follow-Up Period Procedures

As soon as possible after the decision to withdraw was made and provided 30 days have elapsed since the last set of X-rays were collected:

- Radiographs of all joints for which X-rays were obtained at Screening and other at risk joint identified during the study period.

Procedures to be conducted at the Early Termination Follow-Up period in clinic visits and telephone visits are described in Sections 6.16.1.1, 6.16.1.2, 6.16.1.3, 6.16.1.4, and 6.16.1.5.

6.16.1.1. Early Termination Follow-Up Period Visit 1 (8 weeks after last dose received)

- WOMAC (all subscales);
- Patient's Global Assessment of Osteoarthritis;
- Survey of Autonomic Symptoms (SAS);
- WPAI:OA;
- EQ-5D-5L;
- Weight;
- Vital signs after sitting for at least 5 minutes, (blood pressure and heart rate);
- Orthostatic blood pressure (supine/standing) measurement;
- ECG (12-lead);
- General physical examination;
- Musculoskeletal physical examination;
- Neurologic exam/Neuropathy Impairment Score (referral for neurological consult if required);
- Review/Update of concomitant medication;
- Assess compliance with IRT and review diary entries (weekly joint pain, rescue medication and NSAID use entries);

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- Adverse events review;
- Dispense rescue medication;
- Serum and plasma retention samples;
- Serum pregnancy test for female subjects of childbearing potential;
- Blood sample for Anti-Drug Antibody assessment;
- Blood samples for PK and PD (NGF) analyses;
- Blood and urine samples for biomarkers (fasting if possible; urine sample should not be taken from the 1st void of the day, See Section 7.6);
- Reminder to female subjects of child-bearing potential to continue with contraceptive requirements.

6.16.1.2. Early Termination Follow-Up Period Phone Visit (12 weeks post last dose)

- Review/Update of concomitant medication
- Review subject compliance with weekly diary and review entries (weekly joint pain, rescue medication and NSAID use entries);
- Adverse events review;
- Female subjects of child-bearing potential to be reminded of contraceptive requirements.

6.16.1.3. Early Termination Follow-Up Period Visit 2 (16 weeks after last dose received)

- WOMAC (all subscales);
- Patient's Global Assessment of Osteoarthritis;
- Survey of Autonomic Symptoms (SAS);
- Health Care Resource Utilization (HCRU);
- Musculoskeletal physical examination;
- Neurologic exam/Neuropathy Impairment Score (referral for neurological consult if required);
- Review/Update of concomitant medication;

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- Subject diary entries review and compliance review (weekly joint pain, rescue medication and NSAID use entries);
- Vital signs after sitting for at least 5 minutes, (blood pressure and heart rate);
- Orthostatic blood pressure (supine/standing) measurement;
- Clinical laboratory tests (blood chemistry, hematology);
- Serum and plasma retention samples;
- Serum pregnancy test for female subjects of childbearing potential;
- Blood sample for Anti-Drug Antibody assessment;
- Blood samples for PK and PD (NGF) analyses;
- Blood and urine samples for biomarkers (fasting if possible; urine sample should not be taken from the 1st void of the day; see Section 7.6);
- Adverse events review;
- Rescue medication compliance review/Dispense;
- Assignment of standard of care treatment at the discretion of the investigator (Standard of care treatment refers to analgesics approved to relieve the pain of OA; these include opioids, topical analgesics, NSAIDs, capsaicin products, injectable corticosteroids and viscosupplementation (eg, hyaluronan) and are prescribed at the discretion of the investigator. Pre-specified analgesics will be reimbursed by Pfizer).

6.16.1.4. Early Termination Follow-Up Period Phone Visit (20 weeks post last dose)

- Review/Update of concomitant medication;
- Review subject compliance with weekly diary and review entries (weekly joint pain scores), weekly rescue medication and weekly NSAID use entries);
- Adverse events review;
- Schedule subject for follow-up radiographic assessments (X-rays) to occur within 30 days of, and preferably prior to, Early Termination Follow-up period Visit 3.

6.16.1.5. Early Termination Follow-Up Period Visit 3 (24 Weeks after last dose)

- Radiographic assessment (x-rays) of both knees, both hips and both shoulders and any other joint for which a radiograph was obtained at the Screening visit, and other at risk joint identified during the study period (window ±30 days; preferably prior to the visit);

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- Health Care Resource Utilization (HCRU);
- Survey of Autonomic Symptoms (SAS);
- Vital signs after sitting for at least 5 minutes, (blood pressure and heart rate);
- Orthostatic blood pressure (supine/standing) measurement;
- ECG (12-lead);
- Musculoskeletal physical examination;
- Neurologic exam/Neuropathy Impairment Score (referral for neurological consult if required);
- Review/Update of concomitant medication;
- Review weekly diary entries (weekly joint pain scores, weekly rescue medication and weekly NSAID use entries);
- Adverse events review;
- Blood sample for Anti-Drug Antibody assessment.

6.16.2. Procedures for Subjects Undergoing Joint Replacement

Subjects who have undergone or plan to undergo total joint replacement or other arthroplasty procedure during the study will be discontinued from study treatment.

Subjects who undergo total knee, hip or shoulder joint replacement surgery during the study (Double-blind Treatment Period or Follow-up Period) will be followed for 24 weeks after the procedure as part of a separate protocol (Study A4091064; provided the subject consents).

Transition procedures into Study A4091064 are determined by the timing of total joint replacement surgery:

- Subjects who have undergone or plan an immediate total knee, hip or shoulder replacement procedure will be discontinued from the treatment period and enter into Study A4091064. At the discontinuation visit, all End of Treatment (Week 16) and Week 24 procedures should be completed (6.9 and 6.11); unless the subject has already completed the Week 16 and Week 24 visits, in which case only the Week 40 visit procedures should be completed (Section 6.15). In addition, if the Week 16 visit or Week 24 visit was not completed prior to termination, a general physical examination, WPAI:OA, EQ-5D-5L will also be obtained. Applicable Study A4091064 Baseline visit activities should be completed on the same day as the Study A4091056 End of Treatment Visit.

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- Subjects who plan to undergo total joint replacement during the study will be discontinued from the treatment period and entered into Early Termination Follow-up (See Section 6.16) until their joint replacement or other arthroplasty procedure. For these subjects, a complete early termination visit should be conducted prior to the total joint replacement (See Section 6.16) and entrance into Study A4091064. Applicable Study A4091064 baseline visit activities should be completed on the same day as the Study A4091056 early termination visit. Subjects who have not undergone or scheduled total joint replacement surgery within the study treatment or safety follow-up period of this study will not be eligible for the total joint replacement study.

Subjects who undergo other types of joint replacement surgery or arthroplasty during the study should be discontinued from study treatment and complete the protocol specified Safety Follow-up Period but not be entered into Study A4091064 for follow-up.

Female subjects of child-bearing potential will be advised to continue their contraception regimen during a period of 112 days (16 weeks) after the last dose of investigational product.

6.17. Total Joint Replacement Follow-up Protocol

As part of the total joint replacement follow-up protocol, all source documents from the surgical procedure (including any prior orthopedic consultations and pre-operative assessments), immediate post-operative recovery, and follow-up therapy will be collected. In addition, the surgeon will be asked to complete an assessment of procedural difficulty. Imaging studies of the affected joint (such as x-rays and MRI scans) will be collected. Instructions regarding pathology specimens will be provided.

7. ASSESSMENTS

Every effort should be made to ensure that the protocol required tests and procedures are completed as described. However it is anticipated that from time to time there may be circumstances outside of the control of the investigator, that may make it unfeasible to perform a test. In these cases the investigator will take all steps necessary to ensure the safety and well-being of the subject. When a protocol-required test cannot be performed, the investigator will document the reason for this and any corrective and preventive actions which he or she has taken to ensure that normal processes are adhered to as soon as possible. The study team will be informed of these incidents in a timely fashion.

7.1. Subject Diary Assessments

7.1.1. Daily/Weekly Pain Diary Assessments

Diary assessments of pain in the index joint and assessment of the major, non-index joints will be completed by the subjects at approximately the same time each day (or each week). Average pain will be assessed with an 11-point Numeric Rating Scale (NRS) ranging from zero (no pain) to 10 (worst possible pain) captured through IRT. The subjects should describe their pain in the index joint (and non-index joint when applicable) during the past

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24 hours by choosing the appropriate number from 0 to 10. If possible, the subject should conduct the self-assessment in the evening prior to midnight.

Index Joint Pain Assessment

Average pain in the index joint will be assessed by the subject daily from the beginning of the Initial Pain Assessment Period to the Week 16 Visit, followed by weekly during the Safety Follow-Up Period (and weekly during the Early Termination Follow-Up Period, if applicable).

- Example of question when the index joint is the knee:

"Select the number that best describes your average pain in your index (left/right) knee in the past 24 hours"*:

0	1	2	3	4	5	6	7	8	9	10
No Pain										Worst Possible Pain

- Example of question when the index joint is the hip:

"Select the number that best describes your average pain in your index (left/right) hip in the past 24 hours":

0	1	2	3	4	5	6	7	8	9	10
No Pain										Worst Possible Pain

Non-index Joint Pain Assessment

Assessment of pain in the major, non-index joints will be performed weekly beginning at the Initial Pain Assessment Period until the end of the Safety Follow-Up Period (also weekly during the Early Termination Follow-Up Period, if applicable). A major joint is defined as a mobile synovial joint in the limbs such as shoulders, elbows, wrists, hips, knees, ankles and excluding the joints of the toes and hands.

During the Initial Pain Assessment Period, subjects will be asked if they experienced pain in a non-index, major joint in the past 24 hours and if they respond that they did, subjects will be asked to rate the pain in that joint using a 24-hour recall on an 11-point numeric rating scale such as that shown above.

Post-baseline, assessment of pain in the non-index joints will be done weekly by asking subjects if they experienced new or increased pain in non-index major joints via IRT. If a subject responds that he/she has experienced increased pain in a non-index joint, the subject will be asked to rate his/her pain in that joint using a 24-hour recall period on the same 11-point numeric rating scale shown above for the remainder of the study.

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7.1.2. Rescue Medication and Amount

Rescue medication use will be collected daily via IRT from the beginning of the Initial Pain Assessment Period to the Week 16 Visit. The dosage strength of the acetaminophen tablets/caplets/capsules will be captured. The subject should note the number of pills (tablets/caplets/capsules) of rescue medication taken during the last 24 hours.

Following the Week 16 visit and up to Week 40 (or during the Early Termination Follow-Up, for subjects who enter that phase of the study), the use of acetaminophen as rescue medication (number of days and maximum amount used in a day) will be reported weekly via IRT.

7.1.3. Concomitant NSAID Use

Use of over-the counter or prescription NSAID will be collected weekly via IRT from the Initial Pain Assessment Period until the Week 40 visit (or through the Early Termination Follow-Up period for subjects who enter this phase of the study). Subjects will record the number of days of NSAID use in the past week using IRT. Via telephone calls or at clinic visits, sites will interview the subject regarding their NSAID use and record additional information, such as the medication name, dosage and reason for use on the CRF. The investigator or designee should closely monitor for concomitant NSAID use. Subjects reporting concomitant NSAID use will be managed per guidance provided in Section 5.8.1.

7.2. Study Visit Efficacy Assessments

7.2.1. Western Ontario and McMaster Universities Osteoarthritis Index

A copy of the WOMAC[44] Osteoarthritis NRS Index NRS 3.1 English for USA – V5 is presented in Appendix 4. The WOMAC subscales will be recorded by the subjects via IRT at relevant study visits.

7.2.1.1. WOMAC Pain Subscale

The WOMAC Pain subscale questionnaire is to be completed by the subject at Screening, Baseline (Day 1, prior to administration of investigational product) and Weeks 2, 4, 8 (Day 57, prior to administration of investigational product), 12, 16 (End of Treatment visit), and 24 (Follow-Up Period efficacy visit) or at Early Termination (as described in Section 6.16).

The WOMAC Pain subscale questionnaire will be completed for the index knee or hip.

The WOMAC Pain subscale is comprised of 5 questions regarding the amount of pain experienced due to OA in the index joint (selected study knee or hip) in the past 48 hours. The WOMAC Pain subscale is calculated as the mean of the scores from the five individual questions, which may not be a whole (integer) number. The WOMAC Pain subscale NRS scores for each question, and the WOMAC Pain subscale score, range from 0 to 10, with higher scores indicating higher pain.

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7.2.1.2. WOMAC Physical Function Subscale

Subjects will complete the WOMAC Physical Function subscale questionnaire at Screening, Baseline (Day 1, prior to administration of investigational product) and Weeks 2, 4, 8 (Day 57, prior to administration of investigational product), 12, 16 (End of Treatment visit), and 24 (Follow-Up Period efficacy visit) or at Early Termination (as described in Section 6.16).

The WOMAC Physical Function subscale questionnaire will be completed for the index knee or hip.

The WOMAC Physical Function subscale is comprised of 17 questions regarding the degree of difficulty experienced due to arthritis in the index joint (selected study knee or hip) in the past 48 hours. The WOMAC Physical Function subscale is calculated as the mean of the scores from the seventeen individual questions, which may not be a whole (integer) number. The WOMAC Physical Function subscale NRS scores for each question, and the WOMAC Physical Function subscale score, range from 0 to 10 with higher scores indicating worse function. This refers to the subject's ability to move around and perform usual activities of daily living.

7.2.1.3. WOMAC Stiffness Subscale

Subjects will complete the WOMAC Stiffness subscale questionnaire at Screening, Baseline (Day 1, prior to administration of investigational product) and Weeks 2, 4, 8 (Day 57, prior to administration of investigational product), 12, 16 (End of Treatment visit), and 24 (Follow-Up Period efficacy visit) or at Early Termination (as described in Section 6.16).

The WOMAC Stiffness subscale questionnaire will be completed for the index knee or hip.

The WOMAC Stiffness subscale is comprised of 2 questions regarding the amount of stiffness experienced in the index joint (selected study knee or hip) in the past 48 hours. The WOMAC Stiffness subscale is calculated as the mean of the scores from the two individual questions, which may not be a whole (integer) number. The WOMAC Stiffness subscale NRS scores for each question, and the WOMAC Stiffness subscale score, range from 0 to 10 with higher scores indicating more stiffness. Stiffness is defined as a sensation of decreased ease with which the subject moves the index knee.

7.2.2. Patient's Global Assessment of Osteoarthritis

Subjects will complete the Patient's Global Assessment (PGA) of Osteoarthritis questionnaire (Appendix 5) at Baseline (Day 1, prior to administration of investigational product) and Weeks 2, 4, 8 (Day 57, prior to administration of investigational product), 12, 16 (End of Treatment visit), and 24 (Follow-Up Period efficacy visit) or at Early Termination (as described in Section 6.16) on via IRT. Refer to Appendix 5 for more details.

The Patient's Global Assessment of Osteoarthritis questionnaire will be completed for the index knee or hip.

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Subjects who have a knee as index joint will answer the following question: "Considering all the ways your osteoarthritis in your knee affects you, how are you doing today?"

Subjects who have a hip as index joint will answer the following question: "Considering all the ways your osteoarthritis in your hip affects you, how are you doing today?"

Subjects will rate their condition using the following scale:

Grade	Description
1 – Very Good	Asymptomatic and no limitation of normal activities
2 – Good	Mild symptoms and no limitation of normal activities
3 – Fair	Moderate symptoms and limitation of some normal activities
4 – Poor	Severe symptoms and inability to carry out most normal activities
5 – Very Poor	Very severe symptoms which are intolerable and inability to carry out all normal activities

7.2.3. Work Productivity and Activity Impairment Questionnaire - Osteoarthritis – Knee or Hip V2.0 (WPAI:OA)

Subjects will complete the WPAI:OA at Screening and Week 16 (or at early Termination, as described in Section 6.16).

The WPAI-OA Knee or Hip[45] is a 6- item validated questionnaire that assesses the impact of OA on absenteeism, presenteeism, work productivity, and activity impairment. Each subscale score is expressed as an impairment percentage (0-100) where higher numbers indicate greater impairment and less productivity. The WPAI-OA is self-administered by the subject and takes less than 5 minutes to complete (Appendix 6).

7.2.4. EuroQol 5 Dimension (EQ-5D-5L)

The EQ-5D-5L™ will be completed at Baseline (Day 1, prior to administration of investigational product), Week 8 (prior to administration of investigational product) and at Week 16 (or at Early Termination as described in Section 6.16).

The EQ-5D-5L™ is a subject completed questionnaire designed to assess the subject's current health and translate that score into an index value or utility score. Health status is described in terms of 5 dimensions: mobility, self-care, usual activity, pain/discomfort, and anxiety/depression. There are two components to the EQ-5D-5L™: a Health State Profile and an optional visual analog scale (VAS) item. Only the 5 item health state profile will be assessed to calculate a single index value (see Appendix 7). This instrument provides a mechanism for conducting cost-effectiveness and cost-utility analyses.[46]

7.2.5. Assessment of Health Care Resource Utilization (HCRU)

The utilization of health care resources (eg, doctor office visits, hospitalizations, surgeries or procedures, etc; see Appendix 12) during the 3 months prior to Baseline will be assessed by questionnaire at Baseline (Day 1). In addition, Health Care Resource Utilization will be collected at study visits at Week 24 and at Week 40. Subjects will enter responses using IRT.

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7.3. Safety Assessments

Each subject will provide a general medical history as well as a detailed musculoskeletal/joint specific medical history. The information will be recorded on the appropriate CRF(s) at Screening. Information on prior medications (within 30 days of the Screening Visit for non-analgesic medications, any used for pain and other medications for the treatment and relief of symptoms of OA), non-pharmacologic therapies, supplements and concomitant medication use will be collected at Screening and concomitant medication at each scheduled study visit. Information regarding tobacco and alcohol use and dependency will also be collected at Screening.

7.3.1. Physical Examination

7.3.1.1. General Physical Examination

Each subject will undergo a general physical examination at Screening and at Week 16, or at Early Termination (as described in Section 6.16).

7.3.1.2. Musculoskeletal History and Physical Examination

Each subject will also undergo a musculoskeletal physical examination at Screening, Baseline (Day 1), and at Weeks 2, 4, 8, 12, 16, 24 and 40 (or at Early Termination, as described in Section 6.16).

At Screening, the investigator should collect a thorough musculoskeletal history. The investigator should inquire about current and past history of osteoarthritis, ligament tear or rupture, joint surgeries (including arthroscopic procedures), fractures, gout, pseudogout, osteoporosis or osteopenia, joint injuries or other conditions.

At each visit, the investigator will conduct a thorough musculoskeletal physical examination of all major joints. The musculoskeletal physical exam should evaluate the joints for swelling, redness, tenderness, deformity, osteophytes or nodes, crepitus and pain on motion, and will be documented on the CRF. The investigator should also collect subject-reported information on any current joint symptoms including pain, stiffness, and swelling. Any clinically significant change in symptoms or the examination should be reported as an adverse event.

7.3.2. Screening Numeric Pain Scale Rating (NRS)

Average pain in the major joints (shoulders, hips and knees) or any other painful major joint that will be imaged at Screening, will be assessed by the subject at the Screening Visit with an 11-point Numeric Rating Scale (NRS) ranging from zero (no pain) to 10 (worst possible pain) and captured through IRT.

7.3.3. Laboratory Safety Assessments

Blood and urine tests for safety assessments and/or determination of eligibility will be performed as indicated in this table and described in the subsections below:

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Table 3. Blood and Urine tests

Chemistry	Hematology	Other	Urinalysis
Screening, Baseline and Week 24 (or Early Termination): Sodium, potassium, chloride, bicarbonate, glucose (non fasting), Blood Urea Nitrogen (BUN), creatinine, calcium, phosphorus, magnesium, total and direct bilirubin, total protein, albumin, cholesterol, triglycerides, gamma glutamyltransferase (GGT), alanine aminotransferase (ALT), aspartate aminotransferase (AST), lactic dehydrogenase (LDH), alkaline phosphatase, creatine phosphokinase (CPK), and uric acid	Screening, Baseline, and Week 24 (or Early Termination): Complete blood count with differential	Screening only: HbA1c Hepatitis screen (eg, HBsAg, Anti-HCV), HIV test (HIV Ab screen) Urine toxicology screen (eg, for opiates, barbiturates, amphetamines, cocaine, propoxyphene, methadone, phencyclidine, and methaqualone). Serum FSH if applicable Screening, Week 16 and Week 24 (or early termination Visits 1 and 2): Serum Pregnancy Test Baseline and Week 8 (Prior to dosing at Dosing Visits): Urine Pregnancy Test Baseline, Week 16 and Week 24 (or Early Termination Visits 1 and 2): Serum and plasma retention samples	Screening only: pH, protein, glucose, ketones, blood, bilirubin, nitrite, specific gravity and leukocytes. Microscopic analysis performed if abnormalities are present on the above components.
Does not include PK, PD (NGF), ADA and biomarkers (refer to sections below for collection details)			

7.3.3.1. Blood Tests

Blood tests for clinical laboratory testing (chemistry, hematology) will be performed at Screening, Baseline and Week 24 (or at Early Termination, as described in Section 6.16). An unscheduled visit(s) may be necessary for follow-up of abnormal test results.

Serum and plasma retention samples will be collected at Baseline, Week 16 and Week 24 or at Early Termination (as described in Section 6.16).

See Section 7.3.3.3 for sample collected for serum pregnancy test and Section 7.3.3.4 for sample collected for FSH testing.

Blood samples collected for PK, PD (NGF), biomarkers and anti-drug antibody measurements are described in Sections 7.3.9, 7.5, and 7.6.1).

7.3.3.2. Urinalysis and Urine Toxicology Screen

Urinalysis and urine toxicology screen will be performed at Screening only.

See Section 7.3.3.3 for urine pregnancy test.

Urine samples collected for biomarker analyses are described in Section 7.6.2).

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7.3.3.3. Pregnancy Testing

For female subjects of childbearing potential, a serum pregnancy test, with sensitivity of at least 25 mIU/mL, will be performed during the Screening period, reviewed and confirmed as negative. A negative pregnancy result is required before the subject may receive the investigational product. Pregnancy tests will also be done whenever 1 menstrual cycle is missed during the double-blind treatment period (or when potential pregnancy is otherwise suspected), repeated at the Visits specified in the Schedule of Activities (Table 1) or at Early Termination as described in Section 6.16 to confirm the subject has not become pregnant during the study. Pregnancy tests may also be repeated as per request of institutional review boards (IRBs)/ethics committees (ECs) or if required by local regulations.

Qualitative urine pregnancy tests must be sensitive to at least 25 mIU/mL. These tests will be performed prior to dosing with SC investigational product at Baseline (Day 1), and Week 8 and confirmed as negative prior to dosing. Qualitative point-of-service urine pregnancy tests will be conducted with the test kit approved by the sponsor in accordance with instructions provided in its package insert. Subjects who have missed a menstrual period or who show an indeterminant or positive result on the qualitative point-of-service urine test may not further progress in the study until pregnancy is ruled out using further diagnostic testing (eg, a negative quantitative serum pregnancy test conducted at a certified laboratory).

Additional serum pregnancy tests will be conducted at Week 16 (to confirm the subject has not become pregnant during the study period) and at Week 24, which corresponds to 16 weeks after the last dose of SC investigational product was received and more than 5 times the elimination half-life of tanezumab (or at Early Termination, as described in Section 6.16).

In the case of a positive confirmed pregnancy, the subject will be withdrawn from administration of investigational product but may remain in the study.

Refer to Section 8.10 and 8.10.1 for guidance pertaining to exposure during pregnancy and post-natal follow-up.

7.3.3.4. Serum FSH Testing

Female subjects who are not of childbearing potential and who have not had a hysterectomy or bilateral oophorectomy and who have been amenorrheic for at least 1 year with no alternative pathological or physiological cause will under serum FSH testing to confirm post-menopausal status. A serum FSH level within the laboratory's reference range for postmenopausal females is required.

Female subjects who have undergone documented total hysterectomy or bilateral oophorectomy or who have medically confirmed ovarian failure do not require serum FSH testing.

Female subjects who have been amenorrheic less than 1 year will be considered of child-bearing potential.

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Female subjects who are considered of childbearing potential do not require FSH testing.

7.3.4. Vital Signs

Vital signs (systolic blood pressure, diastolic blood pressure, heart rate) will be collected and recorded throughout the study at Screening, Baseline (Day 1, prior to administration of investigational product), and at Weeks 2, 4, 8 (prior to dosing), 12, 16 (End of Treatment visit), 24 and 40, or at Early Termination visit (as described in Section 6.16). Vital signs will be collected after the subject has been in a sitting position for at least five minutes at each noted visit.

7.3.4.1. Orthostatic Blood Pressure Measurement

In addition to sitting vital sign measurements, orthostatic blood pressure measurements will be obtained using a standard manual sphygmomanometer at Screening, Baseline (Day 1), and at Weeks 2, 4, 8, 12, 16 (End of Treatment), 24 and 40 (End of Study; or at Early Termination, as described in Section 6.16). At each of these clinic visits, blood pressure will be assessed in supine and standing positions. Orthostatic blood pressure measurements will be obtained after collection of the sitting vital signs and before any required phlebotomy (and prior to dosing at dosing visits). To minimize chances of orthostatic hypotension related to volume depletion, subjects should be reminded to report for clinic visits well hydrated. In this regard, investigators could consider recommending to subjects that they consume 8-16 ounces (240-480 mL) of water prior to reporting to the clinic for study visits. All orthostatic blood pressure measurements will be recorded in the IRT system.

Supine blood pressure measurement will be obtained after subjects have been in the supine position for a minimum of 10 minutes. To ensure that a stable supine blood pressure measurement is obtained, at least two systolic and diastolic measurements will be performed. If the replicate systolic and diastolic measurements differ by no more than 10 mmHg and 5 mmHg, respectively, the supine blood pressure will be considered to be stable. The mean of the two stable replicate measures will be considered to represent the baseline supine blood pressure (mean systolic and mean diastolic blood pressure) for that visit. Once the supine blood pressure is considered to be stable, subjects will be asked to assume the standing position. After subjects have been in the standing position for 1 minute and 3 minutes, systolic and diastolic blood pressure will be measured and recorded for both time points. If the measurements do not meet the criteria for orthostatic (postural) hypotension, no further measurements are needed. If either the 1 minute standing or the 3 minute standing BP measurements show decreases meeting the criteria shown in Table 4, the sequence of supine and standing measurements should be repeated up to 2 more times.

Refer to Table 4 for the criteria defining orthostatic hypotension and actions that should be taken when orthostatic hypotension criteria are met.

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Table 4. Orthostatic Blood Pressure Changes and Subject Management

Mean Supine Systolic Blood Pressure	Decrease in Blood Pressure Defining Orthostatic (postural) Hypotension	Actions (for both criteria)
≤150 mmHg OR >150 mmHg	≥20 mmHg systolic **or** ≥10 mmHg diastolic ≥30 mmHg systolic **or** ≥15 mmHg diastolic	- Repeat the sequence of measurements (supine and standing) up to 2 times. If either the 1 minute or 3 minute standing BP meets the orthostatic (postural) hypotension criteria, then that sequence is considered positive. If 2 of 2 or 2 of 3 sequences are positive, then orthostatic hypotension is considered confirmed and an adverse event of orthostatic hypotension will be reported. - Refer to Section 7.4.3 for subject management and guidance.

Refer to Section 7.4.3 for guidance on determining which subjects with confirmed orthostatic hypotension will require consultation with a neurologist or cardiologist.

7.3.5. 12-lead Electrocardiogram

A 12-lead ECG will be performed at Screening for determination of ECG-related eligibility, Week 16 (End of Treatment) and Week 40 (End of Study), or at Early Termination as described in Section 6.16). Additional ECGs may be collected during the double-blind period of the study if needed (for cause), at the discretion of the investigator.

A 12–lead ECG should be recorded after subjects have been resting at least 5 minutes in the supine position in a quiet environment. Digital ECG tracings will be performed using equipment from and analyzed by a central ECG laboratory. All standard intervals (PR, QRS, QT, QTcF, QTcB, QTcF, RR intervals and HR) will be collected. The QTc interval reading produced by machine will be listed in the data listings. The QT interval will be manually measured by the central laboratory. The cardiologist at the central ECG laboratory reading the ECGs will be blinded regarding investigational product assignment. In the event a clinically significant ECG abnormality is seen at a visit on an ECG obtained for cause (post-treatment), the investigator should consider evaluation of the subject by a cardiologist.

Investigators will also be alerted of subjects with evidence of the following as a potential indicator of sympathetic nervous system dysfunction:

- Significant bradycardia (heart rate of ≤45 beats per minute (BPM) on an ECG, exclusionary at Screening).
- Heart rate decrease from Screening of ≥25% with resulting heart rate <60 BPM.

Investigators should report adverse events of bradycardia for subjects who meet the ECG criteria listed above. Refer to Section 7.4.3 for additional details pertaining to subject evaluation and dosing in subjects with sympathetic function adverse events.

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7.3.6. Survey of Autonomic Symptoms

The Survey of Autonomic Symptoms (SAS) is a validated, easily administered instrument to measure autonomic symptoms that has been proposed to be valuable in assessing neuropathic autonomic symptoms in clinical trials (refer to Appendix 11).

Subjects will complete the SAS prior to SC dosing at Screening, Week 24 and at Week 40 (End of Study visit, or at Early Termination, as described in Section 6.16).

7.3.7. Radiographic Assessments

Radiographic assessments (x-rays) of both knees, both hips and both shoulders will be obtained at Screening and at the End of Study visit (or at Early Termination, as described in Section 6.16). Other major joints exhibiting signs or symptoms suggestive of osteoarthritis should also be imaged. A major joint is defined as a mobile synovial joint in the limbs such as shoulders, elbows, wrists, hips, knees, ankles and excluding the joints of the toes and hands. Any joint imaged at Screening or other at risk joints identified during the study period should also be imaged at the End of Study visit (or at Early Termination, as described in Section 6.16).

A central radiology reader (Central Reader) will review the radiology images for assessment of eligibility including determination and identification of exclusionary joint conditions. It is recommended that the radiographs required at Screening be obtained at least two weeks prior to the beginning of the IPAP to permit central radiology review of the images and to establish subject eligibility for initial dosing in the study. Subjects will not be permitted to start dosing in the study until the Screening radiographs are reviewed and eligibility is established. Radiographs required for the Week 40 visit may be conducted within 30 days of the visit (ie, before or after the visit), but preferably prior to the Week 40 visit and if possible, no more than 14 days after Week 40.

For subjects who are discontinued prior to the Week 40 visit, follow-up radiographs of both knees, both hips and both shoulders (and any other joint imaged at Screening or other at risk joints identified during the study period) should be performed as soon as possible (refer to Section 6.16.1) after the decision to withdraw from the study has been made, provided at least 30 days have passed since the last set of X-rays were collected. A final set of follow-up radiographs should be obtained 24 weeks (Early Termination Visit 3, Section 6.16.1.5) after the last dose of study treatment was administered.

The X-ray technologists, in addition to their professional training and certifications, will be trained in performing the radiographic protocols for the knees, hips, and shoulders for this study and given approval by Pfizer or its representative to perform study X-rays. To facilitate reproducibility and accuracy of joint space width measurement in the knees and hips, a semi-automated software and positioning frame standardized subject and joint positioning protocol will be utilized. The Core Imaging Laboratory will be responsible for working with the sites to ensure quality, standardization and reproducibility of the radiographic images/assessments made at the Screening and follow-up time-points. Additional details regarding the required X-rays will be provided in a site imaging manual.

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Central radiology readers (Central Readers) will be board certified radiologists or have the international equivalent as musculoskeletal radiologists. The Central Readers will be governed by an imaging atlas and an imaging Charter which includes a specific description of the scope of their responsibilities. Central Readers will review the radiology images at Screening for assessment of eligibility (including determination of Kellgren-Lawrence Grade) and identification of exclusionary joint conditions such as rapidly progressive osteoarthritis, atrophic or hypotrophic osteoarthritis, subchondral insufficiency fractures (spontaneous osteonecrosis of the knee [SPONK]), primary osteonecrosis and pathological fractures. After randomization, the Central Reader will review radiology images for diagnosis of joint conditions that would warrant further evaluation by the Adjudication Committee such as possible or probable rapidly progressive osteoarthritis, subchondral insufficiency fractures (spontaneous osteonecrosis of the knee [SPONK]), primary osteonecrosis or pathological fracture.

For subjects who are identified with a possible or probable joint event (ie, rapidly progressive osteoarthritis, subchondral insufficiency fractures, spontaneous osteonecrosis of the knee (SPONK), primary osteonecrosis or pathological fracture) and subjects undergoing total joint replacement for any reason, all images and other source documentation will be provided to the blinded tanezumab Adjudication Committee for review and adjudication of the event. The Adjudication Committee's assessment of the event will represent the final classification of the event.

7.3.7.1. Radiation Exposure

The International Commission on Radiation Protection (ICRP) has developed and applied the ALARA principle in developing guidelines that balance the benefits of radiation exposures against possible risks. This principle states that human exposures to radiation should be "**A**s **L**ow **A**s **R**easonably **A**chievable (ALARA), with economic and social considerations taken into account."

Within the context of medical and research exposures, this is usually taken to mean that each individual should receive no more radiation than is necessary to obtain reliable information and that no more research participants should be irradiated than is necessary to answer a particular scientific question.

Radiograph	Annual Effective Dose (mSv)
Knee	0.024 mSV
Hip	1.9 mSV
Shoulder	0.04 mSV
Total	1.964 mSV

The average subject exposure per body part per year is shown in the table above. The total effective dose per subject in this study is expected to be approximately 2.0 mSv. Subject exposure (table above) in this study should not exceed the annual effective dose from natural background dose of approximately 3.0 mSv/year.[47,48,49,50] In some cases, it is expected that a repeat image of a joint may be performed around quality of the x-ray images.

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7.3.8. Neurologic Examination

Neurologic examinations will be performed by a designated physician at Screening, Baseline, and Weeks 2, 4, 8, 12, 16, 24 and 40, (or at Early Termination, as described in Section 6.16) and the Neuropathy Impairment Score (NIS) will be completed at these time points based on this neurological exam (see Appendix 7). Neurologic examination will assess strength of groups of muscles of the head and neck, upper limbs and lower limbs, deep tendon reflexes and sensation (tactile, vibration, joint position sense and pin prick) of index fingers and great toes in order to complete the NIS. The NIS is a standardized instrument which has been tested in both healthy subjects and subjects with neuropathy and which has been used to evaluate subjects for signs of peripheral neuropathy in clinical trials.[51] Investigators and other designated physicians performing the neurologic evaluations are required to attend a training session for neurological exam in order to apply consistency across sites. The neurological exams must be performed in a controlled and consistent manner and by the same examiner when possible.

A neurologic evaluation should be performed by a consulting neurologist if any of the following occurs:

- If an adverse event suggestive of new or worsening peripheral neuropathy or an adverse event of abnormal peripheral sensation (eg, allodynia, burning sensation, carpal tunnel syndrome, dysesthesia, hyperesthesia, hyperpathia, hypoesthesia, neuralgia, neuritis, neuropathy peripheral, pallanesthesia, paresthesia, peripheral sensory neuropathy, sensory disturbance, sensory loss, sciatica, tarsal tunnel syndrome) is reported as: 1) a serious adverse event or 2) an adverse event which has resulted in the subject being withdrawn from the study, or 3) an adverse event ongoing at the end of the subject's participation in the study, or 4) an adverse event of severe intensity.

- A new or worsened clinically significant abnormality on the neurologic exam should be reported as an adverse event and may result in a neurologic evaluation/consult further to the guidance above.

- A neurological adverse event which is non-neuropathic (eg, stroke, seizure) but which the investigator considers medically important should also result in a neurological consultation.

In these cases, a neurologic evaluation should be obtained as soon as possible after these signs and symptoms are known. The results of the neurological consultation will be recorded on the appropriate CRF. Adverse events will be reported where applicable as described in Section 8.

7.3.9. Anti-Drug Antibody Testing

Blood samples for the assessment of ADA against tanezumab (anti-tanezumab antibodies) will be collected at Baseline (Day 1; predose) and Weeks 8 (predose), 16, 24 and 40 (or at Early Termination, as described in Section 6.16). Specifically if subject terminates prior to

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Week 16, ADA will be determined at approximately Weeks 8, 16 and 24 weeks after the last SC dose was administered.

Instructions regarding sample processing (eg, sample volumes, tube types, storage temperatures) will be provided in the laboratory manual.

Samples will be analyzed using a validated analytical method in compliance with Pfizer Standard Operating Procedures.

Samples may be used for further evaluation of the bioanalytical method. These data will be used for internal exploratory purposes and will not be included in the clinical report.

The shipment address and assay lab contact information will be provided to the investigator site prior to initiation of the trial.

7.4. Triggered Events and Subject-Level Stopping Rules

The following rules will apply to individual subjects at the time of the second injection of SC investigational product.

7.4.1. Dysesthesia/Allodynia

Transient, resolved dysesthesia/allodynia: Administer SC investigational product as planned as long as the condition has resolved before the next scheduled dose of investigational product.

Unresolved dysesthesia/allodynia: Withhold the SC investigational product for a maximum of 14 days beyond the planned dosing day to allow for resolution of the adverse event. If the dysesthesia/allodynia has not resolved within the 14-day period after the scheduled dosing date, the subject will not receive the second dose of investigational product and will enter the Early Termination Follow-Up Period (see Section 6.16).

7.4.2. Hypersensitivity or Injection Site Reactions

If a severe hypersensitivity reaction or severe injection reaction occurs following the first administration of SC investigational product, investigational product should be discontinued immediately and no further administration of SC investigational product will be allowed. Subjects experiencing these types of reactions will enter the Early Termination Follow-Up Period (see Section 6.16).

Severe hypersensitivity reactions are defined as those causing anaphylaxis. Severe injection site reactions are defined as those in which ulceration or severe necrosis occurs.

7.4.3. Orthostatic Hypotension and Sympathetic Function Adverse Events

Blood pressure changes meeting the pre-specified criteria for orthostatic hypotension and confirmed as described in Section 7.3.4.1 will be designated as confirmed orthostatic hypotension episode and should be reported as an adverse event whether or not the subject had accompanying symptoms.

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Confirmed episodes of orthostatic hypotension: If a confirmed episode of orthostatic hypotension occurs (as defined in Section 7.3.4.1) it should be reported as an adverse event and the subject should be further evaluated described below to determine below to determine if a neurology or cardiology consultation should be obtained and/or whether further treatment with investigational product should occur. Figure 1 provides a flow diagram for the processes described below.

1. If no apparent medical cause (eg, dehydration, illness, medications) is identified at the time the orthostatic hypotension criterion is met and the subject is symptomatic, the subject should be further evaluated for the presence of sympathetic autonomic neuropathy by a cardiologist or neurologist as soon as possible. See "Sympathetic function adverse events" below for decisions regarding subject management and continued dosing with investigational product.

2. If an apparent medical cause is identified at the time the orthostatic hypotension criterion is met or if the subject is asymptomatic, the subject should have a repeat assessment of orthostatic hypotension performed at least 1 week later but not more than 4 weeks later. During this time the investigator should attempt to address the underlying medical cause of the orthostatic hypotension. If confirmed orthostatic hypotension (as defined in Section 7.3.4.1) is present at the follow up visit, the subject should be further evaluated for the presence of sympathetic autonomic neuropathy by a cardiologist or neurologist as soon as possible. See "Sympathetic function adverse events" below for decisions regarding subject management and repeat dosing.

Sympathetic function adverse events: Subjects reporting adverse events (any seriousness or severity) with preferred terms of bradycardia (see Section 7.3.5 for ECG criteria for bradycardia), syncope, orthostatic hypotension (as described above and in boxes C and E of flow diagram Figure 1), anhidrosis or hypohidrosis should be further evaluated for the presence of sympathetic autonomic neuropathy by a cardiologist or neurologist as soon as possible.

The investigator should determine the appropriate type of consultation (neurology or cardiology) depending on symptom presentation and the investigator's assessment as to the specialist best able to evaluate the subject. Pfizer will provide a guidance document which outlines appropriate recommendations regarding tests to consider for subject work-up.

These subjects should not be dosed with SC investigational product until the absence of sympathetic autonomic neuropathy has been confirmed. Subjects who are not deemed to have a sympathetic autonomic neuropathy based on this evaluation can continue the study provided no more than 12 weeks have elapsed since the last dose of SC treatment treatment (Boxes H and K of flow diagram Figure 1). However, if the subject is still symptomatic with bradycardia, syncope, orthostatic hypotension, anhidrosis or hypohidrosis up to 12 weeks after the last dose of SC treatment, s/he should not receive additional investigational product; even if a sympathetic autonomic neuropathy has not been confirmed (Boxes J and L of flow diagram Figure 1), and will enter the Early Termination Follow-up period (refer to

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Section 6.16.1). Subjects found to have a sympathetic autonomic neuropathy (Boxes I and L of flow diagram Figure 1) should not be receive additional investigational product and will enter the Early Termination Follow-up period (refer to Section 6.16.1).

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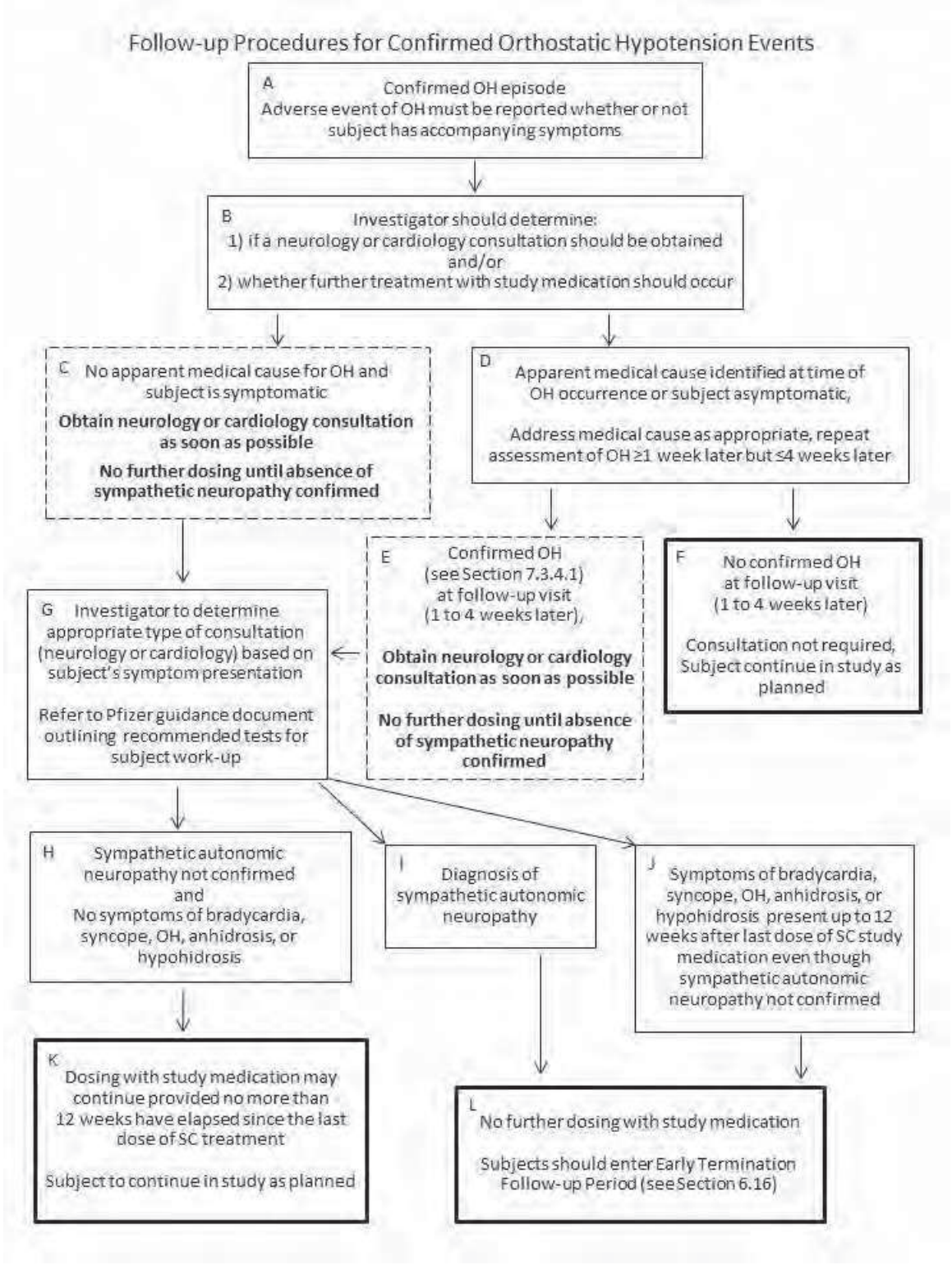
Figure 1. Follow-up Procedures for Confirmed Orthostatic Hypotension Events



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7.4.4. Evaluation and Follow-up for Increased, Severe Persistent Joint Pain

Average daily pain in the index joint will be assessed with an 11-point numeric rating scale (0 to 10) and collected via IRT beginning in the IPAP and through Week 16, followed by weekly assessments during the Follow-Up period. In addition, on a weekly basis beginning at the Initial Pain Assessment Period and through Week 40 of the study, the subject will also be asked if he/she has experienced new or increased pain in a major, non-index joint such as knee, hip or shoulder. If a subject responds that he/she has experienced new or increased pain in a major, non-index joint (post-baseline), the subject will be asked to rate his/her pain in that joint on the same 11-point numeric rating scale, using a 24-hour recall (see Section 7.1) and will continue reporting pain in that joint for the remainder of the study.

Joint pain scores recorded electronically will be monitored by site staff to identify subject who have a pattern of severe pain over several days or a rapid increase in pain. Subjects who record increased pain scores of severe intensity (eg, a score of 7-10 out of 10 on a numerical rating scale) in a knee, hip, shoulder or other major joint which is persistent for at least 2 weeks despite treatment with analgesic medication should be evaluated radiographically. An earlier evaluation of the subject can be made at the discretion of the investigator.

At each study visit, systematic site review of the WOMAC Pain Scores, electronically recorded pain scores, and relevant spontaneously reported adverse events will be implemented. In addition, adverse events of joint pain, joint swelling, joint injury/accidents, fractures or worsening of osteoarthritis symptoms in index or non-index joints will be evaluated by the site personnel. An assessment of the subjects' general health and major joints for any changes in their osteoarthritis status will be carried out.

Musculoskeletal physical exam findings, review of reported musculoskeletal adverse events, and in-clinic efficacy assessments will be recorded on specific case report forms for each study visit.

Subjects meeting the criteria for increased severe or persistent pain or with other clinically significant findings based on the assessment of the Investigator are considered to have a joint(s) at risk and must have radiographs (x-rays) of the joint(s) obtained and sent to the central reader for assessment. MRI scans will not be required but should be obtained if warranted for diagnostic purposes. If warranted, the subject will be referred to an orthopedic surgeon for evaluation.

Radiology (and any MRI) images collected as part of follow-up procedures for reports of increased severe or persistent pain or clinically significant findings of the investigator will be assessed by the Central Reader for possible or probable events of rapidly progressive osteoarthritis, subchondral insufficiency fractures (spontaneous osteonecrosis of the knee [SPONK]), primary osteonecrosis, or pathological fracture.

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7.4.5. Central Reader and Subject-Level Stopping Criteria for Joint Safety Events

Subjects identified through the measures described above (in Section 7.4.4) who are determined by the Central Reader to have possible or probable rapidly progressive osteoarthritis, subchondral insufficiency fractures (spontaneous osteonecrosis of the knee [SPONK]), primary osteonecrosis, or pathological fracture, will be withdrawn from treatment and enter the Early Termination Follow-Up period (see Section 6.16).

The Central Reader will review the radiology images on an ongoing basis and provide assessments to the investigator and Pfizer. For subjects who are identified with a possible or probable event described above and for subjects undergoing total joint replacement for any reason, all images and other source documentation will be provided to the blinded tanezumab Adjudication Committee for review and adjudication of the event. The Adjudication Committee's assessment of the event will represent the final classification of the event.

Subjects with adverse event reports of rapidly progressive osteoarthritis (type 1 or type 2), subchondral insufficiency fractures (spontaneous osteonecrosis of the knee [SPONK]), primary osteonecrosis, or pathological fracture, will be withdrawn from treatment and enter the Early Termination Follow up period (refer to Section 6.16).

In addition to Subject-Level Stopping Criteria for Joint Safety Events, this study will also employ Protocol-Level Stopping Criteria. Protocol-Level Stopping Criteria for Joint Safety Events are described in Section 9.6.1.3.

7.4.6. Procedures for Subjects Undergoing Joint Replacement

Subjects who have undergone or plan to undergo total joint replacement or other arthroplasty procedure during the study will be discontinued from study treatment. Follow up procedures for these subjects are described in Section 6.16.2. In addition, subjects who undergo total knee, hip or shoulder joint replacement surgery during the study (Treatment Period or Follow up Period) will be followed for 24 weeks after the procedure as part of a separate protocol, provided the subject consents (see Section 6.17).

7.5. Pharmacokinetic (PK) and Pharmacodynamic (PD)

7.5.1. Plasma for Analysis of Tanezumab

Blood samples for the assessment of the pharmacokinetics of tanezumab will be collected pre-dose at Baseline (Day 1) and at Weeks 2 (in a subset of subjects), 4 (in a subset of subjects), 8 (pre-dose), 12 (in a subset of subjects), 16, and 24 (or at Early Termination, as described in Section 6.16). Specifically, if subject terminates prior to Week 16, PK will be determined at approximately 8 and 16 weeks after the last SC dose was administered.

Instructions regarding sample collection processing (eg, sample volumes, tube types, storage temperatures) will be provided in the laboratory manual.

Samples will be analyzed using a validated analytical method in compliance with Pfizer standard operating procedures.

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Samples may be used for further evaluation of the bioanalytical method. These data will be used for internal exploratory purposes and will not be included in the clinical report.

The shipment address and assay lab contact information will be provided to the investigator site prior to initiation of the trial.

7.5.2. Nerve Growth Factor (NGF) for Pharmacodynamic Analyses

Blood samples for the assessment of total and free NGF (if assay is available) concentrations will be collected pre-dose at Baseline (Day 1) and at Weeks 2 (in a subset of subjects), 4 (in a subset of subjects), 8 (pre-dose), 12 (in a subset of subjects), 16, and 24 (or at Early Termination, as described in Section 6.16). Specifically, if subject terminates prior to Week 16, NGF will be determined at approximately 8 and 16 weeks after the last SC dose was administered.

NGF can exist in different forms including but not limited to NGF bound to drug or not bound to drug, NGF bound to soluble p75, and proNGF. Blood volume collected may limit the number of NGF assessments to 3 to 4 NGF endpoints including a measure of total NGF (sum of all NGF forms). The final set of NGF forms that will be measured will depend on the availability of the analytical assay that can reliably measure the different NGF concentrations.

Instructions regarding sample collection and processing (eg, sample volumes, tube types, storage temperatures) will be provided in the laboratory manual.

Samples will be analyzed using a validated analytical method in compliance with Pfizer standard operating procedures.

NGF samples may be used for further evaluation of the bioanalytical methods used for measuring NGF. These data will be used for internal exploratory purposes and will not be included in the clinical report.

The shipment address and assay lab contact information will be provided to the investigator site prior to initiation of the study.

7.6. Biomarkers

7.6.1. Serum Biomarkers

Blood samples for the assessment of biomarkers that can be modulated by the osteoarthritis condition will be collected pre-dose at Baseline (Day 1), at Week 8 (pre-dose) and at Week 24 or Early Termination, as described in Section 6.16. Specifically, if subject terminates prior to Week 16, serum biomarkers will be determined at approximately 8 weeks after the last SC dose was administered and at approximately 16 weeks after the last SC dose was administered.

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If possible, the samples should be obtained following a fasting period of at least 8 hours and at approximately the same time at each study visit to control for variations in the biomarkers related to prandial status and diurnal cycle. Fasting status should be recorded on the eCRF.

Currently the measurement of 12 biomarkers is planned: CCI [redacted], CCI [redacted] CCI [redacted] CCI [redacted] CCI [redacted], CCI [redacted], CCI [redacted] CCI [redacted] CCI [redacted], CCI [redacted] CCI [redacted]. This selection of biomarkers could change due to blood volume limitations and/or assay performance issues. Possibly OA biomarkers different from the ones listed could be added or substituted if considered informative to further understand the osteoarthritis condition.

Instructions regarding sample collection and processing (eg, sample volumes, tube types, storage temperatures) will be provided in the laboratory manual.

Samples will be analyzed using validated analytical methods in compliance with Pfizer standard operating procedures.

Biomarker samples may be used for further evaluation of biomarkers other than the ones listed that could improve the understanding of the safety and efficacy profile of tanezumab. These data will be used for internal exploratory purposes and will not be included in the clinical report.

The shipment address will be provided to the investigator site prior to initiation of the study.

7.6.2. Urine Biomarkers

Urine samples for the assessment of the cartilage biomarker CCI [redacted] will be collected pre-dose at Baseline (Day 1), Week 8 (pre-dose) and at Week 24 or Early Termination, as described in Section 6.16. Specifically, if subject terminates prior to Week 16, urine biomarkers will be determined at approximately 8 weeks after the last SC dose was administered and at approximately 16 weeks after the last SC dose was administered.

The samples should not be obtained from the first void of the day and should be obtained at approximately the same time at each study visit to control for diurnal variations in the biomarkers, and preferably following a fasting period of at least 8 hours. Fasting status should be recorded on the eCRF. Sites will provide collection containers and storage instructions for subjects doing home collection.

Instructions regarding sample processing (eg, sample volumes, tube types, storage temperatures) will be provided in the laboratory manual.

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Samples will be analyzed using a validated analytical method in compliance with Pfizer standard operating procedures.

Biomarker samples may be used for further evaluation of biomarkers other than the ones listed that could improve the understanding of the safety and efficacy profile of tanezumab. These data will be used for internal exploratory purposes and will not be included in the clinical report.

The shipment address and assay lab contact information will be provided to the investigator site prior to initiation of the study.

7.7. Banked Biospecimens

7.7.1. Markers of Drug Response

Studying the variation in genetic markers and other biomarkers may help to explain some of the variability in response seen with some drugs among different individuals. This is referred to as pharmacogenomic/biomarker research. Comparing the deoxyribonucleic acid (DNA), ribonucleic acid (RNA), protein, and metabolite variation patterns of subjects who respond well and those who respond poorly to treatment may help to better define the most appropriate group of subjects in which to target a given treatment. Collecting biospecimens for exploratory pharmacogenomic/biomarker analyses and retaining them in the Pfizer BioBank makes it possible to better understand the drug's mechanism of action and to seek explanations for differences in, for example, exposure, efficacy, tolerability, or safety not anticipated prior to the beginning of the study.

Providing these biospecimens is a required study activity for study sites and subjects, unless prohibited as such by local regulations or ethics committee decision.

To protect subjects' confidentiality, the banked biospecimens and data generated from them will be coded with the subject's study identification (ID) number. Samples will be kept in a facility accessible only by swiping a badge. Data will be stored on password-protected computer systems. The key between the code and the subject's personal identifiers will be held at the study site; the researchers using the biospecimens and data generated from them will not have access to the key nor any personally identifying information. Biospecimens will only be used only for the purposes described here and in the informed consent document/subject information sheet; any other uses require additional ethical approval. Unless a time limitation is required by local regulations or ethical requirements, biospecimens will be stored indefinitely to allow for future research on the topics described here, including research conducted during the lengthy drug development process and also post-marketing research. Subjects can withdraw their consent for the use of their biospecimens at any time by making a request to the investigator, in which event any remaining biospecimen will be destroyed; data already generated from the biospecimens will continue to be stored to protect the integrity of existing analyses. It is very unlikely that results generated from the biospecimens will have any clinical, diagnostic, or therapeutic implications for the individual study participants. Subjects are notified in the informed consent document/subject information sheet that their results will not be given to them,

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unless required by local laws or regulations, in which case results will be returned via the investigator. Results will not be provided to family members or other physicians; nor will they be recorded in the subject's medical record. There is no intention to contact subjects after completion of the clinical study.

A 4 mL blood biospecimen **Prep D1 (K_2ethylenediaminetetraacetic acid (EDTA) whole blood collection optimized for DNA analysis)** will be collected at the Baseline visit to be retained for potential pharmacogenomic/biomarker analyses related to drug response, unless prohibited by local regulations or ethics committee decision. For example, putative safety biomarkers, drug metabolizing enzyme genes, drug transport protein genes, or genes thought to be related to the mechanism of drug action may be examined.

The banked biospecimen will be collected from all subjects unless prohibited by local regulations or ethics committee decision. Detailed collection, processing, storage and shipment instructions are provided in the central laboratory manual.

It is possible that the use of these biospecimens may result in commercially viable products. Subjects will be advised in the informed consent document/subject information sheet that they will not be compensated in this event.

7.7.2. Additional Research

Unless prohibited by local regulations or ethics committee decision, subjects will be asked to indicate on the consent form whether they will allow the banked biospecimens to also be used for the following research:

- Investigations of the disease under study in the clinical study, and related conditions.
- Biospecimens may be used as controls. This includes use in case-control studies of diseases for which Pfizer is researching drug therapies; use in characterizing the natural variation amongst people in genes, RNA, proteins, and metabolites; and use in developing new technologies related to pharmacogenomics/biomarkers.

Subjects need not provide additional biospecimens for the uses described in Section 7.7.2; the biospecimens specified in the Markers of Drug Response Section 7.7.1 will be used. Subjects may still participate in the clinical study if they elect not to allow their banked biospecimens to be used for the additional purposes described in this Section.

8. ADVERSE EVENT REPORTING

8.1. Adverse Events

All observed or volunteered AEs regardless of treatment group or suspected causal relationship to the investigational product(s) will be reported as described in the following sections.

For all AEs, the investigator must pursue and obtain information adequate both to determine the outcome of the AE and to assess whether it meets the criteria for classification as a

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serious adverse event (SAE) requiring immediate notification to Pfizer or its designated representative. For all AEs, sufficient information should be obtained by the investigator to determine the causality of the AE. The investigator is required to assess causality. Follow-up by the investigator may be required until the event or its sequelae resolve or stabilize at a level acceptable to the investigator, and Pfizer concurs with that assessment.

As part of ongoing safety reviews conducted by the Sponsor, any non-serious AE that is determined by the Sponsor to be serious will be reported by the Sponsor as an SAE. To assist in the determination of case seriousness further information may be requested from the investigator to provide clarity and understanding of the event in the context of the clinical study.

8.2. Reporting Period

For SAEs, the active reporting period to Pfizer or its designated representative begins from the time that the subject provides informed consent, which is obtained prior to the subject's participation in the study, ie, prior to undergoing any study-related procedure and/or receiving investigational product, through the end of the safety Follow-Up period or through and including 112 calendar days after the subject's last administration of the subcutaneous investigational product if the subject refuses the protocol defined Follow-Up period.

SAEs occurring to a subject after the active reporting period has ended should be reported to the Sponsor if the investigator becomes aware of them; at a minimum, all serious adverse events that the investigator believes have at least a reasonable possibility of being related to investigational product are to be reported to the Sponsor.

AEs (serious and non-serious) should be recorded on the CRF from the time the subject has taken at least 1 dose of investigational product through the subject's last visit.

8.3. Definition of an Adverse Event

An AE is any untoward medical occurrence in a clinical investigation subject administered a product or medical device; the event need not necessarily have a causal relationship with the treatment or usage. Examples of AEs include but are not limited to:

- Abnormal test findings;
- Clinically significant symptoms and signs;
- Changes in physical examination findings;
- Hypersensitivity;
- Progression/worsening of underlying disease;
- Drug abuse;
- Drug dependency.

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Additionally, they may include the signs or symptoms resulting from:

- Drug overdose;
- Drug withdrawal;
- Drug misuse;
- Drug interactions;
- Extravasation;
- Exposure during pregnancy (EDP);
- Exposure via breastfeeding;
- Medication error;
- Occupational exposure.

8.4. Medication Errors

Medication errors may result, in this study, from the administration or consumption of the wrong product, by the wrong subject, at the wrong time, or at the wrong dosage strength. Such medication errors occurring to a study participant are to be captured on the medication error CRF which is a specific version of the AE page, and on the SAE form when appropriate. In the event of medication dosing error, the sponsor should be notified immediately.

Medication errors are reportable irrespective of the presence of an associated AE/SAE, including:

- Medication errors involving subject exposure to the investigational product.
- Potential medication errors or uses outside of what is foreseen in the protocol that do or do not involve the participating subject.

Whether or not the medication error is accompanied by an adverse event, as determined by the investigator, the medication error should be captured on the medication error version of the AE page and, if applicable, any associated AE(s) are captured on an AE CRF page.

8.5. Abnormal Test Findings

The criteria for determining whether an abnormal objective test finding should be reported as an AE are as follows:

- Test result is associated with accompanying symptoms; and/or

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- Test result requires additional diagnostic testing or medical/surgical intervention; and/or

- Test result leads to a change in study dosing or discontinuation from the study, significant additional concomitant drug treatment, or other therapy; and/or

- Test result is considered to be an AE by the investigator or sponsor.

Merely repeating an abnormal test, in the absence of any of the above conditions, does not constitute an AE. Any abnormal test result that is determined to be an error does not require reporting as an AE.

8.6. Serious Adverse Events

A serious adverse event is any untoward medical occurrence at any dose that:

- Results in death;

- Is life-threatening (immediate risk of death);

- Requires inpatient hospitalization or prolongation of existing hospitalization;

- Results in persistent or significant disability/incapacity (substantial disruption of the ability to conduct normal life functions);

- Results in congenital anomaly/birth defect.

Medical and scientific judgment is exercised in determining whether an event is an important medical event. An important medical event may not be immediately life-threatening and/or result in death or hospitalization. However, if it is determined that the event may jeopardize the subject or may require intervention to prevent one of the other AE outcomes, the important medical event should be reported as serious.

Examples of such events are intensive treatment in an emergency room or at home for allergic bronchospasm; blood dyscrasias or convulsions that do not result in hospitalization; or development of drug dependency or drug abuse.

Medical device complaints may meet the SAE reporting requirement criteria (see section on Medical Device Complaint Reporting Requirements). An incident is any malfunction (ie, the failure of a device to meet its performance specifications or to perform as intended; performance specifications include all claims made in the labeling for the device) that, directly or indirectly, might lead to or might have led to the death of a subject, or user, or of other persons, or to a serious deterioration in their state of health.

A serious injury that can cause a serious deterioration in state of health can include:

- a life-threatening illness, even if temporary in nature;

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- a permanent impairment of a body function or permanent damage to a body structure;
- a condition necessitating medical or surgical intervention to prevent the above 2 bulleted items

  Examples: clinically relevant increase in the duration of a surgical procedure, a condition that requires hospitalization or significant prolongation of existing hospitalization;
- any indirect harm as a consequence of an incorrect diagnostic or in vitro diagnostic device test results when used within the manufacturer's instructions for use;
- fetal distress, fetal death, or any congenital abnormality or birth defects.

8.6.1. Protocol-Specified Serious Adverse Events

There are no protocol-specified SAEs in this study. All SAEs will be reported by the investigator as described in previous sections, and will be handled as SAEs in the safety database (see Section on Serious Adverse Event Reporting Requirements).

8.6.2. Potential Cases of Drug-Induced Liver Injury

Abnormal values in aspartate aminotransferase (AST) and/or alanine aminotransferase (ALT) levels concurrent with abnormal elevations in total bilirubin level that meet the criteria outlined below in the absence of other causes of liver injury are considered potential cases of drug-induced liver injury (potential Hy's law cases) and should always be considered important medical events.

The threshold of laboratory abnormalities for a potential case of drug-induced liver injury depends on the subject's individual baseline values and underlying conditions. Subjects who present with the following laboratory abnormalities should be evaluated further to definitively determine the etiology of the abnormal laboratory values:

- Subjects with AST or ALT and total bilirubin baseline values within the normal range who subsequently present with AST or ALT values ≥3 times the upper limit of normal (X ULN) concurrent with a total bilirubin value ≥2 X ULN with no evidence of hemolysis and an alkaline phosphatase value ≤2 X ULN or not available.
- For subjects with preexisting ALT **OR** AST **OR** total bilirubin values above the ULN, the following threshold values should be used in the definition mentioned above:
  - For subjects with pre-existing AST or ALT baseline values above the normal range: AST or ALT values ≥2 times the baseline values and ≥3 X ULN, or ≥8 X ULN (whichever is smaller).

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Concurrent with

- For subjects with pre-existing values of total bilirubin above the normal range: Total bilirubin increased from baseline by an amount of at least 1 time the upper limit of normal or if the value reaches ≥3 times the upper limit of normal (whichever is smaller).

The subject should return to the investigational site and be evaluated as soon as possible, preferably within 48 hours from awareness of the abnormal results. This evaluation should include laboratory tests, detailed history, and physical assessment.

In addition to repeating measurements of AST and ALT, laboratory tests should include albumin, creatine kinase, total bilirubin, direct and indirect bilirubin, gamma-glutamyl transferase, prothrombin time (PT)/international normalized ratio (INR), and alkaline phosphatase. A detailed history, including relevant information, such as review of ethanol, acetaminophen, recreational drug and supplement consumption, family history, occupational exposure, sexual history, travel history, history of contact with a jaundiced person, surgery, blood transfusion, history of liver or allergic disease, and work exposure, should be collected. Further testing for acute hepatitis A, B, or C infection and liver imaging (eg, biliary tract) may be warranted. All cases confirmed on repeat testing as meeting the laboratory criteria defined above, with no other cause for liver function test (LFT) abnormalities identified at the time should be considered potential Hy's law cases irrespective of availability of all the results of the investigations performed to determine etiology of the abnormal LFTs. Such potential Hy's Law cases should be reported as SAEs.

8.7. Hospitalization

Hospitalization is defined as any initial admission (even less than 24 hours) in a hospital or equivalent healthcare facility or any prolongation of an existing admission. Admission also includes transfer within the hospital to an acute/intensive care unit (eg, from the psychiatric wing to a medical floor, medical floor to a coronary care unit, or neurological floor to a tuberculosis unit). An emergency room visit does not necessarily constitute a hospitalization; however, the event leading to the emergency room visit should be assessed for medical importance.

Hospitalization does not include the following:

- Rehabilitation facilities;
- Hospice facilities;
- Respite care (eg, caregiver relief);
- Skilled nursing facilities;
- Nursing homes;

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- Same day surgeries (as outpatient/same day/ambulatory procedures).

Hospitalization or prolongation of hospitalization in the absence of a precipitating, clinical AE is not in itself an SAE. Examples include:

- Admission for treatment of a preexisting condition not associated with the development of a new AE or with a worsening of the preexisting condition (eg, for work-up of persistent pre-treatment laboratory abnormality);
- Social admission (eg, subject has no place to sleep);
- Administrative admission (eg, for yearly physical examination);
- Protocol-specified admission during a study (eg, for a procedure required by the study protocol);
- Optional admission not associated with a precipitating clinical AE (eg, for elective cosmetic surgery);
- Hospitalization for observation without a medical AE;
- Pre-planned treatments or surgical procedures. These should be noted in the baseline documentation for the entire protocol and/or for the individual subject.

Diagnostic and therapeutic non-invasive and invasive procedures, such as surgery, should not be reported as AEs. However, the medical condition for which the procedure was performed should be reported if it meets the definition of an AE. For example, an acute appendicitis that begins during the AE reporting period should be reported as the AE, and the resulting appendectomy should be recorded as treatment of the AE.

8.8. Severity Assessment

If required on the AE CRFs, the investigator will use the adjectives MILD, MODERATE, or SEVERE to describe the maximum intensity of the AE. For purposes of consistency, these intensity grades are defined as follows:	
MILD	Does not interfere with subject's usual function.
MODERATE	Interferes to some extent with subject's usual function.
SEVERE	Interferes significantly with subject's usual function.

Note the distinction between the severity and the seriousness of an AE. A severe event is not necessarily an SAE. For example, a headache may be severe (interferes significantly with the subject's usual function) but would not be classified as serious unless it met one of the criteria for SAEs, listed above.

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8.9. Causality Assessment

The investigator's assessment of causality must be provided for all AEs (serious and non-serious); the investigator must record the causal relationship in the CRF, as appropriate, and report such an assessment in accordance with the SAE reporting requirements if applicable. An investigator's causality assessment is the determination of whether there exists a reasonable possibility that the investigational product caused or contributed to an AE; generally the facts (evidence) or arguments to suggest a causal relationship should be provided. If the investigator does not know whether or not the investigational product caused the event, then the event will be handled as "related to investigational product" for reporting purposes, as defined by the Sponsor (see Section on Reporting Requirements). If the investigator's causality assessment is "unknown but not related to investigational product", this should be clearly documented on study records.

In addition, if the investigator determines that an SAE is associated with study procedures, the investigator must record this causal relationship in the source documents and CRF, as appropriate, and report such an assessment in accordance with the SAE reporting requirements, if applicable.

8.10. Exposure During Pregnancy

For both unapproved/unlicensed products and for marketed products, an exposure during pregnancy occurs if:

1. A female becomes, or is found to be, pregnant either while receiving or having been exposed (eg, because of treatment or environmental exposure) to the investigational product; or the female becomes, or is found to be pregnant after discontinuing and/or being exposed to the investigational product;

   An example of environmental exposure would be a case involving direct contact with a Pfizer product in a pregnant woman (eg, a nurse reports that she is pregnant and has been exposed to chemotherapeutic products).

2. A male subject has been exposed (eg, because of treatment or environmental exposure) to the investigational product prior to or around the time of conception and/or is exposed during his partner's pregnancy.

If a study subject or study subject's partner becomes or is found to be pregnant during the study subject's treatment with the investigational product, the investigator must submit this information to the Pfizer Drug Safety Unit on an SAE Report Form and Exposure During Pregnancy (EDP) supplemental form, regardless of whether an SAE has occurred. In addition, the investigator must submit information regarding environmental exposure to a Pfizer product in a pregnant woman (eg, a subject reports that she is pregnant and has been exposed to a cytotoxic product by inhalation or spillage) using the EDP Supplemental form. This must be done irrespective of whether an AE has occurred and within 24 hours of awareness of the exposure. The information submitted should include the anticipated date of delivery (see below for information related to termination of pregnancy).

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Follow-up is conducted to obtain general information on the pregnancy and its outcome for all EDP reports with an unknown outcome. The investigator will follow the pregnancy until completion (or until pregnancy termination) and notify Pfizer of the outcome as a follow up to the initial EDP Supplemental form. In the case of a live birth, the structural integrity of the neonate can be assessed at the time of birth. In the event of a termination, the reason(s) for termination should be specified and, if clinically possible, the structural integrity of the terminated fetus should be assessed by gross visual inspection (unless pre-procedure test findings are conclusive for a congenital anomaly and the findings are reported).

If the outcome of the pregnancy meets the criteria for an SAE (ie, ectopic pregnancy, spontaneous abortion, intrauterine fetal demise, neonatal death, or congenital anomaly [in a live born baby, a terminated fetus, an intrauterine fetal demise, or a neonatal death]), the investigator should follow the procedures for reporting SAEs.

Additional information about pregnancy outcomes that are reported as SAEs follows:

- Spontaneous abortion includes miscarriage and missed abortion;
- Neonatal deaths that occur within 1 month of birth should be reported, without regard to causality, as SAEs. In addition, infant deaths after 1 month should be reported as serious adverse events when the investigator assesses the infant death as related or possibly related to the exposure to investigational product.

Additional information regarding the EDP may be requested by the investigator. Further follow-up of birth outcomes will be handled on a case-by-case basis (eg, follow-up on preterm infants to identify developmental delays). In the case of paternal exposure, the investigator will provide the study subject with the Pregnant Partner Release of Information Form to deliver to his partner. The Investigator must document in the source document that the subject was given the Pregnant Partner Release of Information Form to provide to his partner.

8.10.1. Additional Postnatal Development Follow-Up

The investigator will be asked to assist with collection of assessments of postnatal development as part of a separate protocol. Participation in that protocol is optional and will require that the subject review, agree and sign a separate informed consent document specific to that study, explaining the details of the post-partum follow-up for the subject and the newborn to participate in these assessments of postnatal development.

8.11. Occupational Exposure

An occupational exposure occurs when during the performance of job duties, a person (whether a healthcare professional or otherwise) gets in unplanned direct contact with the product, which may or may not lead to the occurrence of an adverse event.

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An occupational exposure is reported to the drug safety unit within 24 hours of investigator's awareness, using the SAE Report form, regardless of whether there is an associated AE/SAE. Since the information does not pertain to a subject enrolled in the study, the information is not reported on a CRF, however a copy of the completed SAE Report form is maintained in the investigator file.

8.12. Withdrawal Due to Adverse Events (See Also Section 6.16 Subject Withdrawal)

Withdrawal due to AEs should be distinguished from withdrawal due to other causes, according to the definition of AE noted earlier, and recorded on the appropriate AE CRF page.

When a subject withdraws because of an SAE, the SAE must be reported in accordance with the reporting requirements defined below.

8.13. Eliciting Adverse Event Information

The investigator is to report all directly observed AEs and all AEs spontaneously reported by the study subject or legally acceptable representative. In addition, each study subject or legally acceptable representative will be questioned about AEs.

8.14. Reporting Requirements

Each AE is to be assessed to determine if it meets the criteria for SAEs. If an SAE occurs, expedited reporting will follow local and international regulations, as appropriate.

8.14.1. Serious Adverse Event Reporting Requirements

If an SAE occurs, Pfizer is to be notified within 24 hours of investigator awareness of the event.

In particular, if the SAE is fatal or life-threatening, notification to Pfizer must be made immediately, irrespective of the extent of available AE information. This timeframe also applies to additional new information (follow-up) on previously forwarded SAE reports as well as to the initial and follow-up reporting of exposure during pregnancy, exposure via breastfeeding and occupational exposure cases.

In the rare event that the investigator does not become aware of the occurrence of an SAE immediately (eg, if an outpatient study subject initially seeks treatment elsewhere), the investigator is to report the event within 24 hours after learning of it and document the time of his/her first awareness of the AE.

For all SAEs, the investigator is obligated to pursue and provide information to Pfizer in accordance with the timeframes for reporting specified above. In addition, an investigator may be requested by Pfizer to obtain specific additional follow-up information in an expedited fashion. This information collected for SAEs is more detailed than that captured on the AE CRF. In general, this will include a description of the AE in sufficient detail to allow for a complete medical assessment of the case and independent determination of possible causality. Information on other possible causes of the event, such as concomitant medications, vaccines, and/or illnesses must be provided. In the case of a subject death, a

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summary of available autopsy findings must be submitted as soon as possible to Pfizer or its designated representative.

8.14.2. Non-Serious Adverse Event Reporting Requirements

All AEs will be reported on the AE page(s) of the CRF. It should be noted that the form for collection of SAE information is not the same as the AE CRF. Where the same data are collected, the forms must be completed in a consistent manner. For example, the same AE term should be used on both forms. AEs should be reported using concise medical terminology on the CRFs as well as on the form for collection of SAE information.

8.14.3. Medical Device Complaint Reporting Requirements

All medical device complaints regardless of whether the medical device complaint is associated with an AE will be collected on the applicable pages within the CRF. This includes potential incidents or malfunctions associated with the use of a medical device product. An incident or malfunction is an event that might have led to death or serious deterioration in health, or if it occurred again might have led to death or serious deterioration in health.

Pfizer is to be notified of all medical device complaints within 24 hours of the investigator's awareness of the event.

Refer to the Pharmacy Manual for procedures for forwarding medical device complaints not associated with an SAE to Pfizer.

8.14.4. Sponsor's Reporting Requirements to Regulatory Authorities

Adverse event reporting, including suspected unexpected serious adverse reactions, will be carried out in accordance with applicable local regulations.

9. DATA ANALYSIS/STATISTICAL METHODS

Detailed methodology for summary and statistical analyses of the data collected in this study will be documented in a Statistical Analysis Plan, which will be maintained by the sponsor. This document may modify the plans outlined in the protocol; however, any major modifications of the primary endpoint definition and/or its analysis will also be reflected in a protocol amendment.

9.1. Sample Size Determination

A sample size of approximately 230 subjects per treatment group is needed to provide approximately 90% power to achieve statistical significance (at the 5% two-sided level) for the two comparisons of tanezumab 2.5 and 2.5/5 mg SC versus placebo, over all three co-primary endpoints. The total sample size will be approximately 690 subjects.

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The assumed treatment differences for this calculation comes from a combined analysis of Studies A4091011 and A4091014, using an ANCOVA model for the change from Baseline to Week 16 with Baseline Observation Carried Forward (BOCF) used for missing data. Under this model the difference between tanezumab 2.5 mg and placebo were approximately -1.0 for WOMAC Pain and Physical Function subscales, and -0.32 for the Patient's Global Assessment of Osteoarthritis. The within-group standard deviations were 2.73, 2.58 and 0.92 for WOMAC Pain, WOMAC Physical Function and Patient's Global Assessment of Osteoarthritis, respectively. The correlation of the change from Baseline to Week 16 value for WOMAC Pain versus Physical Function was 0.93. The correlations between Patient's Global Assessment of Osteoarthritis and WOMAC Pain and Physical Function were approximately 0.68.

Using the treatment differences, variability and correlations given above, the sample size of 230 subjects per group would give 90% power to show significant treatment differences using the two-sided 5% significance, for the two tanezumab versus placebo comparisons over all three co-primary efficacy endpoints.

9.2. Efficacy Analysis

9.2.1. Analysis of Primary Endpoint

The primary efficacy population will be the ITT population, defined as all randomized subjects who received SC investigational product (either tanezumab or matching placebo). The primary analysis will use multiple imputation methods for missing data at Week 16. Details of the multiple imputation procedure are given below. All treatment comparisons will use the two-sided 5% significance level.

The co-primary efficacy endpoints will be analyzed using an ANCOVA model, with model terms for Baseline score, Baseline diary average pain, index joint (knee or hip), Kellgren-Lawrence grade, and treatment group, and study site as a random effect. The assessment of significance for the tanezumab SC versus placebo treatment contrasts will use a step-down testing strategy within each of the co-primary efficacy endpoints defined as first testing tanezumab 2.5/5 mg versus placebo, and if statistically significant ($p \leq 0.05$) to then test tanezumab 2.5 mg versus placebo. Finally, a tanezumab treatment group is declared as superior to placebo if the corresponding treatment contrast is significant over all three co-primary endpoints. This testing procedure will maintain the Type I error to 5% or less within each of the co-primary efficacy endpoints, and to less than 5% for all three co-primary efficacy endpoints. An additional (main effects ANCOVA) analysis for each of the co-primary efficacy endpoints will use a per-protocol analysis set, which will exclude subjects who are major protocol deviators.

The primary analysis of the co-primary endpoints will use multiple imputation for missing data, to account for uncertainty around the subject response. The basis for imputing missing values will be dependent on the reasons for missing data. For subjects with missing data due to discontinuation prior to Week 16 for lack of efficacy or for an adverse event or death, imputation will be based on sampling from a normal distribution using a mean value equal to the subject's Baseline efficacy value and the standard deviation (over all treatment groups) of

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the observed efficacy data at Week 16. For subjects with missing data for any other reason, imputation will be based on sampling from a normal distribution using a mean value of the subject's last observed efficacy value and standard deviation (over all treatment groups) of the observed efficacy data at Week 16. Imputed values for the Patient's Global Assessment of Osteoarthritis will be rounded to integer values from 1 to 5. Imputed values for WOMAC Pain and Physical Function will be truncated at 0 and 10. One hundred imputation samples will be used, and the ANCOVA model described above will be used for each imputation dataset. The final results will be calculated using the combined sets of results from each imputation dataset analysis.[52]

Additional analyses will explore the sensitivity of results to the effect of missing data on the co-primary efficacy endpoints. The first analysis will use the same main effects ANCOVA model as described above, but with Last Observation Carried Forward (LOCF) for missing data. The second analysis will use the same main effects ANCOVA model as described above, but with BOCF for missing data. The third analysis will use Mixed Model for Repeated Measurements (MMRM) utilizing all observed data up to and including Week 16.

All analyses will show estimates of the treatment group response and treatment group differences of each tanezumab group versus placebo, with corresponding standard errors of the mean, and 95% confidence intervals (and p-values for treatment differences).

9.2.2. Analysis of Secondary Endpoints

Secondary endpoints will examine the change from Baseline to additional timepoints in the WOMAC Pain and Physical Function subscales, and the Patient's Global Assessment of Osteoarthritis, using the multiple imputation for missing data procedure and analysis described above. Other secondary endpoints include the WOMAC Stiffness subscale, WOMAC Average score and WOMAC Pain subscale items (Pain When Walking on a Flat Surface, and Pain When Going Up or Down Stairs), all conducted for the change from Baseline to Weeks 2, 4, 8, 12 and 16. Analysis of Average Pain in the index joint will be conducted for the change from Baseline to Weeks 1, 2, 3, 4, 6, 8, 10, 12 and 16. The analysis of these endpoints will use the same ANCOVA analysis as described above for the co-primary endpoints, with multiple imputation for missing data.

Subject response endpoints of the OMERACT-OARSI responder criteria, improvement in the WOMAC Pain ≥30, 50, 70 and 90%, WOMAC Physical Function ≥30, 50, 70 and 90%, and improvement in the Patient's Global Assessment of Osteoarthritis ≥2 will be analyzed for change from Baseline to Weeks 2, 4, 8, 12 and 16, using logistic regression for binary data, with model terms for baseline WOMAC Pain subscale score, WOMAC Physical Function subscale score, or Patient's Global Assessment score, Baseline diary average pain, index joint, Kellgren-Lawrence grade, and treatment group. Imputation for missing data will use both LOCF and BOCF, where imputation with BOCF will lead to the subject being assessed as a non-responder for the response endpoint at a particular timepoint. In addition, in order to closely match the primary imputation analysis, a mixed BOCF/LOCF imputation for response endpoints will be used. In this analysis BOCF imputation (ie, a subject would be a non-responder) would be used for missing data due to discontinuation for reasons of

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lack of efficacy, adverse event or death up to the timepoint of interest, and LOCF imputation would be used for missing data for any other reason.

The change from Baseline in the Patient's Global Assessment of Osteoarthritis to Weeks 2, 4, 8, 12 and 16 will also be analyzed using Cochran-Mantel-Haenszel (CMH) test, stratified by the combinations of the two stratification factors. Changes by each level of improvement will be summarized. For this analysis imputation for missing data will used mixed BOCF/LOCF, as well as BOCF and LOCF separately. If there are too few subjects in any stratification combination group (defined as <15 subjects in any stratification factor) then an unstratified test will be performed.

The incidence and number of days of use of rescue medication, will be analyzed for Weeks 2, 4, 8, 12, 16 and 24 and the amount of rescue medication use per week will be analyzed for Weeks 2, 4, 8, 12, and 16. The incidence of use of rescue medication will be analyzed using logistic regression for binary data, with model terms for Baseline WOMAC Pain subscale score, Baseline diary average pain, index joint, Kellgren-Lawrence grade, and treatment group. The number of days and amount of rescue medication (mg dosage of acetaminophen) will be analyzed using the Negative Binomial model, with model terms of Baseline WOMAC Pain subscale score, Baseline diary average pain score, index joint, Kellgren-Lawrence grade, and treatment group. Estimated levels of rescue medication use will be shown for each treatment group, and the ratio (with 95% CI) for comparisons versus placebo will be shown. Imputation for missing rescue medication data will use LOCF only.

The incidence of and time to withdrawal due to lack of efficacy will also be analyzed for discontinuation up to Week 16 (end of treatment period). The time to discontinuation will be analyzed using the log-rank test, with Kaplan-Meier estimates of the time to discontinuation shown for selected percentiles, dependent on the level of discontinuation. The expectation is that these would be the 1st, 2nd, 5th, 10th and 25th percentiles. Other percentiles may be shown if the level of discontinuation due to lack of efficacy as calculated using Kaplan-Meier procedure is sufficiently large. The analysis of the incidence of discontinuation due to lack of efficacy will be made using logistic regression for binary data, with model terms for Baseline WOMAC Pain subscale score, Baseline diary average pain score, index joint, Kellgren-Lawrence grade, and treatment group.

Cumulative WOMAC Pain response and WOMAC Physical Function at Week 16 using response definitions from a reduction of >0% to =100% (in steps of 10%) will be summarized, using mixed BOCF/LOCF (as described above), and also LOCF and BOCF imputation for WOMAC Pain and WOMAC physical function. Imputation with BOCF for subjects with missing data at that timepoint will lead to the subjects being assessed as non-responders for the response endpoint.

A table showing number and percentage of subjects will summarize the response for each dimension (item) for the EQ-5D-5L™ at Baseline and Week 8 and Week 16. These summary tables will be shown by treatment group. In addition, for each treatment and for each time point assessed, descriptive statistics (mean, standard deviations, median, number of

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subjects) will characterize the five-item health status profile on the EQ-5D-5L™ in terms of the health utility score, and the EQ-Visual Analog Scale (EQ-VAS).

The HCRU data will be reported as outlined in the SAP.

Summaries of the change from Baseline to Week 16 in the WPAI:OA impairment scores will be shown by treatment group.

All endpoints up to Week 24 will be summarized (where available), and endpoints up to Week 16 will be analyzed.

The proportion of subjects who meet a WOMAC Pain response definition at Week 16 will be examined in the cohort of subjects who had a WOMAC Pain response to treatment at Week 8 and the cohort of subjects who did not have a WOMAC Pain response at Week 8. Treatment comparisons will be made within each cohort for tanezumab 2.5/5 mg versus placebo and tanezumab 2.5 mg versus placebo. A descriptive comparison will also be made between the treatment groups of tanezumab 2.5/5 mg versus tanezumab 2.5 mg. These analyses will be produced for the WOMAC Pain response levels of 30% and 50%, and other response definitions.

9.3. Safety Analysis

Adverse events, concomitant medications, laboratory safety tests, physical and neurological examinations, vital signs, ECGs, the anti-drug antibody test will be collected for each subject during the study according to the Schedule of Assessments. Standard safety reporting tables will summarize and list the safety data.

Selected adverse events of interest and common adverse events will be summarized using Risk Differences between each tanezumab group and placebo, together with 95% confidence interval, using exact methods. In addition, significance testing will be performed for tanezumab versus placebo comparisons using exact methods for the adverse events of interest. There will be no multiplicity adjustment for these significance tests.

Separate adverse event summaries by treatment group for adverse events of decreased sympathetic function will be conducted. More specifically, adverse events with the following preferred terms will be considered to represent adverse events of decreased sympathetic function: Blood pressure orthostatic decreased, bradycardia, dizziness postural, heart rate decreased, orthostatic hypotension, presyncope, sinus bradycardia, syncope, anhidrosis, hypohidrosis, abdominal discomfort, diarrhea, early satiety, fecal incontinence, nausea, vomiting, ejaculation delay, ejaculation disorder, ejaculation failure, hypertonic bladder, micturition urgency, nocturia, urinary frequency, urinary hesitation, urinary incontinence, respiratory distress and respiratory failure. If necessary, this list of preferred terms may be adjusted for updates made to the MEDICAL DICTIONARY FOR DRUG REGULATORY AFFAIRS (MedDRA) dictionary versions used for reporting.

In addition to summaries of adverse events considered to represent adverse events of decreased sympathetic function noted above, adverse events of syncope, bradycardia,

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orthostatic hypotension, anhidrosis, or hypohidrosis are designated as adverse events of interest that will be reviewed by the unblinded E-DMC (See Section 9.6).

Incidence of orthostatic hypotension using postural changes in blood pressure, in addition to mean changes in postural blood pressure will be summarized.

The Survey of Autonomic Symptoms (SAS) scores will be summarized by treatment group for the total number of symptoms reported and total impact score. The summary will be shown by visit, and for the change from Baseline.

The NIS is the sum of scores over all 37 items from both the Left and Right side. The change from baseline to each post-baseline visit in the NIS (using LOCF for missing data), and to both the Last and Worst change from Baseline (over all post-Baseline visits) will be summarized, and analyzed using Cochran-Mantel-Haenszel test. The NIS data, the neurological consultation data and the conclusion from neurological examination data will be reported.

The neurological consultation data will be summarized all subjects, and for subjects with adverse events of abnormal peripheral sensation, which are described in the adverse event section above. The "conclusion from the neurological examination" data will be summarized for each timepoint, and then a summary of the final assessment over all neurological examinations for each subject.

The incidence of subjects with any of the joint safety adjudication outcomes of rapidly progressive osteoarthritis (type 1 and type 2), subchondral insufficiency fracture (or SPONK), primary osteonecrosis, or pathological fracture, and for occurrence of total joint replacement will be shown by number of subjects treated and years of exposure (treatment plus follow up periods), for individual treatment groups and differences between tanezumab treatment groups and the placebo treatment group.

9.3.1. Anti-Tanezumab Antibodies (ADA)

The following assessments of ADA formation will be made:

- For each tanezumab dose arm, a listing of anti-drug antibody test results sorted by subject, dose and nominal time post-dose. The listing of results will also include the actual times post-dose.
- For each tanezumab dose arm, the proportion of subjects who develop anti-tanezumab will be summarized for each dose.
- Individual subjects with positive ADA results will be evaluated for potential ADA impact on the individual's PK, efficacy and safety profile.

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9.4. Analysis of Other Endpoints

9.4.1. Pharmacokinetic Data

Tanezumab concentrations will be measured to support the development of a SC population PK model that allows for the prediction of the tanezumab concentration over time in individuals. In addition tanezumab concentrations will be measured to inform the immunogenicity profile of tanezumab.

PK samples at Weeks 2, 4 and 12 will only be collected in approximately 30% of subjects randomized at selected sites.

The following reporting of PK data will be done:

- A listing of all plasma tanezumab concentrations sorted by subject, dose and nominal time post dose. The listing of concentrations will also include the actual times post dose.

- A descriptive summary of the plasma tanezumab concentrations based on nominal time post dose for each dose.

9.4.2. Pharmacodynamic (NGF) Data

NGF samples at Weeks 2, 4 and 12 will be collected in the same subset of subjects for whom PK samples are collected.

NGF data analysis will be conducted according to the NGF data analysis plan.

9.4.3. Biomarker Data

Biomarker data analysis will be conducted according to the tanezumab biomarker analysis plan.

9.5. External Adjudication Committee

A blinded Adjudication Committee consisting of external experts in orthopedic surgery, rheumatology, orthopedic pathology, or radiology with expertise in subjects with end stage osteoarthritis and osteonecrosis will be convened. The Adjudication Committee will have written operating procedures and a Charter, including a specific description of the scope of their responsibilities. In general, the Adjudication Committee will be asked to review all possible or probable joint-related safety events identified by the Central Reader (refer to Section 7.4.5), total joint replacement as well as investigator-reported adverse events of osteonecrosis, rapidly progressive osteoarthritis, subchondral insufficiency fracture (spontaneous osteonecrosis of the knee [SPONK]) or pathologic fracture. Adverse events related to joint safety that the investigator or sponsor considers medically important may also be reviewed by the Adjudication Committee. These will include, but will not be limited to events identified for adjudication by the Central Reader (see Section 7.3.7).

Prior to the Adjudication Committee's review of a given event, the Committee will be provided with blinded, available source documentation of progress reports from the

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investigator, orthopedic consult reports, operative reports, radiology reports, pathology reports, x-ray images, MRI images, and pathology specimens for review. Copies of all relevant clinical information including the items listed above should be provided to Pfizer or its designee for review by the external Adjudication Committee. Copies of the information should include the study number, site number and subject number, but it should not include the subject's name or initials.

The external DMC will be provided with a blinded summary of the Adjudication Committee's review of events after each review meeting.

9.6. Data Monitoring Committee

This study will use an external data monitoring committee (E-DMC).

An independent, E-DMC has been reinstituted for the tanezumab clinical program. This committee will be composed of at least one rheumatologist, neurologist, statistician, and epidemiologist. The E-DMC will review unblinded safety data including (but not limited to) adverse events and serious adverse events on a regular basis throughout the trial. Adverse events of syncope, bradycardia, orthostatic hypotension, anhidrosis or hypohidrosis along with other adverse events that are possibly related to the sympathetic nervous system will be monitored by the E-DMC during review of unblinded safety data. The E-DMC will have written operating procedures and a Charter, including a specific description of the scope of their responsibilities.

The E-DMC will be responsible for ongoing monitoring of the safety of subjects in the study according to the charter. If the blinded Adjudication Committee identifies adjudicated events of rapidly progressive osteoarthritis type 2, subchondral insufficiency fracture (spontaneous osteonecrosis of the knee [SPONK]), primary osteonecrosis or pathological fracture, occurring at a rate that could trigger the protocol-based stopping criteria, an urgent, ad hoc assessment of the events will be made by the E-DMC.

The recommendations made by the E-DMC to alter the conduct of the study will be forwarded to Pfizer for final decision. Pfizer will forward such decisions which may include summaries of aggregate analyses of endpoint events and of safety data which are not endpoints, to regulatory authorities, as appropriate.

Pfizer Standard Operating Procedures regarding periodic safety reviews by the study team and the Tanezumab Risk Management Committee will be followed. This committee will be composed of members inside and outside the immediate study team who will review blinded safety data from individual studies as well as data pooled across the studies on an ongoing basis. A safety review plan will be in place governing the frequency and extent of safety review.

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9.6.1. Protocol-Level Rules for Dosing Suspension/Safety Assessment

9.6.1.1. Serious Adverse Events

Tanezumab safety will be reviewed at two levels; blinded data reviews by Pfizer and unblinded reviews by the E-DMC. The E-DMC will review unblinded safety data including adverse events and serious adverse events on a regular basis throughout the course of these studies. Pfizer performs blinded review of all serious adverse event data (including those serious adverse events specified below) and a cumulative review on a monthly basis. If blinded review notes a pre-specified serious adverse event occurring at a rate that could trigger the protocol-based dosing suspension rule (ie, at least 3 or more cases of a given pre-specified serious adverse event), an urgent, ad hoc assessment by the E-DMC will be conducted. The E-DMC will determine whether a protocol-based dosing suspension rule should be triggered. At the individual protocol-level, if a given pre-specified serious adverse event is reported in 3 more subjects in any individual tanezumab treatment group than for placebo-treated subjects, the protocol-based rule for dosing suspension will be triggered.

The pre-specified serious adverse events are:

- Sudden cardiac death or cardiac death;
- Acute renal failure;
- Anaphylactic shock or severe anaphylactic reaction;
- Neuropathic joint or neuropathic arthropathy (ie, Charcot joint);
- Peripheral neuropathy confirmed with objective findings such as treatment-emergent abnormalities on neurologic examination, nerve conduction abnormalities or biopsy findings consistent with peripheral neuropathy.
- One of the events related to sympathetic dysfunction (orthostatic hypotension, bradycardia, syncope, anhidrosis, or hypohidrosis).

If a protocol-based rule for dosing suspension is triggered, it will result in suspension of further dosing of subjects in the study until a decision is reached regarding whether it is safe to resume dosing in some or all treatment groups or whether the study should be terminated completely. This decision will be made by the Sponsor in consultation with the tanezumab E-DMC.

If the protocol-based stopping rule is triggered, the E-DMC will consider the implications of this action on a program-level basis and formulate a recommendation whether it is safe to continue dosing (for some or all treatment groups) in other ongoing tanezumab clinical studies. Decisions regarding stopping treatment in other ongoing tanezumab clinical studies will be made by the Sponsor in consultation with the E-DMC.

Factors that may be considered in making this decision in relation to serious adverse events or adjudicated clinically significant adverse events include:

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- Consideration of relationship of investigational product to the adverse event;
- Consideration of whether similar adverse events are occurring in other tanezumab studies with similar subject populations;
- Dosage of tanezumab (2.5 mg or 5 mg) and distribution of adverse events across tanezumab dose arms;
- Possible differences in the baseline demographics between treatment groups;
- Use of concomitant medications;
- Possible differences in baseline medical history and/or co-morbidities;
- Duration of therapy (0-6 months, 6-12 months).

9.6.1.2. Events Consistent with Hy's Law

If two events are reported which are consistent with Hy's Law in tanezumab-treated subjects, irrespective of dose across all ongoing osteoarthritis and chronic low back pain studies, dosing will be temporarily suspended in all studies until the relationship to investigational product is established for the given events which were consistent with Hy's Law. If two events consistent with Hy's Law are considered to be related to treatment with tanezumab or the cause cannot be determined, all dosing in the tanezumab osteoarthritis and chronic low back pain program may be stopped. The E-DMC will determine whether the dosing suspension should be triggered. Subsequently the E-DMC will formulate a recommendation whether all studies should be permanently terminated. Decisions regarding permanently stopping treatment and terminating studies will be made by the Sponsor in consultation with the E-DMC.

9.6.1.3. Joint Safety Events

If the blinded Adjudication Committee identifies adjudicated events of rapidly progressive osteoarthritis type 2, subchondral insufficiency fractures (or spontaneous osteonecrosis of the knee [SPONK]), primary osteonecrosis, or pathological fracture, occurring at a rate that could trigger the protocol-based stopping criteria, an urgent, ad hoc assessment of the events will be made by the E-DMC.

The protocol (or treatment group) stopping rule will be based on the assessment of the number of subjects with adjudicated events of interest (rapidly progressive osteoarthritis type 2, subchondral insufficiency fractures (or spontaneous osteonecrosis of the knee [SPONK]), primary osteonecrosis, or pathological fracture) during the course of the study. Assuming the rate of adjudicated events in the placebo group is no more than CCI [redacted], if adjudicated events of interest are reported in CCI [redacted] in any tanezumab treatment group than for the placebo treatment group, a treatment-group or protocol-based stopping rule will be triggered. If the rate of events in the placebo treatment group is higher than CCI [redacted] the appropriate threshold number of events for

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the stopping rule will be reassessed. If the protocol-based stopping rule is triggered, the E-DMC will formulate a recommendation whether it is safe to continue dosing in some or all treatment groups or whether the study should be terminated completely. This decision will be made by Pfizer in consultation with the E-DMC.

10. QUALITY CONTROL AND QUALITY ASSURANCE

Pfizer or its agent will conduct periodic monitoring visits during study conduct to ensure that the protocol and Good Clinical Practices (GCPs) are being followed. The monitors may review source documents to confirm that the data recorded on CRFs are accurate. The investigator and institution will allow Pfizer monitors/auditors or its agents and appropriate regulatory authorities direct access to source documents to perform this verification. This verification may also occur after study completion.

During study conduct and/or after study completion, the study site may be subject to review by the Institutional Review Board (IRB)/Independent Ethics Committee (IEC), and/or to quality assurance audits performed by Pfizer, or companies working with or on behalf of Pfizer, and/or to inspection by appropriate regulatory authorities.

The investigator(s) will notify Pfizer or its agents immediately of any regulatory inspection notification in relation to the study. Furthermore, the investigator will cooperate with Pfizer or its agents to prepare the study site for the inspection and will allow Pfizer or its agent, whenever feasible, to be present during the inspection. The investigator will promptly provide copies of the inspection findings to Pfizer or its agent. Before response submission to the regulatory authorities, the investigator will provide Pfizer or its agents with an opportunity to review and comment on responses to any such findings.

It is important that the investigator(s) and their relevant personnel are available during the monitoring visits and possible audits or inspections and that sufficient time is devoted to the process.

11. DATA HANDLING AND RECORD KEEPING

11.1. Case Report Forms/Electronic Data Record

As used in this protocol, the term CRF should be understood to refer to either a paper form or an electronic data record or both, depending on the data collection method used in this study.

A CRF is required and should be completed for each included subject. The completed original CRFs are the sole property of Pfizer and should not be made available in any form to third parties, except for authorized representatives of Pfizer or appropriate regulatory authorities, without written permission from Pfizer.

The investigator has ultimate responsibility for the collection and reporting of all clinical, safety and laboratory data entered on the CRFs and any other data collection forms (source documents) and ensuring that they are accurate, authentic/original, attributable, complete, consistent, legible, timely (contemporaneous), enduring and available when required. The CRFs must be signed by the investigator or by an authorized staff member to attest that the

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data contained on the CRFs is true. Any corrections to entries made in the CRFs, source documents must be dated, initialed and explained (if necessary) and should not obscure the original entry.

In most cases, the source documents are the hospital's or the physician's subject chart. In these cases data collected on the CRFs must match the data in those charts.

In some cases, the CRF, or part of the CRF, may also serve as source documents. In these cases, a document should be available at the investigative site as well as at Pfizer and clearly identify those data that will be recorded in the CRF, and for which the CRF will stand as the source document.

11.2. Record Retention

To enable evaluations and/or audits from regulatory authorities or Pfizer, the investigator agrees to keep records, including the identity of all participating subjects (sufficient information to link records, eg, CRFs and hospital records), all original signed informed consent documents, copies of all CRFs, safety reporting forms, source documents, and detailed records of treatment disposition, and adequate documentation of relevant correspondence (eg, letters, meeting minutes, telephone calls reports). The records should be retained by the investigator according to ICH guidelines, local regulations, or as specified in the Clinical Study Agreement (CSA), whichever is longer.

If the investigator becomes unable for any reason to continue to retain study records for the required period (eg, retirement, relocation), Pfizer should be prospectively notified. The study records must be transferred to a designee acceptable to Pfizer, such as another investigator, another institution, or to an independent third party arranged by Pfizer. investigator records must be kept for a minimum of 15 years after completion or discontinuation of the study or for longer if required by applicable local regulations.

The investigator must obtain Pfizer's written permission before disposing of any records, even if retention requirements have been met.

12. ETHICS

12.1. Institutional Review Board (IRB)/Independent Ethics Committee (IEC)

It is the responsibility of the investigator to have prospective approval of the study protocol, protocol amendments, informed consent documents, and other relevant documents, eg, recruitment advertisements, if applicable, from the IRB/IEC. All correspondence with the IRB/IEC should be retained in the Investigator File. Copies of IRB/IEC approvals should be forwarded to Pfizer.

The only circumstance in which an amendment may be initiated prior to IRB/IEC approval is where the change is necessary to eliminate apparent immediate hazards to the subjects. In that event, the investigator must notify the IRB/IEC and Pfizer in writing immediately after the implementation.

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12.2. Ethical Conduct of the Study

The study will be conducted in accordance with legal and regulatory requirements, as well as the general principles set forth in the International Ethical Guidelines for Biomedical Research Involving Human Subjects (Council for International Organizations of Medical Sciences 2002), Guidelines for GCP (ICH 1996), and the Declaration of Helsinki (World Medical Association 1996 & 2008).

In addition, the study will be conducted in accordance with the protocol, the ICH guideline on GCP, and applicable local regulatory requirements and laws.

12.3. Subject Information and Consent

All parties will ensure protection of subject personal data and will not include subject names on any sponsor forms, reports, publications, or in any other disclosures, except where required by laws.

When study data are compiled for transfer to Pfizer and other authorized parties, subject names, address, and other identifiable data will be replaced by a numerical code based on a numbering system provided by Pfizer in order to de-identify study subjects. The study site will maintain a confidential list of subjects who participated in the study, linking each subject's numerical code to his or her actual identity. In case of data transfer, Pfizer will maintain high standards of confidentiality and protection of subject's personal data consistent with applicable privacy laws.

The informed consent document must be in compliance with ICH GCP, local regulatory requirements, and legal requirements, including applicable privacy laws.

The informed consent document(s) used during the informed consent process must be reviewed and approved by the sponsor, approved by the IRB/IEC before use, and available for inspection.

The investigator must ensure that each study subject, or his/her legal representative, is fully informed about the nature and objectives of the study and possible risks associated with participation.

Whenever consent is obtained from a subject's legally acceptable representative, the subject's assent (affirmative agreement) must subsequently be obtained when the subject has the capacity to provide assent, as determined by the IRB/EC. If the investigator determines that a subject's decisional capacity is so limited he/she cannot reasonably be consulted, then, as permitted by the IRB/EC and consistent with local regulatory and legal requirements, the subject's assent may be waived with source documentation of the reason assent was not obtained. If the study subject does not provide his or her own consent, the source documents must record why the subject did not provide consent (eg, minor, decisionally impaired adult), how the investigator determined that the person signing the consent was the subject's legally acceptable representative, the consent signer's relationship to the study subject (eg, parent, spouse), and that the subject's assent was obtained, or waived. If assent is obtained verbally it must be documented in the source documents.

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The investigator, or a person designated by the investigator, will obtain written informed consent from each subject or the subject's legal representative before any study-specific activity is performed. The investigator will retain the original of each subject's signed consent document.

12.4. Subject Recruitment

Advertisements approved by IRBs/ECs and investigator databases may be used as recruitment procedures.

Pfizer will have an opportunity to review and approve the content of any study recruitment materials directed to potential study subjects before such materials are used.

12.5. Reporting of Safety Issues and Serious Breaches of the Protocol or ICH GCP

In the event of any prohibition or restriction imposed (ie, clinical hold) by an applicable Competent Authority in any area of the World, or if the investigator is aware of any new information which might influence the evaluation of the benefits and risks of the investigational product, Pfizer should be informed immediately.

In addition, the investigator will inform Pfizer immediately of any urgent safety measures taken by the investigator to protect the study subjects against any immediate hazard, and of any serious breaches of this protocol or of ICH GCP that the investigator becomes aware of.

13. DEFINITION OF END OF TRIAL

13.1. End of Trial in a Member State

End of Trial in a Member State of the European Union is defined as the time at which it is deemed that sufficient subjects have been recruited and completed the study as stated in the regulatory application (ie, Clinical Trial Application (CTA)) and ethics application in the Member State. Poor recruitment (recruiting less than the anticipated number in the CTA) by a Member State is not a reason for premature termination but is considered a normal conclusion to the study in that Member State.

13.2. End of Trial in all Other Participating Countries

End of Trial in all other participating countries is defined as database lock.

14. SPONSOR DISCONTINUATION CRITERIA

Premature termination of this study may occur because of a regulatory authority decision, change in opinion of the IRB/EC, or investigational product safety problems, or at the discretion of Pfizer. In addition, Pfizer retains the right to discontinue development of tanezumab at any time.

If a study is prematurely terminated or discontinued, Pfizer will promptly notify the investigator. After notification, the investigator must contact all participating subjects and the hospital pharmacy (if applicable) within 1 week. As directed by Pfizer, all study materials must be collected and all CRFs completed to the greatest extent possible.

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15. PUBLICATION OF STUDY RESULTS

15.1. Communication of Results by Pfizer

Pfizer fulfills its commitment to publicly disclose clinical trial results through posting the results of studies on www.clinicaltrials.gov (ClinicalTrials.gov), www.pfizer.com, and/or the the European Clinical Trials Database (EudraCT), and/or www.pfizer.com, and other public registries in accordance with applicable local laws/regulations.

In all cases, study results are reported by Pfizer in an objective, accurate, balanced, and complete manner and are reported regardless of the outcome of the study or the country in which the study was conducted.

www.clinicaltrials.gov

Pfizer posts clinical trial US Basic Results on www.clinicaltrials.gov for Pfizer-sponsored interventional studies conducted in patients that evaluate the safety and/or efficacy of a Pfizer product, regardless of the geographical location in which the study is conducted. US Basic Results are submitted for posting within 1 year of the primary completion date for studies in adult populations or within 6 months of the primary completion date for studies in pediatric populations.

Primary completion date is defined as the date that the final subject was examined or received an intervention for the purposes of final collection of data for the primary outcome, whether the clinical study concluded according to the pre-specified protocol or was terminated.

EudraCT

Pfizer posts EU Basic Results on EudraCT for all Pfizer-sponsored interventional studies that are in scope of EU requirements. EU Basic Results are submitted for posting within 1 year of the primary completion date for studies in adult populations or within 6 months of the primary completion date for studies in pediatric populations.

www.pfizer.com

Pfizer posts Public Disclosure Synopses (clinical study report synopses in which any data that could be used to identify individual patients has been removed) on www.pfizer.com for Pfizer-sponsored interventional studies at the same time the US Basic Results document is posted on www.clinicaltrials.gov.

15.2. Publications by Investigators

Pfizer supports the exercise of academic freedom and has no objection to publication by principal investigator of the results of the study based on collected or generated by principal investigator, whether or not the results are favorable to the Pfizer product. However, to ensure against inadvertent disclosure of Confidential Information or unprotected Inventions, the investigator will provide Pfizer an opportunity to review any proposed publication or

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other type of disclosure of the results of the study (collectively, "Publication") before it is submitted or otherwise disclosed.

The investigator will provide any publication to Pfizer at least 30 days before they are submitted for publication or otherwise disclosed. If any patent action is required to protect intellectual property rights, investigator agrees to delay the disclosure for a period not to exceed an additional 60 days.

Investigator will, on request, remove any previously undisclosed Confidential Information before disclosure, except for any study- or Pfizer product-related information necessary to the appropriate scientific presentation or understanding of the study results.

If the study is part of a multicenter study, the investigator agrees that the first publication is to be a joint publication covering all study sites, and that any subsequent publications by the principal investigator will reference that primary publication. However, if a joint manuscript has not been submitted for publication within 12 months of completion or termination of the study at all participating sites, the investigator is free to publish separately, subject to the other requirements of this section.

For all publications relating to the study, the institution will comply with recognized ethical standards concerning publications and authorship, including Section II - "Ethical Considerations in the Conduct and Reporting of Research" of the Uniform Requirements for Manuscripts Submitted to Biomedical Journals, http://www.icmje.org/index.html#authorship, established by the International Committee of Medical Journal Editors.

Publication of study results is also provided for in the CSA between Pfizer and the institution. In this section entitled Publications by Investigators, the defined terms shall have the meanings given to them in the CSA.

If there is any conflict between the CSA and any Attachments to it, the terms of the CSA control. If there is any conflict between this protocol and the CSA, this protocol will control as to any issue regarding treatment of study subjects, and the CSA will control as to all other issues.

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Appendix 1. American College of Rheumatology (ACR) Classification Criteria for Osteoarthritis

1986 OA Knee Criteria[41]

Clinical and radiographic criteria for classification of idiopathic osteoarthritis of the knee.

Meets criteria 1, 2 and 3:

1. Knee pain;
2. Presence of at least 1 of the following 3:
    - Age greater than 50 years;
    - Morning stiffness less than 30 minutes in duration;
    - Crepitus.
3. Presence of osteophytes on x-ray.

OA Hip Criteria[53]

Combined clinical (history, physical examination, laboratory) and radiographic criteria for osteoarthritis of the hip, traditional format.

1. Hip pain;

   AND

2. At least 2 of the 3 following features:
    - Erythrocyte sedimentation rate (ESR) less than 20 mm/hour;
    - Radiographic femoral or acetabular osteophytes;
    - Radiographic joint space narrowing (superior, axial, and/or medial).

Because the presence of osteophytes on x-ray is a protocol requirement (defined by a Kellgren-Lawrence x-ray Grade of ≥2 in inclusion criteria #3), protocol defined requirement for diagnosis of OA of the Hip will be the presence of hip pain, presence of osteophytes on x-ray and either an ESR<20 mm/hour OR joint space narrowing on x-ray.

ESR testing may be conducted at the local laboratory.

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Appendix 2. American Society of Anesthesiologists (ASA) Physical Status Classification

ASA Physical Status Classification

The ASA physical status classification system is used for assessing the fitness of patients before surgery. In 1963 the American Society of Anesthesiologists (ASA) adopted the five-category physical status classification system;[54] a sixth category was later added (http://www.asahq.org/Home/For-Members/Clinical-Information/ASA-Physical-Status-Classification-System).

1. A normal healthy patient.
2. A patient with mild systemic disease.
3. A patient with severe systemic disease.
4. A patient with severe systemic disease that is a constant threat to life.
5. A moribund patient who is not expected to survive without the operation.
6. A declared brain-dead patients whose organs are being removed for donor purposes.

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Appendix 3. Half-Lives of Prohibited Prior and Concomitant Medications

Half-Lives of NSAIDs and Other Analgesics

Use of any analgesic except acetaminophen is prohibited throughout the study, beginning 48 hours prior to the start of the Initial Pain Assessment Period or at the period of time prior to the start of the Initial Pain Assessment Period that is at least 5 times the half-life of the particular analgesic used, whichever is greater. Note that a stable regimen of aspirin taken for cardiac prophylaxis at a dose of ≤325 mg/day is permitted throughout the study.

These lists are not all-inclusive. The Physician's Desk Reference provides half-life information.

HALF-LIVES OF NSAIDs AND OTHER ANALGESICS		
Analgesic	**Half-life (hours)**	**Minimum Washout Period**
Aspirin >325 mg/day	0.25	2 days
Azapropazone	15.0	4 days
Bromfenac	1.3-3.1	2 days
Capsaicin (cream, ointments, patches)	2.0	2 days
Carprofen	12.0	3 days
Celecoxib	11.0	3 days
Codeine	3.5	2 days
Diclofenac	1.1	2 days
Diclofenac gels	1.9	2 days
Diclofenac/misoprostol	2.4-9.0	2 days
Diflunisal	13.0	3 days
Dipyrone	2.0-5.0	2 days
Etodolac	6.0	2 days
Fenbufen	11.0	3 days
Fentanyl	Contact study clinician	
Fenoprofen	2.5	2 days
Flufenamic acid	1.4	2 days
Flurbiprofen	3.8	2 days
Hydrocodone	4.5	2 days
Hydromorphone	3.0	2 days
Ibuprofen	2.1	2 days
Indomethacin	4.6	2 days
Ketoprofen	1.8	2 days
Ketorolac	4.0-9.0	2 days
Lidocaine patch or EMLA (lidocaine/prilocaine)	2.0	2 days
Meclofenamate	2.0-4.0	2 days
Mefenamic acid	2.0	2 days

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HALF-LIVES OF NSAIDs AND OTHER ANALGESICS		
Meloxicam	16.0 to 20.0	5 days
Meperidine	3.7	2 days
Mexiletine	6.0-17.0	4 days
Morphine	2.0	2 days
Nabumetone	26.0	6 days
Naproxen	14.0	3 days
Oxaprofen	40.0-50.0	11 days
Oxaprozin	58.0	13 days
Oxycodone	3.2	2 days
Oxycodone CR	8.0	2 days
Oxymorphone	7.3-9.4	2 days
Phenylbutazone	68.0	15 days
Piroxicam	57.0	12 days
Pirprofen	3.8	2 days
Propoxyphene	12.0	3 days
Salicylates	2.0-15.0	4 days
Sulindac	14	3 days
Suprofen	2.5	2 days
Tapentadol	4	2 days
Tenoxicam	60.0	13 days
Tiaprofenic acid	3.0	2 days
Tolmetin	1.0	2 days
Tramadol	5.9	2 days

Corticosteroids The following use of oral or intramuscular corticosteroids is prohibited through Week 24) within 30 days prior to the Initial Pain Assessment Period or, 2) at the period of time prior to the start of the Initial Pain Assessment Period that is at least 5 times the half-life of the particular corticosteroid used, whichever is greater or, 3) the anticipated need to start such during the study. Intra-articular injection of corticosteroids within 12 weeks to the index joint or 30 days to any other joint prior to the Initial Pain Assessment Period is PROHIBITED. Topical, inhaled and intranasal corticosteroids are PERMITTED.

Corticosteroids	
Generic	Brand
betamethasone	Celestone, Soluspan
Cortisone	
dexamethasone	Decadron, Dexacort, Turbinaire
fludrocortisone	Florinet
hydrocortisone	A-hydroCort, Cortef, Hydrocortone, Solu-Cortef
methylprednisolone	Medrol, Solu-Medrol
Prednisolone	Orapred, Prelone
Prednisone	Cortan, Deltasone, Medicorten

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Hyaluronic Acid

Intra-articular hyaluronic acid injection to the index knee is prohibited within 30 days (or within 18 weeks for long-acting formulations such as Synvisc) of the Initial Pain Assessment Period and throughout the study.

Biologicals

Use of biologicals is prohibited within 3 months of the Initial Pain Assessment Period and during the study.

The following lists are provided for your reference but may not be all-inclusive. Refer to the Physician's Desk Reference for exclusion determination of a particular agent.

TNFα Inhibitors	
Generic	**Brand**
Adalimumab	Humira
Etanercept	Enbrel
Infliximab	Remicade

Use of live attenuated vaccines (with the exception of Flumist® Influenza Virus Vaccine Live, Intranasal or other inhaled live attenuated influenza vaccines and Pneumovax) is prohibited within 3 months of Initial Pain Assessment Period and during the study.

The following lists are provided for your reference but may not be all-inclusive. Refer to the Physician's Desk Reference for exclusion determination of a particular agent.

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Live Attenuated Vaccines	
Generic	**Brand**
BCG (for tuberculosis)	Not available in the US
Herpes zoster vaccine	Zostavax
Influenza, intranasal	FluMist
Measles	Attenuvax
Measles, Mumps, and Rubella (MMR)	MMR
Mumps	Mumpsvax
Oral poliovirus vaccine, oral	OPV (no longer available in the US)
Rotavirus, oral	RotaTeq
Rubella	Meruvax II
Smallpox	Dryvax (Not commercially available in the US)
Typhoid, oral	Vivotif Berna
Varicella zoster	Varivax
Yellow fever	YF-VAX

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Appendix 4. Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC)

WOMAC Osteoarthritis Index NRS3.1

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WOMAC NRS 3.1 – English for USA – V5 Page 1 of 6

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WOMAC Osteoarthritis Index NRS3.1

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WOMAC Osteoarthritis Index NRS3.1

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WOMAC Osteoarthritis Index NRS3.1

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Appendix 5. Patient's Global Assessment of Osteoarthritis

Patient's Global Assessment of Osteoarthritis – Knee

Subjects will answer the following question when the knee is selected as index ioint:

Considering all the ways your osteoarthritis in your knee affects you, how are you doing today?

Subjects will rate their condition using the following scale:

Grade	Description
1 – Very Good	Asymptomatic and no limitation of normal activities
2 – Good	Mild symptoms and no limitation of normal activities
3 – Fair	Moderate symptoms and limitation of some normal activities
4 – Poor	Severe symptoms and inability to carry out most normal activities
5 – Very Poor	Very severe symptoms which are intolerable and inability to carry out all normal activities

Patient's Global Assessment of Osteoarthritis – Hip

Subjects will answer the following question when the hip is selected as index joint:

Considering all the ways your osteoarthritis in your hip affects you, how are you doing today?

Subjects will rate their condition using the following scale:

Grade	Description
1 – Very Good	Asymptomatic and no limitation of normal activities
2 – Good	Mild symptoms and no limitation of normal activities
3 – Fair	Moderate symptoms and limitation of some normal activities
4 – Poor	Severe symptoms and inability to carry out most normal activities
5 – Very Poor	Very severe symptoms which are intolerable and inability to carry out all normal activities

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Appendix 6. Work Productivity and Activity Impairment Questionnaire: Osteoarthritis of the Knee or Hip V2.0 (WPAI:OA)

Work Productivity and Activity Impairment Questionnaire: Osteoarthritis of the Knee or Hip V2.0 (WPAI:OA)

The following questions ask about the effect of your osteoarthritis of the knee or hip on your ability to work and perform regular activities. *Please fill in the blanks or circle a number, as indicated.*

1. Are you currently employed (working for pay)? _____ NO ___ YES
   If NO, check "NO" and skip to question 6.

The next questions are about the past seven days, not including today.

2. During the past seven days, how many hours did you miss from work because of problems associated with your osteoarthritis of the knee or hip? *Include hours you missed on sick days, times you went in late, left early, etc., because of your* osteoarthritis of the knee or hip. *Do not include time you missed to participate in this study.*

   _____ HOURS

3. During the past seven days, how many hours did you miss from work because of any other reason, such as vacation, holidays, time off to participate in this study?

   _____HOURS

4. During the past seven days, how many hours did you actually work?

   _____HOURS *(If "0", skip to question 6.)*

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5. During the past seven days, how much did your osteoarthritis of the knee or hip affect your productivity while you were working?

Think about days you were limited in the amount or kind of work you could do, days you accomplished less than you would like, or days you could not do your work as carefully as usual. If osteoarthritis of the knee or hip affected your work only a little, choose a low number. Choose a high number if osteoarthritis of the knee or hip affected your work a great deal.

Consider only how much osteoarthritis of the knee or hip affected productivity while you were working.

Osteoarthritis of the knee or hip had no effect on my work	0	1	2	3	4	5	6	7	8	9	10	Osteoarthritis of the knee or hip completely prevented me from working

CIRCLE A NUMBER

6. During the past seven days, how much did your osteoarthritis of the knee or hip affect your ability to do your regular daily activities, other than work at a job?

By regular activities, we mean the usual activities you do, such as work around the house, shopping, childcare, exercising, studying, etc. Think about times you were limited in the amount or kind of activities you could do and times you accomplished less than you would like. If osteoarthritis of the knee or hip affected your activities only a little, choose a low number. Choose a high number if osteoarthritis of the knee or hip affected your activities a great deal.

Consider only how much osteoarthritis of the knee or hip affected your ability to do your regular daily activities, other than work at a job.

Osteoarthritis of the knee or hip had no effect on my daily activities	0	1	2	3	4	5	6	7	8	9	10	Osteoarthritis of the knee or hip completely prevented me from doing my daily activities

CIRCLE A NUMBER

WPAI:OA V2.0 (US English)

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Appendix 7. EuroQol 5 Dimension (EQ-5D-5L)

By placing a check mark in one box in each group below, please indicate which statements best describe your own health state today.

Mobility

- I have no problems walking ☐
- I have slight problems walking ☐
- I have moderate problems walking ☐
- I have severe problems walking ☐
- I am unable to walk ☐

Self-Care

- I have no problems washing or dressing myself ☐
- I have slight problems washing or dressing myself ☐
- I have moderate problems washing or dressing myself ☐
- I have severe problems washing or dressing myself ☐
- I am unable to wash or dress myself ☐

Usual Activities (e.g. work, study, housework, family or leisure activities)

- I have no problems doing my usual activities ☐
- I have slight problems doing my usual activities ☐
- I have moderate problems doing my usual activities ☐
- I have severe problems doing my usual activities ☐
- I am unable to do my usual activities ☐

Pain/Discomfort

- I have no pain or discomfort ☐
- I have slight pain or discomfort ☐
- I have moderate pain or discomfort ☐
- I have severe pain or discomfort ☐
- I have extreme pain or discomfort ☐

Anxiety/Depression

- I am not anxious or depressed ☐
- I am slightly anxious or depressed ☐
- I am moderately anxious or depressed ☐
- I am severely anxious or depressed ☐
- I am extremely anxious or depressed ☐

Continued to next page

SAMPLE

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We would like to know how good or bad your health is TODAY.

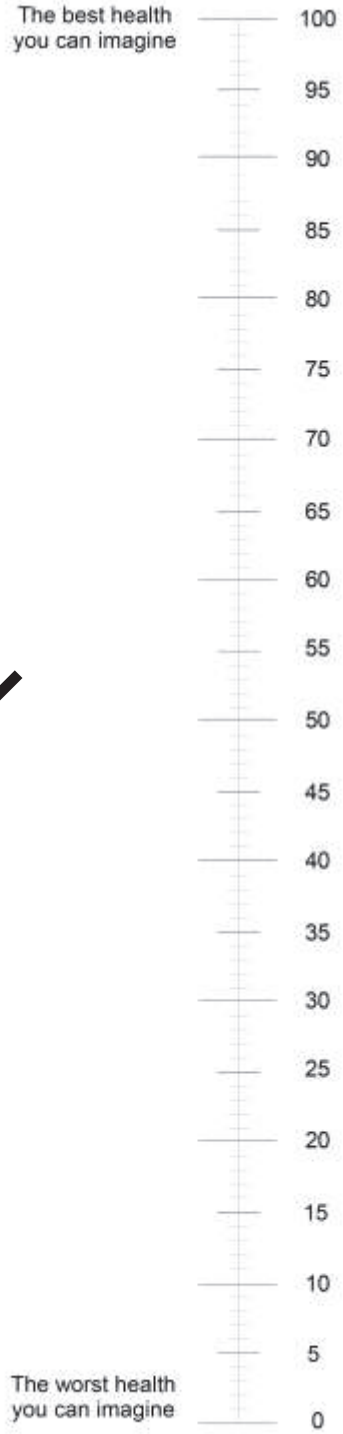
This scale is numbered from 0 to 100.

100 means the best health you can imagine.

0 means the worst health you can imagine.

Mark an X on the scale to indicate how your health is TODAY.

Now, please write the number you marked on the scale in the box below.



Health State: ___

SAMPLE

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Appendix 8. Neuropathy Impairment Score (NIS) Sample

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NEUROPATHY IMPAIRMENT SCORE (NIS)

OBJECTIVE: To provide a single score of neuropathic deficits and subset scores: cranial nerve, muscle weakness, reflexes and sensation. Abnormalities are abstracted from a neurologic examination in which all of the assessments are made.

SCORING: The examiner scores deficits by what he (she) considers to be normal considering test, anatomical site, age, gender, height, weight, and physical fitness.

4

SCORING, MUSCLE WEAKNESS	
0 = NORMAL	3.25 = MOVE AGAINST GRAVITY
1 = 25% WEAK	3.5 = MOVEMENT, GRAVITY ELIMINATED
2 = 50% WEAK	3.75 = MUSCLE FLICKER, NO MOVEMENT
3 = 75% WEAK	4 = PARALYSIS

	RIGHT									LEFT								
Cranial Nerves	N/A	0	1	2	3	3.25	3.5	3.75	4	N/A	0	1	2	3	3.25	3.5	3.75	4
1. 3rd Nerve	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
2. 6th Nerve	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
3. Facial weakness	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
4. Palate weakness	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
5. Tongue weakness	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
Muscle Weakness	N/A	0	1	2	3	3.25	3.5	3.75	4	N/A	0	1	2	3	3.25	3.5	3.75	4
6. Respiratory	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
7. Neck flexion	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
8. Shoulder abduction	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
9. Elbow flexion	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
10. Brachioradialis	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
11. Elbow extension	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
12. Wrist flexion	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
13. Wrist extension	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
14. Finger flexion	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
15. Finger spread	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
16. Thumb abduction	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
	N/A	0	1	2	3	3.25	3.5	3.75	4	N/A	0	1	2	3	3.25	3.5	3.75	4
17. Hip flexion	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
18. Hip extension	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
19. Knee flexion	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
20. Knee extension	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
21. Ankle dorsiflexors	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
22. Ankle plantar flexors	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
23. Toe extensors	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
24. Toe flexors	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○

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NEUROPATHY IMPAIRMENT SCORE (NIS)

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For patients 50-69 years old, ankle reflexes which are decreased are graded 0 and when absent are graded 1.
For patients ≥ 70 years, absent ankle reflexes are graded 0.

SCORING, REFLEXES
0 = NORMAL; 1 = DECREASED; 2 = ABSENT

	RIGHT				LEFT			
Reflexes	N/A	0	1	2	N/A	0	1	2
25. Biceps brachii	○	○	○	○	○	○	○	○
26. Triceps brachii	○	○	○	○	○	○	○	○
27. Brachioradialis	○	○	○	○	○	○	○	○
28. Quadriceps femoris	○	○	○	○	○	○	○	○
29. Triceps surae	○	○	○	○	○	○	○	○

Touch-pressure, pin-prick and vibration sensation are tested on the dorsal surface, at the base of the nail, of the terminal phalanx of the index finger and great toe. Touch-pressure is assessed with long fiber cotton wool. Pin-prick is assessed with straight pins. Vibration sensation is tested with a 165 hz tuning fork (V. Mueller, Chicago, length 25 cm, made from 1/2" x 1 1/4 " stock; 165 hz with counterweights). Joint motion is tested by moving the terminal phalanx of the index finger and great toe.

SCORING, SENSATION
0 = NORMAL; 1 = DECREASED; 2 = ABSENT

	RIGHT				LEFT			
Sensation - I. Finger	N/A	0	1	2	N/A	0	1	2
30. Touch pressure	○	○	○	○	○	○	○	○
31. Pin-prick	○	○	○	○	○	○	○	○
32. Vibration	○	○	○	○	○	○	○	○
33. Joint position	○	○	○	○	○	○	○	○
Sensation - G. Toe	N/A	0	1	2	N/A	0	1	2
34. Touch pressure	○	○	○	○	○	○	○	○
35. Pin-prick	○	○	○	○	○	○	○	○
36. Vibration	○	○	○	○	○	○	○	○
37. Joint position	○	○	○	○	○	○	○	○

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Appendix 9. Patient Health Questionnaire (PHQ-9)

Administration of the PHQ-9 is not mandatory but may be used by the investigator to assess the severity of depression. The severity score is the sum of questions 1 through 9 only. A score of 15 or higher on questions 1 through 9 indicates severe depression. If used the PHQ-9 should be stored in the subject file. The results of this instrument will not be entered into a database, nor will it be analyzed.

PATIENT HEALTH QUESTIONNAIRE (PHQ-9):

Over the last 2 weeks, how often have you been bothered by the following problems?	Not at all (0)	Several days (1)	More than half the days (2)	Nearly every day (3)
1. Little interest or pleasure in doing things	☐	☐	☐	☐
2. Feeling down, depressed, or hopeless	☐	☐	☐	☐
3. Trouble falling or staying asleep, or sleeping too much	☐	☐	☐	☐
4. Feeling tired or having little energy	☐	☐	☐	☐
5. Poor appetite or overeating	☐	☐	☐	☐
6. Feeling bad about yourself--or that you are a failure or have let yourself or your family down	☐	☐	☐	☐
7. Trouble concentrating on things, such as reading the newspaper or watching television	☐	☐	☐	☐
8. Moving or speaking so slowly that other people could have noticed? Or the opposite--being so fidgety or restless that you have been moving around a lot more than usual	☐	☐	☐	☐
9. Thoughts that you would be better off dead or of hurting yourself in some way	☐	☐	☐	☐

Total Score: ___

If you checked off any problems, how difficult have these problems made it for you to do your work, take care of things at home, or get along with other people?

Not difficult at all	Somewhat difficult	Very difficult	Extremely difficult
☐	☐	☐	☐

PHQ-9 is adapted from PRIME MD TODAY, developed by Drs Robert L. Spitzer, Janet B.W. Williams, Kurt Kroenke, and colleagues, with an educational grant from Pfizer Inc. For research information, contact Dr Kroenke at kkroenke@regenstrief.org. Use of the PHQ-9 may only be made in accordance with the Terms of Use available of http://www.pfizer.com. Copyright ©1999 Pfizer Inc. All rights reserved. PRIME MD TODAY is a trademark of Pfizer Inc.

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Appendix 10. Adjudication Categories

Adjudication Category	Adjudicated Condition
1	Primary Osteonecrosis
2	Worsening Osteoarthritis
2a	Rapidly Progressive Osteoarthritis (type-1 or type-2)
2b	Normal progression of osteoarthritis
2c	Not enough information to distinguish between rapidly progressive osteoarthritis and normal progression of osteoarthritis
3	Subchondral insufficiency fracture
4	Pathologic fracture
5	Other (with diagnosis specified)
6	Not enough information to specify a diagnosis

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Appendix 11. Survey of Autonomic Symptoms (SAS)[55]

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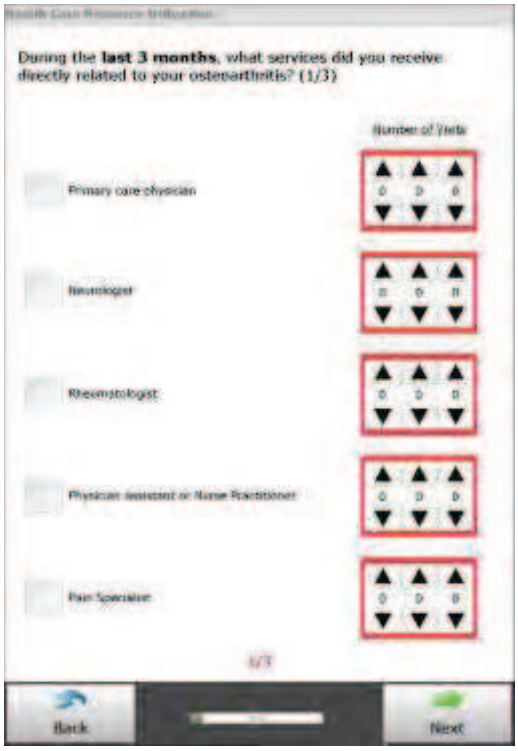
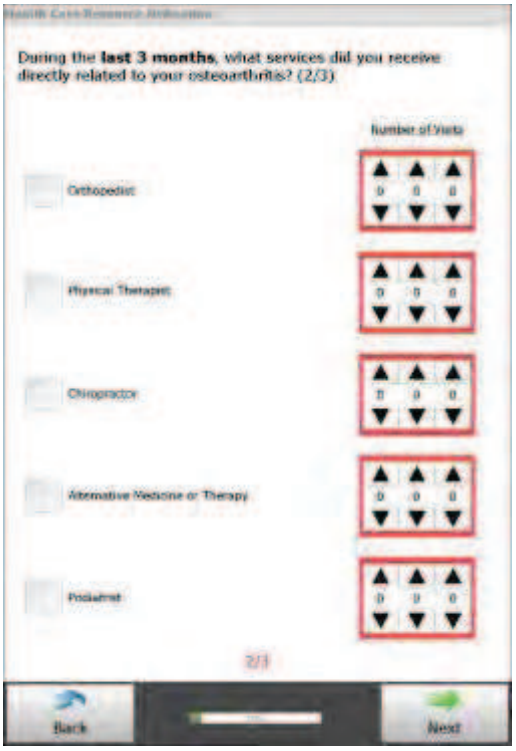
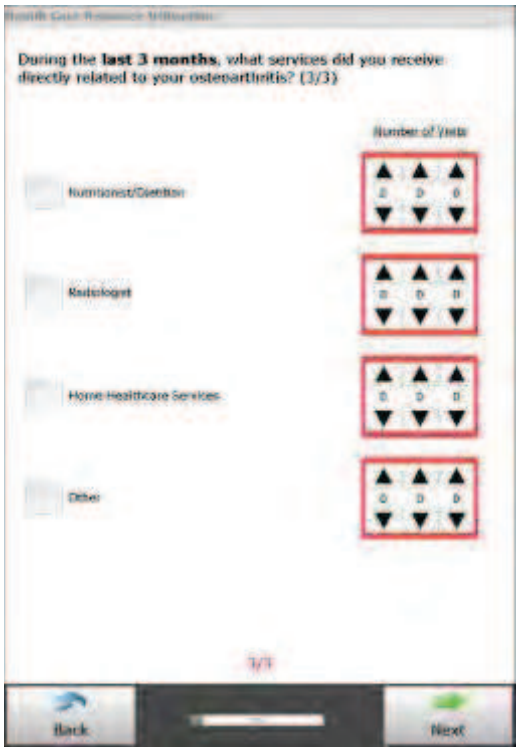
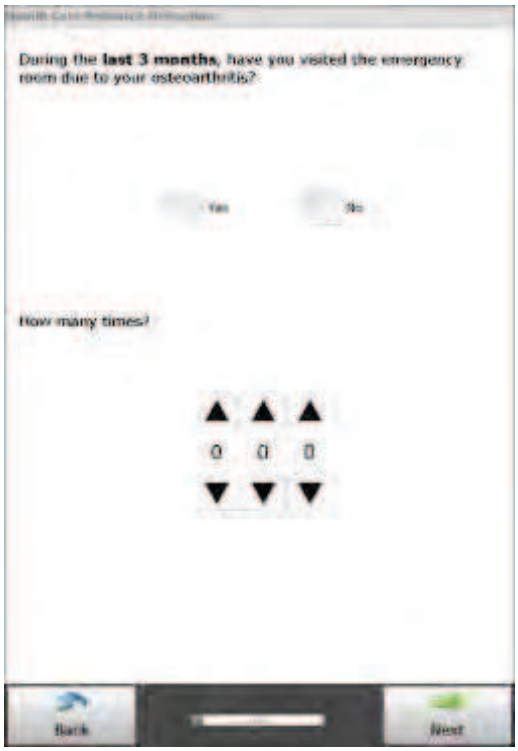
Appendix 12. Health Care Resource Utilization

Example with 3 month recall period.









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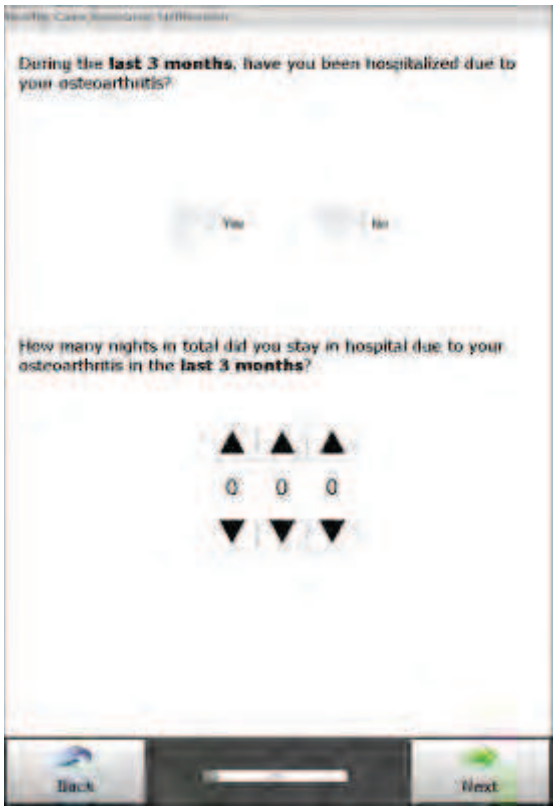

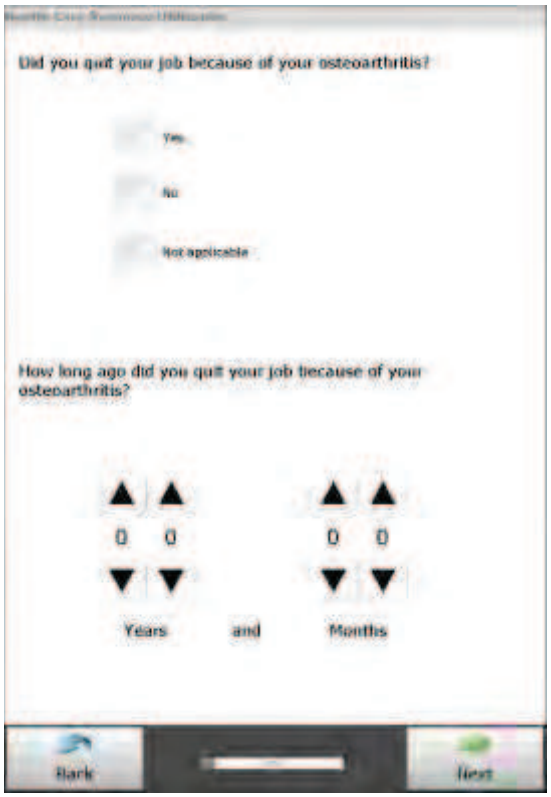

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Appendix 13. Abbreviations

This is a list of abbreviations that may or may not be used in the protocol.

Abbreviation	Term
Ab	antibody
ACR	American College of Rheumatology
ADA	anti-drug antibody
Ag	antigen
AE	adverse event
AEMPS	Agency on Medicinal Products and Medical Devices
ALT	alanine aminotransferase
ANC	absolute neutrophil count
ANCOVA	analysis of covariance
ASA	American Society of Anesthesiologists
AST	aspartate aminotransferase
BL	baseline
BMI	body mass index
BOCF	baseline observation carried forward
BP	blood pressure
BAP	Baseline Assessment Period
BPI	Brief Pain Inventory
BUN	blood urea nitrogen
CCI [redacted]	CCI [redacted]
CCI [redacted]	CCI [redacted]
CMH	Cochran-Mantel-Haenszel
CPK	Creatinine Phosphokinase
CRF	case report form
CRA	Clinical Research Associate
CCI [redacted]	CCI [redacted]
CSA	clinical study agreement
CT	computerized tomography
CTA	clinical trial application
CTS	carpal tunnel syndrome
CCI [redacted]	CCI [redacted]
CCI [redacted]	CCI [redacted]
DAAAP	Division of Analgesia, Anesthetic, and Addiction Products
DAI	Drug Administration Instructions
DNA	deoxyribonucleic acid
DU	dispensable unit
EC	ethics committee
ECG	electrocardiogram
ECOG	Eastern Cooperative Oncology Group
E-DMC	External Data Monitoring Committee

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Abbreviation	Term
EDP	exposure during pregnancy
EDTA	edetic acid (ethylenediaminetetraacetic acid)
EQ-5D-5L	EuroQol 5 Dimension 5 Level
ER	extended release
ESR	erythrocyte sedimentation rate
EudraCT	European Clinical Trials Database
FDA	Food and Drug Administration (United States)
FDAAA	Food and Drug Administration Amendments Act (United States)
FSH	follicle-stimulating hormone
GCP	Good Clinical Practice
GGT	gamma glutamyltransferase
HBsAG	hepatitis B surface antigen
CCI	CCI [redacted]
HCRU	Health Care Resource Utilization
HCV	hepatitis C virus
HIV	human immunodeficiency virus
HR	heart rate
CCI	CCI [redacted]
ICH	International Conference on Harmonisation
ICRP	International Commission on Radiation Protection
CCI	CCI [redacted]
ID	identification
IND	Investigational New Drug application
INR	International Normalized Ratio
IgG	immunoglobulin G
IgG2	immunoglobulin G Type 2
IRB	institutional review board
IL	interleukin
CCI	CCI [redacted]
IPAP	Initial Pain Assessment Period
IR	immediate release
IRT	interactive response technology
ITT	intent to treat
IUD	intrauterine device
IV	intravenous
IWRS	interactive web response system
KL	Kellgren-Lawrence
LDH	lactate dehydrogenase
LFT	liver function test
LOCF	Last Observation Carried Forward
LSLV	last subject last visit
LSMean	least squared mean
MAb 911	murine precursor antibody to tanezumab

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Abbreviation	Term
MDA	Multi-Domain Average
MHRA	Healthcare products Regulatory Agency
CCI	CCI
MMR	measles, mumps and rubella
MMRM	Mixed Model for Repeated Measurements
MRI	magnetic resonance imaging
mSv	millisievert
N/A	not applicable
NGF	nerve growth factor
NGFI	nerve growth factor inhibitor
NIS	Neuropathy Impairment Score
NRS	numeric rating scale
NSC	Neuropathy Symptom Change
NSAID	non-steroidal anti-inflammatory drug
NYHA	New York Heart Association
OA	osteoarthritis
CC	CCI
OMERACT-OARSI	Outcomes Measures in Arthritis Clinical Trials-Osteoarthritis Research Society International
OTC	over-the-counter
PCD	primary completion date
PD	pharmacodynamic
PEI	Paul Ehrlich Institute
PFS	pre-filled syringe
PGA	Patient's Global Assessment
PHQ-9	Patient Health Questionnaire
CCI	CCI
PK	pharmacokinetic
PT	prothrombin time
PTT	partial thromboplastin time
QT	in electrocardiography, the time corresponding to the beginning of depolarization to repolarization of the ventricles
QTc	in electrocardiography, the time corresponding to the beginning of depolarization to repolarization of the ventricles, corrected for heart rate
QTcB	QT corrected for heart rate using Bazett's formula
QTcF	QT corrected for heart rate using Fridericia's formula
RNA	ribonucleic acid
RPOA	rapidly-progressive osteoarthritis
SAE	serious adverse event
SAPS	Self-Administered Patient Satisfaction Scale
SAS	Survey of Autonomic Symptoms
SC	subcutaneous

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Abbreviation	Term
SNRI	serotonin norepinephrine reuptake inhibitor
SOA	schedule of activities
CCI	CCI
SPADI	Shoulder Pain and Disability Index
SPONK	spontaneous osteonecrosis of the knee
SRSD	single reference safety document
Tan	tanezumab
TNF-α	tumor necrosis factor alpha
trkA	tropomyosin receptor kinase A
ULN	upper limit of normal
US	United States
VAS	visual analog scale
WHO	World Health Organization
WOMAC	Western Ontario and McMaster University Osteoarthritis Index
WPAI-OA	Work Productivity and Activity Impairment Questionnaire - Osteoarthritis

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Tanezumab
Protocol A4091056
Statistical Analysis Plan v2_0, 19 June 2018

Note:

Certain information has been redacted from the plan for public disclosure purposes.
Redacted information is as follows:

- **WOMAC® instrument:** The WOMAC® instrument and the content of the user manual (including the 3 statements redacted on pages 28 and 29 of the SAP) are protected by copyright and confidentiality restrictions that do not permit disclosure to a 3rd party.

- **Administrative Error**: Attorney-client privilege (page 35). This sentence contains advice from Pfizer legal counsel and was inadvertently introduced to the document due to administrative error.

- **Protocol Stopping Rules:** The specifics of the stopping rule (page 37) were established pursuant to an agreement with FDA and have not previously been disclosed publicly. They are to remain confidential until such disclosure. The stopping rule was not met and the study was not stopped prematurely.

Protocol A4091056

A Phase 3 Randomized, Double-Blind, Placebo-Controlled, Multicenter Study of the Analgesic Efficacy and Safety of a Dose Titration Regimen for the Subcutaneous Administration of Tanezumab in Subjects with Osteoarthritis of the Hip or Knee

Statistical Analysis Plan
(SAP)

Version: 2.0

Date: 19-JUN-2018

TMF Doc ID: 98.03

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1. VERSION HISTORY

This Statistical Analysis Plan (SAP) for study A4091056 is based on the Protocol Amendment 1 dated 23Sep2015.

Table 1. Summary of Major Changes in SAP Amendments

SAP Version	Change	Rationale
1.0	Not Applicable	Not Applicable
2.0	Throughout	The changes described below reflect updates from blinded data reviews and program decisions for alignment of analysis. Additionally, clarifications, removal of redundant text, and correction of typos have been implemented.
	Section 2.2, 3.5.1, 3.5.2, 6.6.1	clarified that various safety results will be presented by treatment period, safety follow-up period, and overall
	Section 3.2.1	removed Week 24 (off-treatment period) from some categorical endpoints
	Section 3.4, 5.3	clarified baseline diary pain scores are from the 3 most recent days of the 7-day pre-dose period
	Section 3.4.1, 5.2.2	added listing of mis-matches in stratification variables; removed some interaction analyses and sensitivity analysis with no covariates
	Section 3.5.2	clarified summaries of consultations
	Section 3.5.3	simplified total joint replacement and joint event summary descriptions
	Section 3.5.4	removed Survey of Autonomic Symptoms analysis by gender
	Section 4.2	added description of Per Protocol (PP) population process
	Section 4.2.1, 4.2.2	revised PP population criteria

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	Section 4.4, 6.3.3	removed descriptions of biomarker and NGF populations and summaries
	Section 5.1.2	specified WOMAC Pain 50% responders as key secondary endpoint to be tested with the Hochberg procedure
	Section 5.2	specified estimands, description of planned on-treatment efficacy assessment period and definition of on- and off-treatment data windows
	Section 5.2.2, 5.2.3, 5.3	specified MMRM analysis using multiply imputed data, and addition of MMRM analysis using all available data, on- or off-treatment; removed analysis of Week 24 data; added analysis of diary pain data for individual Days 1 through 7; specified WPAI analysis using ANCOVA model instead of CMH test
	Section 5.2.4	removed time to event summaries for joint safety events
	Section 6.6	removed AE patient-year summaries, AE plots, summaries of specific AE start day and duration; added NSAID use summaries
	Section 6.6.1	removed Tier 1 and 2 AE graphs
	Section 6.6.2	removed summary of mean change in postural blood pressure
	Section 6.6.3	clarified neurologic data summaries
	Section 6.6.5	clarified joint safety event summaries
	Appendix 2.1	clarified windows for various data types

2. INTRODUCTION

Note: in this document any text taken directly from the protocol is *italicized*.

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This SAP provides the detailed methodology for summary and statistical analyses of the data collected in study A4091056. This document may modify the plans outlined in the protocol; however, any major modifications of the primary endpoint definition or its analysis will also be reflected in a protocol amendment.

2.1. Study Objectives

2.1.1. Primary Objective

Demonstrate superior efficacy of tanezumab 2.5 mg administered subcutaneously (SC) and tanezumab 2.5 mg SC titration to 5 mg SC versus placebo at Week 16.

2.1.2. Secondary Objectives

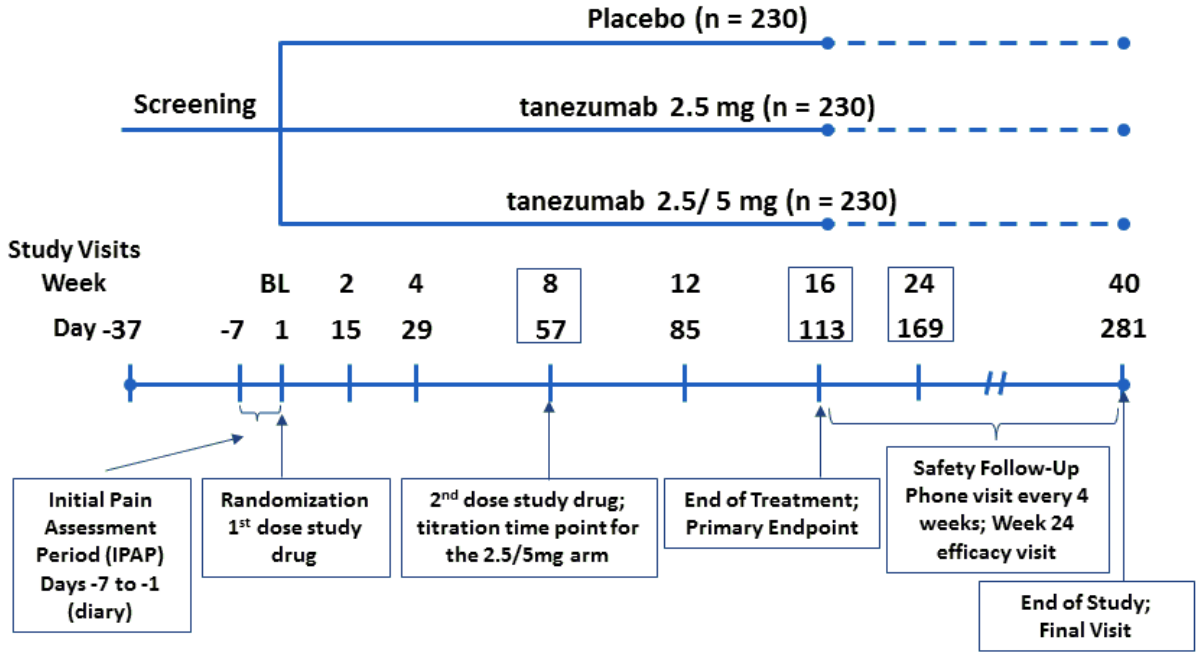
Evaluate treatment benefit (efficacy) of a tanezumab 2.5 mg SC titration to 5 mg SC dosing regimen relative to tanezumab 2.5 mg SC alone (descriptive analyses);

Evaluate the safety of tanezumab 2.5 mg SC and tanezumab 2.5 mg SC titration to 5 mg SC regimens.

2.2. Study Design

The study design is summarized in the diagram below.

Figure 1. Study Design



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This is a randomized, double-blind, placebo-controlled, multicenter, parallel-group, Phase 3 study of the efficacy and safety of tanezumab when administered by SC injection for 16 weeks compared to placebo in subjects with osteoarthritis of the knee or hip. Approximately 690 subjects (approximately 230 per treatment group) will be randomized to one of 3 treatment groups in a 1:1:1 ratio. Subjects will receive a total of two SC injections, separated by 8 weeks:

- *tanezumab 2.5 mg (Day 1) and tanezumab 2.5 mg (Week 8);*
- *tanezumab 2.5 mg (Day 1), and tanezumab 5 mg (Week 8);*
- *placebo to match tanezumab (Day 1) and placebo to match tanezumab (Week 8).*

The randomization will be stratified by the factors of index joint and highest Kellgren-Lawrence grade in the knee and hip joints.

This study is designed with a total (post-randomization) duration of 40 weeks and will consist of three periods: Screening (up to a maximum of 37 days), Double-blind Treatment (16 weeks, 6 in-clinic visits), and Safety Follow-up (24 weeks). The Screening Period (to begin up to 37 days prior to Randomization) includes a Washout Period (lasting a minimum of 2 days for all prohibited pain medications) if required, and an Initial Pain Assessment Period (IPAP; 7 days prior to Randomization/Baseline; minimum 3 days).

The end of treatment period is at Week 16, with the safety follow-up period up to Week 40. The primary time point for efficacy is Week 16. The period of interest for most safety results is the treatment period. Selected safety results will also be provided separately for the safety follow-up period, and some results will be provided for the combined overall study period comprising the treatment and safety follow-up periods.

3. ENDPOINTS AND BASELINE VARIABLES: DEFINITIONS AND CONVENTIONS

3.1. Primary Endpoint(s)

- *Change from Baseline to Week 16 in the WOMAC Pain subscale;*
- *Change from Baseline to Week 16 in the WOMAC Physical Function subscale;*
- *Change from Baseline to Week 16 in the Patient's Global Assessment of Osteoarthritis.*

Baseline values will be those from the Baseline window as described in Appendix 2.1.

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3.2. Secondary Endpoint(s)

3.2.1. Efficacy Measures

- *WOMAC Pain subscale change from Baseline to Weeks 2, 4, 8, 12 and 24;*
- *WOMAC Physical Function subscale change from Baseline to Weeks 2, 4, 8, 12 and 24;*
- *Patient's Global Assessment of Osteoarthritis (5-point Likert scale) change from Baseline to Weeks 2, 4, 8, 12 and 24;*
- *OMERACT-OARSI responder index at Weeks 2, 4, 8, 12, and 16;*

The OMERACT-OARSI responder index at Week X utilizes the change from Baseline to Week X in the WOMAC Pain subscale, the WOMAC Physical Function Subscale and the PGA of Osteoarthritis (using appropriate imputation where necessary for any components). According to this definition, a patient is classified as a responder if, either:

- The improvement from Baseline to Week X was ≥50% and ≥2 points in either the WOMAC Pain or Physical Function subscales, OR
- At least 2 of the following 3 were true:
    - The improvement from Baseline to Week X was ≥20% and ≥1 point in the WOMAC Pain subscale;
    - The improvement from Baseline to Week X was ≥20% and ≥1 point in the WOMAC Physical Function subscale;
    - The improvement from Baseline to Week X ≥20% and ≥1 point in the PGA of Osteoarthritis (note: from the 5-point Likert scale, any change of ≥1 corresponded to a change of ≥20%).
- *Treatment Response: Reduction in the WOMAC Pain subscale of ≥30%, ≥50%, ≥70% and ≥90% at Weeks 2, 4, 8, 12, and 16;*
- *Cumulative distribution of percent change from Baseline in the WOMAC Pain subscale score to Week 16 (endpoint for summary only);*
- *Treatment Response: Reduction in the WOMAC Physical Function subscale of ≥30%, ≥50%, ≥70% and ≥90% at Weeks 2, 4, 8, 12, and 16;*
- *Cumulative distribution of percent change from Baseline in the WOMAC Physical Function subscale score to Week 16 (endpoint for summary only);*
- *Treatment Response: Improvement of ≥2 points in Patient's Global Assessment of Osteoarthritis at Weeks 2, 4, 8, 12, and 16;*

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- *Average pain score in the index joint change from Baseline to Weeks 1, 2, 3, 4, 6, 8, 10, 12, 16, 20 and 24;*
- *WOMAC Stiffness subscale change from Baseline to Weeks 2, 4, 8, 12, 16 and 24;*
- *WOMAC Average score change from Baseline to Weeks 2, 4, 8, 12, 16 and 24;*
- *WOMAC Pain Subscale Item: Pain When Walking on a Flat Surface, change from Baseline to Weeks 2, 4, 8, 12, 16 and 24;*
- *WOMAC Pain Subscale Item: Pain When Going Up or Down Stairs, change from Baseline to Weeks 2, 4, 8, 12, 16 and 24;*
- *Work Productivity and Activity Impairment Questionnaire for Osteoarthritis (WPAI:OA) impairment scores change from Baseline to Week 16.*

The WPAI:OA impairment scores are listed below:

- Percent work time missed due to Osteoarthritis.
- Percent impairment while working due to Osteoarthritis.
- Percent overall work impairment due to Osteoarthritis.
- Percent activity impairment due to Osteoarthritis.

The calculation of these endpoints is described in Appendix 3.

- EuroQol 5 Dimension (EQ-5D-5L™) dimensions and overall health utility score at Baseline, Week 8 and Week 16;

The Baseline and Weeks 8 and 16 responses in the five dimensions (mobility; self-care; usual activities; pain/discomfort; anxiety/depression) and overall health utility score from the EuroQol 5 Dimensions (EQ-5D-5L), and the EQ-VAS will be summarized by treatment group. This summary will use observed data only (no imputation for missing data). The overall health utility score is calculated using the EuroQol value sets, and is described in Appendix 3.

An additional question, called the EQ-VAS asks the patient to rate their health today using a VAS scale from 0 (the worst health you can imagine) to 100 (the best health you can imagine). This will be summarized along with the health utility score.

- *Health Care Resource Utilization at Baseline, Week 24 and Week 40;*
- *Incidence and time to discontinuation due to Lack of Efficacy;*

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- *Usage of rescue medication (incidence, number of days) during Weeks 2, 4, 8, 12, 16 and 24;*
- *Usage of rescue medication (amount taken) during Weeks 2, 4, 8, 12, and 16.*

3.3. Other Endpoints

3.3.1. Pharmacokinetic and Pharmacodynamic Measures

- *Plasma tanezumab concentrations;*
- *Serum NGF assessments;*
- *Serum and urine osteoarthritis biomarker concentrations*.

3.4. Baseline Variables

Baseline is generally defined as the last observation prior to first receipt of study drug, within the baseline window as defined in Appendix 2.1.

For analysis of diary pain intensity scores for the index joint, baseline is defined as the mean average daily Pain NRS score using the last 3 values during the final 7 days of the Initial Pain Assessment Period prior to Randomization/Day 1.

The stratification variables are *index joint* (hip or knee) *and most severe Kellgren-Lawrence grade (of any knee or hip joint)* at study entry (grade 2, 3 or 4).

3.4.1. Covariates

For all models analyzing the continuous primary and secondary efficacy endpoints (except rescue medication) the corresponding Baseline value will be used as a covariate, in addition to Baseline diary average pain. Study site will be fitted as a random effect in the ANCOVA models. The randomization stratification variables of index joint (hip or knee) and highest Kellgren-Lawrence grade (2, 3 or 4) will be included as fixed effects.

A listing of subjects with mis-matches between the stratification variables entered at randomization and the case report form data (including central lab data for Kellgren-Lawrence grade) will be provided. In analysis models, the strata entered at randomization will be used, but for descriptive summarization of the population and identification of subgroups, the strata as indicated on the case report form data will be used.

For the models analyzing the amount and number of days of rescue medication use the model will include terms for Baseline WOMAC Pain, Baseline diary average pain and stratification factors.

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The analysis of the incidence of treatment *discontinuation due to lack of efficacy will* include *… model terms for Baseline WOMAC Pain subscale score, Baseline diary average pain score, index joint, Kellgren-Lawrence grade, and treatment group.*

For categorical/binary response endpoints relating to WOMAC Pain and PGA, the corresponding Baseline WOMAC Pain or PGA value will be used as a covariate in the analysis model, in addition to the stratification parameters of index joint and highest Kellgren-Lawrence grade, as well as Baseline diary average pain. For the OMERACT response endpoint, the Baseline WOMAC Pain subscale score and Baseline diary average pain score will be used as covariates in the analysis model, in addition to the stratification parameters of index joint and highest Kellgren-Lawrence grade.

Additional analyses of the three co-primary endpoints will examine the treatment interactions with Study site and Country.

3.5. Safety Endpoints

- *Adverse Events;*
- *Standard safety assessments (safety laboratory testing [chemistry, hematology], sitting vital signs,* electrocardiogram *(ECG [12-lead]);*
- *Joint safety adjudication outcomes;*
- *Total joint replacements;*
- *Neurologic examination (Neuropathy Impairment Score [NIS]);*
- *Orthostatic (supine/standing) blood pressure assessments;*
- *Survey of Autonomic Symptom scores;*
- *Anti-drug antibody (ADA) assessments;*
- *Physical examinations.*

3.5.1. Adverse Events

An adverse event is considered treatment emergent relative to a given treatment if:

- the event occurs for the first time during the effective duration of treatment and was not seen prior to the start of treatment (for example, during the baseline or run-in period), or
- the event was seen prior to the start of treatment but increased in severity during treatment.

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The effective duration of treatment is determined by the lag time. Any event occurring within the lag time, whether this occurs during a break in treatment or at the end of treatment, is attributed to the corresponding treatment period. An infinite lag will be used for the study, meaning any treatment-emergent AE reported in the database will be included in tables of AEs up to end of study.

The adverse events of Abnormal Peripheral Sensation (APS) are defined in the table below.

Allodynia	Neuritis
Axonal neuropathy	Neuropathy peripheral
Burning sensation	Paraesthesia
Carpal Tunnel Syndrome	Paraesthesia oral
Decreased Vibratory Sense	Peripheral sensorimotor neuropathy
Demyelinating polyneuropathy	Peripheral sensory neuropathy
Dysaesthesia	Polyneuropathy
Formication	Polyneuropathy chronic
Hyperaesthesia	Sensory disturbance
Hyperpathia	Sensory loss
Hypoaesthesia	Thermohypoaesthesia
Hypoaesthesia oral	Sciatica
Intercostal neuralgia	Tarsal Tunnel Syndrome
Neuralgia	

Adverse Events of Sympathetic Nervous System are defined in the table below.

Abdominal discomfort	Micturition urgency
Anal incontinence	Nausea
Anhidrosis	Nocturia
Blood pressure orthostatic decreased	Orthostatic hypotension
Bradycardia	Pollakiuria
Diarrhoea	Presyncope
Dizziness postural	Respiratory distress
Early satiety	Respiratory failure
Ejaculation delayed	Sinus bradycardia
Ejaculation disorder	Syncope
Ejaculation failure	Urinary hesitation
Heart rate decreased	Urinary incontinence
Hypertonic bladder	Vomiting
Hypohidrosis	

A smaller set of the above Adverse Events (to be called AEs of Decreased Sympathetic Function) may also be summarized. These are defined below.

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Anhidrosis	Orthostatic hypotension
Bradycardia	Syncope
Hypohidrosis	

The lists given above may be updated depending on any additional adverse events observed in any tanezumab study. There are a number of summaries based on these groupings of adverse events.

A 3-tier approach will be used to summarize AEs. Under this approach, AEs are classified into 1 of 3 tiers. Different analyses will be performed for different tiers. A description of the three tiers and analyses are given in Section 6.6.1.

All summaries of adverse events will be shown for adverse events that begin or worsen from the first SC dose (treatment-emergent) up to the end of the treatment period. In addition a selection of adverse event tables will be produced for the safety follow-up period and for the whole period up to the end of the study, including the treatment period and safety follow-up period.

3.5.2. Vital Signs

The incidence of orthostatic hypotension at each visit, at any treatment period visit (including unscheduled visits), and at any safety follow-up period visit (including unscheduled visits), will be summarized. The definition of orthostatic hypotension is:

- For patients with Baseline supine systolic Blood Pressure ≤150 mmHg:
  - Reduction in sBP (standing minus supine) ≥20; OR
  - Reduction in dBP (standing minus supine) ≥10.
- For patients with Baseline supine systolic Blood Pressure >150 mmHg:
  - Reduction in sBP (standing minus supine) ≥30; OR
  - Reduction in dBP (standing minus supine) ≥15.

An additional summary will be provided of outcomes of assessments resulting from an incident of orthostatic hypotension or other events of interest, using data from both the CRF database and the consultation database, as appropriate.

3.5.3. Total Joint Replacement and Surgical Endpoints

A summary of adjudication outcomes (including outcomes of rapidly progressive osteoarthritis (type-1 only), rapidly progressive osteoarthritis (type-2 only), rapidly progressive osteoarthritis (type-1 or type-2 combined), subchondral insufficiency fracture,

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primary osteonecrosis, and pathological fracture) and total joint replacements will be provided.

Reporting of total joint replacement events including surgery and recovery will be described in a separate Statistical Analysis Plan for Study 1064, and reported in the 1064 study report. Corresponding data from Studies 1056, 1057 and 1058 will be reported under study 1064, as well as patients who enter study 1064 from studies 1059, 1061, and 1063 due to those studies closing out.

3.5.4. Neurological Endpoints

The Neuropathy Impairment Score (NIS) is the sum of scores over all 37 items from both the Left and Right side. Items 1-24 are scored on a 0-4 scale (0, 1, 2, 3, 3.25, 3.5, 3.75, 4) and items 25-37 are scored on a 0-2 scale (0, 1, 2). The possible range of the NIS is 0-244.

The Survey of Autonomic Symptoms (SAS) is a 12 item (11 for females) questionnaire. From this the total number of symptoms (0-12 for males and 0-11 for females) will be calculated. Where a patient has a symptom, the impact of that symptom is then rated from 1 ('not at all') to 5 ('a lot'). The total impact score is calculated using this 1-5 scale, with 0 assigned where the patient does not have the particular symptom. The range for the total impact score is 0-60 for males and 0-55 for females.

4. ANALYSIS SETS

Data for all subjects will be assessed to determine if subjects meet the criteria for inclusion in each analysis population prior to unblinding and releasing the database and classifications will be documented per standard operating procedures.

If a subject was:

- Randomized but not treated, then that subject will be excluded from all efficacy and safety analyses.
- Treated but not randomized, then by definition that subject will be excluded from the efficacy analyses, but will be reported under the treatment they actually received for all safety analyses.
- Randomized but received incorrect treatment, then that subject will be reported under their randomized treatment group for all efficacy analyses, but will be evaluated on a case-by-case basis for presentation for safety analyses. Decisions will be made before unblinding.

4.1. Full Analysis Set

The intent to treat (ITT) analysis set is the primary analysis set for efficacy analyses. It consists of all randomized subjects who received at least one dose of SC study medication (either tanezumab or placebo SC). This analysis set is used in the presentations of all

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efficacy data, and all data listings, and is labeled as the 'ITT Analysis Set' or 'ITT Population'.

4.2. Per Protocol Analysis Set

The per-protocol (PP) analysis set is the secondary efficacy analysis set. It is defined as all subjects in the ITT analysis set who are not major protocol deviators (which would potentially affect efficacy). The criteria for major protocol deviators are described below in Sections 4.2.1 and 4.2.2. The identification of specific subjects included and excluded (and reason for exclusion) for this analysis set will be made and documented prior to unblinding. Protocol deviations for the PP analysis set will be obtained from the collected list of potentially important protocol deviations, and this list will comprise deviations identified from review of programmed listings and study monitoring. This analysis set is used in a specific sensitivity analysis of the co-primary efficacy endpoints, and is labeled as the 'Per Protocol Population'.

Any other major deviation which is not pre-specified below, but results in a subject being excluded from the PP analysis set, will be specified in the protocol deviations document which is completed prior to unblinding.

The following protocol deviations are defined as 'major' and would exclude a subject from the PP analysis set. These deviation criteria can be split into those assessed prior to randomization relating to the protocol inclusion and exclusion criteria, and those assessed post randomization.

4.2.1. Major Deviations Assessed Prior to Randomization

- Inclusion criteria: #3-5.
- Exclusion criteria: #3, 4 (if any of the following conditions are in the index joint: severe chondrocalcinosis, other arthropathies [eg, rheumatoid arthritis], systemic metabolic bone disease [eg, pseudogout, Paget's disease, metastatic calcifications], primary or metastatic tumor lesions, stress or traumatic fracture), 10, 14, 15, 16 (if index joint was involved), 17.
- Randomization criteria: #1, 3-6. Note, subjects with missing Baseline data for any of the co-primary endpoints would not meet the randomization/inclusion criteria for Baseline co-primary endpoints and so would be defined as a deviation according to these criteria.

4.2.2. Major Deviations Assessed Post-Randomization

- Mismatch in specification of index joint in the CRF vs. electronic tablet for WOMAC data collection.
- Rescue medication taken within 24 hours prior to the Week 16 visit

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- Prohibited medications that could affect pain and function assessments (protocol section 5.8.1) taken (i) within 48 hours prior to Week 16 visit for non-NSAID medications (or any use if long-acting, eg, Synvisc), or (ii) within 48 hours prior to Week 16 visit or within the wash-out period specified by Appendix 3 of the protocol, for NSAID medications.

In addition, unforeseen major protocol deviations may be added to this list. However the final definition of this criteria and the per-protocol population will be made prior to unblinding of this study.

4.3. Safety Analysis Set

The safety analysis set is defined as all subjects treated with tanezumab or placebo SC (see beginning of Section 4 for further details). This analysis set will be labeled as the 'Safety Analysis Set' or 'Safety Population' in the corresponding data analyses and summary presentations.

4.4. Other Analysis Sets

Not applicable.

5. GENERAL METHODOLOGY AND CONVENTIONS

5.1. Hypotheses and Decision Rules

5.1.1. Statistical Hypotheses

The treatment comparisons being made in this study are tanezumab 2.5 mg and 2.5 mg then 5 mg (2.5/5 mg) versus placebo. For these treatment comparisons, the null and alternative hypotheses are shown below (note $\mu_{TREATMENT}$ relates to the mean change from Baseline for the specified treatment group). All tests will be 2-sided.

Null Hypotheses	H0: $\mu_{TANEZUMAB\ 2.5mg} - \mu_{PLACEBO} = 0$
	H0: $\mu_{TANEZUMAB\ 2.5/5mg} - \mu_{PLACEBO} = 0$
Alternative Hypotheses	H1: $\mu_{TANEZUMAB\ 2.5mg} - \mu_{PLACEBO} \neq 0$
	H1: $\mu_{TANEZUMAB\ 2.5/5mg} - \mu_{PLACEBO} \neq 0$

The hypotheses for other types of analyses (e.g. for the binary response endpoints) would be similar to those shown above.

5.1.2. Statistical Decision Rules

The Type I error rate (α-level) used in the assessment of pair-wise treatment comparisons for the primary efficacy endpoints is 5%. The fixed sequence testing strategy of the co-primary endpoints is described below.

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The assessment of significance for the tanezumab versus placebo treatment contrasts will use a step-down testing strategy within each of the co-primary efficacy endpoints. The step-down testing will first test tanezumab 2.5/5 mg versus placebo, and if statistically significant ($p \leq 0.05$) will then test tanezumab 2.5 mg versus placebo. Finally, the particular tanezumab dose group is declared as superior to placebo if the corresponding treatment contrast is significant over all three co-primary endpoints. This testing procedure will maintain the Type I error to 5% or less within each of the co-primary efficacy endpoints, and to less than 5% for all three co-primary efficacy endpoints.

The secondary endpoint of subjects with ≥50% reduction from baseline in WOMAC Pain at Week 16 is identified as a key secondary endpoint and will be included in the testing strategy. If both tanezumab regimens are found to be statistically significantly better than placebo in the primary comparisons described above, the two comparisons of this secondary endpoint (tanezumab 2.5 mg treatment group versus placebo and tanezumab 2.5/5 mg treatment group versus placebo) will be adjusted for multiple comparisons using the Hochberg procedure and an overall significance level of 0.05.

The primary analysis will be that with the multiple imputation approach (see below for details), and thus the overall type I error is controlled for each of the two dose regimens (2.5 mg and 2.5/5 mg) since all three co-primary endpoints need to be significant for a single dose. The overall type I error of the study is also controlled given the step-down testing strategy for each of the primary endpoints and the key secondary endpoint. Control of the type I error rate accounting for multiplicity of contrasts will only apply to the three co-primary endpoints (model with the primary imputation analysis) and the key secondary endpoint.

Regardless of the outcome of the primary and key secondary analyses, the secondary endpoints will be tested. No adjustment for multiple comparisons will be made for the secondary efficacy, and for the safety endpoints. The α-level for each hypothesis test for the secondary and exploratory analyses will be 5%.

5.2. General Methods

Subjects will be randomized at Baseline to one of three treatment groups: placebo SC, tanezumab 2.5 mg SC, or tanezumab 2.5/5 mg SC. These will be labeled as placebo, tanezumab 2.5 mg, and tanezumab 2.5/5 mg for the three treatment groups respectively.

A modified treatment-policy estimands strategy is applied as the main strategy to assess effectiveness of tanezumab. Data collected will be included for efficacy assessment regardless of rescue medication being used or not.

The general study design for efficacy, as depicted below, includes a planned treatment period through the Week 16 visit, and a planned 24-week post-treatment safety follow-up period. Efficacy data planned to be collected during this post-treatment safety follow-up period are intended to have efficacy measures contemporaneous to safety observations during this period. They are not intended to assess treatment effects or compare treatment groups. *All*

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endpoints up to Week 24 will be summarized (where available), and endpoints up to Week 16 will be analyzed.

1056 Study Visits / Analysis Windows

Week	B	2	4	8	12	16	24	40
Day	1	15	29	57	85	113	169	281
completer	x	x	x	x	x	x	x	
drop after Day 1 dose	x	x	x	x ET1		x ET2		

B = Baseline; x = collection of most efficacy endpoints; ET = Early Termination Visit

The method and definition of reporting windows for assigning efficacy data to particular time points is described in Appendix 2.1.

All efficacy assessments during the treatment period are made on the analysis windows defined in Appendix 2.1. Using these windows we find the analysis window for a subject's last subcutaneous (SC) dose. Any data included in a window that is up to 8 weeks from this last SC dose window is 'on-treatment', and any data included in a window that is more than 8 weeks after the last SC dose window is off-treatment. Data in on-treatment analysis windows will be used in summaries and analyses, while data in off-treatment analysis windows will be excluded from all summaries and analyses of treatment period efficacy data, ie, up to Week 16.

For example, the table below shows on-treatment and off-treatment windows for the planned collection visits for the WOMAC data during the treatment period:

Last SC Dose Analysis Window	On-treatment Analysis Window Data	Off-treatment Analysis Window Data
Baseline	Weeks 2, 4, 8	Weeks 12, 16
Week 2	Weeks 2, 4, 8	Weeks 12, 16
Week 4	Weeks 2, 4, 8, 12	Week 16
Week 8	Weeks 2, 4, 8, 12, 16	None
Week 12	Weeks 2, 4, 8, 12, 16	None
Week 16	Weeks 2, 4, 8, 12, 16	None

Efficacy data at Week 24 is planned to be off-treatment so will not be subject to the above handling, ie, all available data in the Week 24 window will be used in summaries.

Efficacy data collected via subject diary (NRS pain scores and rescue medication use) are collected daily or weekly, not at study visits. Diary efficacy data will be considered on-treatment if it is collected up to 12 weeks (84 days) after the last SC dose. Diary efficacy data collected more than 12 weeks (84 days) after the last SC dose will be considered off-

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treatment and excluded from summaries and analyses of treatment period efficacy data, ie, for presentations up to Week 16.

Diary data after Week 16 is planned to be off-treatment so will not be subject to the above handling, ie, all available data in windows after Week 16 will be used in summaries.

A summary of all analyses is given in Appendix 1. In all tables the treatment group ordering will be: placebo, tanezumab 2.5 mg, tanezumab 2.5/5 mg. Unless otherwise specified, efficacy analyses use the ITT analysis set only.

5.2.1. Analyses for Binary Data

Binary response parameters, and the incidence of rescue medication use and treatment discontinuation due to lack of efficacy will be analyzed using logistic regression for binary data, with covariates described in Section 3.4.1. Output will show the number and percentage of subjects in each response category, and odds ratios (with 95% confidence intervals) for the treatment comparisons shown in Section 5.1.1.

Subject response endpoints of the OMERACT-OARSI responder criteria, improvement in the WOMAC Pain ≥30, 50, 70 and 90%, WOMAC Physical Function ≥30, 50, 70 and 90%, and improvement in the Patient's Global Assessment of Osteoarthritis ≥2 will be analyzed for change from Baseline to Weeks 2, 4, 8, 12 and 16, using logistic regression for binary data, with model terms for baseline WOMAC Pain subscale score, WOMAC Physical Function subscale score, or Patient's Global Assessment score (Baseline WOMAC Pain for OMERACT-OARSI responder index)*, Baseline diary average pain, index joint, Kellgren-Lawrence grade, and treatment group. Imputation for missing data will use both LOCF and BOCF, where imputation with BOCF will lead to the subject being assessed as a non-responder for the response endpoint at a particular timepoint. In addition, in order to closely match the primary imputation analysis, a mixed BOCF/LOCF imputation for response endpoints will be used. In this analysis BOCF imputation (ie, a subject would be a non-responder) would be used for missing data due to discontinuation for reasons of lack of efficacy* ('Insufficient Clinical Response' on the End of Treatment Subject Summary Case Report form)*, adverse event or death up to the timepoint of interest, and LOCF imputation would be used for missing data for any other reason.*

The use of BOCF for missing data implies subjects with missing data are included in the analysis as non-responders. Similarly the use of LOCF in the case where subjects have no post-Baseline data (and Baseline would be carried forward) again implies those subjects are included in the analysis as non-responders.

The incidence … of use of rescue medication… will be analyzed for Weeks 2, 4, 8, 12, and *16… The incidence of use of rescue medication will be analyzed using logistic regression for binary data, with model terms for Baseline WOMAC Pain subscale score, Baseline diary average pain, index joint, Kellgren-Lawrence grade, and treatment group. … Estimated levels of rescue medication use will be shown for each treatment group, and the ratio (with*

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95% CI) for comparisons versus placebo will be shown. Imputation for missing rescue medication data will use LOCF only.

The incidence of treatment *withdrawal due to lack of efficacy* ('Insufficient Clinical Response' on the Subject Summary Case Report form) *will also be analyzed for discontinuation up to Week 16 (end of treatment period). The analysis of the incidence of discontinuation due to lack of efficacy will be made using logistic regression for binary data, with model terms for Baseline WOMAC Pain subscale score, Baseline diary average pain score, index joint, Kellgren-Lawrence grade, and treatment group.* Discontinuation in the post-treatment safety follow-up period will not be included in this endpoint for analysis, but will be summarized as part of the safety tables.

Cumulative WOMAC Pain response and WOMAC Physical Function at Week 16 using response definitions from a reduction of >0% to =100% (in steps of 10%) will be summarized, using mixed BOCF/LOCF (as described above), and also LOCF and BOCF imputation for WOMAC Pain and WOMAC physical function. Imputation with BOCF for subjects with missing data at that timepoint will lead to the subjects being assessed as non-responders for the response endpoint.

The proportion of subjects who meet a WOMAC Pain response definition at Week 16 will be examined in the cohort of subjects who had a WOMAC Pain response to treatment at Week 8 and the cohort of subjects who did not have a WOMAC Pain response at Week 8. Treatment comparisons will be made within each cohort for tanezumab 2.5/5 mg versus placebo and tanezumab 2.5 mg versus placebo. A descriptive comparison will also be made between the treatment groups of tanezumab 2.5/5 mg versus tanezumab 2.5 mg. These analyses will be produced for the WOMAC Pain response levels of 30% and 50%, and other response definitions (15%). This is an exploratory analysis as treatment comparisons are not specifically powered to achieve significance within cohorts of subjects.

5.2.2. Analyses for Continuous Data

The co-primary efficacy endpoints will be analyzed using an ANCOVA model, with model terms for Baseline score, Baseline diary average pain, index joint (knee or hip), highest *Kellgren-Lawrence grade, and treatment group, and study site as a random effect. The assessment of significance for the tanezumab SC versus placebo treatment contrasts will use a step-down testing strategy within each of the co-primary efficacy endpoints defined as first testing tanezumab 2.5/5 mg versus placebo, and if statistically significant ($p \leq 0.05$) to then test tanezumab 2.5 mg versus placebo. Finally, a tanezumab treatment group is declared as superior to placebo if the corresponding treatment contrast is significant over all three co-primary endpoints. This testing procedure will maintain the Type I error to 5% or less within each of the co-primary efficacy endpoints, and to less than 5% for all three co-primary efficacy endpoints. An additional (main effects ANCOVA) analysis for each of the co-primary efficacy endpoints will use a per-protocol analysis set, which will exclude subjects who are major protocol deviators.*

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The primary analysis of the co-primary endpoints will use multiple imputation for missing data, to account for uncertainty around the subject response. The basis for imputing missing values will be dependent on the reasons for missing data. For subjects with missing data due to discontinuation prior to Week 16 for lack of efficacy or for an adverse event or death, imputation will be based on sampling from a normal distribution using a mean value equal to the subject's Baseline efficacy value and the standard deviation (over all treatment groups) of the observed efficacy data at Week 16. For subjects with missing data for any other reason, imputation will be based on sampling from a normal distribution using a mean value of the subject's last observed efficacy value and standard deviation (over all treatment groups) of the observed efficacy data at Week 16. Imputed values for the Patient's Global Assessment of Osteoarthritis will be rounded to integer values from 1 to 5. Imputed values for WOMAC Pain and Physical Function will be truncated at 0 and 10. One hundred imputation samples will be used, and the ANCOVA model described above will be used for each imputation dataset. The final results will be calculated using the combined sets of results from each imputation dataset analysis.

The primary analysis set is the Intent to Treat analysis set. These three primary endpoint analyses will be used to assess the primary objective of the study.

The mixed model ANCOVA, with multiple imputation, will also be used with other continuous change from Baseline endpoints for landmark (single time point) analyses. The model will include the covariates described in Section 3.4.1, including study site as a random effect. Estimates of treatment effects and pair-wise treatment comparisons will be based on least squares means (LS means) and 95% confidence intervals (CI) will be provided.

A number of sensitivity analyses will be performed on the primary efficacy endpoints in order to assess the robustness of the conclusions for the primary objective. These relate to the analyses for missing data and the analysis population, the homogeneity of the results across factors that may influence efficacy, and for a secondary analysis of the PGA. The analyses described below will not be subject to the testing strategy described for multiple comparisons of the primary analyses. As such, assessment of all treatment comparisons will be made independent of results over the three co-primary endpoints or the two treatment comparisons for each analysis.

<u>Primary Endpoint Sensitivity Analyses</u>

The ITT analysis set is used in the analyses numbered 2 and 3 below, and Per-Protocol analysis set used in analysis number 1 below.

(1) Per-Protocol Analysis Set

The primary analysis using multiple imputation described above will be repeated, but using the Per-Protocol analysis set in place of the ITT analysis set. This analysis will assess the robustness of the efficacy conclusions to subjects who have more strictly adhered to protocol inclusion and exclusion criteria, and to protocol defined study procedures.

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(2) Alternative Missing Data Analyses

There are four additional analyses that will assess the robustness of the efficacy conclusions to the choice of multiple imputation as the primary method for accounting for missing data.

In the first and second analyses, the primary ANCOVA analysis model described above will be repeated, but using BOCF and LOCF respectively for missing data (note these are single imputation analyses).

The third sensitivity analysis for the primary endpoints will use a mixed model repeated measures analysis using the observed and imputed data up to Week 16 from the primary multiple imputation analysis, with covariate terms for Time (study week, treated as a categorical variable), Treatment Group and Time-by-Treatment interaction, as well as the covariates described in Section 3.4.1. The unstructured covariance will be used in the modeling of the within-subject errors in the analysis. Even though this is a sensitivity analysis for the primary endpoints, estimates for the time points of Weeks 2, 4, 8, and 12, in addition to Week 16 will be shown from this analysis. See Appendix 2.1 for details on windows.

The fourth sensitivity analysis for the primary endpoints will use a mixed model repeated measures analysis using all observed data up to Week 16 (i.e. retrieved dropout), with covariate terms for Time (study week, treated as a categorical variable), Treatment Group and Time-by-Treatment interaction, as well as the covariates described in Section 3.4.1. The unstructured covariance will be used in the modeling of the within-subject errors in the analysis.

A summary of the missing data pattern will be shown for the WOMAC Pain and Physical Function subscales and the PGA over Baseline and Weeks 2, 4, 8, 12, and 16. This summary will show the incidence of subjects with each pattern of observed and missing data over these visits and endpoints. This summary will be shown overall, and split by treatment group.

(3) Interaction Analyses

Interaction analyses will be performed for the co-primary endpoints, exploring the effect of Study site and Country. These analyses will fit the covariate terms described in Section 3.4.1 (except for use of Study site as a covariate in the Country interaction analysis, where Country will be used instead [as a fixed term]), in addition to the interaction term of treatment group by factor.

The interaction of Treatment with Study site will be fitted as a random effect (in addition to Study site itself), with the resulting estimated treatment differences being shown for the largest (pertaining to enrollment) study sites to illustrate the level of consistency of treatment benefit across the larger study sites. The study sites to be examined in this way will be any site with an average of four or more subjects per treatment group within the site, which for this study relates to any site with 12 or more subjects in total. This assessment will be made prior to unblinding, therefore a study site in this group may still have fewer than four subjects

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in one or more of the treatment groups, however that site will still be included in this summary of efficacy of the largest study sites. To aid the interpretation of the treatment-site and treatment-country interactions, a summary of the efficacy data for each co-primary endpoint by treatment group will be shown for the sites with ≥12 subjects and for the countries in this study (USA, Canada, and Puerto Rico [summarized separately from other USA sites]).

Other time points for the primary efficacy measure

The ANCOVA model described above for the co-primary endpoints, using covariates of Baseline score, Baseline diary average pain, Index Joint, Highest KL grade (2, 3 or 4) and Treatment, with Study Site as a random variable, will be used in the analysis of WOMAC Pain, WOMAC Physical Function and PGA for the change from Baseline to Weeks 2, 4, 8, and 12. This analysis will be produced using multiple imputation, and BOCF and LOCF for missing data.

The MMRM analyses described above will also analyze results for the secondary time points of Weeks 2, 4, 8, and 12, for the co-primary efficacy endpoints.

Secondary Endpoint Analyses

Other secondary endpoints include the WOMAC Stiffness subscale, WOMAC Average score and WOMAC Pain subscale items (Pain When Walking on a Flat Surface, and Pain When Going Up or Down Stairs), all conducted for the change from Baseline to Weeks 2, 4, 8, 12 and 16. Analysis of Average Pain in the index joint will be conducted for the change from Baseline to Days 1, 2, 3, 4, 5, 6, and 7, and *to Weeks 1, 2, 3, 4, 6, 8, 10, 12 and 16. The analysis of these endpoints will use the same ANCOVA analysis as described above for the co-primary endpoints, with multiple imputation for missing data*, and using the additional covariate of baseline diary average pain score.

The rescue medication data will be converted to Weekly scores for the week prior to the timepoint of interest. Calculation of the endpoints is described in Appendix 3.

The … number of days of use of rescue medication … will be analyzed for Weeks 2, 4, 8, 12, and *16… and the amount of rescue medication use per week will be analyzed for Weeks 2, 4, 8, 12, and 16. … The number of days and amount of rescue medication (mg dosage of acetaminophen) will be analyzed using the Negative Binomial model, with model terms of Baseline WOMAC Pain subscale score, Baseline diary average pain score, index joint, Kellgren-Lawrence grade, and treatment group.* In this model the error term is defined with a negative binomial distribution, and 'log' is used as the link function. *Estimated levels of rescue medication use will be shown for each treatment group, and the ratio (with 95% CI) for comparisons versus placebo will be shown. Imputation for missing rescue medication data will use LOCF only.* For this analysis, Baseline data will not be carried forward in the case of a post-Baseline observation not being available for use in LOCF.

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A table showing number and percentage of subjects will summarize the response for each dimension (item) for the EQ-5D-5L™ at Baseline and Week 8 and Week 16. These summary tables will be shown by treatment group. In addition, for each treatment and for each time point assessed, descriptive statistics (mean, standard deviations, median, number of subjects) will characterize the five-item health status profile on the EQ-5D-5L™ in terms of the health utility score, and the EQ-Visual Analog Scale (EQ-VAS).

The HCRU data at Baseline, Week 24, and Week 40 will be summarized.

Summaries of the change from Baseline to Week 16 in the WPAI:OA impairment scores will be shown by treatment group. This summary will use observed data only (no imputation for missing data). The calculation of these endpoints is described in Appendix 3.

The summary will show number and % of subjects with a decrease, no change, and an increase in score for the change from Baseline to Week 16 as well as descriptive statistics (mean, standard deviation, median, number of subjects) of the Baseline and change at Week 16. The 4 WPAI parameters will be analyzed using the ANCOVA model described above for the primary endpoint using covariates of the corresponding Baseline score, Baseline diary average pain, Index Joint, Highest KL grade (2, 3 or 4), and Treatment, with Study Site as a random variable.

5.2.3. Analyses for Categorical Data

The change from Baseline in the Patient's Global Assessment of Osteoarthritis to Weeks 2, 4, 8, 12 and 16 will also be analyzed using Cochran-Mantel-Haenszel (CMH) test, stratified by the combinations of the two stratification factors. Changes by each level of improvement will be summarized, as well as any improvement (change<0), and any worsening (change>0). *For this analysis imputation for missing data will used mixed BOCF/LOCF, as well as BOCF and LOCF separately. If there are too few subjects in any stratification combination group (defined as <15 subjects in any stratification factor) then an unstratified test will be performed.* The mixed BOCF/LOCF analysis at Week 16 will provide a sensitivity analysis for the primary analysis of the PGA.

For any analysis using the CMH test, if there are too few subjects in any stratification combination group (defined as <15 subjects in any of the 6 combinations of stratification factors) then an unstratified test will be performed.

The change from Baseline in the NIS will be analyzed using a Cochran-Mantel-Haenszel (CMH) test for 'row mean scores differ', using change from Baseline categories as the scores in the analysis, and stratified by the combined levels of the stratification factors. Output will show number and percentage of subjects whose NIS score worsened (change>0), improved (change<0) or had no change, in addition to the mean (with standard deviation) and median change, and minimum and maximum change. This analysis will be performed for the two treatment comparisons separately, and shown by visit and worst change (largest change from baseline to any post-baseline visit), and by last change (summary statistics only).

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5.2.4. Analyses for Time to Event Data

The … time to treatment *withdrawal due to lack of efficacy will also be analyzed for discontinuation up to Week 16 (end of treatment period). The time to discontinuation will be analyzed using the log-rank test, with Kaplan-Meier estimates of the time to discontinuation shown for selected percentiles, dependent on the level of discontinuation. The expectation is that these would be the 1st, 2nd, 5th, 10th and 25th percentiles*, in addition to the minimum and maximum time to discontinuation. *Other percentiles may be shown if the level of discontinuation due to lack of efficacy as calculated using Kaplan-Meier procedure is sufficiently large.*

A plot of the time to discontinuation (failure) will be shown using the Kaplan-Meier estimates. Only treatment discontinuation up to the end of treatment period (Week 16 visit or early discontinuation) will be used in this analysis. Discontinuation due to lack of efficacy after the end of treatment visit will be included in the standard safety tables. Time to event for discontinued subjects (discontinuing for reasons other than lack of efficacy) prior to the Week 16 visit uses censoring at the time of discontinuation. Imputation of time to event for completed subjects or discontinued subjects (for any reason) post Week 16 visit uses censoring at the Week 16 visit time point.

5.3. Methods to Manage Missing Data

The three co-primary efficacy endpoints are the changes from Baseline to Week 16 in the WOMAC Pain subscale, the WOMAC Physical Function subscale, and the Patient Global Assessment of Osteoarthritis.

The primary analysis of the co-primary endpoints will use multiple imputation for missing data at Week 16 (where the method for imputation will be dependent on the reason for missing data) followed by the ANCOVA analysis with the model described below for the multiple imputed datasets. The imputation strategies are described in the following table. While the table describes the multiple imputation strategy specifically for the Week 16 time point, multiple imputation analysis at other time points will use the same strategy but with the appropriate time point, eg, 'Week 2,' substituted for 'Week 16' in the table below. Efficacy data missing from windows after the Week 16 window, eg, Week 24, will not be imputed for any summary or analysis unless otherwise indicated.

Type of Missing Data	Imputation Method
Missing data resulting from discontinuation due to Death, Adverse Events (AEs) or Insufficient Clinical Response (Lack of Efficacy, LoE) prior to or during the Week 16 visit reporting window*.	Multiple imputations will be created by sampling from a normal distribution based on the subject's baseline score and the standard deviation (over all treatment groups) of the observed efficacy data at Week 16 over all ITT subjects. This is a multiple imputation version of BOCF single imputation method. [Seeds 1, 3, and 5 below]

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Missing data for other reasons, ie, • Subject did not discontinue on or before Week 16 (includes discontinuation for any reason after the end of the Week 16 visit reporting window*) • Subject discontinued for a different reason prior to or during the Week 16 visit reporting window*.	Multiple imputations will be created by sampling from a normal distribution based on the subject's last score and the standard deviation (over all treatment groups) of the observed efficacy data at Week 16 over all ITT subjects. For example if last observation for a subject is at Week 12, then the imputation sample for that subject is created using the subject's Week 12 observation and the standard deviation of the Week 16 observations for all subjects. Note, a subject's last observation may be the Baseline observation. This is a multiple imputation version of LOCF single imputation method. [Seeds 2, 4, and 6 below]

* See Appendix 2.1 for a definition of the reporting windows

The imputation of baseline-like data for subjects with missing data due to discontinuation due to Death, AE or LoE is intended to impute conservative efficacy values for those subjects who discontinue because of a reason that is considered to be a poor outcome for the subject, and so a poor outcome is imputed. For those subjects with missing data that is likely to not be related to treatment group, the intention is that missing data should be imputed based on a 'missing at random' assumption taking into account the subject's previous available data.

One hundred imputed datasets will be used in this analysis. In order to pre-define the analysis (and not to allow the results to change if run again), the following seeds will be used in the creation of the multiple imputed data: WOMAC scores: [1] 1001-1100 and [2] 2001-2100; PGA scores: [3] 3001-3100 and [4] 4001-4100; and diary pain scores: [5] 5001-5100 and [6] 6001-6100. Imputed Week 16 data for the PGA will be rounded to integer scores in the range 1 to 5. Imputed Week 16 data for the WOMAC subscale and Average scores, and for the diary pain scores <0 and >10 will be truncated to 0 and 10, respectively. Imputed Week 16 data for the WOMAC items of Pain when Going Up or Down Stairs and Pain when Walking on a Flat Surface will be rounded to integer scores in the range 0 to 10. The ANCOVA analysis described in Section 5.2.2 (with covariates in Section 3.4.1) will be used for each imputation dataset, and the overall results will be calculated to take account of the variability both within and between imputation datasets using standard methods (Little & Rubin, 2002), which are described in Appendix 3.2.

This analysis will be used for the co-primary efficacy endpoints at Week 16, plus secondary analyses at other time points, and also for a range of secondary efficacy endpoints at all time points up to Week 16. When using the multiple imputation method described above for time points earlier than Week 16, then the reason for missing data is assessed up to the end of the window for that particular time point (see Appendix 2.1).

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Four additional methods will explore the sensitivity of the effect of missing data. The first method of Baseline Observation Carried Forward (BOCF) for missing data at the primary time point of Week 16 will impute the subject's Baseline value for the Week 16 time point, and therefore a zero change from baseline. If a subject's baseline data is also missing then that subject's data remain missing for the post-baseline time point, and the patient would effectively be excluded from the analysis. The second method of Last Observation Carried Forward (LOCF) for missing data at the primary time point of Week 16 will impute the subject's last observed data value for the efficacy endpoint. With LOCF, if a subject is missing all post-baseline efficacy data for a given efficacy endpoint, then baseline will be carried forward (if baseline is also missing then the subject would have no contributing data to be included in the analysis, and again effectively be excluded from the analysis). In both the BOCF and LOCF imputation analyses, the same main effects ANCOVA model as described below will be used. The third method will use Mixed Model for Repeated Measurements (MMRM) utilizing the datasets created by the multiple imputation process up to and including Week 16 (see Appendix 2.1 for details on windows; if multiple observations are within a window, only the single observation selected for analysis by the windowing algorithm will be used in the MMRM analysis). The fourth method will use MMRM utilizing all available data up to and including Week 16, including data considered off-treatment (retrieved dropout).

Analyses of the three co-primary endpoints at secondary time points will use the BOCF and LOCF imputation methods for missing data, and use the same (main effects) ANCOVA model as described for the primary analyses.

The responder endpoints will be analyzed using logistic regression for binary data, using both BOCF and LOCF separately for missing data of the response endpoint at a particular time point. Imputation using BOCF will lead to the subject being assessed as a non-responder. Imputation using LOCF will use the response from the last available assessment. In addition, in order to closely match the primary imputation analysis, a mixed BOCF/LOCF imputation for response endpoints will be used. In this analysis BOCF imputation (i.e. a subject would be a non-responder) would be used for missing data due to discontinuation for reasons of lack of efficacy, adverse event or death up to the time point of interest, and LOCF imputation would be used for missing data for any other reason.

Note, if Baseline is missing then the subject data for the change from Baseline will be set to missing for all efficacy analyses for that parameter. A subject who has a missing baseline score will be missing for the response criteria for endpoints where the response is based on one parameter. The OMERACT-OARSI responder index is based on 3 parameters. It is set to missing if two or three out of these three parameters are missing at baseline (per its definition, a response can be still be achieved if only one parameter is missing, regardless of which one it is).

The individual WOMAC subscales are calculated as the mean of the individual items (5 for Pain, 17 for Physical Function and 2 for Stiffness). CCI [redacted]

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CCI [redacted] The WOMAC Average score is calculated as the mean of the three WOMAC subscale scores of Pain, Physical Function and Stiffness. CCI [redacted] issing WOMAC subscale or WOMAC Average scores will be subject to the imputation method of the analysis as described above.

For the analysis of the rescue medication endpoints, missing data is imputed for daily missing scores first, and then the last available weekly score (after daily missing data is imputed) will be used for subsequent missing weekly scores, as described below. While subjects are still in the study any missing data will be imputed by carrying forward the last recorded daily data up to Week 16 (LOCF daily data). Imputation using the daily data will occur up to the end of the last week when the subject is in the study (see Appendix 2.1 for definitions of the last study day in each week). For example if a subject discontinues on study day 10, then data up to the end of Week 2 will be imputed in this way. The weekly scores for the rescue medication endpoints can then be calculated for each week the subject is in the study. Rescue medication endpoints are summarized and analyzed using LOCF, and so the last weekly score for the rescue medication will be used for LOCF after the subject has discontinued from the study (note, imputation is taken from the last week with non-missing data and not necessarily from the last available study week, e.g. if Week 8 is missing then Week 7 data can be used). The baseline observation will not be carried forward in the case where a post-baseline observation is not available for the LOCF imputation. In the example above, the subject who discontinued in Week 2 (Study Day 10) will have their Week 2 value used as the LOCF value for all Weeks 3-16. The BOCF imputation rule will not be used for the subject because rescue medication is collected during the Initial Pain Assessment Period only (days -7 to -1) and subjects should not be taking rescue medication within 24 hours of the Baseline visit (so part of day -1), therefore Baseline rescue medication use is not an accurate reflection of subjects' true Baseline use of rescue medication. Imputation of weekly diary data after Week 16 will not be performed.

The electronic diary data are a mix of daily and weekly average pain assessments for the index hip or knee, although the recall assessment period is the past 24 hours for both daily and weekly assessments. A weekly mean score will be calculated from the available daily pain scores. Any missing daily pain scores will be left as missing in the weekly pain score calculated. If there are no non-missing observations then the weekly score will be missing. The Baseline mean will be calculated using the last 3 actual values from the last 7 days of the Initial Pain Assessment Period (IPAP). The weekly pain scores (either calculated from the daily scores when available or directly from the weekly pain assessments) will then be utilized for the multiple imputation, and the LOCF and BOCF imputations in the standard way. Note, for the weekly pain score, a pain score being carried forward with LOCF might not be a visit week assessment (eg, carry forward Week 3 for missing Week 4 data even though there is no scheduled Week 3 visit). For the purposes of the imputation analyses, where there is no post-baseline observation available to carry forward, then the baseline score carried forward will be the baseline average pain score, being the mean of the last 3 pain

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scores in the initial pain assessment period (7 days that precede randomization). If there are less than 3 baseline pain scores then the baseline is calculated over the remaining non-missing values.

Missing values in standard summaries of AEs, lab values, vital signs and ECGs will be handled per Pfizer standard algorithms. For the analysis of NIS the baseline observation will not be carried forward in the case where a post-baseline observation is not available for the LOCF imputation.

The Baseline diary average pain is used in the analysis of most endpoints as described in Section 3.4.1. However if a patient has a missing value for this covariate then to avoid exclusion of the patient for the endpoint then a Baseline value will be imputed as the patient's WOMAC Pain subscale score. This imputed value will not be used in the analysis of the Average Pain from the diary, but as a covariate for other endpoints.

6. ANALYSES AND SUMMARIES

A summary of the details of the efficacy analyses is presented in tabular format in Appendix 1.

6.1. Primary Endpoint(s)

See Appendix 1.

6.2. Secondary Endpoint(s)

See Appendix 1.

6.3. Other Endpoint(s)

6.3.1. Pharmacokinetics

The following reporting of PK data will be done using all available data:

- A listing of all plasma tanezumab concentrations sorted by subject, active treatment group and nominal time post dose. The listing of concentrations will also include the actual times post dose.
- A descriptive summary of the plasma tanezumab concentrations based on nominal time post dose for each treatment group.
- Individual plots of plasma tanezumab concentrations against actual time post dose represented in one graph for each treatment group.
- Mean (SD) and median plot of plasma tanezumab concentrations over time using nominal times for each of the treatment groups combined in one graph.

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6.3.2. Pharmacodynamics (NGF)

- Serum samples from a subset of patients will be run in the bioanalytictal assays for the NGF related proteins (total NGF, bound NGF, proNGF, soluble p75).

- The NGF related protein concentrations will be reported in summary tables and figures and will be part of the A4091056 study CSR. The actual data analysis of the NGF related protein concentrations will be described in a separate Statistical Analysis Plan and the results will be reported in a separate report.

- The following reporting of NGF concentration data will be done in the CSR:

- A listing of individual total NGF, bound NGF/total NGF, proNGF/total NGF and soluble p75 concentrations in serum sorted by subject, active treatment group and nominal time post dose. The listing of concentrations will also include the actual times post dose.

- A descriptive summary (mean, SD, SE, 95% CI, median, min, max) of the total NGF, bound NGF/total NGF, proNGF/total NGF and soluble p75 concentrations in serum based on nominal time post dose for each treatment group.

- Individual plots of total NGF, bound NGF/total NGF, proNGF/total NGF and soluble p75 concentrations against actual time post dose for each treatment group.

- Mean (SD) and median plot of total NGF, bound NGF/total NGF, proNGF/total NGF and soluble p75 concentrations over time in serum using nominal times for each of the treatment group combined in one graph.

6.3.3. Osteoarthritis Biomarkers

- Biomarker concentrations will be measured in subjects with any of the joint safety adjudication outcomes of rapidly progressive osteoarthritis (type 1 and type 2), subchondral insufficiency fracture, primary osteonecrosis, or pathological fracture, and for occurrence of total joint replacement. In addition, control subjects (no AEs) to these cases will be analyzed for biomarker concentrations and will be identified based on matching duration of treatment. The biomarker concentrations will be reported in summary tables and figures in a report separate from the A4091056 study CSR. Also, the data analysis of the osteoarthritis biomarker concentrations will be described in a separate Statistical Analysis Plan and the results will be reported in a separate report which will address the biomarker results in a combined fashion across studies A4091056, A4091057, and A4091058.

6.4. Subset Analyses

There are no planned subset analyses.

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6.5. Baseline and Other Summaries and Analyses

The following non-standard baseline tables will be included:

- A summary of baseline characteristics, including index joint, Kellgren-Lawrence grade of the index joint (for subjects with Hip and Knee OA separately then overall), highest Kellgren-Lawrence grade for each subject, WOMAC subscales at Baseline and Screening (for Pain subscale only), diabetes status (from medical history and/or pre-treatment HbA1c≥6.5%), and the PGA at Baseline. This summary will also include a summary of the number of subjects who are ≥75 years old.

6.5.1. Concomitant Medications and Non-Drug Treatments

Summaries of various classes of concomitant medications based on Case Report Form classifications will be provided, eg, treatments for osteoarthritis, non-NSAID and NSAID medications (shown separately).

6.6. Safety Summaries and Analyses

Adverse events, concomitant medications, laboratory safety tests, physical and neurological examinations, vital signs, ECGs, the anti-drug antibody test will be collected for each subject during the study according to the Schedule of Assessments. Standard safety reporting tables will summarize and list the safety data.

Pfizer standard safety data presentations will be made for demography data, discontinuation data, adverse event data, laboratory test data, vital signs data and ECG data.

The following non-standard safety tables will also be included

- Summary of number of subjects treated by country and treatment group.
- Incidence and severity of Adverse Events leading to discontinuation.
- Summary of AEs, Incidence of AEs, Incidence of AEs leading to discontinuation and summary of Serious AEs will be shown for the whole study period (including the safety follow-up period).
- Summary of evidence of neurological examination abnormalities by visit and final assessment, and incidence of neurological findings over consecutive visits. Further details of this summary are given below.
- Summary of final outcome of neurological consultation. Further details of this summary are given below.
- Summary of the Incidence of sympathetic neuropathy based on investigator assessment and, if performed, expert consultant assessment.

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- 'Incidence and severity' tables of treatment-emergent adverse events of Abnormal Peripheral Sensation (APS) and Sympathetic Nervous Function, as defined above. Other adverse events may be added to these groupings if they are observed in this study or other studies in the tanezumab program.

- Summary table and listing of inclusion and exclusion criteria that are not met by subjects who were screened (but not randomized).

- Summary of discontinuation by treatment group and reason, and study week of discontinuation for the treatment period (Weeks 1-2, 3-4, 5-8, 9-12, 13-16, >16 and for the safety follow-up period (Weeks 1-8, 9-16, 17-24, >24).

- A summary of the maximum increase from baseline in the sitting systolic and diastolic blood pressure. The categories used are: (systolic BP) only decreases or no change, >0 to 10, >10-20, >20-30, >30, and (diastolic BP) only decreases or no change, >0 to 10, >10-20, >20.

- A summary of the maximum decrease from baseline in the sitting systolic and diastolic blood pressure. The categories used are: (systolic BP) <-30, -30 to <-20, -20 to <-10, -10 to <0, only increases or no change, and (diastolic BP) <-20, -20 to <-10, -10 to <0, only increases or no change.

- A summary of the change from baseline to last observation in the sitting systolic and diastolic blood pressure. The categories used for these summaries are: (systolic BP) ≤-40, >-40 to -30, >-30 to -20, >-20 to -10, >-10 to 0, >0 to <10, 10-<20, 20-<30, 30-<40, ≥40, and (diastolic BP) ≤-30, >-30 to -20, >-20 to -10, >-10 to 0, >0 to <10, 10-<20, 20-<30, ≥30.

- A summary of incidence of subjects with confirmed orthostatic hypotension, for each visit and any post-baseline incidence of orthostatic hypotension.

- A summary of discontinuation up to End of Treatment period, and up to End of Study period.

- Incidence of musculoskeletal physical examination at screening.

- Summary of the Survey of Autonomic Symptoms (SAS) number of symptoms reported and total symptom impact score, at each visit, and for the change from Baseline score.

- Summary of concomitant medications for Osteoarthritis for non-NSAID and NSAID medications (shown separately).

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- Summary of number of days of NSAID use per dosing interval (eg, Day 1 to Week 8 and Week 8 to Week 16) and for the first 8-week interval in the safety follow-up period. This will show the number and percentage of subjects in an interval who exceeded the limit of 10 days of NSAID use. If an interval exists, the visits will be used to define the interval, otherwise calendar time will be used. A summary of average number of days of NSAID use will be displayed by interval. Also, a summary of the overall number of days of NSAID use from Day 1 to Week 24 will be shown, as well as the number and percentage of subjects who exceeded the limit of 30 days of NSAID use during this interval.

- Summary of failed drug treatments for protocol qualification, with reasons for discontinuation.

6.6.1. Adverse Events

Adverse Events of Abnormal Peripheral Sensation will be summarized.

Separate adverse event summaries by treatment group for adverse events of decreased sympathetic function will be conducted. More specifically, adverse events with the following preferred terms will be considered to represent adverse events of decreased sympathetic function: Blood pressure orthostatic decreased, bradycardia, dizziness postural, heart rate decreased, orthostatic hypotension, presyncope, sinus bradycardia, syncope, anhidrosis, hypohidrosis, abdominal discomfort, diarrhea, early satiety, fecal incontinence, nausea, vomiting, ejaculation delay, ejaculation disorder, ejaculation failure, hypertonic bladder, micturition urgency, nocturia, urinary frequency, urinary hesitation, urinary incontinence, respiratory distress and respiratory failure. If necessary, this list of preferred terms may be adjusted for updates made to the MEDICAL DICTIONARY FOR DRUG REGULATORY AFFAIRS (MedDRA) dictionary versions used for reporting.

In addition to summaries of adverse events considered to represent adverse events of decreased sympathetic function noted above, adverse events of syncope, bradycardia, orthostatic hypotension, anhidrosis, or hypohidrosis are designated as adverse events of interest that will be reviewed by the unblinded E-DMC.

Selected adverse events of interest and common adverse events will be summarized using Risk Differences between each tanezumab group and placebo, together with 95% confidence intervals, using exact methods. In addition, significance testing will be performed for tanezumab versus placebo comparisons using exact methods for the adverse events of interest. There will be no multiplicity adjustment for these significance tests.

For the 3-tier adverse event reporting, tier 1 adverse events are defined in the tanezumab Safety Review Plan, and this definition of tier-1 adverse events for the report of study 1056 tables will be finalized prior to the unblinding of this study.

Tier 2 AEs are those with a frequency of ≥3% in any treatment group and that are not in tier 1.

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Tier 3 AEs are those not in Tier 1 or Tier 2, and will be summarized using standard Pfizer data standards tables, where all Adverse Events will be included (i.e. Tier 3 AEs will not be shown separately).

Adverse events within tier 1 and 2 will be summarized using Risk Differences between each tanezumab group and placebo, together with 95% confidence intervals, using exact methods. Significance tests will be performed for tanezumab versus placebo comparisons using exact methods for the tier 1 AEs. There will be no multiplicity adjustment for these significance tests. These tables will be produced for the comparisons of tanezumab 2.5 mg versus placebo and tanezumab 2.5/5 mg versus placebo.

The following footnote will be used in the Tier 1 AE tables: "P-values and confidence intervals are not adjusted for multiplicity and should be used for screening purpose only. 95% Confidence intervals are provided to help gauge the precision of the estimates for Risk Difference. Risk Difference is computed as 'Tanezumab 2.5 mg versus placebo' and 'Tanezumab 2.5/5 mg versus placebo'. Exact methods are used for 95% confidence intervals.". Similarly the following footnote will be used in the Tier 2 AE tables: "Confidence intervals are not adjusted for multiplicity and should be used for screening purpose only. 95% Confidence intervals are provided to help gauge the precision of the estimates for Risk Difference. Risk Difference is computed as 'Tanezumab 2.5 mg versus placebo' and 'Tanezumab 2.5/5 mg versus placebo'. Exact methods are used for 95% confidence intervals."

CCI [redacted]

> [redacted]It should be recognized that most studies are not designed to reliably demonstrate a causal relationship between the use of a pharmaceutical product and an adverse event or a group of adverse events. Except for select events in unique situations, studies do not employ formal adjudication procedures for the purpose of event classification. As such, safety analysis is generally considered as an exploratory analysis and its purpose is to generate hypotheses for further investigation. The 3-tier approach facilitates this exploratory analysis.[redacted]

All summaries of adverse events will be shown for adverse events that begin or worsen after the first dose of study drug (treatment-emergent) up to the end of the treatment period. In addition a selection of adverse event tables will be produced for the off-study medication safety follow-up period, and some will be produced for the whole period up to the end of the study, including the treatment period and safety follow-up period.

6.6.2. Vital Signs

Incidence of orthostatic hypotension using postural changes in blood pressure…will be summarized.

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6.6.3. Neurological Results

The Neuropathy Impairment Score (*NIS*) *is the sum of scores over all 37 items from both the Left and Right side. The change from baseline to each post-baseline visit in the NIS (using LOCF for missing data), and to both the Last and Worst* (largest) *change from Baseline (over all post-Baseline visits) will be summarized, and analyzed using Cochran-Mantel-Haenszel test* (stratified by the combinations of two stratification factors, last change from Baseline will not be analyzed). *The NIS data, the neurological consultation data and the conclusion from neurological examination data will be reported.*

The neurological consultation data will be summarized for all subjects…. The "conclusion from the neurological examination" data will be summarized for each timepoint, and then a summary of the final assessment over all neurological examinations for each subject will be provided.

The change from Baseline in the NIS for Weeks 2, 4, 8, 12, 16, 24, and 40 will be analyzed using a CMH test (stratified by the combined levels of the stratification factors) with change categories. Missing data will be imputed using LOCF only. For this analysis, Baseline data will not be carried forward in the case of a post-Baseline observation not being available for use in LOCF. An additional analysis will use the change from Baseline to the largest (worst) post-Baseline value.

The "conclusion from the neurological examination" data will be summarized for each time point and the last subject assessment. In addition the persistence of any neurological examination finding will be summarized, showing the incidence of subjects with new or worsened neurological examination abnormalities (both clinically significant only and also for any finding) for 2, 3, 4, and ≥5 consecutive visits.

6.6.4. Immunogenicity

The following assessments of ADA data will be made:

- A listing of individual serum ADA results sorted by treatment group, subject ID and planned visit. The listing will also include the actual test date/times.
- The proportion of subjects who test positive (i.e. develop anti-tanezumab antibodies) and negative will be summarized by treatment group and planned visit. The summary will also include the proportion of subjects who test positive and negative overall in the study.
- Subjects who develop anti-tanezumab antibodies after treatment will be evaluated for the presence of anti-tanezumab neutralizing antibodies, and individual results will be listed.
- Individual subjects with positive ADA results will be evaluated for potential ADA impact on the individual's PK, NGF, efficacy and safety profile.

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6.6.5. Joint Safety Events

The incidence of subjects with any of the joint safety adjudication outcomes of rapidly progressive osteoarthritis (type 1 and type 2), subchondral insufficiency fracture…, primary osteonecrosis, or pathological fracture, and for occurrence of total joint replacement will be shown by number of subjects treated and subject *years of exposure (treatment plus follow up periods), for individual treatment groups.*

For the joint safety event analyses, the observation period is defined as the time from first SC dose to study completion or discontinuation for subjects who did not have an event, or time from first SC dose to the earliest event for subjects who did have at least one event.

7. INTERIM ANALYSES

7.1. Introduction

Interim analysis is planned for safety as described in the following sections.

7.2. Interim Analyses and Summaries

Safety data will be subject to regular and ongoing reporting and review throughout the study. The details of these interim analyses will be documented in a separate Statistical Analysis Plan. Review of the safety data will be by the tanezumab external Data Monitoring Committee (E-DMC).

A blinded Adjudication Committee will be convened and asked to review all possible or probable joint-related safety events identified by the Central Reader (rapidly progressive osteoarthritis (type-1 or type-2), subchondral insufficiency fractures, primary osteonecrosis, or pathological fracture), total joint replacement as well as investigator reported adverse events of osteonecrosis, rapidly progressive osteoarthritis, subchondral insufficiency fracture or pathologic fracture. Adverse events related to joint safety that the investigator or sponsor considers medically important may also be reviewed by the Adjudication Committee. A stopping rule relating to a set of adjudicated outcomes has been defined, and is described below.

If the blinded Adjudication Committee identifies adjudicated events of rapidly progressive osteoarthritis type 2, subchondral insufficiency fractures…, primary osteonecrosis, or pathological fracture, occurring at a rate that could trigger the protocol-based stopping criteria, an urgent, ad hoc assessment of the events will be made by the E-DMC.

The protocol (or treatment group) stopping rule will be based on the assessment of the number of subjects with adjudicated events of interest (rapidly progressive osteoarthritis type 2, subchondral insufficiency fractures…, primary osteonecrosis, or pathological fracture) during the course of the study. Assuming the rate of adjudicated events in the placebo group is no more than CCI [redacted]*, if adjudicated events of interest are reported in* CCI [redacted] *subjects in any tanezumab treatment group than for the placebo treatment group, a treatment-group or protocol-based stopping rule will be triggered. If the rate of events in the placebo treatment group is higher than* CCI [redacted] *the appropriate threshold*

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number of events (subjects) *for the stopping rule will be reassessed. If the protocol-based stopping rule is triggered, the E-DMC will formulate a recommendation whether it is safe to continue dosing in some or all treatment groups or whether the study should be terminated completely. This decision will be made by Pfizer in consultation with the E-DMC.*

A separate set of dosing suspension rules for specified Serious Adverse Events and events consistent with Hy's Law are described in Section 9.6.1 of the protocol.

Programming and review of unblinded outputs will be performed by individuals independent of the study team.

8. REFERENCES

1. EuroQol Group. EuroQol: a new facility for the measurement of health related quality of life. *Health Policy* 1990; 16:199-208.

2. Little RJ & Rubin DB (2002). *Statistical Analysis with Missing Data*. New Jersey: Wiley.

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9. APPENDICES

Appendix 1. SUMMARY OF EFFICACY ANALYSES

Note: BL=Baseline

Endpoint	Analysis Set	Statistical Method	Model/ Covariates	Missing Data	Objective
Change from Baseline to Week 16 in WOMAC Pain subscale	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	Multiple Imputation	Primary Analysis
Change from Baseline to Week 16 in WOMAC Physical Function subscale	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	Multiple Imputation	Primary Analysis
Change from Baseline to Week 16 in Patient Global Assessment of Osteoarthritis	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	Multiple Imputation	Primary Analysis
Change from Baseline to Week 16 in WOMAC Pain subscale	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Study Site, Treatment Group, Study Site x Treatment Group (Study site and interaction as random effects)	Multiple Imputation	Additional (Interaction) Analysis
Change from Baseline to Week 16 in WOMAC Pain subscale	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Country, Treatment Group, Country x Treatment Group.	Multiple Imputation	Additional (Interaction) Analysis
Change from Baseline to Week 16 in WOMAC Physical Function subscale	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Study Site, Treatment Group, Study Site x Treatment Group (Study site and	Multiple Imputation	Additional (Interaction) Analysis

			interaction as random effects)		
Change from Baseline to Week 16 in WOMAC Physical Function subscale	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Country, Treatment Group, Country x Treatment Group.	Multiple Imputation	Additional (Interaction) Analysis
Change from Baseline to Week 16 in Patient Global Assessment of Osteoarthritis	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Study Site, Treatment Group, Study Site x Treatment Group (Study site and interaction as a random effect)	Multiple Imputation	Additional (Interaction) Analysis
Change from Baseline to Week 16 in Patient Global Assessment of Osteoarthritis	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Country, Treatment Group, Country x Treatment Group	Multiple Imputation	Additional (Interaction) Analysis
Change from Baseline to Week 16 in WOMAC Pain subscale	PP	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	Multiple Imputation	Sensitivity Analysis (Per protocol)
Change from Baseline to Week 16 in WOMAC Physical Function subscale	PP	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	Multiple Imputation	Sensitivity Analysis (Per protocol)
Change from Baseline to Week 16 in Patient Global Assessment of Osteoarthritis	PP	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	Multiple Imputation	Sensitivity Analysis (Per protocol)
Change from Baseline to Week 16 in WOMAC Pain subscale	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	BOCF	Sensitivity Analysis
Change from Baseline to Week 16 in WOMAC Physical Function subscale	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	BOCF	Sensitivity Analysis
Change from Baseline to Week 16 in	ITT	ANCOVA	BL Score, BL diary average pain,	BOCF	Sensitivity Analysis

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Patient Global Assessment of Osteoarthritis			Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)		
Change from Baseline to Week 16 in WOMAC Pain subscale	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	LOCF	Sensitivity Analysis
Change from Baseline to Week 16 in WOMAC Physical Function subscale	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	LOCF	Sensitivity Analysis
Change from Baseline to Week 16 in Patient Global Assessment of Osteoarthritis	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	LOCF	Sensitivity Analysis
Change from Baseline to Weeks 2, 4, 8, 12, and 16 in WOMAC Pain subscale	ITT	MMRM	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Time, Treatment Group, Time x Treatment Group (Study site as a random effect)	Multiple Imputation Data	Sensitivity Analysis for Week 16 (Secondary Endpoint Analysis for other time points)
Change from Baseline to Weeks 2, 4, 8, 12, and 16 in WOMAC Physical Function subscale	ITT	MMRM	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Time, Treatment Group, Time x Treatment Group (Study site as a random effect)	Multiple Imputation Data	Sensitivity Analysis for Week 16 (Secondary Endpoint Analysis for other time points)
Change from Baseline to Weeks 2, 4, 8, 12, and 16 in Patient Global Assessment of Osteoarthritis	ITT	MMRM	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Time, Treatment Group, Time x Treatment Group (Study site as a random effect)	Multiple Imputation Data	Sensitivity Analysis for Week 16 (Secondary Endpoint Analysis for other time points)
Change from Baseline to Weeks 2, 4, 8, 12, and 16 in WOMAC Pain subscale	ITT	MMRM	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Time, Treatment Group, Time x Treatment Group (Study site as a random effect)	All Observed Data Including Off-Treatment	Sensitivity Analysis for Week 16 (Secondary Endpoint Analysis for other time points)
Change from Baseline to Weeks 2, 4, 8,	ITT	MMRM	BL Score, BL diary average pain,	All Observed	Sensitivity Analysis for

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12, and 16 in WOMAC Physical Function subscale			Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Time, Treatment Group, Time x Treatment Group (Study site as a random effect)	Data Including Off-Treatment	Week 16 (Secondary Endpoint Analysis for other time points)
Change from Baseline to Weeks 2, 4, 8, 12, and 16 in Patient Global Assessment of Osteoarthritis	ITT	MMRM	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Time, Treatment Group, Time x Treatment Group (Study site as a random effect)	All Observed Data Including Off-Treatment	Sensitivity Analysis for Week 16 (Secondary Endpoint Analysis for other time points)
Change from Baseline to Weeks 2, 4, 8, 12, and 16 in Patient Global Assessment of Osteoarthritis	ITT	CMH test	Treatment Group [1]	Mixed BOCF/LOCF	Sensitivity Analysis for PGA
Change from Baseline to Weeks 2, 4, 8, 12, and 16 in Patient Global Assessment of Osteoarthritis	ITT	CMH test	Treatment Group [1]	BOCF	Sensitivity Analysis for PGA
Change from Baseline to Weeks 2, 4, 8, 12, and 16 in Patient Global Assessment of Osteoarthritis	ITT	CMH test	Treatment Group [1]	LOCF	Sensitivity Analysis for PGA
Change from Baseline to Week 16 in WOMAC Pain subscale, shown by site (sites with n≥12)	ITT	None (summary)	NA	Multiple Imputation	Supportive summary for interaction analysis
Change from Baseline to Week 16 in WOMAC Physical Function subscale, shown by site (sites with n≥12)	ITT	None (summary)	NA	Multiple Imputation	Supportive summary for interaction analysis
Change from Baseline to Week 16 in Patient Global Assessment of Osteoarthritis, shown by site (sites with n≥12)	ITT	None (summary)	NA	Multiple Imputation	Supportive summary for interaction analysis
Change from Baseline to Week 16 in WOMAC Pain subscale, shown by country	ITT	None (summary)	NA	Multiple Imputation	Supportive summary for interaction analysis
Change from Baseline to Week 16 in WOMAC Physical Function subscale, shown by country	ITT	None (summary)	NA	Multiple Imputation	Supportive summary for interaction analysis
Change from Baseline to Week 16 in Patient Global Assessment of	ITT	None (summary)	NA	Multiple Imputation	Supportive summary for interaction analysis

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Osteoarthritis, shown by country					
Missing data pattern for WOMAC Pain subscale for Baseline and Weeks 2, 4, 8, 12, and 16	ITT	None (summary)	NA	Observed Data	Supportive summary for missing data
Missing data pattern for WOMAC Physical Function subscale for Baseline and Weeks 2, 4, 8, 12, and 16	ITT	None (summary)	NA	Observed Data	Supportive summary for missing data
Missing data pattern for Patient Global Assessment of Osteoarthritis for Baseline and Weeks 2, 4, 8, 12, and 16	ITT	None (summary)	NA	Observed Data	Supportive summary for missing data
Change from Baseline to Weeks 2, 4, 8, and 12 in WOMAC Pain subscale	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	Multiple Imputation	Secondary Endpoint Analysis
Change from Baseline to Weeks 2, 4, 8, and 12 in WOMAC Pain subscale	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	BOCF	Secondary Endpoint Analysis
Change from Baseline to Weeks 2, 4, 8, and 12 in WOMAC Pain subscale	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	LOCF	Secondary Endpoint Analysis
Change from Baseline to Weeks 2, 4, 8, and 12 in WOMAC Physical Function subscale	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	Multiple Imputation	Secondary Endpoint Analysis
Change from Baseline to Weeks 2, 4, 8, and 12 in WOMAC Physical Function subscale	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	BOCF	Secondary Endpoint Analysis
Change from Baseline to Weeks 2, 4, 8, and 12 in WOMAC Physical Function subscale	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	LOCF	Secondary Endpoint Analysis

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Change from Baseline to Weeks 2, 4, 8, and 12 in Patient Global Assessment of Osteoarthritis	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	Multiple Imputation	Secondary Endpoint Analysis
Change from Baseline to Weeks 2, 4, 8, and 12 in Patient Global Assessment of Osteoarthritis	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	BOCF	Secondary Endpoint Analysis
Change from Baseline to Weeks 2, 4, 8, and 12 in Patient Global Assessment of Osteoarthritis	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	LOCF	Secondary Endpoint Analysis
The OMERACT-OARSI response at Weeks 2, 4, 8, 12, and 16	ITT	Logistic Regression	BL Score (WOMAC Pain), BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group	Mixed BOCF/LOCF	Secondary Endpoint Analysis
The OMERACT-OARSI response at Weeks 2, 4, 8, 12, and 16	ITT	Logistic Regression	BL Score (WOMAC Pain), BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group	BOCF	Secondary Endpoint Analysis
The OMERACT-OARSI response at Weeks 2, 4, 8, 12, and 16	ITT	Logistic Regression	BL Score (WOMAC Pain), BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group	LOCF	Secondary Endpoint Analysis
Percentage of subjects with reduction of ≥30/50/70/90% from Baseline to Weeks 2, 4, 8, 12, and 16 in the WOMAC Pain subscale	ITT	Logistic Regression	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group	Mixed BOCF/LOCF	Secondary Endpoint Analysis
Percentage of subjects with reduction of ≥30/50/70/90% from Baseline to Weeks 2, 4, 8, 12, and 16 in the WOMAC Pain subscale	ITT	Logistic Regression	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group	BOCF	Secondary Endpoint Analysis
Percentage of subjects with reduction of ≥30/50/70/90% from Baseline to Weeks 2, 4, 8, 12, and 16 in the WOMAC Pain subscale	ITT	Logistic Regression	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group	LOCF	Secondary Endpoint Analysis

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Percentage of subjects with reduction of ≥30/50/70/90% from Baseline to Weeks 2, 4, 8, 12, and 16 in the WOMAC Physical Function subscale	ITT	Logistic Regression	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group	Mixed BOCF/LOCF	Secondary Endpoint Analysis
Percentage of subjects with reduction of ≥30/50/70/90% from Baseline to Weeks 2, 4, 8, 12, and 16 in the WOMAC Physical Function subscale	ITT	Logistic Regression	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group	BOCF	Secondary Endpoint Analysis
Percentage of subjects with reduction of ≥30/50/70/90% from Baseline to Weeks 2, 4, 8, 12, and 16 in the WOMAC Physical Function subscale	ITT	Logistic Regression	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group	LOCF	Secondary Endpoint Analysis
Percentage of subjects with an improvement of ≥2 points from Baseline to Weeks 2, 4, 8, 12, and 16 in the Patient Global Assessment of Osteoarthritis	ITT	Logistic Regression	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group	Mixed BOCF/LOCF	Secondary Endpoint Analysis
Percentage of subjects with an improvement of ≥2 points from Baseline to Weeks 2, 4, 8, 12, and 16 in the Patient Global Assessment of Osteoarthritis	ITT	Logistic Regression	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group	BOCF	Secondary Endpoint Analysis
Percentage of subjects with an improvement of ≥2 points from Baseline to Weeks 2, 4, 8, 12, and 16 in the Patient Global Assessment of Osteoarthritis	ITT	Logistic Regression	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group	LOCF	Secondary Endpoint Analysis
Percentage of subjects with reduction of ≥30/50% from Baseline to Week 16 in the WOMAC Pain subscale, separately for Week 8 Responders and Non-Responders (same definition as Week 16)	ITT	Logistic Regression	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group	Mixed BOCF/LOCF	Exploratory Analysis
Reduction of >0%, ≥10%, to ≥90% (in steps of 10%) and =100% from Baseline to Week 16 in the WOMAC Pain subscale	ITT	None (summary and plot)	NA	Mixed BOCF/LOCF	Summary of secondary endpoint
Reduction of >0%, ≥10%, to ≥90% (in steps of 10%) and =100% from Baseline to Week 16 in the WOMAC Pain subscale	ITT	None (summary)	NA	BOCF	Summary of secondary endpoint

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Reduction of >0%, ≥10%, to ≥90% (in steps of 10%) and =100% from Baseline to Week 16 in the WOMAC Pain subscale	ITT	None (summary)	NA	LOCF	Summary of secondary endpoint
Reduction of >0%, ≥10%, to ≥90% (in steps of 10%) and =100% from Baseline to Week 16 in the WOMAC Physical Function subscale	ITT	None (summary and plot)	NA	Mixed BOCF/LOCF	Summary of secondary endpoint
Reduction of >0%, ≥10%, to ≥90% (in steps of 10%) and =100% from Baseline to Week 16 in the WOMAC Physical Function subscale	ITT	None (summary)	NA	BOCF	Summary of secondary endpoint
Reduction of >0%, ≥10%, to ≥90% (in steps of 10%) and =100% from Baseline to Week 16 in the WOMAC Physical Function subscale	ITT	None (summary)	NA	LOCF	Summary of secondary endpoint
Change from Baseline to Weeks 2, 4, 8, 12, and 16 in the WOMAC Stiffness subscale	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	Multiple Imputation	Secondary Endpoint Analysis
Change from Baseline to Weeks 2, 4, 8, 12, and 16 in the WOMAC Average Score	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	Multiple Imputation	Secondary Endpoint Analysis
Change from Baseline to Weeks 2, 4, 8, 12, and 16 in the WOMAC Item: Pain When Walking on a Flat Surface	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	Multiple Imputation	Secondary Endpoint Analysis
Change from Baseline to Weeks 2, 4, 8, 12, and 16 in the WOMAC Item: Pain When Going Up or Down Stairs	ITT	ANCOVA	BL Score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	Multiple Imputation	Secondary Endpoint Analysis
Change from Baseline to Days 1, 2, 3, 4, 5, 6, 7, and to Weeks 1, 2, 3, 4, 6, 8, 10, 12, and 16 in the weekly average pain score in the index joint	ITT	ANCOVA	BL Score, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (Study site as a random effect)	Multiple Imputation	Secondary Endpoint Analysis

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Time to treatment discontinuation due to lack of efficacy (up to Week 16)	ITT	Log-Rank (with KM estimates)	Treatment Group	Observed	Secondary Endpoint Analysis
Incidence of treatment discontinuation due to lack of efficacy (up to Week 16)	ITT	Logistic Regression	BL WOMAC Pain, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group	Observed	Secondary Endpoint Analysis
Incidence of rescue medication use during Weeks 2, 4, 8, 12, and 16	ITT	Logistic Regression	BL WOMAC Pain, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group	LOCF	Secondary Endpoint Analysis
Number of days of rescue medication use during Weeks 2, 4, 8, 12, and 16	ITT	Negative Binomial Model	BL WOMAC Pain, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group	LOCF	Secondary Endpoint Analysis
Amount (mg) of rescue medication taken during Weeks 2, 4, 8, 12, and 16	ITT	Negative Binomial Model	BL WOMAC Pain, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group	LOCF	Secondary Endpoint Analysis
EQ-5D-5L dimensions (Mobility; Self-care; Usual activities; Pain/Discomfort; Anxiety/Depression), EQ-VAS and Overall Health Utility at Baseline and Weeks 8 and 16	ITT	Summary	NA	Observed	Secondary Endpoint Analysis
WPAI endpoints (% work time missed; % impairment while working; % overall work impairment; % activity impairment) at Week 16	ITT	ANCOVA	BL score, BL diary average pain, Index Joint (Knee/Hip), Highest KL grade (2, 3 or 4), Treatment Group (study site as a random effect)	Observed	Secondary Endpoint Analysis
Healthcare Resource Utilization at Baseline, Weeks 24 and 40	ITT	Descriptive Summary	Treatment group	Observed	Secondary Endpoint Analysis
Change from baseline to Weeks 2, 4, 8, 12, 16, 24, and 40 in the NIS score, and Change from Baseline to Worst post-Baseline NIS score	ITT	CMH test	Treatment Group [1]	LOCF (Weeks 2, 4, 8, 12, 16, 24, & 40), Worst post-baseline score	Safety Endpoint Analysis
Survey of Autonomic Symptoms (SAS), number of symptoms and total impact score (by visit and change from Baseline)	ITT	Summary	NA	Observed	Safety Endpoint Analysis

[1] CMH test will be stratified by the levels of the combined stratification parameters (6 levels). If there are <15 subjects in any combined stratification level then the CMH test will be unstratified

Appendix 2. DATA DERIVATION DETAILS

Appendix 2.1. Definition and Use of Visit Windows in Reporting

Study visits are planned at Screening, Baseline and then at post-baseline Weeks 2, 4, 8, 12, 16, 24, and 40. If a subject discontinues from the trial then there will be an Early Termination Follow-Up period and for those who refuse, ideally, an Early termination visit. To account for this visit and any early or late scheduled visits (compared to the target study days) we define 'windows' to be able to allocate each efficacy observation to a single specific study visit. For the assessments made at each planned study visit (e.g. WOMAC subscales, Patient Global Assessment of Osteoarthritis etc.) these visit windows are shown below. When multiple observations occur in a visit window, the observation closest to the protocol specified target day will be used, noting that the latter will be used in the case of a tie.

Visit	Target Study Day	Window
Screening [1]	Variable (up to 37 days prior to baseline visit)	[No lower limit, Day -8]
Baseline	1 (defined as initial day of study drug administration)	[-7,1]
Week 2	15	[2,22]
Week 4	29	[23,43]
Week 8	57	[44,71]
Week 12	85	[72,99]
Week 16	113	[100,141]
Week 24	169	[142,197]

[1] Only efficacy data collected at screening is WOMAC Pain subscale

One additional window will be created relative to the date of last SC dose for summaries of efficacy data collected beyond the planned treatment period. This window will include data from 16 +/- 4 weeks past the date of the last SC dose. The target day is 113 days after the last SC dose, with a window of [85, 141] days after the last SC dose. If multiple observations occur in this visit window, the observation closest to the specified target day will be used, noting that the latter will be used in the case of a tie.

For the assessments not made at each planned study visit, broader visit windows are shown below. When multiple observations occur in a visit window, the observation closest to the protocol specified target day will be used, noting that the latter will be used in the case of a tie.

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EQ-5D-5L

Visit	Target Study Day	Window
Baseline	1 (defined as initial day of study drug administration)	[-7, 1]
Week 8	57	[2, 85]
Week 16	113	[86, 141]

WPAI: OA

Visit	Target Study Day	Window
Baseline	1 (defined as initial day of study drug administration)	[-7,1]
Week 16	113	[2, 141]

HCRU

Visit	Target Study Day	Window
Baseline	1 (defined as initial day of study drug administration)	[no lower limit, 1]
Week 24	169	[2, 197]
Week 40	281	[198, no upper limit]

For the average pain in the index joint, the data are collected daily via electronic diary up to the end of Week 16, and thereafter Weekly up to Week 40. Data up to Week 16 will be reported as part of the efficacy assessment (summary up to Week 24; analysis up to Week 16).

The Baseline score is the mean of the last 3 non-missing pain scores over study days -7 to -1. If fewer than 3 are available between study days -7 and -1, the baseline will be the mean of the available scores.

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The table below describes the visit days for each week (Weeks 1-16). All available on-treatment diary data in each of the weekly intervals will be used to calculate the mean daily pain score for that study week.

Study Week	Days	Study Week	Days
1	1-7	9	57-63
2	8-14	10	64-70
3	15-21	11	71-77
4	22-28	12	78-84
5	29-35	13	85-91
6	36-42	14	92-98
7	43-49	15	99-105
8	50-56	16	106-112

After the Week 16 visit, pain scores are captured only once a week in the diary. These are grouped in 4-week intervals using visit windows as shown below. If a subject comes in late for a Week 16 visit (or weekly diary is not activated at the visit), and so has daily diary data collected past Day 112, these data will be averaged with any data obtained weekly for any given interval. All available on- or off-treatment data will be used for these windows after the planned treatment period.

Summary Week	Includes Weeks	Days
20	17 - 20	113-140
24	21 - 24	141-168
28	25 - 28	169-196
32	29 - 32	197-224
36	33 - 36	225-252
40	37 – 40	253-280

One additional window will be created relative to the date of last SC dose for summaries of diary pain scores collected beyond the planned treatment period. This window will be identified as 16 Weeks Post Last Dose, and will include the average of all data from 13 to 16 weeks (85 to 112 days)_past the date of the last SC dose. All available on- or off-treatment data will be used for this window after the planned treatment period.

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Appendix 3. STATISTICAL METHODOLOGY DETAILS

Health State Utility of the EQ-5D-5L

The EQ-5D-5L contains five questions that measure the following dimensions: mobility, self-care, usual activities, pain/discomfort, and anxiety/depression. Each of the five dimensions has five levels: (1) no problems; (2), slight problems; (3) moderate problems; (4) severe problems; and (5) extreme problems.

The health utility scores are defined for every possible set of outcome combinations of the five dimensions for the following countries:

- Denmark, France, Germany, Japan, the Netherlands, Spain, Thailand, UK, US and Zimbabwe

It is intended that this study will recruit subjects from the following countries:

- US (including Puerto Rico), Canada.

Some of these may not actually recruit or treat subjects, and other countries may be added. As there is a mismatch between countries where subjects are being recruited and the currently available EQ-5D-5L health utility scoring, we will assign subjects to the following scoring countries based on the following assignments.

EQ-5D Scoring Country	Study Recruitment Country
Denmark	-
France	-
Germany	-
Japan	-
The Netherlands	-
Spain	-
Thailand	-
UK	-
US	US (including Puerto Rico), Canada
Zimbabwe	-

If more EQ-5D-5L utility scores become available or other countries are added, then this assignment may be modified.

The health utility for a patient with no problems in all 5 items is 1 for all countries (except for Zimbabwe where it is 0.9), and is reduced where a patient reports greater levels of problems across the five dimensions. The minimum score in US scoring is -0.109.

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WPAI:OA Endpoints

The tables below summarize the 6 questions of the WPAI:OA questionnaire, and the four endpoints of the effect of impairment on activity and impairment.

Question	Question Wording	Scoring
1	Are you currently employed? [if No skip to question 6]	Yes, No
2	During the past seven days, how many hours did you miss from work due to problems associated with your OA of the knee or hip	number of hours (free text)
3	During the past seven days, how many hours did you miss from work because of any other reason, such as vacation, holidays, time off to participate in this study?	number of hours (free text)
4	During the past seven days, how many hours did you actually work (if '0' skip to Question 6)	number of hours (free text)
5	During the past seven days, how much did your OA of the knee or hip affect productivity while you were working?	0 to 10 scale with 0 being 'No effect on my work' and 10 being 'Completely prevented me from working'
6	During the past seven days, how much did your OA of the knee or hip affect your ability to do your regular daily activities, other than work at a job?	0 to 10 scale with 0 being 'No effect on my daily activities' and 10 being 'Completely prevented me from doing my daily activities'

WPAI endpoint	Calculation
Percent activity impairment due to Osteoarthritis	Q6*10
Percent impairment while working due to osteoarthritis	Q5*10
Percent overall work impairment due to osteoarthritis	$\left\{\frac{Q2}{Q2+Q4}+\left[1-\left(\frac{Q2}{Q2+Q4}\right)\right]\left(\frac{Q5}{10}\right)\right\}\times 100$
Percent work time missed due to Osteoarthritis	$\frac{Q2}{Q2+Q4}*100$

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Healthcare Resource Utilization (example using 3 month recall—8 week recall is also used in study)

Question	Response	Scoring
During the last 3 months, what services did you receive directly related to your osteoarthritis? • Primary Care Physician • Neurologist • Rhematologist • Physician Assistant or Nurse Practioner • Pain Specialist • Orthopedist • Physical Therapist • Chiropractor • Alternative Medicine or Therapy • Podiatrist • Nutritionist/Dietician • Radiologist • Home healthcare services • Other	Number of Visits	Response not selected = 0 Number of visits = 1-999
During the past 3 months, have you visited the emergency room due to your osteoarthritis?	Yes, No	No = 0 Yes = 1
How many times?	Number of visits	0-999
During the past 3 months, have you been hospitalized due to your osteoarthritis?	Yes, No	No = 0 Yes = 1
How many nights in total did you stay in hospital due to your osteoarthritis in the last 3 months?	Number of Nights	0-999 (max should be 92)
Did you use these aids or devices to help you in doing things because of your osteoarthritis in the last 3 months? • Walking Aid • Wheelchair • Devices or utensils to help you dress, eat or bathe • Other	Did not use any aids or devices Never, rarely, sometimes, often, always	Did not use any aids or devices = 0 Device not selected = 0 Never = 1 Rarely = 2 Sometimes = 3 Often = 4 Always = 5

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Did you quit your job because of your osteoarthritis?	Yes, No	No = 0 Yes = 1 Not applicable = 2
How long ago did you quit your job because of your osteoarthritis?	Years and Months	0-99 Years and 0-99 Months (should be max of 11 months)

Rescue Medication Endpoints

Rescue medication data is collected daily using an electronic system up to Week 16, and weekly after Week 16 and up to Week 40. Daily and weekly collected data will be assigned to a specific study week for summary and reporting. The assignment of daily and weekly data to weeks will use the same principle as described above in Appendix 2.1 for the daily and weekly index joint pain data.

The incidence of rescue medication use will look for any incidence in the week of interest (collected through daily or weekly diary data). The number of days of RM use (using daily and weekly data) and the total amount taken (using daily data up to Week 16 only) over the week will be calculated for the assigned week algorithm described above.

Imputation is described in Section 5.3 above. Imputation occurs for daily data up to Week 16 where the subject is in the trial and up to the end of that particular week.

An example of imputation and calculating the three endpoints using the daily diary data is shown below.

Example of calculating rescue medication data from Daily Diary Data (Subject does not discontinue)

In this example, a subject has a Week 2 visit on study day 14 (slightly earlier than the nominal day 15). Study days 8-14 would represent Week 2 data.

Using the Week 2 interval described above for a subject, i.e. study days [8-14], we have the following rescue medication example data.

The amount taken and number of days of rescue medication use is adjusted for the duration of the Weekly interval.

Study Day (Week)	Number of Doses of RM taken [1]	Number of Doses of RM taken [1] with LOCF imputation
8 (Week 2)	2	2
9 (Week 2)	Missing	2 [2]
10 (Week 2)	0	0
11 (Week 2)	1	1

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12 (Week 2)	Missing	1 [2]
13 (Week 2)	2	2
14 (Week 2)	0	0

[1] 500mg tablets of acetaminophen; [2] Using LOCF imputation for missing data

For this subject the following data will be calculated for Week 2:

- Incidence of rescue medication taken in Week 2: Yes. Rescue medication taken on days 8, 9 (imputed), 11, 12 (imputed), 13.

- Number of days of rescue medication use in Week 2: 5. For days 8-14 we have rescue medication taken on days 8, 9 (imputed), 11, 12 (imputed), and 13. The number of days taken for the 7 day period is 5/7*7=5.

- Amount (mg) of rescue medication use in Week 2: For days 8-14 we have the number of doses taken of 2, 2 (imputed), 0, 1, 1 (imputed), 2, and 0. The number of doses taken for the 7 day period is 8/7*7=8, making the amount of acetaminophen dosage of 4000mg.

Example of calculating rescue medication data from Daily Diary Data (Subject discontinues)

In this example, a subject discontinues on study day 62, a few days after a Week 8 visit (which was on study day 60). The Week 5-8 data is calculated as described above (e.g. Week 8 using days [50, 56]). The subject has rescue medication data as shown below.

Study Day (Week)	Number of Doses of RM taken [1]	Number of Doses of RM taken [1] with LOCF imputation
57 (Week 9)	1	1
58 (Week 9)	1	1
59 (Week 9)	Missing	1 [2]
60 (Week 9)	Missing	1 [2]
61 (Week 9)	Missing	1 [2]
62 (Week 9)	Missing	1 [2]
63 (Week 9)	Missing	1 [2]

[1] 500mg tablets of acetaminophen; [2] Using LOCF imputation for missing data

Week 9 is calculated as days 57 to 63. The data up to the end of the last week the subject was in the trial is imputed using LOCF as shown above. Therefore the Week 9 scores are then used to impute the Weekly data for summary and analysis for Weeks 10 to 16.

As above the incidence of rescue medication for Week 9 would be 'Yes'. The number of days of rescue medication use would be 7, and the average dose would be 7/7*7*500=3500mg for this week.

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Appendix 3.1. Further Details of Interim Analyses

Details of the ongoing review of safety data (including joint safety events) are given in a separate statistical analysis plan for the Data Monitoring Committee.

Appendix 3.2. Further Details of the Statistical Methods

A description of the combination of the ANCOVA results from each of the multiple imputed datasets is given below, and taken from Little & Rubin (2002), page 86-7.

In this analysis we have defined the number of imputations (D) to be 100.

The treatment estimates for individual treatment groups and treatment contrasts are defined as θ_i for $i = 1 \ldots D$. The combined estimate is $\bar{\theta}_D = \frac{1}{D}\sum_{i=1}^{D}\theta_i$. The variability of the combined estimate contains components of both Within- (W) and Between- (B) imputation dataset variability. These are shown below:

$$\overline{W}_D = \frac{1}{D}\sum_{i=1}^{D} W_i \text{ and } B_D = \frac{1}{D-1}\sum_{i=1}^{D}\left(\hat{\theta}_i - \bar{\theta}_D\right)^2$$

where W_i is the variance for the parameter θ_i.

The total variance for $\bar{\theta}_D$ is shown below:

$$T_D = \overline{W}_D + \frac{D+1}{D} B_D .$$

The test statistic $\frac{\left(\theta - \bar{\theta}_D\right)}{\sqrt{T_D}}$ has a t-distribution with v^* degrees of freedom, which is defined below:

$$v^* = \left(\frac{1}{v} + \frac{1}{\hat{v}_{obs}}\right),$$

using

$$v = (D-1)\left(1 + \frac{1}{D+1}\frac{\overline{W}_D}{B_D}\right)^2$$

$$\hat{v}_{obs} = \left(1 - \hat{\gamma}_D\right)\left(\frac{v_{com}+1}{v_{com+3}}\right)v_{com}$$

$$\hat{\gamma}_D = \left(1 + \frac{1}{D}\right)\frac{B_D}{T_D}.$$

This distribution can be used to construct the test statistics and 95% confidence intervals for θ.

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Supplementary Online Content

Schnitzer TJ, Easton R, Pang S, et al. Effect of tanezumab on joint pain, physical function, and patient global assessment of osteoarthritis among patients with osteoarthritis of the hip and knee: a randomized clinical trial. *JAMA*. doi:10.1001/jama.2019.8044

eFigure. Study Design and Dosing Schedule
eTable 1. Complete Listing of Prespecified Secondary Efficacy End Points
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eTable 3. Summary of Rescue Medication Use
eTable 4. Summary of Change in Neuropathy Impairment Score

This supplementary material has been provided by the authors to give readers additional information about their work.

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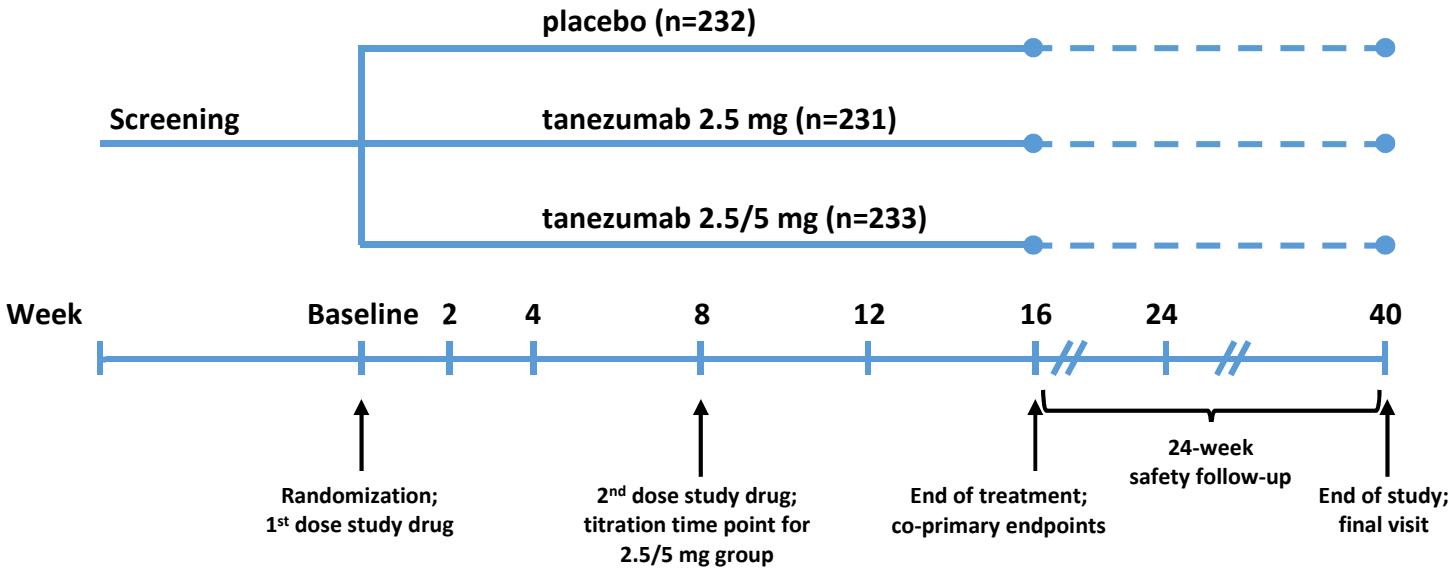
eFigure. Study Design and Dosing Schedule



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eTable 1. Complete Listing of Prespecified Secondary Efficacy End Points

Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) Pain subscale change from baseline to weeks 2, 4, 8, 12, and 24
WOMAC Physical Function subscale change from baseline to weeks 2, 4, 8, 12, and 24
Patient's Global Assessment (PGA) of OA change from baseline to weeks 2, 4, 8, 12, and 24
OMERACT-OARSI responder index at weeks 2, 4, 8, 12, 16, and 24
Treatment Response: reduction in WOMAC Pain score of ≥30%, ≥50%, ≥70%, and ≥90% at weeks 2, 4, 8, 12, 16, and 24
Cumulative distribution of percent change from baseline in WOMAC Pain score to week 16
Treatment Response: reduction in WOMAC Physical Function score of ≥30%, ≥50%, ≥70%, and ≥90% at weeks 2, 4, 8, 12, 16, and 24
Cumulative distribution of percent change from baseline in WOMAC Physical Function score to Week 16
Treatment Response: Improvement of ≥2 in PGA-OA at weeks 2, 4, 8, 12, 16, and 24
Average pain score in the index joint change from baseline to weeks 1, 2, 3, 4, 6, 8, 10, 12, 16, 20, and 24
WOMAC Stiffness score change from baseline to weeks 2, 4, 8, 12, 16, and 24
WOMAC average change from baseline to weeks 2, 4, 8, 12, 16, and 24
WOMAC Pain Subscale item: pain when walking on a flat surface, change from baseline to weeks 2, 4, 8, 12, 16, and 24
WOMAC Pain Subscale item: pain when going up or down stairs, change from baseline to weeks 2, 4, 8, 12, 16, and 24
Work Productivity and Activity Impairment Questionnaire for OA (WPAI:OA) impairment scores change from baseline to week 16
EuroQol 5 Dimension (EQ-5D-5L™) dimensions and overall health utility score at baseline, week 8, and week 16
Health care resource utilization (HCRU) at baseline, week 24, and week 40
Incidence and time to discontinuation due to lack of efficacy
Usage of rescue medication (incidence, number of days) during weeks 2, 4, 8, 12, 16, and 24
Usage of rescue medication (amount taken) during weeks 2, 4, 8, 12, and 16

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eTable 2. Summary of WOMAC Pain Response: Percentage of Patients With a Reduction From Baseline of ≥15%, ≥30%, and ≥50% at Week 16 According to Response at Week 8 (Planned Exploratory Analysis)

	Tanezumab 2.5/5 mg	**Tanezumab 2.5 mg**	**Placebo**
	(n=233)	(n=231)	(n=232)
N (%) of patients			
≥15% Improvement at Week 8			
≥15% Improvement at Week 16	161 (94.2)	160 (94.7)	140 (87.5)
<15% Improvement at Week 16	10 (5.8)	9 (5.3)	20 (12.5)
Odds Ratio (95% CI) versus placebo	2.35 (1.05, 5.23)	2.44 (1.06, 5.59)	
P value	0.037	0.035	
<15% Improvement at Week 8			
≥15% Improvement at Week 16	28 (45.2)	26 (41.9)	24 (33.3)
<15% Improvement at Week 16	34 (54.8)	36 (58.1)	48 (66.7)
Odds Ratio (95% CI) versus placebo	1.71 (0.83, 3.51)	1.40 (0.68, 2.88)	
P value	0.144	0.358	
≥30% Improvement at Week 8			
≥30% Improvement at Week 16	131 (91.6)	132 (92.3)	94 (82.5)
<30% Improvement at Week 16	12 (8.4)	11 (7.7)	20 (17.5)
Odds Ratio (95% CI) versus placebo	2.29 (1.05, 4.99)	2.56 (1.15, 5.73)	
P value	0.037	0.022	
<30% Improvement at Week 8			
≥30% Improvement at Week 16	33 (36.7)	25 (28.4)	33 (28.0)
<30% Improvement at Week 16	57 (63.3)	63 (71.6)	85 (72.0)
Odds Ratio (95% CI) versus placebo	1.54 (0.84, 2.83)	1.13 (0.60, 2.13)	
P value	0.160	0.703	
≥50% Improvement at Week 8			
≥50% Improvement at Week 16	89 (89.9)	99 (90.8)	57 (81.4)
<50% Improvement at Week 16	10 (10.1)	10 (9.2)	13 (18.6)
Odds Ratio (95% CI) versus placebo	1.87 (0.76, 4.65)	2.13 (0.85, 5.32)	
P value	0.175	0.106	
<50% Improvement at Week 8			
≥50% Improvement at Week 16	44 (32.8)	27 (22.1)	31 (19.1)
<50% Improvement at Week 16	90 (67.2)	95 (77.9)	131 (80.9)
Odds Ratio (95% CI) versus placebo	2.16 (1.25, 3.72)	1.22 (0.68, 2.20)	
P value	0.006	0.513	

CI, confidence interval; WOMAC, Western Ontario and McMaster Universities Osteoarthritis Index.
Odds ratio and p values versus placebo using a logistic regression model including baseline WOMAC pain subscale, baseline diary average pain, and classification variables index joint, highest Kellgren-Lawrence grade, and treatment.

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eTable 3. Summary of Rescue Medication Use

	Tanezumab 2.5/5 mg	**Tanezumab 2.5 mg**	**Placebo**
	(n=233)	(n=231)	(n=232)
N (%) patients who took rescue medication			
Week 2 Odds Ratio (95% CI) versus placebo P value	138 (59.2) 0.65 (0.44, 0.96) 0.031	137 (59.6) 0.70 (0.48, 1.04) 0.076	160 (69.0)
Week 4 Odds Ratio (95% CI) versus placebo P value	110 (47.2) 0.54 (0.37, 0.79) 0.001	119 (51.5) 0.65 (0.45, 0.95) 0.026	143 (61.6)
Week 8 Odds Ratio (95% CI) versus placebo P value	117 (50.2) 0.99 (0.69, 1.43) 0.972	110 (47.6) 0.92 (0.64, 1.33) 0.671	117 (50.4)
Week 12 Odds Ratio (95% CI) versus placebo P value	88 (37.8) 0.77 (0.53, 1.12) 0.171	93 (40.3) 0.89 (0.61, 1.29) 0.531	103 (44.4)
Week 16 Odds Ratio (95% CI) versus placebo P value	96 (41.2) 0.92 (0.63, 1.33) 0.656	101 (43.7) 1.06 (0.73, 1.54) 0.748	100 (43.1)
Number of days of rescue medication use per week			
Week 2 LS Mean (95% CI) P value versus placebo	2.02 (1.68, 2.44) 0.094	2.27 (1.88, 2.73) 0.483	2.46 (2.04, 2.97)
Week 4 LS Mean (95% CI) P value versus placebo	1.68 (1.35, 2.09) 0.051	1.93 (1.55, 2.40) 0.337	2.20 (1.76, 2.75)
Week 8 LS Mean (95% CI) P value versus placebo	1.53 (1.21, 1.94) 0.324	1.68 (1.33, 2.13) 0.728	1.77 (1.40, 2.24)
Week 12 LS Mean (95% CI) P value versus placebo	1.12 (0.85, 1.48) 0.139	1.31 (1.00, 1.72) 0.568	1.45 (1.09, 1.93)
Week 16 LS Mean (95% CI) P value versus placebo	1.39 (1.06, 1.82) 0.709	1.57 (1.21, 2.05) 0.728	1.48 (1.12, 1.97)
Amount of rescue medication use per week			
Week 2 LS Mean (95% CI) P value versus placebo	2537.6 (1684.6, 3822.6) 0.354	3139.1 (2085.0, 4726.3) 0.916	3226.0 (2110.9, 4930.2)

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Week 4			
LS Mean (95% CI)	2005.3 (1248.8, 3220.0)	2815.6 (1724.9, 4595.9)	3033.2 (1839.1, 5002.7)
P value versus placebo	0.175	0.807	
Week 8			
LS Mean (95% CI)	2104.2 (1253.0, 3533.7)	2385.9 (1418.0, 4014.5)	2608.6 (1534.1, 4435.8)
P value versus placebo	0.506	0.784	
Week 12			
LS Mean (95% CI)	1633.4 (889.9, 2998.1)	1810.1 (979.5, 3345.1)	2121.2 (1112.0, 4046.2)
P value versus placebo	0.503	0.682	
Week 16			
LS Mean (95% CI)	2166.0 (1180.7, 3973.5)	2846.9 (1525.6, 5312.6)	2303.7 (1219.4, 4352.3)
P value versus placebo	0.873	0.580	

CI, confidence interval LS, least squares.

Incidence of rescue medication use was analyzed using a logistic regression model including baseline WOMAC pain subscale, baseline diary average pain, and classification variables index joint, highest Kellgren-Lawrence grade, and treatment. Number of days and amount of rescue medication use was analyzed using a negative binomial model with model terms of baseline WOMAC Pain subscale score, baseline diary average pain score, index joint, Kellgren Lawrence grade, and treatment group.

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eTable 4. Summary of Change in Neuropathy Impairment Score[a]

	Tanezumab 2.5/5 mg	**Tanezumab 2.5 mg**	**Placebo**
	(n=233)	(n=231)	(n=232)
N (%) of patients[b]			
Week 2			
Worsened[c]	10 (4.5)	16 (7.1)	8 (3.6)
No Change	188 (83.9)	177 (79.0)	189 (85.5)
Improved[d]	26 (11.6)	31 (13.8)	24 (10.9)
P value versus placebo[e]	0.998	0.859	
Week 4			
Worsened	14 (6.1)	17 (7.4)	16 (7.0)
No Change	183 (80.3)	180 (78.6)	185 (81.5)
Improved	31 (13.6)	32 (14.0)	26 (11.5)
P value versus placebo	0.460	0.599	
Week 8			
Worsened	19 (8.3)	24 (10.4)	19 (8.3)
No Change	178 (77.7)	173 (75.2)	180 (78.6)
Improved	32 (14.0)	33 (14.3)	30 (13.1)
P value versus placebo	0.836	0.862	
Week 12			
Worsened	13 (5.7)	17 (7.4)	16 (7.0)
No Change	179 (78.2)	175 (76.1)	186 (81.2)
Improved	37 (16.2)	38 (16.5)	27 (11.8)
P value versus placebo	0.172	0.304	
Week 16			
Worsened	17 (7.4)	21 (9.1)	19 (8.3)
No Change	176 (76.9)	171 (74.3)	176 (76.9)
Improved	36 (15.7)	38 (16.5)	34 (14.8)
P value versus placebo	0.717	0.849	
Week 24			
Worsened	15 (6.6)	25 (10.9)	17 (7.4)
No Change	179 (78.2)	169 (73.5)	174 (76.0)
Improved	35 (15.3)	36 (15.7)	38 (16.6)
P value versus placebo	0.896	0.342	
Week 40			
Worsened	10 (4.4)	20 (8.7)	14 (6.1)
No Change	179 (78.2)	172 (74.8)	182 (79.5)
Improved	40 (17.5)	38 (16.5)	33 (14.4)
P value versus placebo	0.260	0.931	

[a]Mean (SD) scores at baseline were 2.0 (4.8) for tanezumab 2.5mg, 1.8 (4.5) for tanezumab 2.5/5mg, and 1.1 (2.6) for placebo on a scale from 0 to 244, with greater score indicating greater impairment.
[b]The denominator to calculate each percentage is the number of subjects with available data.

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[c]Indicates a change in score greater than 0.
[d]Indicates a change in score less than 0.
[e]Pairwise P value for comparison versus placebo. P-value is based on Cochran-Mantel-Haenszel test for row mean scores differ, using Change from Baseline Category as the score in the analysis, and stratified by the combinations of the two stratification factors (index joint and highest Kellgren-Lawrence grade). A last-observation-carried forward approach to missing data was used.

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Data Sharing Statement

Data
Data available: Yes
Data types: Deidentified participant data
How to access data: CTD@Pfizer.com
When available: beginning date: 05-14-2020

Supporting Documents
Document types: None

Additional Information
Who can access the data: Upon request, and subject to certain criteria, conditions and exceptions (see https://www.pfizer.com/science/clinical-trials/trial-data-and-results for more information), Pfizer will provide access to individual de-identified participant data from Pfizer-sponsored global interventional clinical studies conducted for medicines, vaccines and medical devices (1) for indications that have been approved in the US and/or EU or (2) in programs that have been terminated (ie, development for all indications has been discontinued).
Types of analyses: The de-identified participant data will be made available to researchers whose proposals meet the research criteria and other conditions, and for which an exception does not apply, via a secure portal.
Mechanisms of data availability: The de-identified participant data will be made available to researchers whose proposals meet the research criteria and other conditions, and for which an exception does not apply, via a secure portal. To gain access, data requestors must enter into a data access agreement with Pfizer.
Any additional restrictions: Pfizer will also consider requests for the protocol, data dictionary, and statistical analysis plan. Data may be requested from Pfizer trials 24 months after study completion.